Encyclopaedia

of Derivations

A field-by-field tour from Leibniz to the present

Abderrahim Adrabi

First edition • v0.18

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Origins and Development, with Applications and Worked Examples

Abderrahim Adrabi

Faculty of Sciences and Faculty of Educational Sciences,
Mohammed V University in Rabat, Morocco

Living edition, updated monthly
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The titles of the works in the atlas name **173** kinds of derivation-like map, out of the 200 rows the kind census holds. The 27 rows that are not kinds are fragments of the mining method, and they are listed with their reason in the book's kind ledger. Each number after a name is the number of works whose title names that kind. A title that names three kinds counts once under each of the three, so the numbers do not add up to a number of works. The list was cut at two works, because a name on a single title in the whole atlas is not yet a kind the literature uses. A name appears twice when the literature writes it two ways, joined by a dash and written as one word: *skew derivation* (179) with *skew-derivation* (10), and *lie derivation* (107) with *lie-derivation* (3). Each work counts once, under the spelling its title uses. The order is the order of the census table: by count, highest first. Kinds with the same count are in the order the mining tool met them in the atlas, so the order within a group of equal counts carries no meaning.

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21. superderivation (22).
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23. (θ, ϕ)-derivation (21).
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50. positive derivation (9).
51. twisted derivation (9).
52. (m, n)-jordan derivation (9).
53. hom-derivation (9).
54. point derivation (8).

55. finite derivation (8).
56. vanishing derivation (8).
57. (α, α)-derivation (8).
58. permuting tri-derivation (8).
59. local superderivation (8).
60. (m, n)-derivation (8).
61. nonsingular derivation (7).
62. q-skew derivation (7).
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93. (ϕ, ψ)-derivation (4).
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155. generalized coderivation (2).

156. ternary bi-derivation (2).

157. (α, β)-generalized derivation (2).

158. leibniz-derivation (2).

159. generalized superderivation (2).

160. graded derivation (2).

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163. modular derivation (2).

164. e-derivation (2).

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172. bracket derivation (2).

173. *-reverse derivation (2).

Preface

A derivation is the algebraic form of a derivative. Leibniz wrote the product rule in 1684, and three centuries later the same rule defines a map on any ring: an additive map d with $d(xy) = d(x)y + xd(y)$. Every chapter of this book studies some version of that equation, either the equation itself or a weaker or stronger form of it.

Why a encyclopaedia and not a survey

A survey selects and argues. A encyclopaedia tries to be complete in scope, so that a reader can find where a given question is treated and what is known about it.

This book is exhaustive in scope, not in text. Every flavour of derivation, every class of ring and algebra where derivations act, and every line of development from Leibniz to this year has a place in the plan. The plan is fixed in advance and the text grows into it.

The source rule

No fact, definition, example or application is stated before its source has been read in full from the page. A recalled theorem appears as a theorem-like environment with its located citation, keeping the kind and the wording of the source. Proofs and remarks of the book itself are marked as its own.

This rule has a cost, and the cost is visible. A section whose sources are not yet obtained stays a marked planned stub, and the stub says which source it waits for. A reader who finds a stub knows that the coverage is

planned there. The text is not yet written, which is a different thing from a section that was written and then cut.

This edition has no such stub. Every section is written. Two sources could not be obtained at all. Their sections name the work and say so rather than quoting a theorem from it, and both are listed in the download record of this book. The policy stays in place for later editions, which is why it is stated here.

Where the applications are

A derivation is a rule that obeys the Leibniz product rule. The rule itself goes back to Leibniz [1684-1]. The name was already familiar to Ritt when he built differential algebra on it [1950-1, p. 1]. The same rule appears again in nine fields beyond the differential calculus. The nine names are classical and quantum mechanics, differential geometry, differential algebra, coding, control, group analysis, model theory, algebraic topology and automatic differentiation. Each field is reported in the chapter that owns its notions. Each report names one field, cites the source that names the derivation it rests on, and reads the definition or theorem of that source. The reports do not narrate: they record.

Physics was the first field that asked why a derivation arises. Dirac [1925-2] translated noncommutative kinematics into the language of a Poisson bracket (Section 2.2), and Kadison [1966-1] later asked whether every derivation of an operator algebra comes from inside (Section 19.2). Geometry came next: a tangent vector at a point of a smooth manifold is a derivation of the smooth functions there (Section 17.6). Differential algebra began when Ritt and Kolchin axiomatized the symbol d/dx as a derivation on a field, and the field-of-definition question for the Painlevé equations sits inside that axiomatization (Section 18.2). In positive characteristic the ordinary product rule survives in the divided-power form of Hasse and Schmidt; the higher derivations are the algebraic backbone of Reed–Solomon and Goppa codes (Section 7.2). Control theory discovered that an input–output system is a finitely generated differential field extension, with the input a differential transcendence basis (Section 18.3). On a group the same rule is a derivation of the group ring, and Fox showed the free group carries one derivation per generator (Section 13.2). Ax translated the conjecture of Schanuel about the complex exponential into a statement about a derivation on a differential field. His 1971 theorem is the seed of the Ax–Schanuel inequalities (Section 18.7). Hochschild and cyclic homology

compute derivations as the homology of a chain complex, and on a smooth commutative algebra the answer is the exterior algebra of Kähler differentials (Section 20.6). A computer program that differentiates a function runs the chain rule and the Leibniz rule step by step under automatic differentiation (Section 18.4).

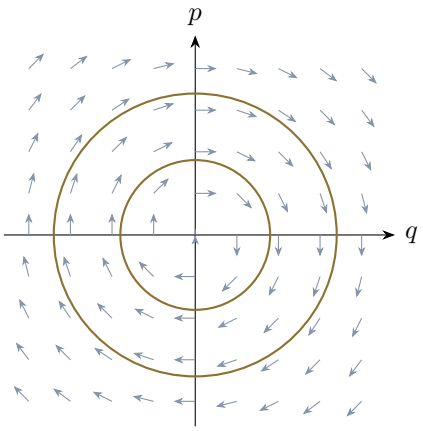


Figure 1: The harmonic oscillator. The derivation $f \mapsto \{f, H\}$ with $H = (p^2+q^2)/2$ is the vector field $(p, -q)$ on the phase plane; its flow is a rotation.

The living edition

The subject grows. Papers on derivations appear every month, and a printed encyclopaedia would be out of date before it reached a reader. This edition is therefore updated monthly: new work is added where it changes a statement, a table entry or an open problem. Appendix A reports what the edition has read, and the version number on the title page names the edition.

How the book was made

The work of reading, checking and drafting was carried out with artificial intelligence under human direction. A person set the scope, fixed the questions and decided what the book would contain. Machine tools assisted the reading of each source, the extraction of its results and the drafting of each section. Every claim was then checked against the page it cites. The rules that govern this are stated in the repository of the book. The records

of what was read are kept there too, so a reader can trace any statement to its source.

The name on the title page is the person who directed the work and answers for it.

How to use this encyclopaedia

The book is organised around eight questions, not around its chapters. A reader who wants one of them answered can go straight to the appendix that indexes it. A reader who wants to follow the subject from its beginnings can read Part I onward.

The eight question families

Section 2.9 fixes the eight questions the book asks of every class of algebra. They are:

1. Which kind of derivation implies which other?
2. When are two definitions equivalent?
3. Is every derivation inner?
4. Is every derivation continuous, and does continuity come for free?
5. Does the algebra have to be commutative?
6. Can all derivations of the class be described?
7. Does a derivation extend or lift to a larger algebra?
8. Where does an expected implication fail?

The chapters answer them in this order wherever the sources allow. Appendix B collects the answers in tables, family by family, so that a reader comparing two classes of algebra can see both at once.

How a chapter is laid out

Each chapter follows the same plan.

The introduction gives the history: why the topic arose, who first asked the question, and what the chapter answers. The sections that follow begin with the definitions as their sources state them, then the theorems as their sources state them, each with the page or numbered item.

Counterexamples come where a family of the last kind fails; they are what answers family 8. An application comes where the notion it uses is defined, so the reader meets the use at the same point as the idea. An open problem comes where the chapter states the question it belongs to, and Appendix B gathers all of them by family and by class.

A recalled theorem keeps the kind of its source: a proposition stays a proposition, and a theorem stays a theorem. The book never states a recalled result again in its own setting. The specialisation is written after the recall, and it refers to the recall by number.

The two appendix chapters

Appendix A is the atlas of the literature. It gives the shape of the bibliography: the works by decade, by language, and by the kind of derivation-like map they treat. It is counted again with each edition, so it records what this edition has read rather than what the field contains.

Appendix B is the collection of question tables. For each of the eight families it gives, class by class, whether the answer is known, false with a counterexample, or open, with the source that settles or asks it.

Reading paths

A reader new to the subject can read Part I in order; it carries the history and the basic theory, and it fixes the question families. A reader who works on rings can take Part II and Part III, which treat the associative classes, and consult Part VI for the analytic ones. A reader who comes from analysis or operator theory can start with Part VI and use Part II for the algebraic background. A reader who wants the current frontier can read Appendix B, which names every open problem with the chapter that states it, then follow those references back.

The living edition

The edition is updated each month. New work is added where it changes a statement, a table entry or an open problem. A work that only confirms what the book already says is not cited for that alone. The version number appears on the title page, so that a citation to this book can name the edition it used.

Notation and conventions

The book keeps one notation throughout. A symbol is defined where it first appears, and the chapters reuse it without giving it a second meaning. This page lists the symbols a reader meets most often, so that a chapter can be opened at any point.

Rings and algebras

R and A are a ring and an algebra; the book says which when the distinction matters. $\mathbb{F}$ is a field, and k is a field when the work is geometric. $\operatorname{char} R$ is the characteristic of R, and $Z(R)$ is its centre. An algebra is associative and unital unless a chapter says otherwise; Chapter 15 drops associativity and says so at each step.

$M_n(R)$ is the ring of $n \times n$ matrices over R. A triangular algebra is written $A = \operatorname{Tri}(A_1, A_2, M)$ with M the bimodule between the two parts. $R[x]$ and $R[x_1, \dots, x_n]$ are polynomial rings, and $k(x)$ is the field of fractions when k is a domain.

Derivations and their relatives

$\operatorname{Der}(A)$ is the set of derivations of A, an additive group and a Lie algebra under the commutator. $\operatorname{Inn}(A)$ is the set of inner derivations, those of the form $x \mapsto [\, a, x \,]$, so that $\operatorname{HH}^1(A, A) \cong \operatorname{Der}(A)/\operatorname{Inn}(A)$ for the classes where that identification holds.

$\operatorname{JDer}(A)$ and $\operatorname{LDer}(A)$ are the Jordan and the Lie derivations. ad_a is the inner derivation $x \mapsto [a, x]$. A derivation is *locally nilpotent* when some

power of it kills each element, and the set of these is written $\mathrm{LND}(A)$.

$\mathrm{HH}^\bullet(A, M)$ is the Hochschild cohomology of A with coefficients in the bimodule M, and $\mathrm{HH}^\bullet(A)$ is the case $M = A$. Low degrees are written out where they are used: the first cohomology group is the derivations modulo the inner ones.

Maps and products

$[x, y] = xy - yx$ is the commutator and $x \circ y = xy + yx$ is the anti-commutator. A map is *additive* when it respects addition, with no linearity assumed over a ring; the book states linearity separately wherever a theorem needs it.

For maps with a parameter, σ and τ are endomorphisms of the algebra, and α, β are scalars or endomorphisms as the chapter says. A σ-derivation satisfies $d(xy) = d(x)y + \sigma(x)d(y)$, and the book gives the defining identity for each twisted kind at the point where it is named.

Continuity and operators

For a Banach algebra, $\|\cdot\|$ is the norm, and a derivation is *bounded* when it is bounded as a linear map. The book asks, for each class, whether continuity has to be assumed or comes for free; the question is fixed in Section 2.9 and answered by class in the chapters.

$\mathcal{B}(H)$ is the algebra of bounded operators on a Hilbert space H, and $\mathcal{K}(H)$ is the ideal of compact operators. A von Neumann algebra is written M.

Conventions

A theorem recalled from a source carries the source in its header, with the page or the numbered item. A result proved in this book is marked as one of its own. A citation after a sentence gives the page on which the cited source says what the sentence attributes to it.

The book uses British spelling in the text and keeps the spelling of a quoted title. Section and theorem numbers are stable within an edition; a later edition may add a section at the end of a chapter, which leaves earlier numbers alone.

Part I

Foundations and history

Chapter 1

Origins of the notion, from Leibniz to Hochschild

A derivation of an algebra A is a linear map $D\colon A \to A$ with

$$D(xy) = D(x)\,y + x\,D(y).$$

The definition is short, but it came late. The product rule is as old as the calculus. For two and a half centuries it was only a property of the derivative of a function. It became a definition only in the 1920s and 1930s. Then a derivation became a map on an abstract algebra, fixed by this one identity.

This chapter tells how that happened. The story has three parts, and Figure 1.1 places its sources in time. We use modern notation throughout. A is an algebra, M a bimodule, K and k are fields, and G is a group. Maps are written on the left. When a source uses other notation, we say so once.

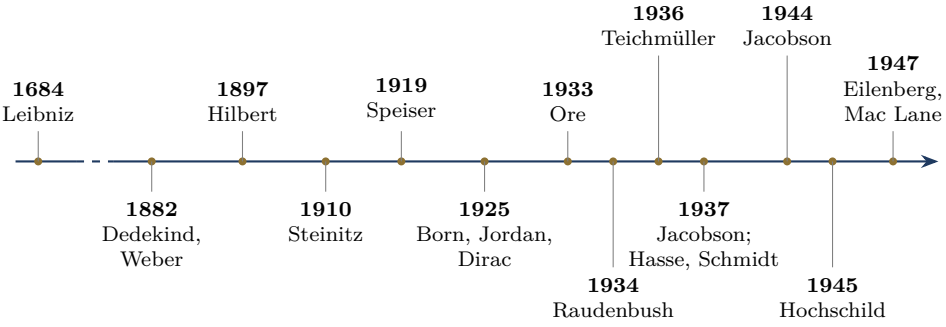


Figure 1.1: The sources of Chapter 1 in order of publication.

The first part stays with commutative rings and fields. It starts in 1684, when Leibniz states the product rule for his differences (Section 1.1). Two centuries later, Dedekind and Weber need a derivative of algebraic functions without any limits. In 1882 they define it by algebra alone, and in 1897 Hilbert writes the different of an algebraic number as a derivative (Section 1.2).

In characteristic p there is a new difficulty. A nonzero polynomial can have zero derivative; for example, x^p does [1910-1, §9, p. 213]. Steinitz meets this in 1910. Later it leads Jacobson to a Galois theory in which derivations take the place of automorphisms (Section 1.3). It also leads Hasse and Schmidt to higher derivations (Section 1.4). Meanwhile Ritt and his students study fields with a derivation. For them such a field is the natural setting for differential equations (Section 1.5).

The second part leaves commutativity behind. In 1925 Born and Jordan keep the product rule for matrices. In the same year Dirac shows that every such operation on quantum variables is a commutator (Section 1.6). In 1933 Ore builds polynomials over a skew field. He finds a product rule twisted by an endomorphism (Section 1.7). In 1937 Jacobson studies the derivations of any algebra. He shows that they form a Lie algebra, and in characteristic p a restricted one (Section 1.8).

The third part shows how these lines meet. Since Hilbert, number theorists had studied maps called crossed homomorphisms. They come from groups that act on fields (Section 1.9). In 1945 Hochschild puts the derivations of an algebra into a cohomology theory. Derivations are the one-cocycles, and inner derivations are the coboundaries (Section 1.10). After this step the derivation is a central object of modern algebra.

Along the way, three facts appear again and again. They are also the first questions of the rest of the book.

The first fact concerns constants. A derivation is fixed by one identity, and its kernel is the set of constants. This kernel carries information about the algebra. Automorphisms have fixed points in the same way. Dedekind, Weber, Steinitz and Jacobson all use this link (Sections 1.2 and 1.3).

The second fact concerns brackets. The commutator of two derivations is again a derivation. So the derivations of any algebra form a Lie algebra. Dirac meets this fact in physics, and Jacobson makes it the base of a theory (Sections 1.6 and 1.8).

The third fact concerns inner derivations. For every element a, the map $x \mapsto ax - xa$ is a derivation. When does every derivation have this form? Dirac asked this for matrices. Hochschild answered it in general with the

group H^1 (Sections 1.9 and 1.10). The question runs through every later chapter.

1.1 The Leibniz rule: Leibniz 1684

Every derivation in this book satisfies the same identity:

$$D(xy) = D(x)\,y + x\,D(y).$$

This rule first appears in print in a paper of Leibniz of October 1684 [1684-1]. The paper is written in Latin. It is the paper that introduced the differential calculus.

Why did Leibniz need such a rule? The title of the paper names the problem. Leibniz wants a method for maxima, minima and tangents "*quae nec fractas nec irrationales quantitates moratur*" (Latin; our translation: "which is not stopped by fractions or by irrational quantities"). The methods of his time first removed all square roots from an equation. Leibniz describes this work as follows:

> "*calculo praestare sublatis irrationalibus, taediosissimae et aliquando insuperabilis operae*"
>
> Our translation from Latin: "to carry out by calculation after removing the irrationals, a most tedious and sometimes insuperable labour"

[1684-1, p. 472]. His answer is a set of rules. Each rule acts on one symbol of a formula at a time. So a square root or a fraction needs no preparation. He tests the rules on the law of refraction, which he finds as a minimum. Then he remarks [1684-1, p. 472]:

> "*Cum alii doctissimi Viri multis ambagibus venati sint, quae hujus calculi peritus tribus lineis imposterum praestabit.*"
>
> Our translation from Latin: "Whereas other most learned men have hunted with many detours for what one skilled in this calculus will henceforth achieve in three lines."

The product rule is one of these rules. It gives the differential of a product from the differentials of its factors. Nothing has to be expanded.

To define his differences, Leibniz draws a curve above a horizontal axis. A point X on the axis has abscissa x, and the curve has height v above it. The tangent to the curve at that point meets the axis at a point B. The

segment BX is called the *subtangent* [1684-1, p. 467]. Leibniz chooses any segment dx. Then he defines dv by the proportion

$$dv : dx = v : BX.$$

He calls dv the *differentia*, that is, the difference of v [1684-1, p. 467]. Figure 1.2 shows the construction.

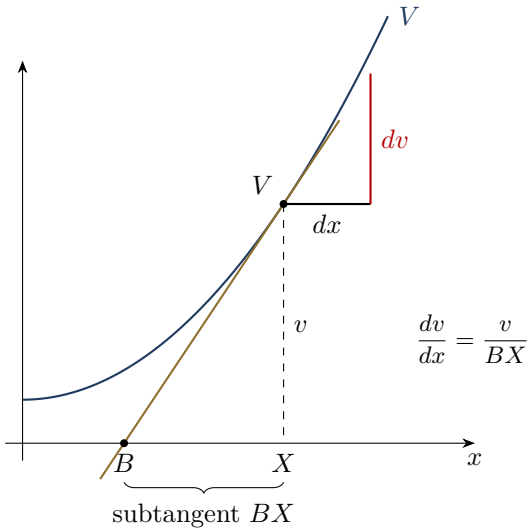


Figure 1.2: The construction of Leibniz [1684-1, p. 467]. The tangent at V meets the axis at B, and BX is the subtangent. The difference dv satisfies $dv : dx = v : BX$.

With this construction, Leibniz states his rules. The first two rules say that constants have zero difference and that d is linear: "*Sit a quantitas data constans, erit da aequalis* 0, *& d ax erit aequ. a dx*" (Latin; our translation: "Let a be a given constant quantity; then da will be equal to 0, and $d\,ax$ will be equal to $a\,dx$"). The difference of a sum is the sum of the differences. The third rule is the product rule: "*Multiplicatio, d xv aequ. x dv + v dx*" (Latin; our translation: "Multiplication: $d\,xv$ equals $x\,dv + v\,dx$") [1684-1, p. 467]. The quotient rule follows on the same page. Figure 1.3 gives the usual picture of the product rule.

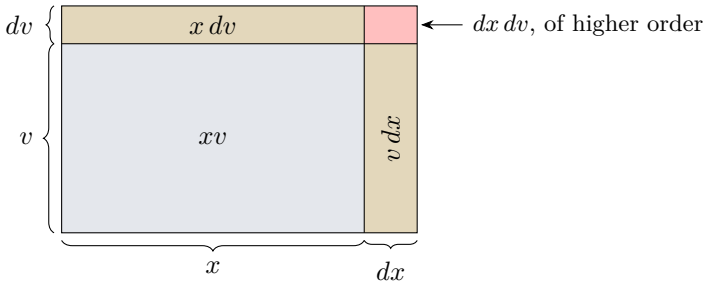


Figure 1.3: The product rule read as area. When x and v grow by dx and dv, the rectangle xv grows by $x\,dv + v\,dx$. The small corner $dx\,dv$ is of higher order.

Three features of this passage return in every later definition. First, the rules are stated for symbols, and no limit appears. Leibniz says that dx is treated in the calculus like y and dy, or like any other letter with its differential [1684-1, p. 467].

Second, constants are exactly the quantities that d sends to zero. Dedekind and Weber stress the same property two centuries later (Section 1.2). Third, the product rule is the only rule that links d to multiplication. So a linear map with this rule behaves like a derivative in every computation made of sums and products.

Today the third feature is the definition itself. Let k be a commutative ring and A a k-algebra. A *derivation* of A is a k-linear map $D\colon A \to A$ with $D(xy) = D(x)y + xD(y)$ for all $x, y \in A$ [1937-2, p. 206]. If A has a unit, then $D(1) = D(1 \cdot 1) = 2D(1)$, so $D(1) = 0$. By linearity, D vanishes on the image of k. Induction on the product rule gives the general Leibniz formula:

$$D^n(xy) = \sum_{i=0}^{n} \binom{n}{i} D^i(x)\, D^{n-i}(y).$$

Jacobson uses this formula in 1937. It shows that in characteristic p the p-th power of a derivation is again a derivation (Section 1.8). Chapter 2 develops the definition and its first consequences.

For two centuries after Leibniz, the product rule stayed a rule of analysis. It entered algebra through the theory of algebraic functions, as the next section shows.

1.2 Differentiation in number and function fields: Dedekind, Weber, Hensel, Hilbert

The derivative of a polynomial did algebraic work long before derivations were defined on rings. It did this work in two places. The first place is the theory of algebraic functions of one variable. The second place is the arithmetic of number fields.

Riemann defined differentials on a Riemann surface by analysis. Dedekind and Weber wanted the same objects without any use of continuity. In number fields, the derivative of a defining polynomial gives a number called the *different* [1897-1, §3, p. 179]. It is defined through the derivative.

Dedekind and Weber gave an answer in 1882. Their memoir is *Theorie der algebraischen Functionen einer Veränderlichen* [1882-1]. It is written in German. It builds the theory of Riemann surfaces again, from the arithmetic of ideals. The introduction states the plan: "*Eine darauf gegründete formale Definition des Differentialquotienten führt sodann zu der Geschlechtszahl und zu einer ganz allgemeinen, eleganten Darstellung der Differentiale erster Gattung*" (German; our translation: "A formal definition of the differential quotient based on this then leads to the genus and to a completely general, elegant representation of the differentials of the first kind") [1882-1, p. 183]. The same page adds a remark. Up to the Riemann–Roch theorem, continuity and power series play no role.

The definition is in Section 23, *Die Differentialquotienten* [1882-1, §23, p. 259]. We follow it in our translation. Let Ω be the field of algebraic functions. Let α and β be elements of Ω. Assume that β is not constant. The two elements satisfy an irreducible equation $F(\alpha, \beta) = 0$. Dedekind and Weber show that Ω contains a function $(d\alpha/d\beta)$. At almost every point, its value is the limit of the difference quotient. A footnote gives a purely algebraic formula:

$$\Big(\frac{d\alpha}{d\beta}\Big) = -\frac{F'(\beta)}{F'(\alpha)}.$$

Here $F'(\alpha)$ and $F'(\beta)$ are the two partial derivatives of F. With this formula they prove the chain rule:

$$\frac{d\alpha}{d\beta} = \frac{d\alpha}{d\gamma}\,\frac{d\gamma}{d\beta}$$

[1882-1, §23, pp. 259–260]. Next they give every element α a differential $d\alpha$. It satisfies $d\alpha = (d\alpha/d\beta)\,d\beta$. They note that "*Die Differentiale der Constanten, und nur diese sind Null zu setzen*" (German; our translation: "the differentials of the constants, and only these, are to be set equal to

zero"). Finally, take any rational relation $F(\alpha, \beta, \gamma, \ldots) = 0$. From it they derive $F'(\alpha)\,d\alpha + F'(\beta)\,d\beta + F'(\gamma)\,d\gamma + \cdots = 0$ [1882-1, §23, p. 261].

In modern language, the construction reads as follows. Fix a nonconstant element z. Then the map $\alpha \mapsto d\alpha/dz$ is a derivation of Ω. Its kernel is exactly the field of constants. The map $\alpha \mapsto d\alpha$ is an early form of the universal derivation (Section 17.1). Dedekind and Weber did not use these words. They needed a derivative defined by algebra alone. With it, they reached the genus and the differentials of the first kind without analysis.

On the side of arithmetic, the derivative appears in the *Zahlbericht* of Hilbert [1897-1]. It is written in German. Let α be an algebraic number. Let $\alpha', \ldots, \alpha^{(m-1)}$ be its conjugates. Hilbert calls the product

$$\delta(\alpha) = (\alpha - \alpha')(\alpha - \alpha'') \cdots (\alpha - \alpha^{(m-1)})$$

the *different* of α. He notes that it is again a number of the field. Indeed, put $f(x) = (x - \alpha)(x - \alpha') \cdots (x - \alpha^{(m-1)})$. Then

$$\delta(\alpha) = \left[\frac{df(x)}{dx}\right]_{x=\alpha}$$

[1897-1, §3, p. 179]. So the different is the derivative of the polynomial at α. The discriminant of α is the product of the squared differences of the conjugates [1897-1, §3, p. 179].

Twenty-six years later, Hecke links the different of a number to the different of the field. He does this in his lectures [1923-1], which are written in German. We follow them in our translation. He calls the ideal of Satz 101 "*die Differente oder das Grundideal des Körpers*" (German; our translation: "the different or the fundamental ideal of the field") (Satz 103). Let ϑ be an integral generator of the field. Let F be its defining polynomial. Then $F'(\vartheta)$ is the different of ϑ. Hecke proves the following. Let α be a number such that α times the different is integral. Then $\alpha = \Phi(\vartheta)/F'(\vartheta)$ for some $\Phi(\vartheta)$ in the ring generated by ϑ [1923-1, §36, pp. 133–134].

Looking back, the two lines have much in common. Both use the derivative as an operation of polynomial algebra. No limit is involved. Both lines also tie the derivative to the branching of an extension. Dedekind and Weber call the ideal of multiple prime factors the "*Verzweigungsideal*" (German; our translation: "branching ideal") [1882-1, §11, p. 226]. They relate it to the differential quotients in Section 23. The different of Hilbert is the derivative of the defining polynomial. The next step is to treat the derivative as an abstract operation on a field, with no polynomial behind it. This is the subject of Section 1.3.

1.3 Derivations of fields and separability: Steinitz, Teichmüller, Jacobson

The fields of the last section have characteristic zero. In characteristic $p > 0$ a nonzero polynomial can have zero derivative. For example, x^p has zero derivative, because the polynomials with zero derivative are exactly the polynomials in x^p [1910-1, §9, p. 213]. This fact first made derivations of fields necessary. This section goes from the formal derivative of Steinitz to the Galois theory of derivations of Jacobson.

The story starts in 1910. The memoir *Algebraische Theorie der Körper* [1910-1] founded the abstract theory of fields. It is written in German. We follow it in our translation. Section 9 is called *Differenzierung*. Let $f(x) = a_0 + a_1 x + \cdots + a_n x^n$ be a polynomial over any field. Steinitz defines its derivative *rein formal* (purely formally):

$$f'(x) = a_1 + 2a_2 x + \cdots + n a_n x^{n-1}.$$

From this formal definition Steinitz derives the sum rule and the product rule for derivatives [1910-1, §9, pp. 209–210]. He then shows the following. In characteristic p, we have $f'(x) = 0$ exactly when f is a polynomial in x^p [1910-1, §9, p. 213].

Steinitz puts this fact to work at once. His Section 13 uses the derivative to separate two kinds of algebraic extension. An irreducible polynomial is *of the first kind* if it has only simple roots [1910-1, §13, p. 231]. An algebraic extension is of the first kind if all its elements are. Over a perfect field every extension is of the first kind. In particular, this holds in characteristic 0. Now let g be irreducible over a field of characteristic p. Suppose that g is not a polynomial in x^p. Then $g'(x) \neq 0$. So g' and g have no common factor. Hence g has only simple roots [1910-1, §13, pp. 231–232]:

> "*Hat zweitens der Exponent f seinen kleinsten Wert 0, so ist $g(x)$ nicht mehr ganze rationale Funktion von x^p, die Ableitung $g'(x)$ verschwindet nicht, kann also mit $g(x)$ keinen gemeinsamen Faktor haben, somit hat die Funktion $g(x)$ lauter einfache Nullstellen, sie ist von erster Art.*"
>
> Our translation from German: "If, secondly, the exponent f has its least value 0, then $g(x)$ is no longer an integral rational function of x^p, the derivative $g'(x)$ does not vanish, and so it can have no common factor with $g(x)$; hence the function $g(x)$ has only simple roots, it is of the first kind."

[1910-1, §13, p. 232]. In present terms: an irreducible polynomial is separable if and only if its derivative is nonzero. The first kind of Steinitz is our separable kind.

The next step came in the 1930s. Hasse had built higher derivatives in algebraic function fields of characteristic p. He used the formulas of analysis for implicit functions (Section 1.4). Teichmüller avoids these formulas. His paper [1936-1] is written in German, and we follow it in our translation. He views the function field as an inseparable extension of a well chosen subfield. From this point of view he introduces all the higher differential quotients. He needs no power series at the primes. He notes that this gives "*eine Differentiationstheorie für die einfachsten inseparablen Oberkörper beliebiger Körper von Primzahlcharakteristik auf rein algebraischer Grundlage*" (German; our translation: "a theory of differentiation for the simplest inseparable extensions of arbitrary fields of prime characteristic on a purely algebraic basis"). He meant it as a tool for his planned work on p-algebras [1936-1, p. 89].

The last step in this line is due to Jacobson, in 1944. A purely inseparable extension has no nontrivial automorphisms. So ordinary Galois theory says nothing about it. Jacobson replaced the automorphisms by derivations [1944-1]. Let K be a field of characteristic p. A finite extension K/k is *purely inseparable of exponent one* when $x^p \in k$ for every $x \in K$. Write $\mathrm{Der}_k(K)$ for the derivations of K that vanish on k.

We use the terms of Jacobson [1937-2, Theorem 1, p. 207]. A set of derivations of K is a *restricted K-Lie ring* if it has three properties. It is a vector space over K. It is closed under commutators. It is closed under p-th powers. It is *finite* if its dimension over K is finite. By Section 1.8, $\mathrm{Der}_k(K)$ is a restricted K-Lie ring.

Theorem 1.1 ([1944-1, p. 645])**.** *Let K/k be a finite, purely inseparable extension of exponent one in characteristic $p > 0$. Then the only elements of K killed by every derivation in $\mathrm{Der}_k(K)$ are the elements of k. Moreover, let Δ be a finite restricted K-Lie ring of derivations of a field K of characteristic p, and let k be its field of constants. Then K/k is finite and purely inseparable of exponent one, and $\Delta = \mathrm{Der}_k(K)$.*

What does Theorem 1.1 say? Jacobson calls the result an order anti-isomorphism [1944-1, p. 645]. It matches two kinds of objects. On one side are certain subfields k of K. On the other side are certain sets of derivations of K. A field k goes to the set $\mathrm{Der}_k(K)$. A set of derivations goes back to its field of constants. A larger field gives a smaller set of derivations.

The theorem has a history in the work of Jacobson himself. He already had a weaker result in his paper of 1937 (Section 1.8). There he matched the fields between k and K with the restricted Lie rings inside $\mathrm{Der}_k(K)$ [1937-2, p. 220]. In 1944 he can start from either side: from the derivations, or from the base field. He compares this step with the work of Artin and Baer in ordinary Galois theory [1944-1, p. 645].

So derivations now play the part of automorphisms. In ordinary Galois theory a group of automorphisms has a fixed field. Here a set of derivations has a field of constants.

Characteristic p caused one more problem, this time for higher derivatives. The next section tells how Hasse and Schmidt solved it.

1.4 Higher derivations in characteristic p: Hasse and Schmidt 1937

In characteristic p the usual higher derivatives also lose information. For example, the p-th derivative of every polynomial is 0. The theory of higher derivations began as an answer to this problem. It began in the study of algebraic function fields.

The solution came from a letter. The paper of Hasse and Schmidt [1937-1] grew out of it, and it is written in German. We follow it in our translation. On 25 May 1936 F. K. Schmidt wrote to Hasse. He was teaching the recent theory of higher differential quotients of Hasse. For his lectures, he had worked it out in a different form. He wanted to introduce the quotients globally ("im Großen") from the start. He did not want to start from the power series at a point [1937-1, p. 215].

Schmidt had assumed that his definition agreed with one given by Teichmüller. Then the paper of Teichmüller appeared. It showed Schmidt that the two definitions differ. Hasse published the result as a free and detailed version of the letter. In some points he changed the argument of Schmidt.

We now turn to the construction itself. The setting is a separably generated algebraic function field K of one variable. Its constant field k is arbitrary. Let x be a separating element of K/k, and let $y \in K$. Let ξ be a copy of x in an auxiliary field. The authors expand y as a power series in powers of $\xi - x$. The coefficients lie in K [1937-1, §2, p. 217].

Definition 1.2 ([1937-1, p. 217])**.** The *differential quotient of order* ν of y with respect to x, written $\dfrac{d^\nu y}{dx^\nu}$, is the coefficient y_ν of $(\xi - x)^\nu$ in this expansion ($\nu = 0, 1, 2, \ldots$).

The authors stress what these quotients are. They are "*Die so definierten Differentialquotienten sind nicht das genaue Äquivalent zu den Differentialquotienten der Analysis, sondern vielmehr zu den Taylorkoeffizienten*" (German; our translation: "not the exact equivalent of the differential quotients of analysis, but rather of the Taylor coefficients") $\frac{1}{\nu!}\frac{d^\nu}{dx^\nu}$ [1937-1, p. 217].

This is the main point of the construction. The division by $\nu!$ is part of the definition. So the definition makes sense in characteristic p.

The quotients depend only on x and y. They do not depend on the auxiliary choices. They satisfy four rules: a sum rule, a rule for constants, a product rule and a power rule [1937-1, p. 218, (1)–(4)]. In the first order they give the derivatives already used in the arithmetic of K/k.

In modern language, the maps $D_\nu(y) = d^\nu y/dx^\nu$ form a *higher derivation* of K over k [1937-1, p. 218]. It is also called a *Hasse–Schmidt derivation* [1937-1, p. 218]. It is a sequence of k-linear maps $D_0 = \mathrm{id}, D_1, D_2, \ldots$ with

$$D_\nu(yz) = \sum_{i+j=\nu} D_i(y)\, D_j(z) \qquad (\nu \geq 0).$$

When k contains the rational numbers, every derivation D gives such a sequence: $D_\nu = D^\nu/\nu!$. In characteristic p we cannot divide by $\nu!$. For this reason the maps D_ν must be given separately. The algebraic theory of higher derivations is the subject of Chapter 7.

The notion earned its place on three counts in the years that followed. Baer had linked derivations to purely inseparable extensions of exponent one in 1927, and Weisfeld generalized his results to extensions of any exponent [1965-1, pp. 307–318]. For algebras, Roy and Sridharan introduce an equivalence among higher derivations and show that its classes correspond to isomorphism classes of certain filtered modules [1968-4, Introduction, pp. 115–116]. And the notion leaves the algebraic setting: Miller asks on a Banach algebra whether every higher derivation has the form $\exp D$, and proves restricted converses [1970-2, pp. 301–331].

Heerema carried the notion into the theory of local rings. His survey of the subject is written to show where higher derivations lead, and what they lead to is the automorphism theory of complete local rings [1970-1, Introduction, p. 1212]. Two years earlier he had defined the convergent

higher derivations on a quasi-local ring, the case where the sequence sums in the topology, and proved the theorems that make them useful [1968-2, Introduction, p. 31].

So far, derivations have served the theory of fields. In the same years, Ritt and his school made a field with a derivation an object of study in its own right.

1.5 Differential rings and fields: Ritt and the New York school

Differential algebra began as an algebraic theory of systems of algebraic differential equations. It started with the Colloquium volume of J. F. Ritt of 1932, *Differential equations from the algebraic standpoint*. Its first abstract definitions appear in a paper of Raudenbush of 1934 [1934-1]. Raudenbush was a member of the school of Ritt.

Raudenbush begins with the problem. He reports two results of Ritt. First, Ritt had introduced the idea of an irreducible system of algebraic differential equations. Second, he had shown that every system is equivalent to a finite set of irreducible systems. Raudenbush wants an abstract ideal theory. The theorem of Ritt should then follow from it [1934-1, p. 361].

The objects of the theory are polynomials in unknowns $y_1, \ldots, y_n$ and their derivatives. Following Ritt, Raudenbush calls them "forms". Their coefficients lie in a differential field.

Definition 1.3 ([1934-1, p. 361])**.** A *differential field* is a field K with a map $\partial\colon K \to K$ such that

$$\partial(a+b) = \partial a + \partial b, \qquad \partial(ab) = \partial(a)\, b + a\, \partial(b) \qquad (a, b \in K).$$

The polynomials in unknowns $y_1, \ldots, y_n$ and all their derivatives, with coefficients in K, form a *differential ring*, and a *differential ideal* is an ideal I of it with $\partial(I) \subseteq I$.

Raudenbush writes a_1 for the derivative of a. We write ∂a.

For abstract differential fields, Raudenbush refers to a work of R. Baer. Its title is *Algebraische Theorie der differentiierbaren Funktionenkörper*. It appeared in the proceedings of the Heidelberg Academy [1934-1, p. 361, footnote].

He also records an example of Ritt. In this example a differential ideal has no finite basis in the ordinary sense. For this reason the paper works with a special class of ideals [1934-1, p. 361].

The theory that grew from here is the subject of Chapter 18. It includes the later work of Kolchin.

With Ritt, the first part of our story ends. All the rings in it are commutative. But in 1925 physicists had already met the product rule on matrices, where the order of the factors matters.

1.6 Derivations in quantum mechanics: Born, Jordan, Dirac 1925

In 1925 physicists found two facts before the algebra researchers did. These facts organise much of this book. First, the product rule still makes sense on a noncommutative algebra, if the order of the factors is kept. Second, on the algebra of all matrices, every map with this rule is a commutator. Both facts came out of the new matrix mechanics.

The first fact is in the paper *Zur Quantenmechanik* [1925-1] of Born and Jordan, written in German. In it, Born and Jordan turn the recent ideas of Heisenberg into a full theory. Their mathematical tool is the calculus of matrices [1925-1, p. 858].

Let the entries of two matrices a and b be functions of a parameter t. Take the derivative of each entry. This gives

$$\frac{d}{dt}(ab) = \dot{a}\,b + a\,\dot{b}.$$

The same rule works for a product of m factors. In each term, the factors stay in their original order [1925-1, (11), (11′), p. 861]. So the derivative with respect to t keeps the product rule.

Born and Jordan also use a second kind of derivative. It is the symbolic derivative $\partial y/\partial x_k$ of a matrix function with respect to one of its matrix variables. This derivative keeps neither the product rule nor the chain rule. They say so directly [1925-1, §2, p. 862]:

> "*Zum Beispiel sind hier die Produktregel der Differentiation oder die Regel für die Differentiation einer Funktion von einer Funktion nicht mehr allgemein in Gültigkeit. Nur dann, wenn alle vorkommenden Matrizen miteinander vertauschbar sind, gelten für diese Differentiation alle Regeln der gewöhnlichen Analysis.*"
>
> Our translation from German: "For example, the product rule of differentiation and the rule for differentiating a function of a function are here no longer valid in general. Only when all the matrices that occur commute with one another do all the rules of ordinary analysis hold for this differentiation."

So on a noncommutative algebra, a derivative that acts on the whole algebra keeps the product rule. The derivative d/dt is an example. A derivative with respect to one of the variables loses it.

The second fact is due to Dirac, who asked the general question [1925-2]. Section 3 of his paper is called *Quantum Differentiation*. There he looks for the most general operation d/dv on quantum variables with two laws. In his words, "We shall now determine the form of the most general quantum operation d/dv that satisfies the laws"

$$\frac{d}{dv}(x+y) = \frac{d}{dv}x + \frac{d}{dv}y, \qquad \frac{d}{dv}(xy) = \frac{d}{dv}x \cdot y + x \cdot \frac{d}{dv}y.$$

He adds: "Note that the order of x and y is preserved in the last equation" [1925-2, §3, p. 645].

Quantum variables are matrices $x = (x(nm))$. Their rows and columns are indexed by the stationary states. By the first law, each entry of dx/dv is a linear combination of the entries of x:

$$dx/dv\,(nm) = \sum_{n'm'} a(nm; n'm')\, x(n'm').$$

Dirac compares the entries on both sides of the second law, for all x and y. Most coefficients must be zero. The rest come from a single matrix a. The result is

$$\frac{dx}{dv} = xa - ax.$$

So every such operation is a commutator with a fixed matrix. Dirac states the result as follows:

> "Thus the most general operation satisfying the laws I and II that one can perform upon a quantum variable is that of taking the difference of its Heisenberg products with some other quantum variable."

[1925-2, §3, pp. 646–647].

He notes that two such derivatives do not commute in general. He also notes a special case. Let a be the diagonal matrix of frequencies. Then the operation becomes the ordinary derivative with respect to time [1925-2, §3, p. 647].

In modern language, Dirac proved a theorem about matrix algebras. Let $A = M_n(\mathbb{C})$ be the algebra of complex $n \times n$ matrices. A linear map $D\colon A \to A$ with $D(xy) = D(x)y + xD(y)$ is a derivation. Dirac shows that $D(x) = xa - ax$ for some $a \in A$. In other words, every derivation of a full matrix algebra is inner. His computation proves this fact in coordinates.

The same fact appears again later in the book, in two forms. One form is the vanishing of H^1 for separable algebras (Section 1.10). The other form is the theorem of Kadison and Sakai for von Neumann algebras (Chapter 19).

Dirac also remarks that $d^2x/du\,dv \neq d^2x/dv\,du$. Behind this remark is the commutator of two derivations. It is again a derivation. Jacobson studies this bracket in Section 1.8.

Before Jacobson, the product rule moved from matrices to polynomials over skew fields. This was the work of Ore, which we follow next.

1.7 Skew polynomials and twisted derivations: Ore 1933

The next step came from polynomials. In the paper *Theory of non-commutative polynomials* [1933-1], Ore builds one theory for polynomials over a field K. The field may be commutative or not. There is only one condition. The degree of a product must be the sum of the degrees of the factors. Ore says that this one theory contains "not only the equations studied in non-commutative algebras but also most of the linear operational equations of analysis as, for instance, linear differential and difference equations" [1933-1, p. 480].

He notes that the theory could also come from the work of Noether and Schmeidler. They studied modules over rings of differential and difference expressions. Ore instead builds the theory directly from the polynomials [1933-1, p. 480].

How should such polynomials multiply? The degree condition fixes the shape of the multiplication. It is enough to define $x \cdot a$ for $a \in K$. The condition leaves only one possibility.

Definition 1.4 ([1933-1, p. 481, (3)])**.** For $a \in K$,

$$x \cdot a = \bar{a}\, x + a',$$

where $\bar{a}$ and a' are elements of K. Ore calls $\bar{a}$ the *conjugate* and a' the *derivative* of a.

The laws of associativity and distributivity give four rules:

$$\overline{a+b} = \bar{a} + \bar{b}, \quad \overline{ab} = \bar{a}\,\bar{b}, \qquad (a+b)' = a' + b', \quad (ab)' = \bar{a}\,b' + a'b$$

[1933-1, p. 482, (4)–(5)]. So the conjugation is an endomorphism of K. The derivative satisfies a product rule twisted by the conjugation.

In modern language, put $\sigma(a) = \bar{a}$. Then $a \mapsto a'$ is a σ-derivation. The ring that Ore built is the skew polynomial ring $K[x; \sigma, \delta]$. It is also called the Ore extension, and the book uses the two names for the same construction. It is the subject of Chapter 6.

Ore had three special cases in mind, and he lists them [1933-1, p. 482].

- Suppose x commutes with the coefficients. Then $\bar{a} = a$ and $a' = 0$. This is the usual polynomial ring.
- Linear differential equations: here K is commutative and $\bar{a} = a$. So $(ab)' = ab' + a'b$, and "the derivative has the ordinary properties of a derivative". Ore remarks that this makes it "possible to introduce the notion of a derivative into non-commutative fields".
- Difference equations: here $a' = 0$, while $\bar{a}$ is usually different from a.

So in 1933 one definition covered three things. It covered the usual derivations of a field, derivations of a skew field, and the shift operators of difference equations. The idea of a derivation twisted by an endomorphism starts here.

Ore studied one derivation at a time. Four years later, Jacobson looked at all the derivations of an algebra together.

1.8 Derivations and Lie algebras: Jacobson 1937 and the structure theory

The first systematic study of derivations of an abstract algebra is a paper of Jacobson from 1937 [1937-2]. Its first sentence states its purpose. It studies "the algebraic properties of the set of operations mapping an algebra on itself and having the formal character of derivation in the field of analytic functions" [1937-2, p. 206].

Jacobson compares his results with the theory of automorphisms. He writes: "Some of the results obtained are analogous to well-known theorems on automorphisms of algebras" [1937-2, p. 206]. In analysis, a derivation is an automorphism taken at an infinitely small scale. His own footnote gives an example. On $k[x]$, the map $\exp D$ acts as $f(x) \mapsto f(x + 1)$ [1937-2, p. 206].

Jacobson asks two questions. What do the derivations of an abstract algebra look like? Which theorems on automorphisms also hold for derivations? He has no analysis to help him. There are no limits and no convergence, and the ground field is arbitrary.

Two facts about derivations have no match for automorphisms. They drive the paper. First, derivations are closed under commutators, so they form a Lie algebra rather than a group. Second, in characteristic p they are also closed under p-th powers.

To answer these questions, Jacobson works with any algebra A over a field k. The algebra need not be commutative. It need not be associative. It need not be finite dimensional.

Definition 1.5 ([1937-2, p. 206])**.** A *derivation* of A is a k-linear map $D\colon A \to A$ with

$$D(xy) = D(x)\,y + x\,D(y) \qquad (x, y \in A).$$

Jacobson wrote maps on the right, as $(xy)D = (xD)y + x(yD)$. This was usual in the algebra of his time. In this book we write maps on the left.

The first properties follow at once from the product rule. The sum of two derivations is a derivation. A scalar multiple of a derivation is a derivation. The product D_1D_2 of two derivations is usually not a derivation. But the commutator $[D_1, D_2] = D_1D_2 - D_2D_1$ is a derivation [1937-2, p. 207].

Repeated use of the product rule gives the Leibniz formula:

$$D^n(xy) = \sum_{i=0}^{n} \binom{n}{i} D^i(x)\,D^{n-i}(y).$$

Now let the characteristic be p. Then the coefficients $\binom{p}{i}$ with $0 < i < p$ are zero. So D^p is again a derivation [1937-2, p. 207, (7)–(8)]. This fact leads to a new notion. A *restricted Lie algebra* is a space of linear maps with two properties. It is closed under commutators. In characteristic p, it is also closed under p-th powers.

Theorem 1.6 ([1937-2, Theorem 1, p. 207])**.** *The derivations of an algebra A over k form a restricted Lie algebra* $\mathrm{Der}(A)$ *of linear maps of A.*

Theorem 1.6 is the reason the word is used: Jacobson calls $\mathrm{Der}(A)$ the derivation algebra of A.

Jacobson also returns to the comparison with automorphisms. The link is the exponential. In a footnote he takes the polynomial ring $k[x]$ over a field of characteristic 0, with the usual derivative $D(f) = f'$. Then $\exp D = 1 + D + D^2/2! + \cdots$ is an automorphism, since $(\exp D)(f)(x) = f(x + 1)$ [1937-2, p. 206, footnote]. Section 2.5 treats this link in general.

The rest of the paper turns to associative algebras of finite order, mainly semisimple ones. Here Jacobson uses the known theory of algebras. He refers the reader to the book *Algebren* of Deuring [1937-2, p. 206].

A recent line of work pairs an algebra with one chosen derivation and studies the pair. Das calls such a pair a *LeibDer pair* when the algebra is a Leibniz algebra and the map is a derivation of it [2021-16, §3, p. 4]. With a skew-symmetric bracket the same construction gives a LieDer pair, so a LeibDer pair is the version without skew-symmetry [2021-16, §3, p. 4].

Definition 1.7 ([2021-16, §3, p. 4])**.** Let $\mathfrak{g}$ be a Leibniz algebra. A linear map $\varphi_{\mathfrak{g}} \colon \mathfrak{g} \to \mathfrak{g}$ is a *derivation* on $\mathfrak{g}$ when

$$\varphi_{\mathfrak{g}}[x, y] = [\varphi_{\mathfrak{g}}(x), y] + [x, \varphi_{\mathfrak{g}}(y)]$$

for all $x, y \in \mathfrak{g}$. A *LeibDer pair* $(\mathfrak{g}, \varphi_{\mathfrak{g}})$ is a Leibniz algebra together with one of its derivations.

Das notes that the derivations in this sense are the one-cocycles of the cochain complex of the Leibniz algebra with coefficients in itself [2021-16, §3, p. 4]. From there the pair carries its own cohomology, and central extensions, deformations and the homotopy version are all read through it [2021-16, p. 1]. For a free Leibniz algebra on a space V, a linear map of V extends to a derivation, and the resulting pair is the free one [2021-16, Proposition 3.4, p. 5].

The derivations of Jacobson map an algebra to itself. Derivations with values in a module have their own history. It began with groups that act on fields, long before 1937.

1.9 Crossed homomorphisms and principal derivations: Hilbert, Speiser, Brauer, Noether, Eilenberg, Mac Lane

The identity $D(gh) = gD(h) + D(g)$ was studied long before derivations of algebras had a cohomology. It was studied for groups that act on fields. There it has the name *crossed homomorphism* [1994-6, Definition, p. 516]. This section follows that line from Hilbert to the textbooks.

The line starts with Theorem 90 of the *Zahlbericht* [1897-1], written in German. It is about a field K that is relatively cyclic over k. Let S be a generator of its Galois group. Hilbert uses symbolic powers $A^{F(S)}$. He takes this notation from Kronecker. The theorem reads: "*Jede ganze oder gebrochene Zahl A in K, deren Relativnorm in Bezug auf k gleich* 1 *ist, wird die symbolische* $(1 - S)$*te Potenz einer gewissen ganzen Zahl B des Körpers K*" (German; our translation: "Every integral or fractional number

A in K whose relative norm with respect to k is equal to 1 becomes the symbolic $(1-S)$-th power of a certain integer B of the field K") [1897-1, §54, Satz 90, p. 272].

We now say this in present terms. A map on a cyclic group is fixed by its value A on the generator. The map is a crossed homomorphism into $K^\times$ exactly when the norm of A is 1. The equation $A = B^{1-S}$ says that the map is principal. The group of crossed homomorphisms is H^1 of the group. Weibel calls this group as old as the other cohomology groups of low degree. Theorem 90 of Hilbert gives it for the Galois group of a cyclic extension [1999-2, p. 806].

In 1919 a more general case appears in a short paper of Speiser [1919-1]. The paper is written in German. It uses neither name. Let K/k be a Galois extension with group G. Speiser gives each $S \in G$ an invertible matrix M_S over K. He asks that these matrices compose by the law $M_S^T M_T = M_{ST}$. Here M_S^T is the matrix obtained from M_S by applying T to its entries [1919-1, p. 2]. In present terms this is a crossed homomorphism from G into the group of invertible matrices over K. His first theorem says that it is principal:

> "*Satz 1: Bilden die Matrizen $M_E, M_A, \ldots$ eine Gruppe nach der obigen Definition, so gibt es in K eine Matrix M derart, daß $M_S = (M^S)^{-1} M$ ist.*"
>
> Our translation from German: "Theorem 1: If the matrices $M_E, M_A, \ldots$ form a group under the above definition, then there is a matrix M in K such that $M_S = (M^S)^{-1} M$."

[1919-1, p. 2]. Speiser then obtains the theorem of Hilbert:

> "*Ein Spezialfall dieses eben bewiesenen Satzes ist Satz 90 im Zahlenbericht von Hilbert*"
>
> Our translation from German: "A special case of the theorem just proved is Theorem 90 in the Zahlbericht of Hilbert"

[1919-1, p. 3]. He cites it as Jahresbericht der Deutschen Mathematiker-Vereinigung 4 (1897), p. 272.

Where does the name come from? It comes from algebras. Weibel writes that the name crossed homomorphism "comes from its role in the study of crossed product algebras" [1999-2, p. 806]. There he cites Brauer and Noether 1927.

We now move to 1947, two years after the paper of Hochschild. Let a group Π act on an abelian group G. Eilenberg and Mac Lane write the

1-cocycle condition as $f(x_1x_2) = f(x_1) + x_1f(x_2)$. They give the names that were already in use:

> "Such functions are known in algebra as crossed homomorphisms."

[1947-1, §3]. They call the 1-coboundaries $f(x) = xg - g$ the principal homomorphisms. Then $H^1(\Pi, G)$ is the group of crossed homomorphisms modulo the principal homomorphisms [1947-1, §3]. So in 1947 these terms were already standard.

Today one object has three names. The modern textbook joins the three sets of terms. Let G be a group and A a left G-module. Weibel defines *a derivation (or crossed homomorphism)* as a map $D\colon G \to A$ with $D(gh) = gD(h) + D(g)$ [1994-6, Definition 6.4.1, p. 174]. He calls the maps $D_a(g) = ga - a$ the *principal derivations*. He proves $H^1(G; A) \cong \mathrm{Der}(G, A)/\,\mathrm{PDer}(G, A)$ [1994-6, Definition 6.4.1, Example 6.4.2, Theorem 6.4.5]. The same book treats a ring R and a bimodule M. There a derivation is also called a crossed homomorphism. The maps $r \mapsto rm - mr$ are the principal derivations [1994-6, §9.2, p. 301]. So three names describe one object: *principal derivation*, *principal crossed homomorphism* and *inner derivation*. They come from three traditions: groups, rings and algebras. Section 1.10 gives the algebra side, Hochschild 1945.

1.10 Derivations as one-cocycles: Hochschild 1945

We now come to the end of the story. In 1945 Hochschild placed derivations inside a cohomology theory. His paper is *On the cohomology groups of an associative algebra* [1945-1].

Hochschild states his aim at once. The paper is "concerned primarily with the connections between the structure of an algebra and the vanishing of its cohomology groups" [1945-1, p. 58]. His model is a pair of results on Lie algebras, the lemmas of Whitehead. The first lemma is about derivations of a semisimple Lie algebra into a module. Hochschild looks for the same kind of result for associative algebras. He finds the answer in the separable algebras. These are exactly the algebras on which every derivation is inner. He also notes a second link. The two-dimensional groups describe the extensions of A. These are the algebras that have A as a homomorphic image [1945-1, p. 58].

His definitions are as follows. Let A be an associative algebra over a field. Let M be an A-bimodule, that is, a two-sided module. An m-cochain

is a linear map from the m-fold tensor power of A into M [1945-1, §1, p. 59]. Hochschild defines a coboundary operator δ. It sends m-cochains to $(m+1)$-cochains [1945-1, §2, Definition 2.1, p. 59]. He credits the coboundary to an unpublished definition of Eilenberg and Mac Lane. He notes that dimensions one and two had appeared before. They appear in the first and second lemmas of Whitehead [1945-1, p. 58].

In the lowest dimensions the definitions are short. A 0-cochain is an element $u \in M$, and

$$(\delta u)(a) = a \cdot u - u \cdot a.$$

For a 1-cochain f,

$$(\delta f)(a_1, a_2) = a_1 \cdot f(a_2) - f(a_1 a_2) + f(a_1) \cdot a_2$$

[1945-1, p. 60]. A cochain f is a cocycle if $\delta f = 0$. It is a coboundary if $f = \delta g$ for some cochain g. The m-dimensional cohomology group $H^m(A, M)$ is the group of m-cocycles modulo the coboundaries [1945-1, p. 60, Definitions 2.2, 2.3].

In modern language, a 1-cocycle is a linear map $f\colon A \to M$ with $f(a_1 a_2) = a_1 f(a_2) + f(a_1) a_2$. So a 1-cocycle is a derivation of A into M. A 1-coboundary is a map $a \mapsto au - ua$. So a 1-coboundary is an inner derivation. Hence $H^1(A, M)$ is the group of derivations modulo the inner ones.

With these definitions, Hochschild reaches his main results. He says that the theory is "in a sense" "degenerate". For every M there is another A-bimodule, written $(A : M)$. For $m \geq 2$ it gives $H^m(A, M) \cong H^{m-1}(A, (A : M))$ [1945-1, Theorem 3.1, p. 60]. So the groups of dimension one already determine all the others. In other words, derivations control the whole theory.

Theorem 1.8 ([1945-1, p. 58 and §4])**.** *An algebra is separable if and only if all its cohomology groups vanish. Equivalently, it is separable if and only if every derivation of it into every two-sided module is inner.*

This theorem extends an earlier result of Hochschild himself. The earlier result needed a non-modular ground field. A footnote places it in Theorems 3.2 and 3.4 of his paper on semisimple algebras and generalized derivations [1945-1, p. 58, footnote 3].

That earlier paper begins with this sentence: "Roughly speaking, a derivation of an algebra is the infinitesimal operation corresponding to an automorphism" [1942-1, p. 677]. The definition there is the one of Jacobson

from 1937. The first example is $\exp(tD)$, which acts on polynomials as a one-parameter group of automorphisms.

So the lines of this chapter meet here. Leibniz gave the product rule. Dedekind, Weber and Steinitz used it inside fields. Born, Jordan, Dirac, Ore and Jacobson carried it to noncommutative algebras. On the side of groups, the same objects are the crossed homomorphisms of Section 1.9. In the work of Hochschild all of these become one notion. The rest of the book starts from here.

Chapter 2

Basic theory and the family of derivations

Chapter 1 followed the product rule from Leibniz to Hochschild. By 1945 the rule was a definition on abstract algebras. This chapter asks what one can do with that definition. Three groups of mathematicians gave answers, and each group studied a different object.

The first group studied differential equations. Ritt wanted a field that comes with a derivative, so that the algebra of such fields could be built like the algebra of ordinary fields [1950-1, p. 1]. His main object was the set of elements with zero derivative, called the constants. Section 2.6 returns to them.

The second group studied Lie algebras. Jacobson saw that the derivations of an algebra are closed under the commutator, so they form a Lie algebra [1937-2, p. 207]. Hochschild then showed what this Lie algebra measures. It is the tangent space at the identity of the group of automorphisms [1942-1, pp. 677–678]. Sections 2.3 and 2.5 tell this part of the story.

The third group studied rings. For them the answer depends on the algebra. In a matrix algebra over a field, every derivation is a commutator with a fixed matrix [2018-7, Corollary 6.2]. In the Weyl algebra this can fail, and it depends on the characteristic of the field [2015-5, p. 48]. Kharchenko built a theory around such differences [1991-5]. Sections 2.4 and 2.7 follow this theory.

The chapter asks one question: what does the set of all derivations tell us about the algebra? We go step by step. First comes the definition. Then we collect all derivations into one object and ask which come from a fixed element. Next we link derivations to automorphisms. After that we study

the elements that a derivation sends to zero, and we ask when a derivation extends to a larger algebra.

The last sections look back. We name the close relatives of a derivation and list the questions that the book asks about each one. The chapter ends with examples from the sources, on matrices and polynomials. Each field where these notions apply is reported in the chapter that owns its notions.

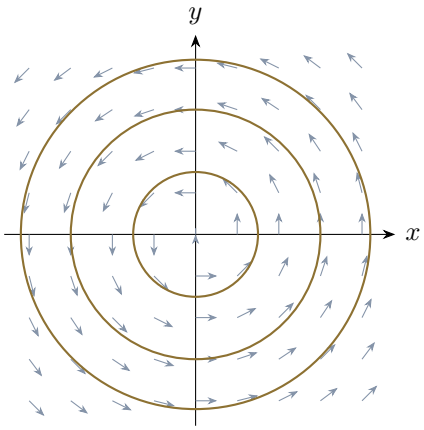


Figure 2.1: The derivation $D = x\,\partial/\partial y - y\,\partial/\partial x$ viewed as a vector field on the plane. On smooth functions, $\exp(tD)$ is the rotation by the angle t. On polynomials the series $\exp(tD)$ does not stop, so it is not defined there. Section 2.5 explains why.

2.1 Definitions, first identities and the Leibniz formula

We start from the product rule of calculus. The polynomial ring has the derivative d/dx, and Section 2.3 gives its derivations. Matrix rings have no derivative that is obvious, and Section 2.4 shows where their derivations come from. So the first task was to say which maps deserve the name, and to find out what follows from the rule alone.

Ritt took the first step for fields [1950-1, p. 1]. He worked in characteristic zero and asked for an operation $a \mapsto a'$ with $(a+b)' = a' + b'$ and $(ab)' = ba' + ab'$. He saw at once that $0' = 0$ and $1' = 0$. He named the elements with zero derivative: "An element with zero for derivative is called a constant" [1950-1, p. 1].

Jacobson took a wider view in 1937 [1937-2, p. 206]. He wanted to study "the set of operations mapping an algebra on itself and having the formal character of derivation in the field of analytic functions" [1937-2, p. 206]. His algebras could be over any field. They did not have to be associative or finite dimensional. So the definition could use only the product rule.

Definition 2.1 ([1937-2, p. 206])**.** Let A be an algebra over a field k. A *derivation* of A is a k-linear map $D\colon A \to A$ with

$$D(xy) = D(x)\,y + x\,D(y) \qquad (x, y \in A).$$

We list two changes that other authors made to this definition. Kharchenko works with rings and writes maps on the right [1991-5, p. 7]. Hochschild lets the values lie in a larger algebra B that contains A [1942-1, p. 688]. A linear map $D\colon A \to B$ with the same rule is then a derivation of A into B [1942-1, Definition 3.1, p. 688]. We use the wider setting when we say so.

We ask next what follows from the definition alone. Jacobson answers with the Leibniz formula, which gives the k-th power of a derivation on a product [1937-2, formula (7), p. 207]:

$$D^k(xy) = \sum_{i=0}^{k} \binom{k}{i} D^{k-i}(x)\,D^i(y).$$

For commutative rings, Freudenburg adds two more rules.

Proposition 2.2 ([2017-7, Proposition 1.6(b), (c), p. 13])**.** *Let B be a commutative ring and $D \in \mathrm{Der}(B)$. For $t \in B$ and $n \geq 1$, $D(t^n) = n\,t^{n-1}D(t)$. If $f, g \in B$ and g is invertible, then $D(fg^{-1}) = g^{-2}(g\,D(f) - f\,D(g))$.*

We call them the power rule and the quotient rule.

We turn to prime characteristic, where the formula takes a special form. Let the field have characteristic p. Then the middle binomial coefficients vanish, so $D^p(xy) = D^p(x)\,y + x\,D^p(y)$. Jacobson states this as the fact that D^p is a derivation [1937-2, formula (8), p. 207]. In characteristic 0 the powers need not be derivations: for $D = d/dx$ on $k[x]$ we get $D^2(x^2) = 2$, while $D^2(x)\,x + x\,D^2(x) = 0$.

We now have a definition and a rule for repeated application. Section 2.3 asks how the derivations of one algebra fit together, and Jacobson answered it in his Theorem 1 [1937-2, p. 207].

2.2 Physics: Poisson brackets and quantum derivations

The oldest application of the product rule is in physics. Dirac posed the question in 1925, in the same paper in which he replaced the Heisenberg matrix commutation by a Poisson bracket [1925-2]. He first writes the two laws that the classical Poisson bracket obeys.

Definition 2.3 ([1925-2, equation after (10), p. 648])**.** Let $[x, y]$ be the Poisson bracket of classical mechanics. The two laws IA and IIa of Dirac read

$$[x + y, z] = [x, z] + [y, z], \qquad [xy, z] = x[y, z] + [y, z]x.$$

The second law is the product rule, so with x fixed the map $y \mapsto [x, y]$ is a derivation on the algebra of classical observables. Dirac then declares that the same algebraic law holds for quantum observables [1925-2, p. 648].

Theorem 2.4 ([1925-2, equation (11), p. 648])**.** *For any two quantum variables x and y,*

$$xy - yx = \frac{ih}{2\pi}\,[x, y],$$

where h denotes the Planck constant and $[x, y]$ on the right is the Poisson bracket of the corresponding classical quantities.

In particular the commutator $x \mapsto xy - yx$ is a scalar multiple of the Poisson bracket, so it is again a derivation. With H the Hamiltonian function of the system, equation (14) of Dirac reads $\dot{x} = [x, H]$ [1925-2, equation (14), p. 651]: time evolution is an inner derivation of the algebra of observables. The question of which derivations of an operator algebra are inner is taken up in Section 19.2.

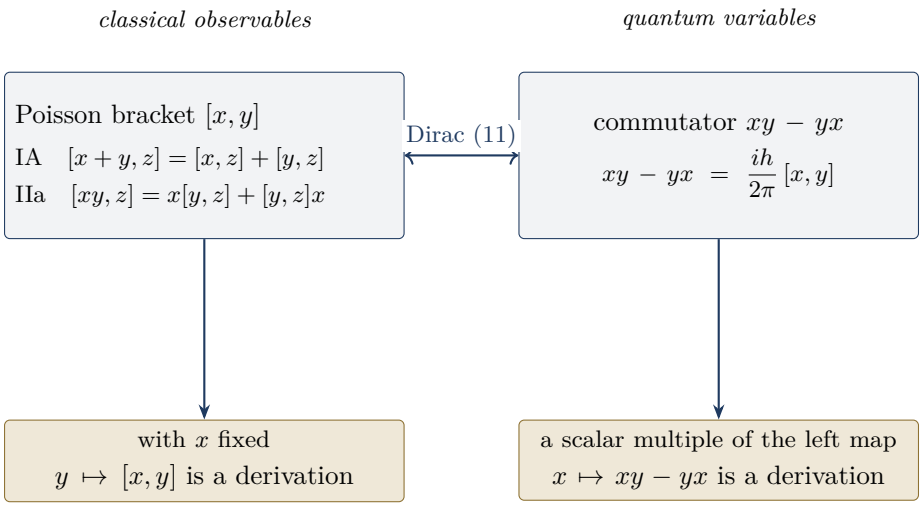


Figure 2.2: The correspondence principle of Dirac. The classical bracket obeys laws IA and IIa [1925-2, equation after (10), p. 648], so with x fixed the map $y \mapsto [x, y]$ is a derivation. Equation (11) [1925-2, equation (11), p. 648] makes the quantum commutator a scalar multiple of that bracket, so it is a derivation too.

The Leibniz rule itself can be obtained rather than assumed. Hosseini derives it in its most general form [2022-5, Abstract, p. 1].

2.3 $\mathrm{Der}(A)$ as a Lie algebra and as a module

Section 2.1 gave the Leibniz formula, which handles one derivation at a time. We ask now whether the product of two derivations is again a derivation. Jacobson checks this by a direct computation: sums, differences and scalar multiples of derivations are derivations, but the product D_1D_2 is not in general [1937-2, p. 207].

The reason is visible in the computation [1937-2, p. 207]. The product D_1D_2 gives the two expected terms and two extra terms:

$$D_1D_2(xy) = D_1D_2(x)\,y + x\,D_1D_2(y) + D_1(x)\,D_2(y) + D_2(x)\,D_1(y).$$

The extra terms are the last two [1937-2, formula (5), p. 207]. They are the same for D_1D_2 and D_2D_1, so they cancel in the commutator $[D_1, D_2] = D_1D_2 - D_2D_1$.

Humphreys records this for any algebra over a field [1972-3, §1.3, p. 4]. The commutator of two derivations is a derivation, and the ordinary product need not be [1972-3, §1.3, p. 4]. We turn to prime rings, which Posner defined [1957-2, p. 1093]. A ring R is *prime* if $xRy = 0$ implies $x = 0$ or $y = 0$. Let R be prime of characteristic not 2. If the product $d_1 d_2$ of two derivations is a derivation, then d_1 or d_2 is zero [1957-2, Theorem 1, p. 1094].

Theorem 2.5 ([1937-2, Theorem 1, p. 207])**.** *The derivations of an algebra A over a field k form a restricted Lie algebra of linear transformations of A.*

The word *restricted* has a meaning in characteristic p. A restricted Lie algebra of linear maps is closed under addition, scalar multiplication, the commutator and the p-th power [1937-2, p. 207]. The formula for D^p of Section 2.1 makes this work. Jacobson calls the Lie algebra of derivations the *derivation algebra* of A. We write it $\mathrm{Der}(A)$.

Theorem 2.5 does not use all the structure of this Lie algebra. Kharchenko studies it for a ring R with centre $Z(R)$ [1991-5, p. 7]. The set $\mathrm{Der}(R)$ is a Lie ring and a module over $Z(R)$ [1991-5, p. 7]. It is not an algebra over $Z(R)$, because the bracket is not linear in the centre. Instead one has the identity $[zD, E] = z[D, E] - E(z)\, D$ for $z \in Z(R)$. Kharchenko writes it with maps on the right [1991-5, identity (1), p. 7].

This identity shows what is missing. Thus the bracket of zD and E has a second term, the derivative of z along E. So the centre takes part in the structure. Every derivation maps $Z(R)$ into itself [1991-5, p. 8]. Restriction therefore gives a map of Lie rings from $\mathrm{Der}(R)$ to $\mathrm{Der}(Z(R))$ [1991-5, p. 8]. Kharchenko calls a Lie ring with these two features a *differential Lie algebra* over the centre [1991-5, Definition 1.2.3, p. 8].

In a commutative ring the picture is simpler. Freudenburg lists the facts we need for a commutative ring B [2017-7, p. 2]. The map bD is a derivation for $b \in B$ and $D \in \mathrm{Der}(B)$, and so is the bracket $[D, E]$ [2017-7, (C.6), p. 2]. The group of automorphisms acts on $\mathrm{Der}(B)$ by conjugation, $\alpha \cdot D = \alpha D \alpha^{-1}$ [2017-7, (C.5), (C.6), p. 2]. Kharchenko gives the same action for any ring: if g is an automorphism, then $g^{-1} D g$ is again a derivation [1991-5, §1.2.6, p. 11].

We take polynomial rings as the first example.

Example 2.6 ([2017-3, p. 1])**.** Let K be a field of characteristic zero and $P_n = K[x_1, \ldots, x_n]$. The Lie algebra of K-derivations of P_n is

$$D_n = \mathrm{Der}_K(P_n) = \bigoplus_{i=1}^{n} P_n \, \partial_i,$$

with the bracket $[\partial, \delta] = \partial\delta - \delta\partial$.

For $n = 1$ the elements of D_1 are the maps $f \mapsto g\, df/dx$.

We now see that this Lie algebra remembers the ring it comes from. Bavula proves that every automorphism of $\mathrm{Der}_k(P_n)$ comes from an automorphism of P_n [2017-3, Theorem 1.1, p. 1].

Theorem 2.7 ([2017-3, Theorem 1.1, p. 1])**.** *Let K be a field of characteristic zero and $P_n = K[x_1, \ldots, x_n]$. The map $\sigma \mapsto (\partial \mapsto \sigma\partial\sigma^{-1})$ is an isomorphism from the group $\mathrm{Aut}_K(P_n)$ of automorphisms of the algebra P_n onto the group $\mathrm{Aut}_{\mathrm{Lie}}(\mathrm{Der}_K(P_n))$ of automorphisms of the Lie algebra of derivations.*

Some derivations are of a special kind: they come from a fixed element of the algebra, or of a larger one, and Hochschild named them in 1942 [1942-1, Definition 3.2, p. 689]. Section 2.4 turns to them.

2.4 Inner, principal and outer derivations; crossed homomorphisms

We ask now which derivations come from the algebra itself. Hochschild answers with a picture from automorphisms [1942-1, p. 678]. Take an invertible element c. We call the map $a \mapsto cac^{-1}$ an inner automorphism. Let $c = \exp(ta_0)$, where t is a parameter. Then the automorphism equals $a + t(a_0a - aa_0)$ plus a term that is small compared with t [1942-1, pp. 678–679]. Its derivative at $t = 0$ is the map $a \mapsto a_0a - aa_0$.

Definition 2.8 ([1942-1, Definition 3.2, p. 689])**.** Let A be an associative algebra. The derivation $a \mapsto a_0a - aa_0$, for a fixed $a_0 \in A$, is an *inner derivation* of A. For a Lie algebra L, the derivation $l \mapsto [l_0, l]$ is an inner derivation of L.

The next example gives the inner derivations of a ring.

Example 2.9 ([2023-12, Example 1.3])**.** Let R be a ring and $a \in R$. The map $\partial_a(x) = xa - ax$ is a derivation of R, the inner derivation with respect to a. It restricts to the zero derivation on the center of R. In particular, if R is commutative, then ∂_a is zero.

A derivation that is not inner is *outer* [1991-5, pp. 9–10]. We state the exact facts for a ring R, following Kharchenko.

The map $b \mapsto \operatorname{ad} b$ sends R to $\operatorname{Der}(R)$ and respects the Lie bracket [1991-5, pp. 9–10]. Its kernel is the centre [1991-5, pp. 9–10]. The image is an ideal, because $[D, \operatorname{ad} b] = \operatorname{ad} D(b)$ for every derivation D. So the outer part is the quotient $\operatorname{Der}(R)/\operatorname{ad} R$.

Humphreys recalls the same fact for a Lie algebra L [1972-3, §5.3, p. 23]. The set $\operatorname{ad} L$ is an ideal in $\operatorname{Der} L$, because $[\delta, \operatorname{ad} x] = \operatorname{ad}(\delta x)$ [1972-3, §5.3, p. 23].

Gupta, Kaur and Singh give a derivation of a matrix algebra that comes from a derivation of the field [2023-12, Example 1.4].

Example 2.10 ([2023-12, Example 1.4])**.** Let (k, d_k) be a differential field. The map δ_c on $M_m(k)$ defined by $\delta_c((a_{ij})) = (d_k(a_{ij}))$ is a derivation of $M_m(k)$, the coordinate-wise derivation.

The same object has other names, and Chapter 1 met them. Let a group G act on an abelian group A. Weibel calls a map D with $D(gh) = gD(h) + D(g)$ a derivation, or crossed homomorphism. The maps $D_a(g) = ga - a$ are the principal derivations [1994-6, Definition 6.4.1, Example 6.4.2]. For a ring R and a bimodule M, the principal derivations are the maps $f_m(r) = rm - mr$. In both settings the group H^1 is the quotient of all derivations by the principal ones [1994-6, Lemma 9.2.1, p. 301].

So inner, principal and outer derivations are three words for one idea. The table shows how the terms match.

Rings and algebras	Groups	Cohomology
derivation	crossed homomorphism	1-cocycle
inner derivation	principal derivation	coboundary
outer derivation class	element of H^1	cohomology class

The table gives a second reading of our question. Asking whether every derivation is inner means asking whether H^1 is zero.

Hochschild settled this for algebras over a field [1945-1, Theorem 4.1]. His theorem says that an algebra is separable if and only if H^1 vanishes for every two-sided module. So an algebra is separable exactly when every derivation of it into every bimodule is inner.

We begin with cases in which every derivation is inner. We take first a central simple algebra, such as the algebra of all $n \times n$ matrices over a field. Brešar, Hanselka, Klep and Volčič prove the following statement [2018-7, Corollary 6.2].

Theorem 2.11 ([2018-7, Corollary 6.2])**.** *Every derivation from a central simple algebra R into an arbitrary R-bimodule M is inner.*

In particular every derivation of a full matrix algebra over a field is inner. We take second a semisimple Lie algebra. Over an algebraically closed field of characteristic 0, such as the complex numbers, every derivation of it is inner [1972-3, §5.3, p. 23].

Some derivations are not inner, and yet we can list them all. Baclawski does this for the incidence algebra of an ordered set [1972-1, p. 351]. Each derivation is the sum of an inner derivation and a second kind, which comes from an additive function [1972-1, Theorem 2, p. 353]. When the ordered set has a smallest or a largest element, every derivation is inner [1972-1, p. 354].

We see next that the answer can also depend on the field. Consider the Weyl algebra of differential operators with polynomial coefficients [2015-5, p. 46]. Over a field of characteristic 0 all its derivations are inner [2015-5, p. 48]. Over a field of prime characteristic some are not [2015-5, p. 48].

We return to H^1 in Chapter 20. So inner derivations are the derivatives of inner automorphisms. Section 2.5 follows this link between derivations and automorphisms in general.

2.5 Derivations and automorphisms: the exponential and one-parameter groups

Section 2.4 ended with a link between derivations and automorphisms. We follow it in general. Hochschild opens his paper of 1942 with the idea: a derivation is "the infinitesimal operation corresponding to an automorphism" [1942-1, p. 677]. His example is the ring of real polynomials with $D = d/dx$. The operator $T_t = \exp(tD)$ sends $P(x)$ to $P(x + t)$, and the maps T_t form a one-parameter group of automorphisms [1942-1, p. 677]. Jacobson had given the same example in a footnote of 1937, with $t = 1$ [1937-2, p. 206].

Hochschild then makes the idea exact [1942-1, p. 678]. Let A be an algebra of finite dimension over the real numbers. Its automorphisms form a Lie group inside the group of invertible linear maps, and its derivations form the corresponding Lie algebra [1942-1, p. 678]. He proves that the

tangent at the identity of a one-parameter group of automorphisms is a derivation [1942-1, p. 678]. He also proves the converse: for each derivation D the maps $\exp(tD)$ are automorphisms [1942-1, p. 678].

We note that the converse needs the exponential series to make sense. A derivation D is *nilpotent* if some power of it is zero [1972-3, §2.3, p. 8]. In characteristic 0 the series is then finite, so it makes sense. Humphreys proves that the exponential of a nilpotent derivation of a Lie algebra is an automorphism of that algebra [1972-3, §2.3, pp. 8–9]. His proof uses the Leibniz formula of Section 2.1. He also shows that conjugation by any automorphism carries such an automorphism to another one of the same kind [1972-3, §2.3, p. 9].

Freudenburg extends the statement to commutative rings [2017-7, p. 12]. Let B be a commutative ring that contains $\mathbb{Q}$. A derivation D of B is *locally nilpotent* if for each $b \in B$ some power $D^n b$ is zero. For such D he defines $\exp D(f) = \sum_{i \geq 0} D^i f / i!$ [2017-7, Section 1.1.9, p. 12].

Proposition 2.12 ([2017-7, Principle 10, p. 25])**.** *Let D be a locally nilpotent derivation of a commutative k-domain B, where k is a field of characteristic zero. Then* $\exp D$ *is an automorphism of B. If E is locally nilpotent and $[D, E] = 0$, then $D + E$ is locally nilpotent and* $\exp(D + E) = \exp D \circ \exp E$.

We also have a converse for commutative rings. An automorphism is *exponential* if it equals $\exp D$ for a locally nilpotent D [2017-7, Section 2.10, p. 66]. Let f be a ring homomorphism of B and set $E = f - I$. Then f is exponential exactly when E is a locally nilpotent f-derivation, and in that case $f = \exp D$ with $D = \log(I + E)$ [2017-7, Proposition 2.57, p. 67].

Kharchenko records how automorphisms act on derivations [1991-5, p. 11]. If g is an automorphism, then $g^{-1}Dg$ is again a derivation [1991-5, p. 11]. Freudenburg adds that this action preserves locally nilpotent derivations [2017-7, Principle 1(d), p. 21]. So the exponential map is compatible with conjugation, and the group of automorphisms acts on both sides of the correspondence.

Humphreys adds a tool for finite dimensional algebras over an algebraically closed field of characteristic 0 [1972-3, §4.2, Lemma B]. The semisimple part and the nilpotent part of a derivation, taken as a linear map, are again derivations [1972-3, §4.2, Lemma B]. So each derivation splits into a part that has an exponential of nilpotent type and a part that is diagonal.

Example 2.13 ([1972-3, §2.3, p. 9])**.** Let F be a field of characteristic 0 and let $L = \mathfrak{sl}(2, F)$, with the standard basis x, y, h. Put

$$\sigma = \exp \operatorname{ad} x \cdot \exp \operatorname{ad}(-y) \cdot \exp \operatorname{ad} x.$$

Then σ is an automorphism of L with $\sigma(x) = -y$, $\sigma(y) = -x$ and $\sigma(h) = -h$. In particular σ has order 2.

Ihara states the same link for a noncommutative algebra [2012-10, p. 192]. Let k be a field of characteristic zero and let $\hat{R} = k\langle\langle a, b\rangle\rangle$ be the algebra of formal power series in two noncommuting variables. The exponential and the logarithm give a bijection between the Lie algebra of nilpotent derivations of $\hat{R}$ and the group of unipotent automorphisms of $\hat{R}$ [2012-10, p. 192]. The algebra $\hat{R}$ is generated freely by a and b, so a derivation of $\hat{R}$ is determined by its values on a and b [2012-10, p. 192].

Example 2.14 ([2012-10, p. 192])**.** Let $\lambda, \alpha, \beta, \gamma, \delta \in k$. For $n \geq 1$ let D_n be the derivation of $\hat{R} = k\langle\langle a, b\rangle\rangle$ with

$$D_n(a) = \lambda a^{n+1}, \qquad D_n(b) = \alpha a^{n+1} + \beta a^n b + \gamma b a^n + \delta b a^{n-1} b.$$

The derivations D_n do not commute with each other in general. The Lie algebra that they generate has the defining equations of the classical Witt algebra [2012-10, p. 192].

We look next at the elements that a derivation sends to zero. Section 2.6 turns to them.

2.6 Constants, kernels and images

We look now at what a derivation leaves fixed. We note an analogy: an automorphism has fixed elements, and a derivation has a matching set. Ritt called the elements with zero derivative *constants* [1950-1, p. 1]. The name fits the derivative of a function, whose constants are the functions with zero slope. Freudenburg calls the same set the *kernel* of D and writes it $\ker D$ [2017-7, p. 2].

We start with a commutative ring B. The kernel is a subring, and it contains the rational numbers when B does [2017-7, (C.3), (C.4), p. 2]. Put $A = \ker D$. Then $D(ab) = aD(b)$ for $a \in A$ and $b \in B$, so D is an A-module endomorphism of B [2017-7, Proposition 1.6(a), p. 13].

We turn to locally nilpotent derivations. Let D be a locally nilpotent derivation of a commutative domain B over a field k of characteristic 0.

Freudenburg proves that $\ker D$ is closed under factors: if $fg \in \ker D$ with $f, g \neq 0$, then $f, g \in \ker D$. He also proves that $k \subseteq \ker D$ and that a nonzero D has a local slice [2017-7, Principle 1, p. 21]. A local slice is an element r with $Dr \neq 0$ and $D^2 r = 0$ [2017-7, p. 3]. A field has no nonzero locally nilpotent derivation [2017-7, Corollary 1.22, p. 22].

We turn now to noncommutative rings, which need other tools. A ring R is prime if $xRy = 0$ forces $x = 0$ or $y = 0$. Posner shows that in a prime ring, if $aD(x) = 0$ for all x, then $a = 0$ or $D = 0$ [1957-2, Lemma 1, p. 1093]. This lemma starts Chapter 3.

Grzeszczuk studies the constants of a nilpotent derivation of a semiprime ring [1993-7, p. 1857].

Theorem 2.15 ([1993-7, p. 1857])**.** *Let d be a nilpotent derivation of a semiprime ring R. Then the subring of constants $\{x \in R \mid d(x) = 0\}$ is non-nilpotent.*

Ferreira recalls that the constants of any Lie algebra of homogeneous derivations of a free associative algebra form a free subalgebra [2002-5, p. 177]. The next example, due to Bergman, has a derivation that is not homogeneous [2002-5, p. 177].

Example 2.16 ([2002-5, p. 177])**.** Let k be a field and let $R = k\langle x, y, z\rangle$ be the free associative algebra of rank 3. Let d be the k-derivation of R with

$$d(x) = xyx + x, \qquad d(y) = yxy - y, \qquad d(z) = x.$$

Bergman gave this derivation as an example whose subalgebra of constants is not free. Kharchenko indicates how to show that $\ker d$ is not a free subalgebra.

In prime characteristic, the constants of a derivation of a free associative algebra are free under some conditions on the derivation [2002-5, p. 177]. For the partial derivatives of a free algebra, the constants form a free subalgebra [2002-5, p. 177].

An automorphism has a similar set, made of the elements that it fixes [2011-7, p. 1365]. Chen calls this set the kernel of $\sigma - 1$ [2011-7, p. 1365].

In a prime ring, one can ask when the kernel of one map lies in the kernel of another [2011-7, p. 1365]. Brešar answered this for algebraic derivations in characteristic 0 [2011-7, p. 1365]. Chuang, Lee and Wong answered it in characteristic $p > 0$ [2011-7, p. 1365]. Chen answers it for automorphisms [2011-7, Theorem 1.1, p. 1366].

We end with the role of constants in structure theory. Kharchenko traces how studies of fixed rings of finite groups of automorphisms led to Galois correspondence theorems for semiprime rings [1991-5, Introduction]. They cover both groups of automorphisms and Lie algebras of derivations.

We have studied the derivations of a given algebra. Section 2.7 asks a different question: when does a derivation of a small algebra extend to a larger one? Chapter 17 returns to the kernel for locally nilpotent derivations.

2.7 Extending derivations

Section 2.6 studied the derivations of one algebra. We ask now when a derivation of a small algebra extends to a larger one. Scheja and Storch show where the question comes from, in the opening remark of their paper "*In algebraischer und analytischer Geometrie ist das Fortsetzen von Derivationen ein wichtiges Hilfsmittel*" (German; our translation: "In algebraic and analytic geometry, extending derivations is an important tool") [1978-4, p. 353].

We begin with uniqueness, which is easy. Let $A \to B$ be an extension of commutative rings. We note that two extensions of the same derivation of A differ by a derivation of B that is zero on A. So an extension is unique if B has no such nonzero derivation [1978-4, §1, p. 354].

We turn to existence. We start with a localization, where the quotient rule decides everything.

Proposition 2.17 ([2017-7, §1.2.4, p. 16])**.** *Let B be a commutative domain over a field k and let S be a multiplicatively closed subset of B that does not contain 0. Every k-derivation D of B extends in a unique way to a derivation of the localization $S^{-1}B$, by the quotient rule.*

We turn to other extensions, which are harder. Scheja and Storch recall a theorem of Seidenberg [1978-4, p. 353]. Their summary says (our translation) that a derivation of a Noetherian domain containing the rational numbers extends to the normalization of the domain. They then study general extensions of rings. They use the classical different and discriminant of a ring extension [1978-4, p. 353]. Chapter 1 tells how Dedekind and Hilbert defined the different (Section 1.2).

We turn to the noncommutative case, which is cleaner. A ring is *semiprime* if it has no nonzero nilpotent ideal [1991-5, Definition 1.1.1, p. 2]. Take a derivation of a semiprime ring R that is defined only on a large

enough ideal. Kharchenko shows that it extends in a unique way to a larger ring of quotients of R [1991-5, Proposition 1.8.1, p. 50]. So we can work inside that ring from then on [1991-5, §1.8.2, p. 51].

We take central simple algebras as a second case. Let k be a field with a derivation, and let A be a central simple algebra over k. Gupta, Kaur and Singh state a theorem of Amitsur in their survey [2023-12, Theorem 3.1]: the derivation of k extends to A, and any two extensions differ by an inner derivation. So one extension fixes all the others up to inner derivations.

A third case puts a twist on both sides of the extension. Over a commutative ring, a (σ, τ)-derivation of an algebra into a bimodule satisfies $D(ab) = D(a)\tau(b) + \sigma(a)D(b)$, with the twists fixed endomorphisms of the algebra [2020-10, p. 2]. Chaudhuri shows that every such map factors through one universal map, and which of the two twists is invertible decides the shape of that map [2020-10, Theorem 3.1, p. 3]. Over the ring of integers of a quadratic field the class is then described completely [2020-10, Theorem 4.2, pp. 6–7]. Section 6.2 returns to these maps.

We have now seen the derivations of one algebra and their extensions. Section 2.8 steps back and names the close relatives of a derivation.

2.8 The kinds of derivation-like maps

We have so far studied one rule, the product rule. We now collect the ways in which later authors changed it.

Some changes keep the rule and move the target. Hochschild allowed the values of D to lie in a larger algebra [1942-1, Definition 3.1, p. 688]. Weibel replaced the algebra by a group, and the product rule by $D(gh) = gD(h) + D(g)$ [1994-6, Definition 6.4.1]. We met both changes in Section 2.4.

We turn to changes that keep the shape of the rule and twist one factor. Section 1.7 met maps δ with $\delta(rs) = \delta(r)s + \sigma(r)\delta(s)$, where σ is a ring map from R to itself [2015-5, p. 47]. Chapter 6 treats these maps and their generalizations.

We cannot show the kinds in one picture, because there are too many and the list grows with the literature. We list them by the change they make to the rule, and we name the chapter that treats each one. Every chapter defines its kinds from the sources, gives the known implications and quotes the counterexamples.

- *A new target.* Derivations into a bimodule or a larger algebra, and crossed homomorphisms (Section 2.4); derivations of groups (Chapter 20); the universal derivation and Kähler differentials (Chapter 17).

- *A twisted factor.* Skew, σ-, α- and (σ, τ)-derivations, and generalized derivations (Chapter 6); semiderivations, left derivations, reverse derivations and multiplicative derivations (Chapter 6); δ-derivations, quasi-derivations and anti-derivations (Chapter 15).

- *A new product.* Jordan derivations, Jordan triple derivations, Jordan n-derivations, Jordan $*$-derivations and Jordan left derivations (Chapter 4); Lie derivations, Lie triple derivations, Lie n-derivations and the nonlinear Lie-type maps (Chapter 5).

- *Maps built from derivations.* Higher derivations and Hasse–Schmidt sequences, biderivations, triderivations, n-derivations and multiderivations (Chapter 7); exponential maps and locally nilpotent derivations (Section 2.5 and Chapter 17).

- *A condition at each point.* Local derivations, 2-local derivations and derivable maps (Chapter 8); commuting and centralizing maps and functional identities (Chapter 9).

- *Kinds named in the titles of the collected literature.* The titles of the papers gathered for this book name further maps, each in at least six titles. Banach and operator algebras (Chapter 19): closed, unbounded, normal, point and double derivations, superderivations, and Hom-derivations. Nonassociative algebras (Chapter 15): ternary derivations. Lie algebras (Chapter 14): prederivations and central derivations. Commutative algebra (Chapter 17): integrable and quadratic derivations. Matrix and triangular algebras (Chapter 10): additive derivations of nest algebras. Functional identities (Chapter 9): commuting and hypercommuting derivations. Polynomial and Weyl algebras (Chapter 12): right derivations. Partial derivations of functions of n variables. Each chapter defines its kinds from the sources it cites.

- *Maps indexed by other maps.* f-derivations and (f, g)-derivations, with their permuting, symmetric and tri-f forms, in lattices, BCI-algebras and subtraction algebras (Chapter 16), and f-derivations on rings and modules (Chapter 6); (α, β)-, (θ, φ)- and (σ, τ)-derivations in their Jordan, Lie and generalized forms (Chapters 4, 5 and 6). Derivations of almost f-algebras (Chapter 19) take their name from a different f: it refers to a Riesz space, not to a map.

- *Further names from the collected titles.* Each occurs in at least six titles. Order bounded, positive, simple, linear, free, ring, almost inner,

generalized inner, orthogonal generalized, locally finite and local triple derivations; Jordan and Lie higher derivations; generalized Jordan left derivations. Each chapter defines those of its setting from the sources it cites.

- *More names from the collected titles.* A second pass over the titles finds names in at least three titles each. Chapter 6: b-generalized, α-generalized, q-skew, generalized semiderivations, bi-semiderivations and multiplicative semiderivations. Chapter 3: (α, α)-, (θ, θ)-, (α, β, γ)- and (σ, τ, ξ)-derivations, (α, β)-reverse derivations, g-derivations, 3-derivations and left biderivations. Chapter 7: symmetric biderivations, permuting triderivations, (m, n)-derivations and k-derivations. Chapter 17: monomial, Weitzenböck, elementary, semisimple, Jacobian, logarithmic, factorizable and r-derivations. Chapter 15: conformal, homogeneous, special and $(n+1)$-ary derivations, and Lie and Jordan superderivations. Chapter 19: continuous, compact, random, hyponormal, self-adjoint and quasi-free derivations. Chapter 14: nonsingular, symplectic and locally finite derivations. Chapter 16: p-derivations and γ-derivations. Chapter 9: two-sided centralizers, super-commuting maps and n-commuting maps. Chapter 20: 2-cocycles. The file lists every name with its count. Each chapter defines its kinds from the sources it cites.

- *A different algebraic setting.* Prime and semiprime rings (Chapter 3). Lie algebras (Chapter 14). Jordan, alternative and Leibniz algebras, superalgebras, Poisson algebras, Hom-Lie algebras and n-ary algebras (Chapter 15). Hopf algebras, with coderivations (Chapter 13). Differential and difference rings (Chapter 18). Banach and operator algebras, with approximate derivations (Chapter 19). Near-rings, semirings, Gamma-rings and lattices (Chapter 16).

Section 2.9 makes precise the one kind of question that the list raises: when does one kind force another?

Further work in the family of derivation-like maps, held in the collection: *Conjugate linear maps from C-algebras into their dual spaces which are ternary derivable at the unit element* [2020-2]. *Infinite dimensional central simple regular algebras with outer derivations* [2020-3].

2.9 The question families of the encyclopaedia

Section 2.8 raised the question of how the relatives of a derivation depend on each other. We ask the same few questions for every relative and every class of algebra. We fix them here, so that each later chapter answers them in the same order.

1. Which kind of derivation implies which other?
2. When are two definitions equivalent?
3. Is every derivation inner? This is the question of H^1, as Section 2.4 showed.
4. Is every derivation continuous, and does continuity come for free?
5. Does the algebra have to be commutative? The theorem of Posner and the functional identities belong here.
6. Can all derivations of the class be described?
7. Does a derivation extend or lift to a larger algebra? This is the question of Section 2.7.
8. Where does an expected implication fail? Counterexamples answer this.

For each question and class, the book records one of three answers: known, false with a counterexample, or open. Appendix B collects the records, and each chapter states the open ones in the section that asks its question. Section 2.10 sorts those problems by the class of algebra they live in.

2.10 Open problems by class of algebra

The chapters state each problem in the section that asks its question. This section keeps the same problems and sorts them by the algebra they live in, so that a reader who works on one class sees the frontier of that class. For each class it records which problem is settled there, which is not, and the chapter that carries the discussion.

The first class is the polynomial algebra $k[x_1, \dots, x_n]$ over a field of characteristic 0, and it holds the deepest problems of the book. Chapter 17 reads them in Section 17.2 and Section 17.4.

Problem 17.7 belongs to this class, because its local rings are the local rings of a polynomial algebra. Biswas, Gurjar and Kolte settle the affine surface case whenever the smooth locus has logarithmic Kodaira dimension at most one [2014-9, Theorem 1.2, p. 270]. Problem 17.26 lives here too, and it stays open for every $n \geq 2$ [2007-2, pp. 1–2]. Problem 12.22 is the Weyl algebra form of the same question, and Belov-Kanel and Kontsevich tie the two stable forms together [2007-2, Theorem 1, p. 3]. Chapter 12 reads the Weyl algebra in Section 12.2.

The finite generation problems of Problems 17.21 and 17.22 also live in this class. The Homogeneous Dependence Problem of Problem 17.21 stays open only in dimension 5 [2017-7, p. 108]. A homogeneous derivation of degree two with full rank is open in every dimension [2017-7, Remark 3.51, p. 112]. The triangularization problem of Problem 17.22 remains generally open [2017-7, p. 288].

The second class is the finite-dimensional algebra over a field. Here the local property holds for some structures and fails for others. Chapter 8 reads the local maps here in Section 8.1.

Hadwin and Li settle the question of Larson for the J-subspace lattices: the space $\operatorname{der}(\operatorname{alg} L)$ is reflexive [2004-5, Theorem 2.15, p. 708]. They also show that $\operatorname{der}(A, M)$ is algebraically 3-reflexive for every Banach A-bimodule M [2004-5, Theorem 2.19, p. 710]. The local property for automorphisms fails even in finite dimension, since $T \mapsto AT^tA^{-1}$ on $M_n(\mathbb{C})$ is a surjective local automorphism that is not an automorphism [2004-5, p. 713]. So Problem 8.6 is answered for the lattices above, and the general algebra of the question stays out of this class.

This class also holds a counterexample for the 2-local property: Zhang and Li built a 2-local derivation on the upper triangular complex 2×2 matrices that is not a derivation [2012-6, p. 15]. So the 2-local question of Problem 8.14 has no uniform answer even among the small algebras of this class.

The third class is the C^*-algebra and its operator relatives. Chapter 8 reads the 2-local notion in Section 8.2, and Chapter 19 reads the continuity question in Section 19.1.

The 2-local problem of Problem 8.14 is settled here for the largest algebras of the class. Ayupov and Kudaybergenov prove that every 2-local derivation of $B(H)$ is a derivation [2012-6, Theorem 2.3, p. 16]. Ayupov and Arzikulov extend this to every semi-finite von Neumann algebra [2014-5, p. 9], and Ayupov and Kudaybergenov extend it to every von Neumann algebra [2015-4, Theorem 2.1, p. 2]. Habibzadeh Fard and Sahleh add the compact operators [2019-7, Corollary 3.4, p. 6]. So the answer is yes for

every algebra of the class that these papers reach. The φ-derivation side of the class has one case that no one has settled [2002-3, Remark 5.3, p. 720].

Problem 2.18 ([2002-3, Remark 5.3, p. 720])**.** Let A be a unital Banach algebra, let φ be an inner automorphism of A, and let Δ be a φ-derivation of A. If a lies in the radical of A and $\Delta^2(a) = 0$, is $\Delta(a)$ quasinilpotent?

Brešar and Villena prove Problem 2.18 for derivations and report that they could not prove it for φ-derivations [2002-3, Remark 5.3, p. 720]. The continuity problem of Problem 19.7 is the other open question of this class, and the authors call it one of the hardest problems of Banach algebra theory [2002-3, p. 710].

The fourth class is the Lie algebra, where the 2-local question has a negative answer in general. Chapter 14 reads the class in Section 14.5.

Ayupov and Yusupov answer Problem 8.14 with yes for the Witt algebra [2020-7, Theorem 4.1, p. 4] and for the positive Witt algebra [2020-7, Theorem 4.4, p. 6]. They answer it with no for the thin Lie algebra, which admits a 2-local derivation that is not a derivation [2020-7, Theorem 4.8, p. 8]. So this class holds a counterexample of the book, and the problem for a general infinite-dimensional Lie algebra stays open.

The fifth class is the prime ring, where the questions about the standard form are answered for most characteristics. Chapter 5 reads the Lie problem in Section 5.1, and Chapter 7 reads the symmetric biderivation in Section 7.5.

Martindale proves the standard form for a primitive ring with a nontrivial idempotent [1964-2, Theorem 2, p. 187]. Brešar extends it to prime rings of characteristic not 2 that do not satisfy S_4 [1993-3, Theorem 5, p. 529]. Lemma 1 of Brešar puts the rings that satisfy S_4 inside $M_2(F)$ over a field F, up to the commutative case [1993-3, Lemma 1, p. 530]. So Problem 5.5 stays open here for characteristic 2 and for a ring that satisfies S_4. The symmetric biderivation of Problem 7.24 stays open for a power above 2 [1990-6, pp. 188–189], and the commuting map of Problem 9.3 stays open for a power above 1 [1997-4, p. 334].

The last class is the triangular algebra, and the problem here comes from the f-derivation. Chapter 10 reads the class in Section 10.2, and Chapter 6 reads the f-derivation side in Section 6.7.

Benkovič proves the standard form for an f-derivation whose polynomial has a nonzero sum of coefficients [2014-7, Theorem 1.1]. The remaining case is Problem 6.66, and it stays open [2014-7, Problem 1.2].

Across the six classes the same split appears. The polynomial class keeps the deepest problems open. The operator classes settle the 2-local

question for their large algebras, and the small ones keep counterexamples: the thin Lie algebra and the upper triangular complex 2×2 matrices. Two problems have no class that settles them, and they are the reflexive space of Larson and the noncommutative conjecture of Singer and Wermer. They are the two that a later chapter is most likely to close first. Each problem stated in the chapters appears here under the class where its algebra lives. The one problem that this section states itself sits with the C^*-algebras. So a change in one class can be read against the question family it belongs to.

Part II

Derivations of associative rings

Chapter 3

Derivations of prime and semiprime rings

Chapter 2 asked what the set of all derivations tells us about an algebra. This chapter gives the first answer for a class of rings. The class is the prime rings, which Posner took as the setting of his paper of 1957 [1957-2, p. 1093].

We start with two theorems of Posner. The first says that the composition of two nonzero derivations of a prime ring is not a derivation [1957-2, p. 1093]. The second says that a nonzero derivation with $[d(a), a]$ central forces the ring to be commutative [1957-2, p. 1093]. Figure 3.1 shows them. Section 3.1 tells how they were proved and simplified.

Herstein proved a commutativity theorem for derivations whose values commute [1978-1, Theorem 2, p. 369]. He also described the elements that commute with all values of a derivation [1979-1, p. 509]. Section 3.2 follows their commutativity theorems. They need a field that does not lie in the ring. Martindale built it in 1969 [1969-2, p. 576], and Section 3.3 describes it.

With this field, Kharchenko gave a general theory of the identities that derivations can satisfy in a prime ring [1978-2, p. 155]. Lanski used it to prove the theorems of Posner and their generalizations in a systematic way [1988-3, p. 276]. Section 3.4 states the theory.

The sections that follow apply it to nilpotent derivations and Engel conditions (Section 3.5), to Lie ideals (Section 3.6), to rings with involution (Section 3.7) and to Jordan ideals (Section 3.8).

The literature on this chapter is large; Appendix A lists it. Chapter 4 and Chapter 5 take up the Jordan and Lie variants of the maps studied

here.

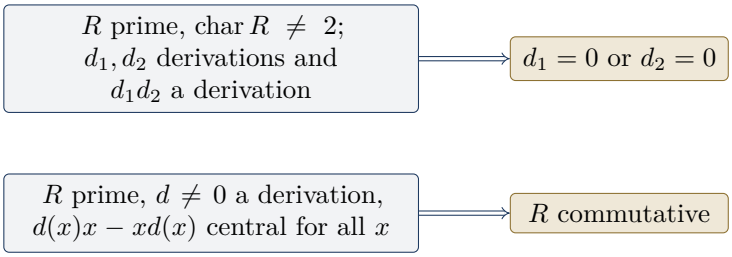


Figure 3.1: The two theorems of Posner [1957-2, p. 1097] on derivations of prime rings, the starting point of the chapter.

3.1 The theorems of Posner

This section tells the story of the paper behind two results used earlier. Chapter 2 used it for the constants and kernels of a derivation, in Section 2.6, and again for the Lie algebra structure on the derivations, in Section 2.3 [1957-2].

The paper proves two theorems about derivations of prime rings [1957-2, p. 1093]. The first one answers a natural question: when is the product of two derivations again a derivation [1957-2, p. 1093]? In a prime ring of characteristic not 2, this happens only when one of the two derivations is zero [1957-2, p. 1093].

The second theorem gives a test for commutativity. Take a prime ring with a nonzero derivation d. Suppose that $ad(a) - d(a)a$ is central for every a. Then the ring is commutative [1957-2, p. 1093].

We define the setting as Posner does [1957-2, p. 1093]. A ring R is *prime* when $xay = 0$ for all $a \in R$ forces $x = 0$ or $y = 0$. Posner calls Lemma 1 known: if d is a derivation of a prime ring and $ad(x) = 0$ for all x, then $a = 0$ or $d = 0$ [1957-2, Lemma 1, p. 1093].

Theorem 3.1 ([1957-2, Theorem 1, p. 1094])**.** *Let R be a prime ring of characteristic not 2, and let d_1 and d_2 be derivations of R such that the composition $d_1 d_2$ is also a derivation. Then at least one of d_1 and d_2 is zero.*

Posner proves this by expanding $d_1 d_2(ab)$ in two ways, which gives $d_2(a)d_1(b) + d_1(a)d_2(b) = 0$ for all a and b [1957-2, p. 1094]. Lemma 1 then finishes the argument [1957-2, p. 1094].

We state the second theorem next.

Theorem 3.2 ([1957-2, Theorem 2, p. 1097])**.** *Let R be a prime ring and let d be a derivation of R such that $ad(a) - d(a)a$ is in the center of R for every $a \in R$. If d is not the zero derivation, then R is commutative.*

The proof of this theorem takes several pages [1957-2, pp. 1094–1100]. Posner first treats the case $ad(a) - d(a)a = 0$, where the ring is commutative or d is zero [1957-2, Lemma 3, pp. 1094–1095]. He then uses a remark of Kaplansky on the Lie ring of derivations [1957-2, Lemma 4, pp. 1096–1097]. Section 2.3 introduced that Lie ring. The last pages treat characteristic 3 and the other characteristics separately [1957-2, pp. 1097–1100].

Mathieu says that the second theorem "has been considered to lie somewhat deeper" than the first [1992-5, p. 601]. He adds that Awtar simplified its proof. He then derives it from the first theorem when the characteristic is not 2.

His argument has three steps [1992-5, pp. 601–602]. Applying the hypothesis to $a+b$ shows that $[a, \delta b]+[b, \delta a]$ is central. A short computation then gives $[a, \delta a] = 0$. So the composition of δ with an inner derivation is a derivation, and Theorem 3.1 applies.

The same argument works on a nonzero ideal I of R [1992-5, p. 601]. It then proves the theorem for a derivation with $[a, \delta a]$ central only for a in I [1992-5, p. 601].

We turn to characteristic 2. Mathieu points to Ahmad for the second theorem [1992-5, pp. 601–602]. For the first theorem, the version of Bergen gives an element c of the extended centroid of R with $\delta_b = c\,\delta$ [1992-5, Remark 2, p. 602]. Section 3.3 explains the extended centroid.

Lanski says that the earlier arguments are "generally ad hoc computations, often lengthy and clever" [1988-3, p. 276]. He replaces them by the differential identities of Kharchenko, which Section 3.4 describes.

With this method he proves both theorems for a noncommutative Lie ideal L of R [1988-3, p. 276]. Let d be a nonzero derivation with $x\,d(x) - d(x)\,x$ central for all $x \in L$. Then R is commutative, or R has characteristic 2 and satisfies the standard identity of degree four. Let d and h be nonzero derivations such that dh is a Lie derivation of L into R. Then the characteristic is 2. Either R satisfies the standard identity of degree four, or $h = dc$ for some c in the extended centroid [1988-3, p. 276].

The theorems above ask for a prime ring [1957-2, p. 1093]. Almahdi, Mamouni and Tamekkante kept the derivation on an arbitrary ring and put the prime hypothesis into an ideal [2020-5, p. 188].

Theorem 3.3 ([2020-5, Theorem 2.2, p. 189])**.** *Let R be a ring, P a prime ideal of R, and d a derivation of R. If $[[x, d(x)], y] \in P$ for all $x, y \in R$, then $d(R) \subseteq P$ or R/P is commutative.*

For a prime ring the zero ideal is prime, and Theorem 3.3 then returns the second theorem of Posner [2020-5, p. 191]. The semiprime case needs a different conclusion [2020-5, p. 192]. Herstein asked whether every minimal prime ideal of a semiprime ring is invariant under every derivation, and Chuang and Lee showed that it need not be [2020-5, p. 192].

Corollary 3.4 ([2020-5, Corollary 2.4, p. 192])**.** *Let R be a semiprime ring and d a centralizing derivation of R. Then R has a minimal prime ideal P such that $d(P) \subseteq P$.*

So one invariant minimal prime ideal always exists. The same argument also tells where the values of the derivation lie [2020-5, p. 192].

Corollary 3.5 ([2020-5, Corollary 2.5, p. 192])**.** *Let R be a semiprime ring and d a centralizing derivation of R. Then $d(R) \subseteq Z(R)$.*

On a semiprime ring the second theorem of Posner gives a central image, and the difference from commutativity matters [2020-5, p. 192]. A semiprime ring can carry a centralizing derivation whose image is central and still not be commutative [2020-5, p. 192], and Corollary 3.5 states exactly this case.

Section 3.2 continues with the commutativity theorems of Herstein and his school [2020-5, p. 192], where that question is pressed further.

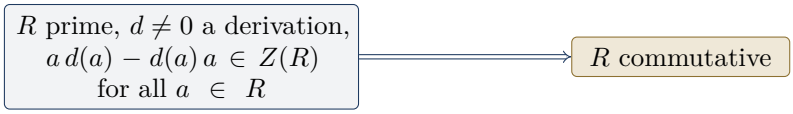


Figure 3.2: The second theorem of Posner [1957-2, Theorem 2, p. 1097]. On a prime ring, a nonzero derivation d with $a\,d(a) - d(a)\,a$ in the center for every a forces the ring to be commutative. Theorem 3.2 states it in full.

3.2 The school of Herstein and the commutativity theorems

Section 3.1 ended with two theorems whose conclusion is that a prime ring is commutative. We ask which other conditions on the values of a derivation

give the same conclusion. Herstein took up this question in two short notes [1978-1, 1979-1].

The first note proves two results about derivations [1978-1, p. 369]. Herstein could not find them in the literature. We state the first result, which holds in any ring.

Theorem 3.6 ([1978-1, Theorem 1, p. 369])**.** *Let R be a ring and let d be a derivation of R with $d^3 \neq 0$. Then the subring generated by all values $d(r)$, for $r \in R$, contains a nonzero ideal of R.*

The condition $d^3 \neq 0$ cannot be dropped, and Herstein gives an example with an inner derivation.

Example 3.7 ([1978-1, p. 369])**.** Let R be a prime ring with a nonzero element a such that $a^2 = 0$. Define

$$d(x) = ax - xa.$$

The subring $B = aR + Ra$ contains $d(R)$, and $d^3 = 0$. Yet B contains no nonzero ideal of R, because $aBa = 0$.

We state the second result, a commutativity theorem for prime rings.

Theorem 3.8 ([1978-1, Theorem 2, p. 369])**.** *Let R be a prime ring and let $d \neq 0$ be a derivation of R such that $d(x)d(y) = d(y)d(x)$ for all $x, y \in R$. If $\operatorname{char} R \neq 2$, then R is a commutative integral domain. If $\operatorname{char} R = 2$, then R is commutative or an order in a simple algebra of dimension four over its center.*

Here the values of d commute with each other. In Theorem 3.2 the condition links $d(a)$ with a itself. Herstein asks whether Theorem 3.8 extends from commuting values to values that satisfy the standard identity of degree k [1978-1, p. 370]. He leaves this as a question.

The second note turns the question around [1979-1, p. 509]. It asks which elements of a prime ring commute with all the values of a nonzero derivation, and Herstein settles it [1979-1, p. 509].

Theorem 3.9 ([1979-1, Theorem, p. 509])**.** *Let R be a prime ring and let $d \neq 0$ be a derivation of R. Suppose that $a \in R$ is such that $ad(x) = d(x)a$ for all $x \in R$. If R is not of characteristic 2, then a is in the center Z of R. If R has characteristic 2, then $a^2 \in Z$, and if $a \notin Z$ then $d(x) = (\lambda a)x - x(\lambda a)$ for all $x \in R$, where λ lies in the extended centroid of R.*

Away from characteristic 2 the answer is simple: a is central [1979-1, p. 509]. In characteristic 2 the derivation is inner, and the element that induces it lies in the extended centroid [1979-1, p. 509]. The extended centroid also appeared in the characteristic 2 case of Section 3.1. Section 3.3 explains where it lives.

Brešar recalls the theorem of Posner: a prime ring with a nonzero centralizing derivation is commutative [1993-1, p. 385]. He adds that Mayne proved the analogous result for centralizing automorphisms [1976-1, p. 113]. A map F of a ring is *centralizing* on a set S of elements if $[s, F(s)]$ is central for all $s \in S$, and *commuting* if $[s, F(s)] = 0$.

Brešar lists many authors who extended both theorems to further maps and sets of elements [1993-1, p. 385]. He then describes all additive maps that are centralizing on a prime ring.

Theorem 3.10 ([1993-1, Theorem A, p. 385])**.** *Let R be a prime ring and let F be an additive map of R into itself that is centralizing on R. If R has characteristic different from 2, or if F is commuting on R, then $F(x) = \lambda x + \zeta(x)$ for $x \in R$, where λ is an element of the extended centroid C of R and ζ is an additive map of R into C.*

Theorem 3.2 is the case of a derivation. Brešar also treats two derivations at once [1993-1, Theorem B, p. 386].

Theorem 3.11 ([1993-1, Theorem B, p. 386])**.** *Let R be a prime ring and let U be a nonzero left ideal of R. Let d and g be derivations of R such that $d(u)u - ug(u)$ is central for all $u \in U$. If $d \neq 0$, then R is commutative.*

Bell and Martindale had proved the case of one derivation when U is a one-sided ideal [1993-1, p. 386].

The extended centroid enters the proofs of Brešar when an identity such as $axb = bxa$ for all x forces $b = \lambda a$ [1993-1, Remark 6, p. 387]. Section 3.3 turns to the ring in which such a λ lives.

3.3 Martindale quotients and the extended centroid

The theorems of Sections 3.1 and 3.2 ended with an element λ of the extended centroid [1969-2, p. 577], which is not in the ring. This section tells where that field comes from.

Martindale built it in 1969 [1969-2]. He wanted one proof for a theorem of Posner on prime rings with a polynomial identity and a theorem of

Amitsur on primitive rings [1969-2, p. 576]. For this he constructs, for any prime ring R, a ring of quotients Q whose center is a field [1969-2, p. 576].

The construction uses the nonzero two-sided ideals of R [1969-2, p. 576]. He takes all R-homomorphisms from such an ideal into R, and calls two of them equivalent if they agree on a nonzero ideal. The classes form a ring Q, and R is a subring of Q because R is prime [1969-2, p. 577].

For each $q \in Q$ there is a nonzero ideal U of R with $qU \subseteq R$ [1969-2, p. 577]. Further, Q is a prime ring [1969-2, p. 577].

Martindale shows that the center C of Q is a field [1969-2, p. 577]. He calls it the *extended centroid* of R, and he calls $S = RC$ the *central closure* of R [1969-2, p. 577].

We state the tool that the previous sections used.

Theorem 3.12 ([1969-2, Theorem 1, p. 577])**.** *Let R be a prime ring with extended centroid C and central closure S. Let $a, b \in S$ be such that $axb = bxa$ for all $x \in R$. Then a and b are C-dependent.*

Herstein uses this result of Martindale [1979-1, p. 510], and so does Brešar [1993-1, p. 387]. Thus it produces the element λ of the earlier theorems. Then an identity $axb = bxa$ forces $b = \lambda a$ with λ in C.

Kharchenko defines the Martindale ring of quotients of a semiprime ring [1991-5, Section 1.4, p. 19].

Let F be the set of ideals of R with zero annihilator. The classes of homomorphisms from ideals in F into R form the left Martindale ring of quotients R_F [1991-5, 1.4.8, p. 21].

Theorem 3.13 ([1991-5, Theorem 1.4.9, p. 22])**.** *Let R be a semiprime ring. The ring R_F has the following properties.*

1. *For every $a \in R_F$ there is an ideal $I_a \in F$ with $I_a a \subseteq R$.*
2. *If $a \in R_F$ and $Ia = 0$ for some $I \in F$, then $a = 0$.*
3. *If $I \in F$ and $\varphi\colon I \to R$ is a homomorphism of left R-modules, then there is $r \in R_F$ with $\varphi(i) = ir$ for all $i \in I$.*

Kharchenko calls the center of R_F the *generalized centroid* of R [1991-5, 1.5.1, p. 24]. For a prime ring it is the extended centroid of Martindale. Lanski describes Q in the same way, as classes of homomorphisms from ideals into R [1988-3, p. 277]. He notes that the center C of Q is a field, and that C is also the centralizer of R in Q.

The quotient ring matters for derivations because derivations pass to it. Section 2.7 recorded that a derivation of a semiprime ring that is defined on an ideal extends in a unique way to a ring of quotients [1991-5, Proposition 1.8.1, p. 50]. Lanski states the same for a prime ring: each derivation has a unique extension to Q [1988-3, p. 277].

So a derivation of R can be studied inside Q, where the extended centroid C is a field. The question that Section 3.4 asks is which derivations are inner in Q, and what identities the values of a derivation can satisfy.

3.4 The theory of Kharchenko: differential identities and inner derivations

Section 3.3 gave a field C and a ring of quotients Q. We ask which identities the values of derivations can satisfy in a prime ring. Kharchenko asked this question in 1978 [1978-2, p. 155].

By a *differential identity* Kharchenko means an expression in which there appear, besides unknowns and ring operations, elements of the ring and derivations viewed as unary operations [1978-2, p. 155].

Some of these identities hold in every ring. Kharchenko lists five forms of them [1978-2, p. 155]. Three are the rule for a derivation of a product, the formula of an inner derivation, and the formula for the commutator of two derivations [1978-2, p. 155]. Two more are the rule for the p-th power of a derivation in characteristic p, and the rules for coefficients that are central [1978-2, p. 155]. Every generalized identity is also a differential identity [1978-2, p. 155].

We state the main result: nothing else occurs.

Theorem 3.14 ([1978-2, p. 155])**.** *Let R be a prime ring. Every multilinear differential identity of R, with coefficients in the ring of quotients of R, is a consequence of identities of the five forms above and of generalized identities satisfied in R.*

Let a prime ring satisfy a differential identity that is not a consequence of the five forms. Then it satisfies a nontrivial generalized identity [1978-2, p. 155]. By the theorem of Martindale, its central closure is a primitive ring with nonzero socle [1978-2, p. 155]. Its skew field is finite dimensional over the center.

The book of Kharchenko gives the same result in the language of the previous section. It shows that a differential identity of a prime ring R,

written in reduced form, yields a generalized identity on the ring of quotients R_F [1991-5, Theorem 2.4.1, pp. 112–113].

A derivation of R is *inner for the quotient ring* if its extension to Q is an inner derivation of Q [1978-2, p. 156]. The book proves a lemma on such derivations [1991-5, Lemma 2.4.2, p. 113].

Theorem 3.15 ([1991-5, Lemma 2.4.2, p. 113])**.** *Let R be a prime ring that satisfies a nontrivial generalized identity, and let C be its generalized centroid. Then every derivation of R that is zero on C is inner for the ring of quotients.*

Thus two derivations of R that agree on C differ by a derivation that is inner for Q [1991-5, p. 114]. So the derivations that R induces on C show how independent the derivations of R are [1991-5, p. 114].

In characteristic zero, every algebraic derivation is inner for the ring of quotients [1978-2, p. 155]. Section 3.5 returns to algebraic derivations.

Kharchenko adds two results in his introduction [1978-2, p. 155]. He shows that the outer derivations generate the universal enveloping algebra of their Lie algebra. He also shows that every restricted Lie algebra is the algebra of outer derivations of some prime ring.

Lanski used this method for the theorems of Posner [1988-3, p. 276]. He shows that the derivations involved are inner. Then the ring satisfies a generalized polynomial identity. The theorem of Martindale reduces the problem to matrix rings over a field [1988-3, p. 276].

Section 3.5 applies this reduction to Engel conditions and nilpotent derivations.

A derivation can be constrained by a polynomial rather than a product. Lee and Shiue take a polynomial that is not central-valued and two derivations d and δ whose action on its values differs by a central element, and prove that one of the derivations is then central [1998-6, Abstract].

3.5 Engel conditions, nilpotent and algebraic derivations

Section 3.4 reduced many problems to the case of an inner derivation. This section asks what a derivation does when its values are nilpotent, and when it satisfies an Engel condition.

Giambruno and Herstein recall that, in a prime ring R, if $a \in R$ and $(ax - xa)^n = 0$ for all $x \in R$ with n fixed, then a is central [1981-3, p. 199].

In other words, an inner derivation with nilpotent values is zero [1981-3, p. 199].

They ask whether this is special to inner derivations [1981-3, p. 199]. The question became a conjecture: a derivation d of a prime ring with $d(x)^n = 0$ for all x is zero [1981-3, p. 199]. They prove the conjecture, and they prove it for semiprime rings [1981-3, p. 199].

Theorem 3.16 ([1981-3, p. 199])**.** *Let R be a semiprime ring and let d be a derivation of R such that $d(x)^n = 0$ for all $x \in R$, where $n > 1$ is a fixed integer. Then $d = 0$.*

The proof is by induction on n [1981-3, p. 200]. It uses a theorem of Levitzki [1981-3, p. 200]. A left ideal in which every element has m-th power zero, for a fixed m, is zero [1981-3, Remark 4, p. 200].

The key step shows that d maps the left annihilator of an element into itself [1981-3, Lemma 1, p. 200].

We turn to a nilpotent derivation. For $d^2 = 0$, the theorem of Posner on compositions gives $d = 0$ in a prime ring of characteristic not 2 [1983-4, p. 179]. Herstein notes that the same holds in a semiprime ring [1979-1, p. 509].

Herstein replaced the zero value by a central power, starting from the inner case. For an inner derivation $d(x) = ax - xa$ with a not central, if $d(x)^n$ is central for all x, then R is an order in a four-dimensional simple algebra [1982-2, p. 163]. He then extended this from inner derivations to arbitrary ones [1982-2, p. 163].

Theorem 3.17 ([1982-2, p. 163])**.** *Let R be a prime ring with center Z and let $d \neq 0$ be a derivation of R such that $d(x)^n \in Z$ for all $x \in R$, for a fixed n. Then R is commutative, or R is an order in a four-dimensional simple algebra.*

The proof reduces the ring to a simple one and then to a division ring, and it uses the result of Giambruno and Herstein when the center is zero [1982-2, p. 163]. Two lemmas on multilinear polynomials carry the reduction [1982-2, Lemmas 1 and 2, p. 163]. The critical case is a simple ring that has a zero-divisor; there Herstein shows that the ring is the ring of 2×2 matrices over a field [1982-2, Theorem 3, p. 169]. A field extension then puts the standard identity of degree four on the ring [1982-2, p. 170].

Martindale and Miers studied the powers of a derivation in general [1983-4, p. 179]. They ask when a power of an inner derivation is inner, and when the powers of two derivations commute. They describe the elements a

and b of a prime ring with $(\operatorname{ad} a)^n = \operatorname{ad} b$, and the derivations d and δ with $[d^n(x), \delta^n(y)] = 0$ for all x and y [1983-4, p. 179]. They recall that, in a simple ring, if $(\operatorname{ad} a)^n = 0$ then there is a scalar λ with $(a - \lambda)^{[(n+1)/2]} = 0$ [1983-4, p. 179].

We turn to the Engel condition, which replaces the power of a value by a repeated commutator. Put $[y, x]_1 = yx - xy$ and $[y, x]_k = [[y, x]_{k-1}, x]$.

The theorem of Posner is the case that $[D(x), x]$ is central [1993-9, p. 731]. Vukman showed that R is commutative if $[D(x), x]_2 = 0$ and the characteristic is not 2 [1993-9, p. 731].

For the general condition Lanski used differential identities [1993-9, p. 731]. For an ideal the answer is complete.

Theorem 3.18 ([1993-9, Theorem 1, p. 732])**.** *Let R be a prime ring and let D be a nonzero derivation of R. If I is a nonzero ideal of R and $[D(r), r]_k = 0$ for all $r \in I$, with $k > 0$ fixed, then R is commutative.*

The proof uses a dichotomy of Kharchenko: either $[y, x]_k$ is an identity of R or D is inner for the ring of quotients [1993-9, p. 732].

The first case gives a polynomial identity, and the second case reduces to a matrix computation [1993-9, p. 732].

For a noncommutative Lie ideal the conclusion changes.

Theorem 3.19 ([1993-9, Abstract, p. 731])**.** *Let R be a prime ring, L a noncommutative Lie ideal of R, and D a nonzero derivation of R. If $[D(x), x]_k = 0$ for all $x \in L$, with k fixed, then* $\operatorname{char} R = 2$ *and R embeds in the matrices $M_2(F)$ over a field F.*

Lanski also studies derivations that are nilpotent on other sets of elements [1992-4, p. 1427]. The sets are Lie ideals, right ideals, and symmetric or skew elements [1992-4, p. 1427].

He shows that such a derivation is inner in the quotient ring [1992-4, p. 1427]. He also shows that it comes from a nilpotent element of about half the index [1992-4, p. 1427].

Section 3.6 asks which sets of elements of a ring can carry such conditions.

3.6 Derivations acting on ideals and Lie ideals

Section 3.5 studied a derivation on the whole ring or on an ideal. We now ask how a derivation acts on a Lie ideal. A *Lie ideal* of R is an additive subgroup U with $[U, R] \subseteq U$ [1988-3, p. 280].

Herstein began to study the Jordan and Lie ideals of a simple ring in the early 1950s [1986-1, p. 14]. The work was then carried to prime and semiprime rings, and Lanski was a major force in it [1986-1, p. 14].

Bergen, Herstein and Kerr took up derivations and Lie ideals together [1981-2, p. 259]. They measure the size of $d(U)$ in two ways [1981-2, p. 259]. The first way is the centralizer of $d(U)$: the bigger $d(U)$ is, the smaller its centralizer should be [1981-2, p. 259]. The other way asks whether the subring generated by $d(U)$ contains a nonzero ideal of R [1981-2, p. 259].

They work in a prime ring R of characteristic not 2, with center Z [1981-2, p. 260]. Their first tool turns a Lie ideal into an ideal.

Theorem 3.20 ([1981-2, Lemma 1, p. 260])**.** *Let R be a prime ring of characteristic not 2 and let $U \not\subseteq Z$ be a Lie ideal of R. Then there is an ideal M of R with $[M, R] \subseteq U$ and $[M, R] \not\subseteq Z$.*

We state the first theorem, on the case $d^2(U) = 0$.

Theorem 3.21 ([1981-2, Theorem 1, p. 261])**.** *Let R be a prime ring of characteristic not 2, let $d \neq 0$ be a derivation of R, and let U be a Lie ideal of R with $d^2(U) = 0$. Then $U \subseteq Z$.*

We state the second theorem, on the centralizer of $d(U)$.

Theorem 3.22 ([1981-2, Theorem 2, p. 262])**.** *Let R be a prime ring of characteristic not 2, let $U \not\subseteq Z$ be a Lie ideal of R, and let $d \neq 0$ be a derivation of R. Then the centralizer of $d(U)$ in R is Z.*

We state the third theorem, on the size of $d(U)$.

Theorem 3.23 ([1981-2, Theorem 3, p. 265])**.** *Let R be a prime ring of characteristic not 2, let $U \not\subseteq Z$ be a Lie ideal of R, and let d be a derivation of R with $d^3 \neq 0$. Then the subring generated by $d(U)$ contains a nonzero ideal of R.*

We state the last theorem, on the product of two derivations.

Theorem 3.24 ([1981-2, Theorem 4, p. 266])**.** *Let R be a prime ring of characteristic not 2 and let $U \not\subseteq Z$ be a Lie ideal of R. Let δ and d be derivations of R with $\delta d(U) = 0$. Then $d = 0$ or $\delta = 0$.*

Theorem 3.21 is a special case of Theorem 3.24 with $\delta = d$ [1981-2, p. 266]. Several authors proved that $d = 0$ or $h = 0$ when the product of two derivations vanishes on a Lie ideal or lies in the center [1988-3, p. 275].

Lanski removed the restriction on the characteristic [1988-3, p. 276]. He treats a noncommutative Lie ideal in every characteristic. He notes one exception, a ring of characteristic 2 that satisfies the standard identity of degree four. Section 3.1 stated these results.

Awtar had already put the hypothesis of the second theorem of Posner on a Lie ideal and on a Jordan ideal [1973-1, p. 67]. He takes a nonzero derivation d of a prime ring and asks that $[u, d(u)]$ lie in the center for every u in the ideal [1973-1, p. 68]. A Jordan ideal is an additive subgroup U with $u \circ r \in U$ for all $u \in U$ and $r \in R$, where $u \circ r = ur + ru$ [1973-1, p. 71].

Theorem 3.25 ([1973-1, Theorem 1, p. 68])**.** *Let R be a prime ring of characteristic different from 2 and 3, let d be a nonzero derivation of R, and let U be a Lie ideal of R with $[u, d(u)] \in Z$ for all $u \in U$. Then $U \subseteq Z$.*

Awtar notes that the exclusion of characteristic 3 does not enter the proof after the first step [1973-1, p. 71]. So he states the characteristic 3 case separately for a square closed Lie ideal, one with $u^2 \in U$ for all $u \in U$.

Theorem 3.26 ([1973-1, Theorem 2, p. 71])**.** *Let R be a prime ring of characteristic 3 and d a nonzero derivation of R. If U is a Lie ideal of R with $[u, d(u)] \in Z$ and $u^2 \in U$ for all $u \in U$, then $U \subseteq Z$.*

The same conclusion holds for a Jordan ideal away from characteristic 2 [1973-1, p. 71].

Theorem 3.27 ([1973-1, Theorem 3, p. 71])**.** *Let R be a prime ring of characteristic not 2, let d be a nonzero derivation of R, and let U be a Jordan ideal of R with $[u, d(u)] \in Z$ for all $u \in U$. Then $U \subseteq Z$.*

In characteristic 2 the Lie and Jordan ideals coincide, and Awtar proves a weaker statement for a Lie ideal that is also a subring [1973-1, p. 72].

Theorem 3.28 ([1973-1, Theorem 4, p. 72])**.** *Let R be a prime ring of characteristic 2, let d be a nonzero derivation of R, and let U be a Lie ideal and a subring of R with $[u, d(u)] \in Z$ for all $u \in U$. Then U is commutative.*

The condition on Z describes the size of the set of values. Brešar and Vukman measured that set directly [1993-5, p. 133]. They take the subring generated by all the elements $[D(x), x]$, and they prove that it contains a one-sided ideal [1993-5, p. 133].

Theorem 3.29 ([1993-5, p. 133])**.** *Let R be a noncommutative prime ring of characteristic not 2, and let D be a nonzero derivation of R. Then U,*

the subring of R generated by all $[D(x), x]$ for $x \in R$, contains a nonzero left ideal of R and a nonzero right ideal of R.

A noncommutative prime ring has no nonzero central one-sided ideal, so the theorem strengthens the statement of Posner that U is not contained in the center [1993-5, p. 133]. The proof rests on a lemma about two derivations.

Lemma 3.30 ([1993-5, Lemma 1, p. 134])**.** *Let R be a prime ring of characteristic not 2. If there exist nonzero derivations D and G of R such that $G([D(x), x]) = 0$ for all $x \in R$, then R is commutative.*

The lemma fails in characteristic 2, and the authors give an example with an inner derivation [1993-5, p. 134]. They close with a question: does U contain a nonzero two-sided ideal [1993-5, p. 141]?

Gölbaşı and Kaya carried the same questions to a generalized derivation [2006-4, p. 862]. On a prime ring of characteristic not 2 they show that the values control the Lie ideal in three ways [2006-4, Theorem 1, p. 864]. The first two results are the following.

Theorem 3.31 ([2006-4, Theorem 1, p. 864])**.** *Let R be a prime ring of characteristic not 2, let U be a nonzero Lie ideal of R, and let f be a generalized derivation of R associated with a derivation d. If $f^2(U) = 0$, then $U \subseteq Z$.*

Their lemma that a nonzero f cannot vanish on a noncentral Lie ideal stands before it [2006-4, Lemma 5, p. 863]. They also treat a map that acts as a homomorphism on the Lie ideal, under the extra condition $u^2 \in U$ for all $u \in U$ [2006-4, p. 864].

Theorem 3.32 ([2006-4, Theorem 2, p. 864])**.** *Let R be a prime ring of characteristic not 2, let U be a Lie ideal of R with $u^2 \in U$ for all $u \in U$, and let f be a generalized derivation of R associated with a derivation d. If $f(uv) = f(u)f(v)$ for all $u, v \in U$, then $U \subseteq Z$.*

The same conclusion holds when f acts as an antihomomorphism on U, and when $f(uv) = f(vu)$ for all $u, v \in U$ [2006-4, Theorems 3 and 4, p. 864].

A Jordan left derivation is the map of Definition 6.53 with $D(u^2) = 2uD(u)$. Ashraf and Rehman put that identity on a square closed Lie ideal [2000-1, p. 200]. Their map d is additive on the whole ring and the identity is assumed only on the Lie ideal, not on all of R [2000-1, p. 205].

Theorem 3.33 ([2000-1, p. 205]). *Let R be a 2-torsion free prime ring and let U be a Lie ideal of R with $u^2 \in U$ for all $u \in U$. If $d\colon R \to R$ is an additive map with $d(u^2) = 2ud(u)$ for all $u \in U$, then $d(uv) = ud(v) + vd(u)$ for all $u, v \in U$.*

So on such a Lie ideal a Jordan left derivation obeys the full left rule. The proof uses the square closed condition at every step, since the sums and products it forms must stay in U [2000-1, p. 205].

Sandhu and Rehman replace the prime ring by a ring with a semiprime ideal and put the central values into that ideal [2025-18, p. 1]. Let T be a semiprime ideal of a ring R with $T \subsetneq I$ for a nonzero ideal I. A derivation d is *T-commuting* on a set K when $[d(x), x] \in T$ for all $x \in K$ [2025-18, p. 1].

Theorem 3.34 ([2025-18, Theorem 2.1, p. 2]). *Let R be a ring with a nonzero ideal I and a semiprime ideal T such that $T \subsetneq I$. If $\operatorname{char}(R/T) \neq 2$ and $[[d(x), x], x] \in T$ for all $x \in I$, then d is T-commuting on I.*

The authors note that their argument is shorter than the one of Vukman for the same identity on a prime ring [2025-18, p. 2]. Two corollaries recover the known statements over a prime ring [2025-18, p. 4]. The second of them concerns a noncommutative prime ring of characteristic not 2 on which $x \mapsto [d(x), x]$ is centralizing. There $d = 0$, and the exclusion of characteristic 3 from a theorem of Vukman is dropped [2025-18, Corollary 2.7, p. 6]. Their next theorem puts the values in the center of the quotient R/T instead of in T, and reaches the same conclusion [2025-18, Theorem 2.5, p. 5].

Section 3.7 asks the same question for the skew elements of a ring with involution, which form a Lie subring.

A Lie derivation can be tested on the products that vanish. Qi, Cui and Hou characterize the additive ξ-Lie derivations of a prime algebra by their action on ξ-Lie zero products [2011-12, pp. 669–682].

Further work in the prime-ring line, held in the collection: *Automorphisms and generalized skew derivations which are strong commutativity preserving on polynomials in prime and semiprime rings* [2016-2]. *Nonlinear strong commutativity preserving maps on skew elements of prime rings with involution* [2012-2]. *Structure of a quotient ring RP with generalized derivations acting on the prime ideal P and some applications* [2021-7]. *On certain functional equations related to Jordan -derivations in semiprime -rings and standard operator algebras* [2018-3]. *Identities involving b -generalized skew derivations in prime rings acting as Jordan derivation* [2024-3]. *Generalized derivations with skew-commutativity conditions on polynomials in prime*

rings [2019-2]. *Engel type identities with derivations for right ideals in prime rings* [2006-2]. *Differential identities involving three generalized derivations on prime rings and Banach algebras* [nd-1]. *Commuting ordinary differential operators with polynomial coefficients and automorphisms of the first Weyl algebra* [2016-3]. *Commutativity theorems for -prime rings with differential identities on Jordan ideals* [2012-1]. *Certain functional identities involving a pair of generalized skew derivations with nilpotent values on Lie ideals* [2023-2]. *Action of three generalized derivations on Lie ideals in prime rings with n-Engel condition* [2026-1]. *Multiplicative generalized-derivations acting on left sided ideals with annihilator conditions in semiprime rings* [2024-4]. *Identities involving skew Lie product and a pair of generalized derivations in prime rings with involution* [2021-2]. *Editorial From the editor-in-chief Comments on Jordan derivations of prime rings with characteristic two and Entropy-preserving maps on quantum states* [2017-1]. *b-Generalized skew derivations acting as a Jordan derivation on multilinear polynomials in prime rings* [2023-1]. *Strong commutativity and Engel condition preserving maps in prime and semiprime rings* [2012-5]. *Strong 3-skew commutativity preserving maps on prime rings with involution* [2023-4]. *Strong 2-skew Commutativity Preserving Maps on Prime Rings with Involution* [2019-3]. *Some Identities Related to Semiprime Ideal of Rings with Multiplicative Generalized Derivations* [2024-7]. *Product of generalized derivations of order 2 with derivations acting on multilinear polynomials with centralizing conditions* [2024-6]. *Prime ideals in differential operator rings and crossed products of infinite groups* [1987-1]. *Power-central values and Engel conditions in prime rings with generalized skew derivations* [2021-5]. *On skew Jordan product and generalized derivations in prime rings with involution* [2024-5]. *On prime and semiprime rings with generalized derivations and non-commutative Banach algebras* [2016-7]. *On n-commuting and n-skew-commuting maps with generalized derivations in prime and semiprime rings* [2011-1]. *On functional equations related to derivations in semiprime rings and standard operator algebras* [2012-4]. *On Jordan ideals with generalized left derivations in 3-prime near-rings* [2023-3]. *Nilpotent inner derivations of the skew elements of prime rings with involution* [1991-1]. *Jordan derivations on a Lie ideal of a semiprime ring and their applications in Banach algebras* [2016-5].

3.7 Rings with involution and *-derivations

Section 3.6 ended with the skew elements of a ring with involution. An *involution* is an anti-automorphism of period at most 2 [1986-1, p. 14]. The symmetric elements $s^* = s$ form a Jordan subring, and the skew elements $k^* = -k$ form a Lie subring [1986-1, p. 14]. We ask what a derivation does on these sets.

Herstein took the commuting condition of Section 3.2 to this setting [1982-1, p. 356]. He asks that $d(s)d(t) = d(t)d(s)$ only for symmetric s and t [1982-1, p. 356].

The full result of the earlier section cannot hold here [1982-1, p. 356].

Example 3.35 ([1982-1, p. 356])**.** Let R be the ring of 2×2 matrices over a field. For the symplectic involution, all symmetric elements are central, so the values of any derivation on symmetric elements commute. For the transpose, take

$$d(x) = xe_{11} - e_{11}x.$$

Then $d(x)d(y) = d(y)d(x)$ for symmetric x and y.

So Herstein proves a weaker conclusion.

Theorem 3.36 ([1982-1, Theorem 2, p. 366])**.** *Let R be a prime ring with involution, of characteristic not 2, and let $d \neq 0$ be a derivation of R with $d(s)d(t) = d(t)d(s)$ for all symmetric s and t. Then R is commutative, or R is an order in a simple algebra of dimension four.*

Lanski studied Lie ideals and derivations in rings with involution [1977-3, p. 449]. Let L be a Lie ideal. Suppose that the commutator of an element with L is central. Then the element or L is central, unless the ring is an order in a simple algebra of dimension at most four [1977-3, p. 449]. This holds for the symmetric part or the skew part of L [1977-3, p. 449].

Lanski applies this to derivations [1977-3, p. 449]. Suppose a derivation sends the symmetric or skew part of L into the center. Then the same three cases occur [1977-3, p. 449].

Theorem 3.37 ([1977-3, Theorem 9, p. 459])**.** *Let R be a prime ring of characteristic not 2 with an involution, let D be a derivation of R, and let L be a Lie ideal with $L^* = L$ such that $D(L)$ lies in the skew elements or in the symmetric elements. Then $L \subseteq Z$, or $D(R) = 0$, or R satisfies the standard identity of degree four.*

Section 3.4 has a version with involution. Every generalized identity with derivations and the involution follows from the generalized identities with involution that R satisfies [1985-1, p. 765]. This gives a general setting for the result of Herstein [1985-1, p. 765].

In characteristic zero, no outer derivation of R acts algebraically on the symmetric elements [1985-1, p. 765]. Suppose the images of the symmetric elements under the derivations satisfy a polynomial relation. Then R satisfies a generalized polynomial identity [1985-1, p. 765].

An involution is of the *second kind* when its symmetric elements meet the center in a nonzero set [2019-9, p. 1950001-1]. Idrissi and Oukhtite take the commuting condition of Section 3.2 and put the commutator of an element with its image in place of the element itself [2019-9, p. 1950001-4].

Theorem 3.38 ([2019-9, Theorem 1, p. 1950001-4])**.** *Let $(R, *)$ be a 2-torsion free prime ring with an involution of the second kind, and let F be a nonzero generalized derivation of R associated with a derivation d. Then R is commutative if and only if $F([x, x^*]) \in Z(R)$ for all $x \in R$.*

They argue the same way for the Jordan product, and again the ring has to commute [2019-9, p. 1950001-6].

Theorem 3.39 ([2019-9, Theorem 2, p. 1950001-6])**.** *Let $(R, *)$ be a 2-torsion free prime ring with an involution of the second kind. If R admits a nonzero generalized derivation F associated with a derivation d, then R is commutative if and only if $F(x \circ x^*) \in Z(R)$ for all $x \in R$.*

A corollary replaces the condition $F([x, x^*]) \in Z(R)$ by $F([x, y]) \pm [x, y] \in Z(R)$ for all x and y, and adds that the three statements are equivalent [2019-9, Corollary 1, p. 1950001-5]. The authors also note that an earlier result for a square closed Lie ideal fails here. When the ideal is the whole ring, that result would force the ring to be commutative [2019-9, p. 1950001-5].

Section 3.8 turns from Lie ideals to Jordan ideals.

Nonlinearity does not weaken the conclusion. On a prime ring with involution whose skew elements are K, a map f with $[f(x), f(y)] = [x, y]$ for all $x, y \in K$ is additive and preserves the commutator [2012-3, Abstract].

3.8 Jordan ideals and the Jordan structure of a prime ring

Section 3.7 met the symmetric elements, which form a Jordan subring. This section follows the Jordan side of the story. A ring A becomes a Jordan

algebra under the product $a \circ b = ab + ba$ [2005-3, p. 1].

An additive subgroup J of R is a *Jordan ideal* if $u \circ r \in J$ for all $u \in J$ and $r \in R$ [2011-11, p. 415]. The standard problem is to describe Jordan ideals in terms of ordinary ideals [2005-3, p. 1]. One asks for conditions under which every nonzero Jordan ideal is an ideal, or at least contains a nonzero ideal [2005-3, p. 1]. The problem has been studied for more than fifty years, and its basic results are due to Herstein [2005-3, p. 1].

Brešar, Fošner and Fošner give a short route to such results, based on an identity that links the Jordan product and the Lie product [2005-3, p. 1].

Theorem 3.40 ([2005-3, Theorem 2.1, p. 2])**.** *Let A be an associative algebra over a commutative unital ring that contains $\frac{1}{2}$, and let J be a Jordan ideal of A. Then J contains the set* $\mathrm{Alg}([A, A])\, J$*. In particular, J contains the ideal* $\mathrm{Id}([[A, A], A])\, J$*.*

We turn to derivations on a ring with involution. Such a ring is *$*$-prime* if $aRb = aRb^* = 0$ implies $a = 0$ or $b = 0$ [2011-11, p. 415].

A prime ring with an involution is $*$-prime [2011-11, p. 415]. The converse fails [2011-11, p. 415]. For example, the ring $R \times R^o$ with the exchange involution is $*$-prime and not prime.

Oukhtite extended the second theorem of Posner to Jordan ideals in such a ring [2011-11, p. 415].

Theorem 3.41 ([2011-11, Theorem 1, p. 416])**.** *Let $(R, *)$ be a 2-torsion free ring with involution, let J be a nonzero $*$-Jordan ideal of R, and let d be a nonzero derivation that is centralizing on J. If R is $*$-prime, then R is commutative.*

Oukhtite and Mamouni study derivations that satisfy other algebraic identities on Jordan ideals [2012-12, p. 341]. They extend several results for derivations of prime rings to Jordan ideals [2012-12, p. 341]. They also give an example that shows that the hypothesis that R is $*$-prime cannot be dropped [2012-12, p. 341].

The same authors put a generalized derivation on a Jordan ideal and compare its values with the products of the ideal [2015-11, p. 1]. They work in a 2-torsion free prime ring. The condition is that a difference of the form $F(xy) - xy$, $F(xy) - yx$, $F(x)F(y) - xy$ or $F(x)F(y) - yx$ be central for all x and y in the ideal [2015-11, p. 1].

Theorem 3.42 ([2015-11, Theorem 1, p. 3])**.** *Let R be a 2-torsion free prime ring and J a nonzero Jordan ideal of R. If R admits a generalized derivation*

F associated with a nonzero derivation d such that $F(xy) - xy \in Z(R)$ for all $x, y \in J$, then R is commutative.

The other three conditions give the same conclusion [2015-11, Theorems 2 and 3, pp. 3–5]. Each proof first shows that the Jordan ideal meets the center, and then reduces the identity to one on the whole ring [2015-11, pp. 3–4].

A weaker condition allows the Jordan identity to fail by a central element. An additive map δ of R into the right Martindale quotient ring is a *weak Jordan derivation* [2020-16, p. 1] when $\delta(x^2) - x\delta(x) - \delta(x)x$ lies in the extended centroid for every $x \in R$. Lin proves that on a prime ring with $\dim_C RC > 4$ such a map is a derivation when the characteristic is not two [2020-16, Theorem 1.2]. When the characteristic is two, it is a derivation plus a map into the centroid [2020-16, Theorem 1.2]. So the relaxation buys exactly one centroid-valued summand.

Theorems 3.2, 3.8, 3.36 and 3.41 have a conclusion of one kind: the ring is commutative, or an order in a simple algebra of dimension four. Each application of such a result sits in the chapter that owns its notions.

Chapter 4

Jordan derivations and Jordan-type maps

Chapter 3 ended with Jordan ideals of a prime ring. This chapter asks what the Jordan product $a \circ b = ab + ba$ does to the product rule. Herstein defines a Jordan derivation as a derivation of the Jordan ring of an associative ring, and he credits the idea to Kaplansky [1957-1, p. 1104]. Every derivation is a Jordan derivation, and Herstein asks about the converse [1957-1, p. 1104].

Section 4.1 gives his answer for prime rings, with the extension of Cusack [1975-1, p. 323]. Section 4.2 follows the semiprime case. Brešar proved the semiprime theorem in 1988 [1988-2, p. 1003]. In 2005 he showed that a Jordan derivation into a bimodule is a derivation on a part of the ring [2005-2, p. 411]. The implications between the three kinds of map are shown in Figure 4.1.

Three families of maps change the rule. Jordan triple derivations state it for xyx [2008-5, p. 416]. Fošner and Ilišević show that on a 2-torsion free semiprime ring a generalized Jordan triple derivation is a generalized derivation [2008-5, Corollary 4.2, p. 424]. Generalized Jordan derivations use a second map [2020-12, p. 1]. The (m, n)-Jordan derivations weight the two terms [2014-18, p. 1881]. Section 4.3 treats these.

Section 4.4 looks at particular algebras, among them triangular algebras [2007-5, p. 1]. Section 4.5 relates the problem to Jordan homomorphisms, which Brešar and Zelmanov survey [2025-4, p. 203]. Section 4.6 ends the chapter with maps defined by an involution [2008-5, p. 416] and with Jordan left derivations [1997-5, p. 1].

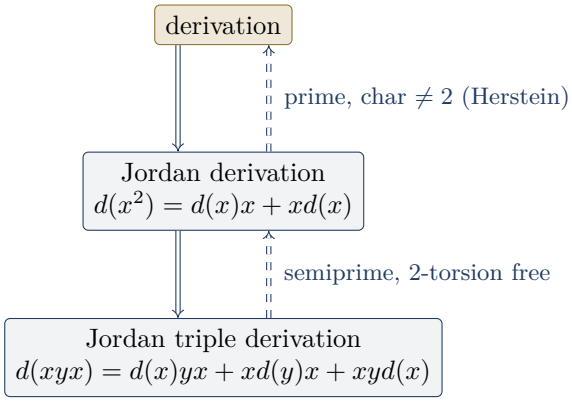


Figure 4.1: Every derivation is a Jordan derivation and every Jordan derivation is a Jordan triple derivation. The dashed converses hold under the conditions shown.

4.1 The theorem of Herstein and its proofs

Chapter 3 ended with Jordan ideals of a prime ring. We now ask what the Jordan product $a \circ b = ab + ba$ does to the product rule itself.

Herstein begins with this product. From an associative ring A one builds the Jordan ring of A by defining $a \circ b = ab + ba$ [1957-1, p. 1104]. A *Jordan derivation* of A is a derivation of this Jordan ring. It is an additive map $'$ with $(a \circ b)' = a' \circ b + a \circ b'$ [1957-1, p. 1104]. Written out, this is $(ab+ba)' = a'b+ab'+b'a+ba'$ [1957-1, p. 1104]. Every derivation is a Jordan derivation. Herstein asks about the converse. He records that Kaplansky first raised the idea of a Jordan derivation in conversation [1957-1, p. 1104, footnote].

The obvious rival is a map $*$ with $(ab)^* = b^*a + ba^*$, which Herstein calls a *reverse derivation* [1957-1, pp. 1104–1105]. Such a map would be a Jordan derivation. Herstein first shows that it hardly ever occurs.

Theorem 4.1 ([1957-1, Theorem 2.1, p. 1105])**.** *Let A be a prime ring and suppose that $*$ is a nonzero reverse derivation of A. Then A is a commutative integral domain and $*$ is an ordinary derivation of A.*

The main result needs two lemmas. In Section 3 of his paper Herstein assumes that A is a prime ring of characteristic different from 2.

Lemma 4.2 ([1957-1, Lemma 3.1, p. 1106])**.** *If $'$ is a Jordan derivation of A, then for all $a, b \in A$, $(aba)' = a'ba + ab'a + aba'$.*

The next lemma records the key identity of the proof.

Lemma 4.3 ([1957-1, Lemma 3.3, p. 1107]). *For all $a, b \in A$, $((ab)' - a'b - ab')(ab - ba) = 0$.*

Herstein proves Lemma 4.3 by expanding $(ab(ab) + (ab)ba)'$ in two ways [1957-1, p. 1107]. He then treats commuting and noncommuting pairs a, b separately [1957-1, pp. 1107–1110]. The fact that A is prime finishes the argument.

Theorem 4.4 ([1957-1, Theorem 3.1, p. 1110]). *If A is a prime ring of characteristic different from 2, then any Jordan derivation of A is an ordinary derivation of A.*

Herstein notes that the characteristic is used in only two places: in the proof of Lemma 4.2 and in the commutative case [1957-1, p. 1110]. So he changes the definition of a Jordan derivation. He asks that $(a+b)' = a' + b'$, $(a^2)' = aa' + a'a$ and $(aba)' = a'ba + ab'a + aba'$. For characteristic different from 2 the third condition follows from the first two [1957-1, p. 1110]. With this definition he treats characteristic 2.

Theorem 4.5 ([1957-1, Theorem 4.1, p. 1110]). *Let A be a prime ring of characteristic 2. Then, if A is not a commutative integral domain, any Jordan derivation of A is an ordinary derivation.*

Eighteen years later Cusack dropped the hypothesis that the ring is prime. He works in a ring R in which $2x = 0$ implies $x = 0$, and he wants to enlarge the class of rings for which the converse holds [1975-1, p. 321]. For a Jordan derivation D he sets $d(a, b) = D(ab) - aDb - (Da)b$ [1975-1, p. 321]. His main step is that $[a, b] \circ d(a, b)$ and $[[a, b], r, d(a, b)]$ vanish [1975-1, Lemma 2, p. 322]. He gets this from the same identities for Jordan homomorphisms [1975-1, Lemma 1, p. 322].

Theorem 4.6 ([1975-1, Theorem 4, pp. 322–323]). *Let D be a Jordan derivation on a ring R in which $2x = 0$ implies $x = 0$. Then for all a and b in R, $D(ab) - aDb - (Da)b$ is in L, the Baer lower radical of R.*

A ring is semiprime when its lower radical is zero [1975-1, p. 322]. Cusack states two consequences, the first in his setting.

Corollary 4.7 ([1975-1, Corollary 5, p. 323]). *If R is semiprime, then D is a derivation.*

The next theorem states the second consequence, for a ring with a commutator that is not a zero divisor.

Theorem 4.8 ([1975-1, Theorem 6, p. 323])**.** *Let R be a ring where $2x = 0$ implies $x = 0$ and which has a commutator which is not a zero divisor. Then every Jordan derivation on R is a derivation.*

Cusack also shows that some hypothesis is needed. Take a ring with an element x such that $axa = 0$ for all a but $axb \neq 0$ for some a, b. Then $Da = ax$ is a Jordan derivation that is not a derivation [1975-1, §3, p. 324]. Now take a ring with $a^2 = 0$ for all a and $ab \neq 0$ for some a, b. The Jordan product is then zero, so every additive map is a Jordan derivation, and the identity map is not a derivation [1975-1, §3, p. 324]. He leaves open the search for necessary conditions [1975-1, p. 324].

Finally, a short proof of Theorem 4.4 appeared in 1988. Brešar and Vukman write a^b for $D(ab) - D(a)b - aD(b)$ [1988-1, p. 321]. For a Jordan derivation D, they note that $b^a = -a^b$ [1988-1, p. 321]. Their key identity follows.

Theorem 4.9 ([1988-1, Theorem 3, p. 322])**.** *Let R be a ring of characteristic not two, and let $D\colon R \to R$ be a Jordan derivation. In this case for all $a, b, r \in R$ we have $a^b r(ab - ba) + (ab - ba)ra^b = 0$.*

They prove it by expanding $abrba + barab$ in two ways [1988-1, p. 322]. With Lemma 3.10 of the *Topics in ring theory* of Herstein they obtain Theorem 4.4 [1988-1, p. 322].

Section 4.2 follows the semiprime case further.

4.2 Cusack, Brešar and the semiprime case

Section 4.1 gave the theorems of Herstein and Cusack. Brešar returned to the problem twice. In 1988 he extended the theorem of Herstein to semiprime rings [1988-2, p. 1003]. In 2005 he studied Jordan derivations into a bimodule [2005-2, p. 411]. He showed that such a map is a derivation on a part of the ring [2005-2, p. 411].

The 1988 paper begins with its definitions. All rings are associative [1988-2, p. 1003].

Definition 4.10 ([1988-2, Preliminaries, p. 1003])**.** A ring R is said to be 2-torsion free if $2a = 0$ with $a \in R$ implies $a = 0$. A ring R is called a prime ring if $aRb = (0)$ implies $a = 0$ or $b = 0$. A ring R is called a semiprime

ring if $aRa = (0)$ implies $a = 0$. An additive mapping $'\colon R \to R$ is called a Jordan derivation if $(a^2)' = a'a + aa'$ holds for all $a \in R$.

Herstein proved that every Jordan derivation on a 2-torsion free prime ring is a derivation [1988-2, p. 1003]. Brešar extends this to semiprime rings [1988-2, p. 1003].

Theorem 4.11 ([1988-2, Theorem 1, p. 1003])**.** *Let R be a 2-torsion free semiprime ring and let $'\colon R \to R$ be a Jordan derivation. In this case, $'$ is a derivation.*

The proof takes several steps. Brešar follows Herstein and writes a^b for $(ab)' - a'b - ab'$ [1988-2, p. 1004]. The first step is an identity, which he records for completeness [1988-2, p. 1004].

Theorem 4.12 ([1988-2, Theorem 3, p. 1004])**.** *Let R be a 2-torsion free ring and let $'\colon R \to R$ be a Jordan derivation. In this case for all elements $a, b, x \in R$, $a^b x[a, b] + [a, b]xa^b = 0$ holds.*

A second lemma on pairs a, b supplies the rest.

Lemma 4.13 ([1988-2, Lemma 4, p. 1004])**.** *Let R be a 2-torsion free semiprime ring and let $a, b \in R$. If $axb + bxa = 0$ holds for all $x \in R$, then $axb = bxa = 0$ for all $x \in R$.*

Brešar proves Lemma 4.13 by using the hypothesis three times to reach $2(axb)y(axb) = 0$ [1988-2, p. 1004]. He then combines the lemma with Theorem 4.12 in several steps. These show that each a^b is central with $(a^b)^3 = 0$, so $a^b = 0$, since the center of a semiprime ring has no nonzero nilpotent elements [1988-2, p. 1005]. Two consequences are stated [1988-2, pp. 1005–1006].

Corollary 4.14 ([1988-2, Corollary 5, p. 1005])**.** *Let R be a 2-torsion free semisimple ring and let $'\colon R \to R$ be a Jordan derivation. In this case $'$ is a derivation.*

The second consequence concerns continuity.

Theorem 4.15 ([1988-2, Theorem 6, p. 1006])**.** *Let A be a Banach algebra and let $'\colon A \to A$ be a linear Jordan derivation. In this case $'$ is continuous.*

Brešar combines Corollary 4.14 with the theorem of Johnson and Sinclair that every linear derivation on a semisimple Banach algebra is continuous [1988-2, p. 1006]. This answers a question of Sinclair [1988-2, p. 1006].

The second paper allows values in a bimodule [2005-2, p. 411].

Definition 4.16 ([2005-2, Section 1, p. 411])**.** Let A be a ring and let M be an A-bimodule. An additive map $d\colon A \to M$ is called a Jordan derivation if $d(x^2) = d(x)x + xd(x)$ and $d(xyx) = d(x)yx + xd(y)x + xyd(x)$ for all $x, y \in A$.

When M is 2-torsion free, the second condition follows from the first [2005-2, p. 414]. Brešar measures the failure of the product rule by a map $\{x, y\} = d(xy) - d(x)y - xd(y)$. The map d is a derivation exactly when $\{A, A\} = 0$ [2005-2, p. 414]. The map $\{x, y\}$ satisfies the three identities of the next theorem [2005-2, p. 414]. For a ring A he sets $A^{(0)} = A$ and $A^{(n+1)} = [A^{(n)}, A^{(n)}]$ [2005-2, p. 412]. He writes $R(S)$ and $I(S)$ for the subring and the ideal of A generated by a subset S [2005-2, p. 412].

Theorem 4.17 ([2005-2, Theorem 2.1, p. 413])**.** *Let A be a ring and let M be an A-bimodule. If $\{\,.\,,\,.\,\}\colon A \times A \to M$ is a biadditive map such that $\{x, x\} = 0$, $\{xy, x\} + \{x, y\}x = 0$ and $\{xy, z\} + \{x, y\}z = \{x, yz\} + x\{y, z\}$ for all $x, y, z \in A$, then $\{R(A^{(2)}), A\} = 0$ (and hence $\{I(A^{(3)}), A\} = 0$).*

Applied to the map $\{x, y\}$ above, this gives the main result of the paper.

Theorem 4.18 ([2005-2, Theorem 3.1, p. 414])**.** *If d is a Jordan derivation from a ring A into an A-bimodule M, then $d(ux) = d(u)x + ud(x)$ for all $x \in A$ and all $u \in R(A^{(2)})$ (and so, in particular, for all $u \in I(A^{(3)})$).*

The theorem has an immediate consequence [2005-2, p. 414].

Corollary 4.19 ([2005-2, Corollary 3.2, p. 414])**.** *If a ring A is such that $R(A^{(2)}) = A$ (in particular, if $I(A^{(3)}) = A$), then every Jordan derivation from A into an arbitrary A-bimodule is a derivation.*

Brešar remarks that the theorem gives no information when A is commutative or close to it [2005-2, p. 411]. For a 2-torsion free semiprime ring he shows that $I(A^{(n)})$ is essential exactly when $Z(A)$ contains no nonzero ideal of A [2005-2, Lemma 3.7, p. 416]. This gives the next statement.

Corollary 4.20 ([2005-2, Corollary 3.8, p. 416])**.** *Let A be a 2-torsion free semiprime ring and let d be a Jordan derivation from A into an A-bimodule M. (i) If $Z(A)$ does not contain nonzero ideals of A, then the restriction of d to some essential ideal of A is a derivation. (ii) Suppose that M satisfies the following two conditions: (a) for every essential ideal E of A and every $m \in M$, $Em = 0$ implies $m = 0$, and (b) $cm = mc$ for all $c \in Z(A)$ and all $m \in M$. Then d is a derivation.*

Both conditions of part (ii) hold when $M = A$, so part (ii) extends the theorem of Cusack [2005-2, p. 416]. Brešar adds that the arguments of Cusack and of a later proof do not give (ii), since both compute with d^2 [2005-2, p. 416]. The next statement concerns the maximal ring of quotients [2005-2, Corollary 3.9, p. 417].

Corollary 4.21 ([2005-2, Corollary 3.9, p. 417])**.** *Let A be a 2-torsion free semiprime ring and let Q be the maximal left (or right) ring of quotients of A. Then every Jordan derivation from A into Q is a derivation.*

For a simple ring Brešar states a definitive conclusion [2005-2, p. 417].

Corollary 4.22 ([2005-2, Corollary 3.10, p. 417])**.** *Let A be a noncommutative simple ring with* $\operatorname{char}(A) \neq 2$*. Then every Jordan derivation from A into any A-bimodule is a derivation.*

Brešar also gives examples of proper Jordan derivations, that is, Jordan derivations that are not derivations [2005-2, p. 411]. One example changes the second example of Cusack [2005-2, §4.2, p. 419]. Let A be the 3-dimensional algebra over a field F generated by a and b with $a^2 = b^2 = 0$ and $ab = -ba \neq 0$ [2005-2, §4.2, p. 419]. Then $A^{(2)} = 0$ and $x \circ y = 0$ for all $x, y \in A$, so every linear map from A to itself is a Jordan derivation [2005-2, §4.2, p. 419]. Theorem 4.18 therefore gives no information here. The map $d(\lambda a + \mu b + \nu ab) = \nu a$ is not a derivation, and it has the following property [2005-2, §4.2, p. 419].

Example 4.23 ([2005-2, Example 4.2, p. 419])**.** There exists a 3-dimensional noncommutative algebra A admitting a proper Jordan derivation $d\colon A \to A$ such that for every $u \neq 0$ in A there is $x \in A$ satisfying $d(ux) \neq d(u)x + ud(x)$.

Section 4.3 turns to maps that satisfy a triple version of the product rule.

4.3 Jordan triple, generalized Jordan and Jordan n-derivations

Section 4.2 showed that a Jordan derivation on a 2-torsion free semiprime ring is a derivation. Three families of maps change the rule $d(x^2) = d(x)x + xd(x)$. Jordan triple derivations state a rule for xyx [2008-5, p. 416].

Generalized Jordan derivations use a second map [2020-12, p. 1]. The (m, n)-Jordan derivations of Vukman weight the two terms [2014-18, p. 1881]. This section takes each family in turn.

Fošner and Ilišević work with a ring R and an R-bimodule M [2008-5, p. 416]. They define the following maps.

Definition 4.24 ([2008-5, Section 1, p. 416])**.** Let R be a ring and let M be an R-bimodule. A Jordan triple derivation $\delta\colon R \to M$ is an additive mapping satisfying $\delta(xyx) = \delta(x)yx + x\delta(y)x + xy\delta(x)$ for all $x, y \in R$. An additive mapping $E\colon R \to M$ is called a generalized derivation if there exists a derivation $\delta\colon R \to M$ such that $E(xy) = E(x)y + x\delta(y)$ for all $x, y \in R$, and it is called a generalized Jordan derivation if there is a Jordan derivation $\delta\colon R \to M$ such that $E(x^2) = E(x)x + x\delta(x)$ holds for all $x \in R$. An additive mapping $E\colon R \to M$ is called a generalized Jordan triple derivation if there exists a Jordan triple derivation $\delta\colon R \to M$ such that $E(xyx) = E(x)yx + x\delta(y)x + xy\delta(x)$ for all $x, y \in R$.

Every derivation is a Jordan derivation, and every generalized derivation is a generalized Jordan derivation [2008-5, p. 416]. Let the ring be 2-torsion free. Then every Jordan derivation is a Jordan triple derivation [2008-5, p. 416]. Also every generalized Jordan derivation is a generalized Jordan triple derivation with the same related map [2008-5, p. 416]. For semiprime rings Fošner and Ilišević study a pair (E, F) of additive maps [2008-5, p. 423].

Theorem 4.25 ([2008-5, Theorem 4.1, p. 424])**.** *Let R be a 2-torsion free semiprime ring and let $E, F\colon R \to R$ be additive mappings. Then the following conditions are mutually equivalent: (a) for all $x, y \in R$, $E(xyx) = E(x)yx + xF(y)x + xyF(x)$ and $F(xyx) = F(x)yx + xF(y)x + xyF(x)$; (b) for all $x, y, z \in R$, $E(xyz) = E(x)yz + xF(y)z + xyF(z)$ and $F(xyz) = F(x)yz + xF(y)z + xyF(z)$; (c) for all $x, y \in R$, $E(xy) = E(x)y + xF(y)$ and $F(xy) = F(x)y + xF(y)$; (d) for all $x \in R$, $E(x^2) = E(x)x + xF(x)$ and $F(x^2) = F(x)x + xF(x)$.*

A direct consequence is the answer for generalized Jordan triple derivations [2008-5, p. 424]. It proves a conjecture posed in an earlier paper [2008-5, p. 424].

Corollary 4.26 ([2008-5, Corollary 4.2, p. 424])**.** *Let R be a 2-torsion free semiprime ring. Then any generalized Jordan triple derivation on R is a generalized derivation.*

Setting $F = 0$ in Theorem 4.25 gives a statement about left centralizers, which are the maps T with $T(xy) = T(x)y$ [2008-5, p. 424].

Corollary 4.27 ([2008-5, Corollary 4.3, p. 424])**.** *Let R be a 2-torsion free semiprime ring and let $T\colon R \to R$ be an additive mapping. Then the following conditions are mutually equivalent: (a) $T(xyx) = T(x)yx$ $(x, y \in R)$; (b) $T(xyz) = T(x)yz$ $(x, y, z \in R)$; (c) $T(xy) = T(x)y$ $(x, y \in R)$; (d) $T(x^2) = T(x)x$ $(x \in R)$.*

Ferreira, Guzzo and Ferreira define a generalized Jordan derivation δ with a relating Jordan derivation τ [2020-12, p. 1]. Jing and Lu introduced this notion [2020-12, p. 2]. They stated the conjecture that every generalized Jordan derivation on a 2-torsion free semiprime ring is a generalized derivation [2020-12, p. 2]. Ferreira, Guzzo and Ferreira prove it when the ring has a nontrivial idempotent with mild conditions [2020-12, p. 2]. The conditions follow the statement on the same page [2020-12, p. 2].

Theorem 4.28 ([2020-12, Theorem 2.1, p. 2])**.** *Let R be a 2-torsion free semiprime unity ring and δ be an additive generalized Jordan derivation from R into itself. If there exist an idempotent e so that $e \neq 0$, $e \neq 1$ in R, then δ is an additive generalized derivation.*

The proof rests on the Peirce decomposition $R = R_{11} \oplus R_{12} \oplus R_{21} \oplus R_{22}$ with respect to e [2020-12, p. 2]. It also rests on a lemma of Jing and Lu [2020-12, p. 3].

Lemma 4.29 ([2020-12, Lemma 2.1, p. 3])**.** *For all $a, b, c \in R$, the following statements hold: (i) $\delta(ab+ba) = \delta(a)b+a\tau(b)+\delta(b)a+b\tau(a)$; (ii) $\delta(aba) = \delta(a)ba + a\tau(b)a + ab\tau(a)$; (iii) $\delta(abc + cba) = \delta(a)bc + a\tau(b)c + ab\tau(c) + \delta(c)ba + c\tau(b)a + cb\tau(a)$.*

Majeed and Özel recall that Vukman introduced the third family [2014-18, p. 1881]. Let $m \geq 1$ and $n \geq 1$ be fixed integers with $m \neq n$, and let A be an algebra [2014-18, p. 1881]. A nonzero additive mapping $\delta\colon A \to A$ with $(m+n)\delta(x^2) = 2m\delta(x)x + 2nx\delta(x)$ for all $x \in A$ is called an (m, n)-Jordan derivation [2014-18, p. 1881]. Their first lemma shows how such a map acts on idempotents.

Lemma 4.30 ([2014-18, Lemma 2.1, p. 1882])**.** *Let A be a unital algebra. If δ is an (m, n)-Jordan derivation from A into itself, then for each idempotent $P \in A$, $(m+n)\delta(P) = 2m\delta(P)P + 2nP\delta(P)$.*

Majeed and Özel then work in algebras of operators that leave the members of a commutative subspace lattice invariant [2014-18, p. 1881].

The additive form of each of these notions has a common setting in axial algebras [2025-10, p. 1]. An axial algebra carries a distinguished set of idempotents whose products follow a fusion rule [2025-10, Definition 2.3, p. 5]. A Jordan algebra with an idempotent is an example, and so is the Norton–Sakuma algebra of type $2A$ [2025-10, Example 4.2, p. 8]. Kawai, Guzzo and Ferreira measure how far a multiplicative map is from being additive by a nullifying function [2025-10, Definition 3.5, p. 6]. This function is symmetric in its finitely many arguments, and it vanishes on a repeated sum exactly when it vanishes on the summands [2025-10, Definition 3.5, p. 6]. The authors prove that it vanishes on a $J(\alpha)$-algebra under three conditions of Martindale type [2025-10, Theorem 4.1, p. 8]. Then four kinds of map are additive. They are the n-multiplicative isomorphisms, the n-multiplicative derivations, the surjective elementary maps, and, when the characteristic is not 2, the surjective Jordan elementary maps [2025-10, Corollary 4.1, p. 10]. The same conclusion holds for $M(\alpha, \beta)$-axial algebras under five conditions [2025-10, Theorem 5.1, p. 12].

On a 2-torsion free semiprime ring the Jordan triple condition collapses to the triple condition. Liu and Shiue show that every Jordan triple (θ, φ)-derivation is a (θ, φ)-derivation, and deduce the same for the generalized Jordan triple maps [2007-7, Abstract].

The comparison holds on a Lie ideal too. Hongan, Rehman and Al-Omary prove that on a 2-torsion free semiprime ring every Jordan triple derivation on a Lie ideal is a derivation on it, and the same with "generalized" throughout [2011-8, Abstract].

Further work in the Jordan-type line, held in the collection: *Continuous linear maps on reflexive algebras behaving like Jordan left derivations at idempotent-product elements* [2013-2]. *Nonlinear Jordan derivable mappings of generalized matrix algebras by Lie product square-zero elements* [2021-3]. *Jordan higher derivable maps on triangular algebras by commutative zero products* [2016-6].

4.4 Jordan derivations of specific algebras

Section 4.2 gave conditions on a ring that force a Jordan derivation to be a derivation. Cheng and Jing study triangular algebras [2007-5, p. 1]. Brešar surveys a construction of Benkovič for upper triangular matrices [2005-2, §4.1, p. 418]. He also treats C^*-algebras [2005-2, Section 5, p. 421].

Let A and B be algebras over a commutative ring R [2007-5, p. 2]. Let M be an (A, B)-bimodule that is faithful as a left A-module and as a right B-module [2007-5, p. 2]. These data give the triangular algebra $T = \mathrm{Tri}(A, M, B)$, whose elements are the matrices $\begin{pmatrix} a & m \\ 0 & b \end{pmatrix}$ [2007-5, p. 2]. Zhang and Yu showed that every Jordan derivation of a triangular algebra is a derivation [2007-5, p. 1]. Cheng and Jing state it as follows [2007-5, p. 1].

Theorem 4.31 ([2007-5, Theorem 1.1, p. 1])**.** *Let A, B be unital algebras over a 2-torsion free commutative ring R, and M be a unital (A, B)-bimodule that is faithful as a left A-module and also as a right B-module. Then every Jordan derivation from the triangular algebra* $\mathrm{Tri}(A, M, B)$ *into itself is a derivation.*

The proof of Zhang and Yu needs both algebras to have a unit [2007-5, p. 1]. Cheng and Jing do not require a unit in either algebra [2007-5, p. 1]. They assume the following property of A and B [2007-5, Section 2, p. 2]: (P) if $a \in A$ (respectively B) and $xay + yax = 0$ for all $x, y \in A$ (respectively B), then $a = 0$.

Theorem 4.32 ([2007-5, Theorem 2.10, p. 5])**.** *Let A and B be two algebras over a 2-torsion free commutative ring R with the property: (P) Suppose that $a \in A$ (respectively B). If $xay + yax = 0$ holds for all $x, y \in A$ (respectively B), then $a = 0$. Let M be a faithful (A, B)-bimodule and T be the triangular algebra* $\mathrm{Tri}(A, M, B)$*. Then every Jordan derivation δ on T into itself is a derivation.*

Cheng and Jing note that Theorem 4.31 is the special case where both algebras are unital [2007-5, Remark 2.11, p. 5]. Their proof is a sequence of lemmas. The first shows that δ is a derivation on each of the three parts T_{11}, T_{12} and T_{22} of T [2007-5, pp. 4–5].

Now let F be a field with $\mathrm{char}(F) \neq 2$, and let $T_n(F)$ be the algebra of upper triangular $n \times n$ matrices over F [2005-2, Section 4, p. 418]. Brešar describes a construction of Benkovič [2005-2, §4.1, p. 418]. It produces maps d with $d(xy) = d(y)x + yd(x)$ for all x, y, which are called antiderivations [2005-2, §4.1, p. 418]. The construction gives a proper Jordan derivation on $T_n(F)$ into a bimodule [2005-2, §4.1, p. 418].

Example 4.33 ([2005-2, Example 4.1, p. 418])**.** Let $A = T_n(F)$, $n \geqslant 2$. Then there exists an A-bimodule M and a proper Jordan derivation (in fact,

an antiderivation) $d\colon A \to M$ such that $d(ue) \neq d(u)e + ud(e)$ for some $u \in A^{(1)}$ and some idempotent $e \in A$.

Brešar adds that the main result of Benkovič says that every Jordan derivation from $T_n(F)$ into a $T_n(F)$-bimodule is the sum of a derivation and an antiderivation [2005-2, p. 419]. So Jordan derivations of $T_n(F)$ act as derivations on a large part of the algebra [2005-2, p. 419].

For operator algebras, Brešar poses a question: is every Jordan derivation from a C^*-algebra A into a Banach A-bimodule a derivation [2005-2, Section 5, p. 421]? He calls it an intriguing open question [2005-2, Section 5, p. 421]. Johnson proved that a continuous Jordan derivation from a C^*-algebra into a Banach bimodule is a derivation [2005-2, p. 421]. For additive maps Brešar gives a definitive answer on the algebras he calls CM-algebras [2005-2, p. 421].

Theorem 4.34 ([2005-2, Theorem 5.1, p. 422])**.** *Let A be a unital* CM*-algebra. Then there exists a proper additive Jordan derivation from A into some unital A-bimodule if and only if A contains an ideal of codimension* 1*.*

Brešar derives the theorem of Johnson from Theorem 4.34 [2005-2, p. 422]. He also notes that the result of Johnson does not extend from CM-algebras to semisimple Banach algebras, because a primitive Banach algebra can have $T_2(\mathbb{C})$ as a quotient [2005-2, p. 422].

The C^*-algebra question of this section is open, and this section states it as a problem.

Problem 4.35 ([2005-2, Section 5, p. 421])**.** Let A be a C^*-algebra and let M be a Banach A-bimodule. Is every Jordan derivation from A into M a derivation?

Brešar calls this an intriguing open question [2005-2, Section 5, p. 421]. Johnson had proved it for continuous Jordan derivations, and Brešar answers it for the algebras he calls CM-algebras [2005-2, p. 421]. The statement of Theorem 4.34 is the answer on that class. Over a commutative ring with unity the torsion hypothesis can be dropped [2018-9, Theorem 2.1, p. 3]. Ghosh and Prakash prove that every Jordan derivation of the upper triangular matrix algebra $T_n(C)$ into itself is a derivation, with no condition on the ring [2018-9, Theorem 2.1, p. 3]. Their proof computes the image of each matrix unit and compares coefficients [2018-9, pp. 3–4]. The same method shows that every derivation of $T_n(C)$ is inner [2018-9, Theorem 2.2, p. 5]. For the 2-element field the result covers a ring that is not 2-torsion

free [2018-9, Corollary 2.1, p. 4]. The full matrix algebra behaves in the same way [2018-9, Theorem 3.1, p. 5]. Every Jordan derivation of $M_n(C)$ into itself is a derivation, and every derivation of $M_n(C)$ is inner [2018-9, Theorem 3.2, p. 8].

The passage from rings to semirings keeps the triangular shape, and it weakens the conclusion. Every derivation of the upper triangular matrix semiring keeps all entries of a matrix except those of one principal dense submatrix, and it sets those entries to zero [2018-17, Theorem 1, p. 5]. Vladeva works over an additively idempotent semiring, where the sum of two equal terms adds nothing [2018-17, Proposition 1, p. 3]. In that setting the Jordan product of a matrix with a partial identity matrix equals the ordinary product, so the map $A \mapsto A \circ D_k$ is a derivation [2018-17, Proposition 1, p. 3]. The paper counts the derivations that arise this way [2018-17, Theorem 1, p. 5].

On matrix rings the Jordan condition can be tested through products alone. Ghahramani characterizes the Jordan derivations of a matrix ring by their action on zero products [2015-8, Abstract].

For prime rings the generalized notion has its own theory. Jing and Lu initiate the study of generalized Jordan derivations and generalized Jordan triple derivations on prime rings and on standard operator algebras [2003-2, Abstract, p. 605].

4.5 Relation with Jordan homomorphisms

The Jordan derivations of Section 4.2 have a twin among the maps between algebras. Brešar and Zelmanov survey the history of this twin [2025-4, Abstract, p. 203]. They assume that algebras are defined over a field F with $\operatorname{char}(F) \neq 2$ unless stated otherwise [2025-4, Section 1, p. 203].

Definition 4.36 ([2025-4, Section 1, p. 203])**.** Let A and B be associative algebras. A linear map $\varphi\colon A \to B$ is called a Jordan homomorphism if $\varphi(x^2) = \varphi(x)^2$ for every $x \in A$.

Homomorphisms and antihomomorphisms are examples [2025-4, p. 204]. An antihomomorphism is a linear map φ with $\varphi(xy) = \varphi(y)\varphi(x)$ for all $x, y \in A$ [2025-4, p. 204]. The classical problem asks whether Jordan homomorphisms can be expressed by homomorphisms and antihomomorphisms [2025-4, p. 204]. Jordan homomorphisms satisfy the rules in the next proposition [2025-4, p. 209].

Proposition 4.37 ([2025-4, Proposition 4.1, p. 209])**.** *Every Jordan homomorphism φ defined on a special Jordan algebra J satisfies (a) $\varphi(xy + yx) = \varphi(x)\varphi(y) + \varphi(y)\varphi(x)$, (b) $\varphi(x^n) = \varphi(x)^n$, $n \in \mathbb{N}$, (c) $\varphi(xyx) = \varphi(x)\varphi(y)\varphi(x)$, (d) $\varphi(xyz + zyx) = \varphi(x)\varphi(y)\varphi(z) + \varphi(z)\varphi(y)\varphi(x)$, (e) $\varphi([[x, y], z]) = [[\varphi(x), \varphi(y)], \varphi(z)]$, (f) $\varphi([x, y]^2) = [\varphi(x), \varphi(y)]^2$ for all $x, y, z \in J$.*

Ancochea first studied these maps, under the name semiautomorphisms [2025-4, p. 211]. His second paper is devoted to the first theorem [2025-4, p. 211].

Theorem 4.38 ([2025-4, Theorem 5.1, p. 211])**.** *A Jordan automorphism of a finite-dimensional simple algebra A is either an automorphism or an antiautomorphism.*

Hua proved the case of division algebras in 1949 [2025-4, p. 211].

Theorem 4.39 ([2025-4, Theorem 5.2, p. 211])**.** *A Jordan homomorphism from a division algebra D to itself is either a homomorphism or an antihomomorphism.*

Jacobson and Rickart gave the name Jordan homomorphism and improved these results [2025-4, p. 211]. Their first theorem extends the theorem of Hua to domains [2025-4, p. 211].

Theorem 4.40 ([2025-4, Theorem 5.3, p. 211])**.** *A Jordan homomorphism from any algebra A to a domain B is either a homomorphism or an antihomomorphism.*

Their second theorem puts conditions on the domain algebra instead of the target [2025-4, p. 211].

Theorem 4.41 ([2025-4, Theorem 5.4, pp. 211–212])**.** *Let S be a unital associative algebra, let $A = M_n(S)$ with $n \geq 2$, and let B be any associative algebra. For every Jordan homomorphism $\varphi\colon A \to B$ there exists a central idempotent $e \in \langle\varphi(A)\rangle$ such that $x \mapsto e\varphi(x)$ is a homomorphism and $x \mapsto (1 - e)\varphi(x)$ is an antihomomorphism.*

Herstein proved the case of a surjective map onto a prime algebra [2025-4, p. 212]. He needed a condition on the characteristic, which Smiley later removed [2025-4, p. 212].

Theorem 4.42 ([2025-4, Theorem 5.6, p. 212])**.** *A surjective Jordan homomorphism from any algebra A onto a prime algebra B is either a homomorphism or an antihomomorphism.*

The prime hypothesis of Theorem 4.42 is the setting of Chapter 3. The theorem answers the homomorphism form of the problem of Martindale of Section 5.1: the two kinds of map named there return here as the homomorphism and the antihomomorphism cases.

A Jordan derivation $\delta\colon A \to A$ gives the Jordan homomorphism $\varphi\colon A \to M_2(A)$ with $\varphi(x) = \begin{pmatrix} x & \delta(x) \\ 0 & x \end{pmatrix}$ [2025-4, Section 11, p. 224]. Results on Jordan homomorphisms can sometimes give results on Jordan derivations in this way [2025-4, p. 224]. Herstein proved that Jordan derivations of prime algebras are derivations, in parallel with Theorem 4.42 [2025-4, p. 224]. The semiprime case reads as follows [2025-4, p. 225].

Theorem 4.43 ([2025-4, Theorem 11.1, p. 225])**.** *A Jordan derivation of a semiprime algebra is a derivation.*

Brešar and Zelmanov credit this theorem to Cusack, and say that it was later proved again and extended by Brešar [2025-4, p. 225]. These are the results of Section 4.2. The survey adds a second theorem of this kind, which is a special case of the main result of Brešar [2025-4, p. 225].

Theorem 4.44 ([2025-4, Theorem 11.2, p. 225])**.** *Let A be an associative algebra and let U be the ideal of A generated by $[[A,A],[A,A]]$ and $[[[A,A],[A,A]],[[A,A],[A,A]]]$. Every Jordan derivation $\delta\colon A \to A$ satisfies $\delta(ux) = \delta(u)x + u\delta(x)$ for all $u \in U$ and $x \in A$.*

4.6 Jordan *-derivations and Jordan left derivations

Section 4.3 ended with the triple form of the product rule. Two further changes of the rule remain. Fošner and Ilišević use an involution [2008-5, p. 416]. Vukman studies Jordan left derivations [1997-5, p. 1].

Fošner and Ilišević start from a $*$-ring, that is, a ring R with an involution $*$ [2008-5, Section 1, p. 415]. An involution is an additive mapping with $(a^*)^* = a$ and $(ab)^* = b^*a^*$ for all $a, b \in R$ [2008-5, p. 415].

Definition 4.45 ([2008-5, Section 1, p. 416])**.** Let R be a $*$-ring and let M be an R-bimodule. An additive mapping $\delta\colon R \to M$ is called a Jordan

$*$-derivation if $\delta(x^2) = \delta(x)x^* + x\delta(x)$ holds for all $x \in R$. A Jordan triple $*$-derivation is an additive mapping $\delta\colon R \to M$ with the property $\delta(xyx) = \delta(x)y^*x^* + x\delta(y)x^* + xy\delta(x)$ for all $x, y \in R$.

On a 2-torsion free $*$-ring every Jordan $*$-derivation is a Jordan triple $*$-derivation [2008-5, p. 417]. For real Banach $*$-algebras with identity the converse holds [2008-5, p. 417]. A recent paper proved the converse for 6-torsion free semiprime $*$-rings [2008-5, p. 417]. Fošner and Ilišević extend this result to 2-torsion free semiprime $*$-rings [2008-5, p. 417]. They rest on a lemma whose proof is found in an earlier paper [2008-5, p. 425].

Lemma 4.46 ([2008-5, Lemma 5.1, p. 425])**.** *If R is a semiprime $*$-ring, then $yxy^* = 0$ for all $y \in R$ (where $x \in R$) implies $x = 0$.*

The lemma gives the main result of the section.

Theorem 4.47 ([2008-5, Theorem 5.2, p. 425])**.** *Let R be a 2-torsion free semiprime $*$-ring and let $\delta\colon R \to R$ be an additive mapping. Then the following conditions are mutually equivalent: (a) δ is a Jordan $*$-derivation; (b) $\delta(xyx) = \delta(x)y^*x^* + x\delta(y)x^* + xy\delta(x)$ $(x, y \in R)$.*

The generalized version uses a second map E that is related to δ [2008-5, p. 417]. An additive mapping E is called a generalized Jordan $*$-derivation if a Jordan $*$-derivation δ exists with $E(x^2) = E(x)x^* + x\delta(x)$ [2008-5, p. 417].

Theorem 4.48 ([2008-5, Theorem 5.4, p. 426])**.** *Let R be a 2-torsion free semiprime $*$-ring and let $E, \delta\colon R \to R$ be additive mappings. Then the following conditions are mutually equivalent: (a) for all $x, y \in R$, $E(xyx) = E(x)y^*x^* + x\delta(y)x^* + xy\delta(x)$ and $\delta(xyx) = \delta(x)y^*x^* + x\delta(y)x^* + xy\delta(x)$; (b) for all $x \in R$, $E(x^2) = E(x)x^* + x\delta(x)$ and $\delta(x^2) = \delta(x)x^* + x\delta(x)$.*

Let R be an associative ring and $D\colon R \to R$ an additive mapping [1997-5, p. 1].

Definition 4.49 ([1997-5, p. 1])**.** D is called a derivation (respectively a left derivation) if $D(xy) = D(x)y + xD(y)$ (respectively $D(xy) = xD(y) + yD(x)$) holds for all $x, y \in R$. And D is called a Jordan derivation (respectively a left Jordan derivation) in case $D(x^2) = D(x)x + xD(x)$ (respectively $D(x^2) = 2xD(x)$) for all $x \in R$.

Vukman notes that a left derivation of a noncommutative prime ring is zero [1997-5, p. 1]. He also records a theorem of Brešar and himself [1997-5, p. 1]. It says that a nonzero Jordan left derivation on a prime ring

of characteristic different from 2 and 3 forces the ring to be commutative [1997-5, p. 1]. Vukman states three lemmas for his theorems [1997-5, p. 2]. The first holds in any 2-torsion free ring.

Lemma 4.50 ([1997-5, Lemma 3, p. 2])**.** *Let R be a 2-torsion free ring. If $D\colon R \to R$ is a Jordan left derivation then for all $x, y \in R$: (a) $D(xy+yx) = 2xD(y) + 2yD(x)$, (b) $D(xyx) = x^2D(y) + 3xyD(x) - yxD(x)$.*

The second lemma treats prime rings [1997-5, p. 2].

Lemma 4.51 ([1997-5, Lemma 5, p. 2])**.** *Let R a noncommutative prime ring of characteristic different from two, and let $D\colon R \to R$ be a left Jordan derivation. In this case $D = 0$.*

Vukman combines these with a result of Giambruno and Herstein on nilpotent derivations [1997-5, p. 1]. His first theorem treats semiprime rings.

Theorem 4.52 ([1997-5, Theorem 1, p. 1])**.** *Let R be a 2-torsion free semiprime ring and $D\colon R \to R$ a Jordan left derivation. If there exists a positive integer n such that $D(x)^n = 0$ for all $x \in R$, then $D = 0$.*

Deng treats the left Jordan condition on its own, without the prime-ring setting Vukman uses, and the map it governs is the one the definition above names [1992-2, Abstract].

The Jordan left condition can be imposed at a product rather than at every element. On a Banach algebra a continuous linear map satisfying $2a\delta(b) + 2b\delta(a) = \delta(z)$ whenever $ab + ba = z$ is characterized for $z = 1$, and then for z a nontrivial idempotent of a reflexive algebra $\operatorname{Alg} L$ [2013-1, Abstract].

Strong commutativity preservation has a skew form. For two generalized skew derivations F and G of a prime ring with the same associated automorphism α, if $[F(x), \alpha(y)] = G([x, y])$ on the values of a non-central polynomial, then F is a scalar multiple of α plus a suitable map [2016-1, Abstract].

Chapter 5

Lie derivations and Lie-type maps

Every associative ring carries a second product next to its own: the commutator $[a, b] = ab - ba$. A map that obeys the product rule for the commutator is a Lie derivation, and Martindale began the theory in 1964 with a single question [1964-2, p. 183]. Every derivation is a Lie derivation, and so is every sum of a derivation and a central map that sends commutators to zero. The question is whether there is anything else.

This chapter follows the answers class by class. Martindale gave the first one for primitive rings with a nontrivial idempotent (Section 5.1). Brešar settled the question for noncommutative prime rings through his theory of functional identities (Section 5.2). Iterating the bracket gives Lie triple and n-Lie derivations (Section 5.3). Dropping linearity gives the nonlinear maps of Section 5.4. Section 5.5 collects the answers on the algebras where the question can be asked in explicit terms, from $B(X)$ to matrix and nest algebras.

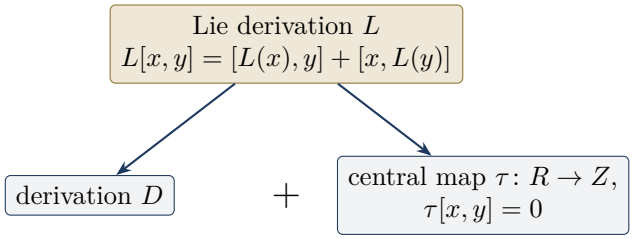


Figure 5.1: The question of Martindale: is every Lie derivation the sum of a derivation D and a central map τ that vanishes on commutators?

5.1 The problem of Martindale and the standard form

Chapter 4 followed the Jordan product $a \circ b = ab + ba$. We now follow the other product that an associative ring carries, the commutator $[a, b] = ab - ba$. A map that satisfies the product rule for the commutator is a Lie derivation [1964-2, p. 183]. Martindale began the theory with the definition [1964-2, p. 183].

Definition 5.1 ([1964-2, p. 183])**.** Let A be an associative ring. An additive mapping $a \mapsto a'$ of A into itself is called a *Lie derivation* of A if for every pair $a, b \in A$, $[a, b]' = [a', b] + [a, b']$.

Every derivation is a Lie derivation. So is the sum of a derivation and an additive map into the center that sends commutators to zero. The problem, set by Martindale in his first paragraph, is whether these two pieces exhaust the Lie derivations [1964-2, p. 183]. Herstein had studied Lie derivations of the classical noncommutative simple rings; Martindale extends the theory to rings whose nonzero two-sided ideals have a nonzero right annihilator, and he notes that primitive rings are precisely such rings [1964-2, p. 183].

His first result extends the map itself.

Theorem 5.2 ([1964-2, Theorem 1, p. 186])**.** *Let A be a primitive ring containing a nontrivial idempotent, but not a domain, with characteristic different from 2. Then every Lie derivation of A can be uniquely extended to an ordinary derivation of an associative algebra A^* containing A.*

The theorem is stated for a primitive ring A that contains a nontrivial idempotent, in characteristic not two [1964-2, pp. 183–184]. That is the setting the decomposition needs, and the paper states it there. The ring of Martindale quotients is a different construction, built in 1969 [1969-2, p. 576]; Chapter 3 uses it for semiprime rings, and this theorem does not need it.

Theorem 5.3 ([1964-2, Theorem 2, p. 187])**.** *Let A and A^* be as in Theorem 5.2. Every Lie derivation d of A is of the form $d = d' + \tau$, where d' is the unique extension of d to an ordinary derivation of A^*, and τ is an additive mapping of A into the center of A^*. The map τ sends commutators into zero.*

The form $d =$ derivation$+$central map vanishing on commutators is the answer the chapter keeps meeting. The functional identities of Section 5.2 later settle when the central map really occurs. Martindale closes with simple rings, where the map into the center stays inside the ring [1964-2, p. 187].

Remark 5.4 ([1964-2, p. 187]). Let A be a simple ring characteristic not 2 with minimal one-sided ideals. Then every Lie derivation of A is of the form $d + \tau$. Here d is an ordinary derivation of A into itself, and τ is additive. The map τ sends A into the center of A and commutators into zero.

Remark 5.4 settles the simple case that has minimal one-sided ideals, and the general case is the problem of Herstein, still open.

Problem 5.5 ([1961-1, Problem 3, p. 529])**.** Let R be a simple ring. Is every Lie derivation of R an ordinary derivation plus an additive map into the center?

Herstein raised the question for a simple ring, and he recorded that Kaplansky had settled the case of matrix units of size at least 3 [1961-1, Problem 3, p. 529]. Martindale proved his theorem for primitive rings with a nontrivial idempotent and said that it falls far short of a general solution of that conjecture [1964-2, p. 183]. Theorem 5.3 gives the form for a primitive ring with a nontrivial idempotent [1964-2, Theorem 2, p. 187]. His closing remark covers the simple case: the derivation then maps R into itself and the additive map goes into the center of R [1964-2, p. 187].

Brešar extends the form to every prime ring of characteristic not 2 that does not satisfy the identity S_4 [1993-3, Theorem 5, p. 529]. His Lemma 1 lists the rings that do satisfy S_4: they are the commutative rings and the

subrings of $M_2(F)$ for a field F [1993-3, Lemma 1, p. 530]. He records that the assumption on S_4 may be removable [1993-3, p. 528]. So Problem 5.5 stays open in characteristic 2 and for the rings that satisfy S_4.

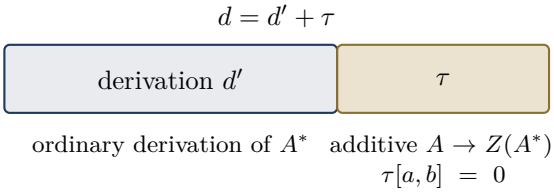


Figure 5.2: The standard form of Martindale [1964-2, Theorem 2, p. 187]. A Lie derivation d splits as $d = d' + \tau$, where d' is an ordinary derivation of A^* and τ is an additive map into the center of A^* that sends commutators to zero. Theorem 5.3 states it in full.

5.2 The solution of Brešar through functional identities

Section 5.1 gave the form of a Lie derivation on a primitive ring with a nontrivial idempotent. Brešar settled the question for all noncommutative prime rings with his functional identities [1993-3, p. 525]. He works in a noncommutative prime ring R of characteristic not 2 and not 3 [1993-3, p. 525]. By Q he denotes the Martindale right ring of quotients of R, and by C its center [1993-3, p. 525]. This is the maximal right ring of quotients of Chapter 3, and its center C is the extended centroid. The pair $(Q)_C$, the central closure of R, measures the algebraic degree of R through the standard identities S_{2n} [1993-3, p. 526].

The opening result is about commuting traces of biadditive mappings [1993-3, Theorem 1, p. 527].

Theorem 5.6 ([1993-3, Theorem 1, p. 527])**.** *Let R be a prime ring of characteristic not 2, and let $q\colon R \to R$ be a trace of a biadditive mapping. Suppose that q is commuting. If R does not satisfy S_4, then q is of the form*

$$q(x) = \lambda x^2 + \mu(x)x + \nu(x), \qquad x \in R,$$

where λ is an element in C, μ is an additive mapping of R into C, and ν is a mapping of R into C.

From the commuting trace Brešar passes to maps on the square.

The trace idea carries to algebras where the prime-ring argument has no grip. For an n-linear map q on an algebra A, the trace is $T_q(a) = q(a, \ldots, a)$. It is *proper* when it has the form $T_q(a) = za^n + \mu_1(a)a^{n-1} + \cdots + \mu_n(a)$, with z central and each μ_i a central-valued trace [2015-12, p. 2]. Xiao, Chen and Zhang prove that on a 2-torsion free triangular algebra whose bimodule is loyal, every centralizing trace of a bilinear map is proper [2015-12, Theorem 3.4]. That is the triangular algebra form of the result Brešar proves for a prime ring.

Theorem 5.7 ([1993-3, Theorem 4, p. 529])**.** *Let R be a prime ring of characteristic different from 2 and 3. Suppose that an additive mapping $d\colon R \to R$ satisfies*

$$[d(x^2) - d(x)x - xd(x),\, x] = 0, \qquad x \in R.$$

If R does not satisfy S_4, then d is of the form

$$d(x) = \gamma x + \delta(x) + \zeta(x), \qquad x \in R,$$

where γ is an element in C, δ is a derivation of R into its central closure, and ζ is an additive mapping of R into C.

Together these give the Lie derivation theorem of the paper.

Theorem 5.8 ([1993-3, Theorem 5, p. 529])**.** *Let R be a prime ring of characteristic not 2. Let d be a Lie derivation of R. If R does not satisfy S_4, then d is of the form $\delta + \tau$, where δ is a derivation of R into its central closure and τ is an additive mapping of R into C sending commutators to zero.*

The question Martindale set for primitive rings is thereby settled for every prime ring of characteristic not 2, with no idempotent hypothesis [1993-3, Theorem 5, p. 529]. Theorem 5.8 even frees it from the characteristic-3 restriction of Theorem 5.7. The one assumption is that the ring does not satisfy the standard identity S_4 [1993-3, Theorem 5, p. 529]. Brešar records it as an open question whether that assumption can be removed [1993-3, p. 528]. His Lemma 1 names the exceptional class exactly: a prime ring satisfies S_4 precisely when it is commutative or embeds in the 2×2 matrices over a field [1993-3, Lemma 1, p. 530].

5.3 Lie triple and Lie n-derivations

Sections 5.1 and 5.2 treated the commutator of two elements. The next kind iterates the bracket one step further. Miers defined it for a von Neumann algebra and proved that the same kind of decomposition as for Lie derivations holds [1978-3, p. 57].

Definition 5.9 ([1978-3, p. 57])**.** Let M be a von Neumann algebra. An associative derivation of M is an additive map $d\colon M \to M$ satisfying $d(xy) = d(x)y + xd(y)$. A Lie derivation of M is an additive map $L\colon M \to M$ satisfying $L([x,y]) = [L(x),y] + [x,L(y)]$. Generalizing both of these definitions, a *Lie triple derivation* of M is an additive map $T\colon M \to M$ with the property that

$$T[[x,y],z] = [[T(x),y],z]+[[x,T(y)],z]+[[x,y],T(z)] \qquad \text{for all } x,y,z \in M.$$

Miers proves his main result together with the Lie derivation form of Theorem 5.8 for von Neumann algebras.

Theorem 5.10 ([1978-3, abstract and p. 58])**.** *If $L\colon M \to M$ is a Lie derivation of the von Neumann algebra M, then there exists an associative derivation $d\colon M \to M$ and an additive map $t\colon M \to \operatorname{center}(M)$ such that $d(x) + t(x) = L(x)$ for all $x \in M$. If $T\colon M \to M$ is a Lie triple derivation of the von Neumann algebra M, then there exists an associative derivation $d\colon M \to M$ and a linear map $\lambda\colon M \to \operatorname{center}(M)$ with $\lambda[[x,y],z] = 0$ for all $x,y,z \in M$ such that $d(x) + \lambda(x) = T(x)$ for all $x \in M$.*

Miers adds a form that respects the involution.

Theorem 5.11 ([1978-3, Theorem 4, p. 60])**.** *If T is a Lie $*$-triple derivation of M, so that $T(x^*) = T(x)^*$, then there exists an associative $*$-derivation d and a linear map $\lambda\colon M \to Z(M)$ with $\lambda[[x,y],z] = 0$ for all $x,y,z \in M$ and $\lambda(x^*) = -\lambda(x)^*$ such that $T(x) = d(x) + \lambda(x)$.*

The next kind after the triple derivation maps into n-fold commutators: an *n-Lie derivation* is a map of the algebra, taken in the operator setting by Abdullaev [1992-1]. His paper states the theorem for von Neumann algebras. No copy of it has been obtained, so this book names the notion and cites the source without quoting a statement from it. The later literature on n-Lie derivations of matrix and triangular algebras is recorded in Chapter 16.

On a Lie triple system the Jordan condition is enough. Najati shows under some conditions that every Jordan triple (ϕ,ψ)-derivation of a Lie

triple system is a (ϕ, ψ)-derivation, and in particular that every Jordan triple θ-derivation is a θ-derivation [2009-8, Abstract, p. 143].

Approximation is a separate question on Lie triple systems. Kim and Lee study approximate homomorphisms and approximate derivations on a normed Lie triple system [2014-15, Abstract].

5.4 Nonlinear and nonadditive Lie-type maps

The maps of Sections 5.1 to 5.3 are additive or linear. The next step drops even additivity. Bell and Daif began this line with a definition and with two commutativity theorems [1994-1, p. 443]: a map $f\colon R \to R$ is *strong commutativity preserving* on a subset S when $[f(x), f(y)] = [x, y]$ for all $x, y \in S$.

Theorem 5.12 ([1994-1, Theorem 1, p. 444])**.** *Let R be a semiprime ring and U a nonzero right ideal. If R admits a derivation D which is strong commutativity preserving on U, then U is contained in the center of R.*

The second theorem of the pair keeps the map and replaces the derivation by an endomorphism [1994-1, p. 446].

Theorem 5.13 ([1994-1, Theorem 2, p. 446])**.** *Let R be a prime ring and U an essential right ideal of R. If R admits a nonidentity endomorphism T which is strong commutativity preserving on U, then R is commutative.*

Once such a map is allowed to be only additive on the whole ring, and involves the extended centroid, the answers stop being commutativity theorems and become form theorems. Brešar and Miers gave the first one [1994-2, p. 457].

Theorem 5.14 ([1994-2, Theorem 1, p. 458])**.** *Let R be a semiprime ring with extended centroid C. Suppose that an additive map $f\colon R \to R$ satisfies $[f(x), f(y)] = [x, y]$ for all $x, y \in R$. Then f is of the form $f(x) = \lambda x + \xi(x)$, where $\lambda \in C$, $\lambda^2 = 1$, and ξ is an additive map of R into C.*

For prime rings with a nontrivial idempotent and surjective maps, no additivity is needed: Qi and Hou obtained the same form with $\lambda = \pm 1$ [2010-7, p. 2791].

Theorem 5.15 ([2010-7, Theorem 2.1, p. 2791])**.** *Let R be a unital prime ring containing a nontrivial idempotent. Assume that $\Phi\colon R \to R$ is a surjective map. Then Φ is strong commutativity preserving if and only if*

there exist a map $f\colon R \to Z(R)$ *and an element* $\lambda \in \{1, -1\}$ *such that* $\Phi(A) = \lambda A + f(A)$ *for all* $A \in R$.

The factor von Neumann algebra case of their theorem is the scalar special case, with $Z(R)$ replaced by the scalar operators [2010-7, p. 2795]. Bai and Du then removed the prime hypothesis, at the price of asking the idempotent to control the ring on both sides [2014-6, p. 733].

Theorem 5.16 ([2014-6, Theorem 1, p. 733])**.** *Let* R *be a unital ring having an idempotent element* e *which satisfies*

(i) $aRe = 0$ *implies* $a = 0$,

(ii) $aR(1 - e) = 0$ *implies* $a = 0$.

Assume that $f\colon R \to R$ *is a surjective map. Then* f *satisfies* $[f(x), f(y)] = [x, y]$ *for all* $x, y \in R$ *if and only if* f *is of the form* $f(x) = \alpha x + \xi(x)$ *for all* $x \in R$*, where* $\alpha \in Z(R)$*,* $\alpha^2 = 1$*, and* ξ *is a map from* R *into* $Z(R)$.

The power of the idempotent hypothesis is visible in the operator case [2014-6, p. 741]. A von Neumann algebra with no central summands of type I_1 carries a core-free projection, and that projection plays the role of the idempotent of Theorem 5.16 [2014-6, p. 741].

Corollary 5.17 ([2014-6, Corollary 2, pp. 734–735])**.** *Let* M *be a von Neumann algebra with no central summands of type* I_1*. Assume that* $f\colon M \to M$ *is a surjective map. Then* f *satisfies* $[f(x), f(y)] = [x, y]$ *for all* $x, y \in M$ *if and only if* f *is of the form* $f(x) = \alpha x + \xi(x)$ *for all* $x \in M$*, where* $\alpha \in Z(M)$*,* $\alpha^2 = I$*, and* ξ *is a map from* M *into* $Z(M)$.

The pattern of the first four sections survives the loss of additivity: again f equals a scalar multiple, here αx with $\alpha^2 = 1$, plus a central remainder $\xi(x)$ [2014-6, p. 733]. The line runs from a commutativity theorem to the same standard form that Chapter 4 and Section 5.1 met for additive maps.

A $*$-algebra carries one more product of this kind, and the maps of this section have an analogue for it [2026-2, p. 127].

Definition 5.18 ([2026-2, p. 127])**.** Let E be a $*$-algebra over the complex field $\mathbb{C}$. For $U, V \in E$ the *left bi-skew Lie product* is $U \diamond V = U^*V - V^*U$. The product of three elements is read as the left product applied twice, and the proof of Lemma 11 records the identity $U^* \diamond V \diamond \frac{I}{2} = -UV + V^*U^*$ for all $U, V \in E$ [2026-2, p. 136].

A family of maps that obeys the product rule of a higher derivation along this product is a bi-skew Lie triple higher derivation [2026-2, p. 128]. The sequence of Definition 7.1 is the model for the name [2026-2, p. 128]. Ali, Ali and Tasleem prove that no additivity has to be assumed on a prime $*$-algebra that has a nontrivial projection [2026-2, Main Theorem, p. 128].

Theorem 5.19 ([2026-2, Main Theorem, p. 128])**.** *Let E be a prime $*$-algebra with unity I and a nontrivial projection P. Let $\Delta = \{\zeta_n\}_{n\in\mathbb{N}}$ be a family of maps $\zeta_n\colon E \to E$, not necessarily linear, with $\zeta_0 = \mathrm{id}_E$, such that*

$$\zeta_n(U \diamond V \diamond W) = \sum_{p+q+r=n} \zeta_p(U) \diamond \zeta_q(V) \diamond \zeta_r(W)$$

for all $U, V, W \in E$ and all $n \in \mathbb{N}$. Then Δ is an additive $*$-higher derivation of E, provided $\zeta_n(\beta I/2)$ is self-adjoint for $\beta \in \{1, i\}$ and every n.

The conclusion in full is the last lemma of the paper [2026-2, p. 136].

Lemma 5.20 ([2026-2, Lemma 11, p. 136])**.** *Under the hypotheses of Theorem 5.19, each ζ_n is additive, and $\zeta_n(UV) = \sum_{p+q=n} \zeta_p(U)\zeta_q(V)$ and $\zeta_n(U^*) = \zeta_n(U)^*$ hold for all $U, V \in E$ and all $n \in \mathbb{N}$.*

So a map that respects the triple bi-skew product at every order is additive and multiplicative at every order [2026-2, p. 136]. The sequence of maps and its own product rule are the subject of Section 7.1.

5.5 Lie derivations of specific algebras

The sections so far moved through general classes of rings and algebras. On a concrete algebra the question has a concrete answer: does the standard form hold there, with what remainder, and what does the remainder look like. The two answers this section records are the algebra of bounded operators on a Banach space and the algebras built around an idempotent.

Lu and Jing settle the question of Martindale for $B(X)$, the algebra of all bounded linear operators on a complex Banach space X of dimension greater than 2 [2010-5, p. 91]. An additive map $\tau\colon B(X) \to B(X)$ is called *centralized* when $\tau([A, B]) \in \mathbb{C}I$ for all $A, B \in B(X)$ [2010-5, p. 91].

Theorem 5.21 ([2010-5, Theorem 2.1, p. 92])**.** *Let X be a Banach space with $\dim X > 2$, and δ be a Lie derivation on $B(X)$. Then there exists an operator $T \in B(X)$, and a linear map $\tau\colon B(X) \to \mathbb{C}I$ vanishing on each commutator, such that $\delta(A) = AT - TA + \tau(A)$ for all $A \in B(X)$.*

Two points mark the answer. Every Lie derivation of $B(X)$ is inner up to the scalar remainder, and the remainder is finer than the general theory asks. It need not be given by a linear functional $A \mapsto f(A)I$ on the subspace spanned by the commutators [2010-5, p. 91]. With the second point Lu and Jing answer in the negative a question of Brešar [2010-5, p. 92].

Benkovič gives the idempotent answer for linear maps over a commutative ring R. The algebra A is unital, 2-torsion free, with an idempotent $e \neq 0, 1$ such that $exe \cdot eA(1-e) = \{0\} = (1-e)Ae \cdot exe$ implies $exe = 0$ and $eA(1-e) \cdot (1-e)x(1-e) = \{0\} = (1-e)x(1-e) \cdot (1-e)Ae$ implies $(1-e)x(1-e) = 0$ [2015-6, p. 141]. A Jordan derivation $\delta\colon A \to A$ is called *singular* for the Peirce decomposition when $\delta(eAe + (1-e)A(1-e)) = 0$, $\delta(eA(1-e)) \subseteq (1-e)Ae$ and $\delta((1-e)Ae) \subseteq eA(1-e)$ [2015-6, p. 144]. Such a map is a Lie triple derivation that no derivation plus central map can produce, so it enters the final form [2015-6, p. 144].

Theorem 5.22 ([2015-6, Theorem 5.1, p. 159])**.** *Let A be a unital 2-torsion free algebra with a nontrivial idempotent e satisfying the annihilation condition above. Assume one of the conditions* (A), (B), (C), (D)*, on the corners eAe and $(1-e)A(1-e)$: generation by idempotents and commutators, or the equality of the corner centers with $Z(A)e$ and $Z(A)(1-e)$, and assume one of the conditions* (1) *to* (4)*, the absence of central ideals in a corner or a prescribed form of $Z(A)$. Then any Lie triple derivation $D\colon A \to A$ is of the form $D = d + \delta + \gamma$, where $d\colon A \to A$ is a derivation, $\delta\colon A \to A$ is a singular Jordan derivation and $\gamma\colon A \to Z(A)$ is a linear map that vanishes on $[[A, A], A]$ [2015-6, p. 159].*

For Lie derivations the singular part disappears.

Theorem 5.23 ([2015-6, Theorem 5.3, p. 160])**.** *Under the analogous hypotheses, a Lie derivation $D\colon A \to A$ is of the form $D = d + \gamma$, where $d\colon A \to A$ is a derivation and $\gamma\colon A \to Z(A)$ is a linear map that vanishes on $[A, A]$ [2015-6, p. 160].*

The applications cover the standard families [2015-6, pp. 161–163].

Corollary 5.24 ([2015-6, Corollary 5.5, p. 161])**.** *Let $A = M_n(A)$, $n \geq 3$, be a matrix algebra over a 2-torsion free unital algebra A. Then every Lie triple derivation $D\colon A \to A$ is of the form $D = d+\gamma$, where d is a derivation and $\gamma\colon A \to Z(A)$ vanishes on $[[A, A], A]$.*

The second family is the nest algebras.

Corollary 5.25 ([2015-6, Corollary 5.12, p. 163]). *Let $\mathcal{N}$ be a nest of a complex Hilbert space H, $\dim H \geq 2$. Then every Lie triple derivation D of the nest algebra $\mathcal{T}(\mathcal{N})$ is of the form $D = d + \gamma$, where d is a derivation and $\gamma\colon \mathcal{T}(\mathcal{N}) \to \mathbb{C}I$ vanishes on $[[\mathcal{A}, \mathcal{A}], \mathcal{A}]$. In particular, every Lie derivation of a nest algebra has the standard form.*

The dimension 2 case can fail. Benkovič recalls an example from his earlier work with Eremita of a triangular algebra $\mathcal{T}_2(A)$ over a $\mathbb{Z}_2$-graded algebra. On it, a Lie triple derivation is not the sum of a derivation and a central linear mapping vanishing on the second commutator [2015-6, p. 163]. On the same algebra every Lie derivation is still of the standard form [2015-6, p. 164].

On a nest algebra of a Banach space the same conclusion holds for every Lie-type derivation, with no dimension bound [2017-17, Theorem 3.1, p. 6]. Zhang and Wei work with the $(n-1)$-th commutator $p_n(x_1, \ldots, x_n) = [\cdots[[x_1, x_2], x_3], \ldots, x_n]$ and call a linear map L a Lie n-derivation when the Leibniz rule holds for p_n [2017-17, p. 2]. Their theorem reads as follows [2017-17, Theorem 3.1, p. 6].

Theorem 5.26 ([2017-17, Theorem 3.1, p. 6]). *Let X be a Banach space, $\mathcal{N}$ a non-trivial nest on X and $\operatorname{Alg}\mathcal{N}$ the nest algebra associated with $\mathcal{N}$. Then $L\colon \operatorname{Alg}\mathcal{N} \to B(X)$ is a Lie n-derivation if and only if there exist a derivation $D\colon \operatorname{Alg}\mathcal{N} \to B(X)$ and a linear map $H\colon \operatorname{Alg}\mathcal{N} \to \mathbb{C}I$ that vanishes on each $(n-1)$-th commutator, such that $L = D + H$. If L is continuous, then D and H are continuous as well.*

So every Lie-type derivation of a nest algebra on a Banach space has the standard form [2017-17, p. 3]. The authors prove the theorem for the algebra itself and for subalgebras that meet six conditions on their rank-one operators [2017-17, pp. 6–7]. The nest algebra meets them [2017-17, p. 7].

5.6 Lie triple derivations across the classes of algebras

A Lie triple derivation satisfies the triple rule and need not be a Lie derivation, because the two differ by a map into the centre that vanishes on brackets. Section 5.3 stated the rule and gave the decomposition on von Neumann algebras. This section carries the same question to the concrete classes, where the answer has one shape: an inner or standard part plus a

central defect. What changes between the classes is which derivation counts as standard and whether the defect can be written down.

Take the algebra $T_n(C)$ of upper triangular matrices over a commutative unital ring, with a 2-torsion free unital bimodule M. Every Lie triple derivation d of $T_n(C)$ into M is a sum of three terms,

$$d = \Delta + \delta + \tau.$$

Here Δ is a derivation and δ is an antiderivation that vanishes on the diagonal matrices. The third term τ carries $T_n(C)$ into the centre of M and sends every commutator to zero [2011-6, Theorem 2.4, p. 823]. The three terms are visible separately in the abstract, which names the standard Lie derivation and the antiderivation and leaves the central term to the theorem. The antiderivation appears because the algebra is triangular.

The nest algebra case is known in both directions. Take a nest N of projections on a complex Hilbert space and a weak* closed algebra B containing the nest algebra $T(N)$. A linear map δ of $T(N)$ into B is a Lie triple derivation if and only if

$$\delta(A) = [S, A] + f(A)I \qquad \text{for all } A \in T(N),$$

where $S \in B$ and f is a linear functional that vanishes on every second commutator $[[X, Y], Z]$ [2007-8, Theorem 1.1, p. 882]. The if-and-only-if is the content: the inner part plus that functional is the whole class rather than one description among several.

The same algebras were treated without linearity. Write $\operatorname{Alg} N$ for the nest algebra of a nest on a Banach space, and suppose that a nontrivial element of the nest is complemented. Let Φ be a Lie triple derivable map of $\operatorname{Alg} N$ into $B(X)$. Then for all A, B in the algebra there is a scalar $\lambda_{A,B}$ with $\Phi(A + B) = \Phi(A) + \Phi(B) + \lambda_{A,B}I$. Moreover $\Phi = D + \tau$, where D is an additive derivation. The map τ carries the algebra into the scalar multiples of the identity, has the same additive defect, and vanishes on second commutators [2013-11, Abstract, p. 653]. On the same algebra a Jordan derivable map is an additive derivation, with no defect at all [2013-11, Abstract, p. 653]. The two derivable conditions therefore differ on nest algebras. On rings the Jordan condition is the weaker one, so this is the case worth naming.

A TUHF algebra is a triangular algebra of Hilbert-space operators. For a continuous Lie triple derivation L of one there is an associative derivation D with $L = D + \lambda$. The map λ carries the algebra into its centre and sends

every bracket of operators to zero [2005-5, Abstract, p. 399]. That is the triangular shape again, restricted to the continuous maps.

Full matrix algebras give the same shape without linearity. On $M_n(R)$ over a 2-torsion free commutative ring with identity, where $n \geq 2$, every nonlinear Lie triple derivation is a sum of three terms: an inner derivation, an additive induced derivation, and a functional that annihilates all second commutators [2013-13, Abstract, p. 77].

A ring with a nontrivial idempotent gives the same decomposition, linearity apart. Under a mild condition, a Lie triple derivable map Φ of such a ring into itself satisfies $\Phi(A+B) = \Phi(A) + \Phi(B) + z_{A,B}$ for a central element $z_{A,B}$ depending on the pair. On a prime ring of characteristic not two, $\Phi = D + \tau$, where D is an additive derivation into the central closure and τ carries the ring into the extended centroid with the same additive defect [2014-17, Abstract].

Three of the four results say the same thing. Each is a standard derivation together with a map into the centre that vanishes on brackets. The fourth says the two derivable conditions differ. The section exists so that a reader who meets the triple rule in one class can see what it becomes in the others.

Part III

Enlarged classes and local maps

Chapter 6

Generalized, skew and twisted derivations

A derivation follows the product rule. Several lines of work weaken that rule, and each weakening starts from a different question. The first line began with a question about norms. Brešar asked how far the composition of two derivations sits from a wider class of maps, and he called that class the generalized derivations [1991-2, p. 89]. A second line keeps the rule but turns it: multiplication on one side is twisted by an endomorphism, and the twisted derivations arise. Each of these classes then developed a life of its own, with its own Posner-type questions, its own commutativity theorems, and its own operator examples.

This chapter follows the lines in turn. Section 6.1 reports the generalized derivations, from the norm estimate where they were named to the algebraic study of their products and commutator identities. The sections after it take up the twists.

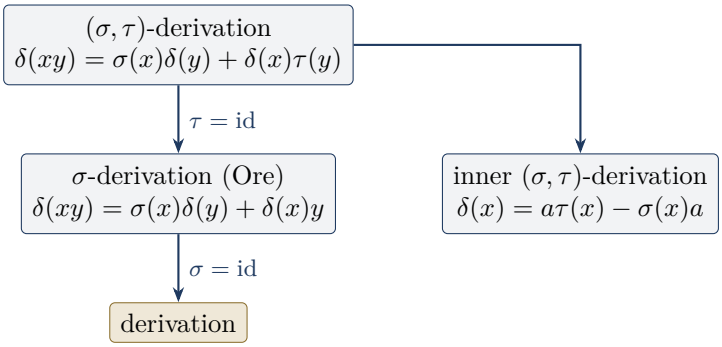


Figure 6.1: Twisted derivations and their specializations. With $\tau = \mathrm{id}$ one gets the σ-derivations of Ore; with $\sigma = \tau = \mathrm{id}$, ordinary derivations.

6.1 Generalized derivations

A classical theorem of Posner concerns rings of characteristic not 2. In a prime ring, the composition of two derivations is again a derivation only if one of the factors is zero [1957-2, Theorem 2]. Estimating how close such a composition lies to a bigger class of maps made Brešar name that class in 1991 [1991-2, p. 89].

Definition 6.1 ([1991-2, p. 89])**.** Let A be a ring. An additive mapping $\delta\colon A \to A$ will be called a *generalized derivation* if there exists a derivation h of A such that $\delta(xy) = \delta(x)y + xh(y)$ for all $x, y \in A$.

A generalized derivation with $h = 0$ is a left multiplier, an additive map with $f(xy) = f(x)y$ [1998-5, p. 1148]. The two-sided special case comes from operator theory [1998-5, p. 1147].

Definition 6.2 ([1998-5, p. 1147])**.** Let A be an algebra. A map of the form $x \mapsto ax + xb$, where a and b are fixed elements of A, is called a *generalized inner derivation* (or inner generalized derivation).

Such maps extend the inner derivations $x \mapsto ax - xa$, and in operator algebras they form a class of elementary operators [1998-5, p. 1147]. Every generalized inner derivation is a generalized derivation: for $f(x) = ax + xb$ one has $f(xy) = f(x)y + xh(y)$ with the inner derivation $h(y) = yb - by$ [1998-5, p. 1147]. The concept thus covers derivations and generalized inner derivations at once [1998-5, p. 1148].

On a semiprime ring the map h of the definition is forced to be a derivation, and on a unital ring it is forced by the value at 1 [1991-2, Remarks 1 and 2, p. 90].

Remark 6.3 ([1991-2, Remark 1, p. 90]). Let A be a ring, $h\colon A \to A$ be any function, and $\delta\colon A \to A$ be an additive mapping satisfying $\delta(xy) = \delta(x)y + xh(y)$ for all $x, y \in A$. Computing $\delta(xyz)$ in two ways gives $x(h(yz) - h(y)z - yh(z)) = 0$ for all $x, y, z \in A$, and computing $\delta(x(y+z))$ in two ways gives $x(h(y+z) - h(y) - h(z)) = 0$. Thus, if A has the property that $Aa = 0$ implies $a = 0$, in particular if A is semiprime, then h is a derivation.

The unital case gives the relation between all generalized derivations and the inner ones [1991-2, p. 90].

Remark 6.4 ([1991-2, Remark 2, p. 90]). Suppose a ring A has a unit element 1 and take $x = 1$ in the identity. Then $\delta(y) = \delta(1)y + h(y)$ for all $y \in A$. Hence every generalized derivation of A is an inner generalized derivation if and only if every derivation of A is inner.

The theorem of Posner survives the enlargement of the class, in a form Brešar noted in the same paper [1991-2, Remark 3, pp. 90–91]. If A is a prime ring of characteristic not 2, and the composition $d_1 d_2$ of two derivations is a generalized derivation of A, then $d_1 = 0$ or $d_2 = 0$. The question when a product of two generalized derivations is again one is the first main result of the study of Hvala. He set out to begin the algebraic study of these maps [1998-5, p. 1147].

Theorem 6.5 ([1998-5, Theorem 1, p. 1149]). *Let R be a prime ring,* $\operatorname{char} R \neq 2$, *and let $f_1, f_2\colon R \to R$ be generalized derivations. The product $f_3 = f_1 f_2$ is again a generalized derivation if and only if one of the following possibilities holds:*

(i) there exists $\gamma \in C$ such that either $f_1(x) = \gamma x$ or $f_2(x) = \gamma x$;

(ii) there exist $a, b \in Q_r(R_c)$ such that $f_1(x) = xa$ and $f_2(x) = xb$;

(iii) there exist $a, b \in Q_r(R_c)$ such that $f_1(x) = ax$ and $f_2(x) = bx$;

(iv) there exist $a, b \in Q_r(R_c)$ and $\lambda, \mu \in C$ such that $f_1(x) = ax + xb$ and $f_2(x) = \lambda x + \mu(ax - xb)$.

Here C is the extended centroid of R, $R_c = RC$ the central closure, and $Q_r(R_c)$ its right Martindale ring of quotients [1998-5, p. 1148]. Applied

to $f_1 = f_2$, the theorem leaves only a one-sided multiplication [1998-5, Corollary 5, p. 1151].

Corollary 6.6 ([1998-5, Corollary 5, p. 1151]). *Let $f\colon R \to R$ be a generalized derivation of a prime ring R with $\operatorname{char} R \neq 2$. Then f^2 is a generalized derivation if and only if there exists $a \in Q_r(R_c)$ such that either $f(x) = ax$ for all $x \in R$, or $f(x) = xa$ for all $x \in R$. Moreover, $f^2 = 0$ if and only if f is of this form and $a^2 = 0$.*

Two more results of the same paper extend theorems known for derivations to the larger class [1998-5, pp. 1148, 1153, 1159]. The theorem of Lanski that derivations commuting pointwise are dependent moves to generalized derivations [1998-5, p. 1148].

Theorem 6.7 ([1998-5, Theorem 2, p. 1153]). *Let R be a noncommutative ring, $\operatorname{char} R \neq 2$, and let $f_1, f_2\colon R \to R$ be nonzero generalized derivations satisfying $[f_1(x), f_2(x)] = 0$ for all $x \in R$. Then there exists $\lambda \in C$ such that $f_1(x) = \lambda f_2(x)$ for all $x \in R$.*

Giambruno and Herstein proved that a derivation with $d(x)^n = 0$ everywhere is zero [1981-3]. Hvala extended this to generalized derivations [1998-5, p. 1148].

Theorem 6.8 ([1998-5, Theorem 3, p. 1159]). *Let $n > 1$ be a fixed integer and R a prime ring of characteristic not equal to $2, \ldots, n-1$. If $f\colon R \to R$ is a generalized derivation such that $f(x)^n = 0$ for all $x \in R$, then $f = 0$.*

Wei carried the same identity from a prime ring to a semiprime one [2004-9, p. 453]. He also removed the restriction on the characteristic and put the condition on an essential ideal [2004-9, p. 453]. The route runs through the left Martindale quotient ring, to which every generalized derivation of an essential ideal extends [2004-9, Theorem 2, p. 455]. On a prime ring the nilpotent values still force the map to vanish [2004-9, Theorem 5, p. 457].

Theorem 6.9 ([2004-9, Theorem 12, p. 460]). *Let R be a semiprime ring and μ a generalized derivation on an essential ideal I of R. Suppose that $\mu(x)^n = 0$ for all $x \in I$, where n is a fixed positive integer. Then $\mu = 0$.*

The proof extends μ to the quotient ring and then reads the values in the stalks of a sheaf, each of which is a prime ring [2004-9, p. 460]. Theorem 5 then applies in each stalk [2004-9, p. 460].

After the structural theory, a second line asks commutativity questions [2007-12, pp. 1–2]. Ashraf, Ali and Rani studied a prime ring with a generalized derivation F whose associated derivation d satisfies identities such as $d(x) \circ F(y) = 0$ on a nonzero ideal [2007-12, p. 1]; Huang then carried these conditions from an ideal to a Lie ideal [2007-12, p. 2].

Theorem 6.10 ([2007-12, Theorem 3.1, p. 3])**.** *Let R be a prime ring with $\operatorname{char} R \neq 2$, U a Lie ideal such that $u^2 \in U$ for all $u \in U$, and F a generalized derivation associated with $d \neq 0$. If $d(x) \circ F(y) = 0$ for all $x, y \in U$, then $U \subseteq Z(R)$.*

The same conclusion holds when the mixed anticommutator is replaced by a commutator that matches the bracket on U [2007-12, p. 5].

Theorem 6.11 ([2007-12, Theorem 3.8, p. 5])**.** *Let R be a prime ring with $\operatorname{char} R \neq 2$, U a Lie ideal such that $u^2 \in U$ for all $u \in U$, and F a generalized derivation associated with $d \neq 0$. If $[d(x), F(y)] = [x, y]$ for all $x, y \in U$, then $U \subseteq Z(R)$.*

Dhara carried the same identities to a semiprime ring and weakened the position of the central values [2010-3, p. 1]. Over a semiprime ring with a nonzero ideal I, he proves that a nonzero generalized derivation with $F([x, y]) = \pm[x, y]$ on I forces a nonzero central ideal [2010-3, Theorem 2.4, p. 2]. Over a prime ring the ring itself has to commute [2010-3, Corollary 2.5, p. 3].

Theorem 6.12 ([2010-3, Theorem 2.4, p. 2])**.** *Let R be a semiprime ring, I a nonzero ideal of R, and F a generalized derivation of R associated with a derivation d with $d(I) \neq 0$. If $F([x, y]) = \pm[x, y]$ for all $x, y \in I$, then R contains a nonzero central ideal.*

The anticommutator gives the same result [2010-3, Theorem 2.6, p. 3]. The two cases in which the difference $F([x, y]) \pm [x, y]$ lies in the center rather than vanishing are kept apart, because they give a weaker conclusion [2010-3, Theorem 2.8, p. 4].

The pattern of the section is the one the whole encyclopaedia follows. A bigger class of maps is named when a derivation question needs more than derivations, and each classical theorem about derivations is then asked again inside it. The twists of the next sections extend in a different direction. The two notions of the class can share one map, and the shared map raises a continuity question [2023-13, p. 1]. Hosseini and Park call a linear map a strongly generalized derivation of order n when its value on a product is

a sum of n terms $E_k(a)F_k(b) + G_k(a)H_k(b)$ [2023-13, Definition 2.1, p. 3]. The class contains the derivations, the generalized derivations, the ternary derivations and the generalized (σ, τ)-derivations [2025-9, p. 2]. On a unital C^*-algebra a strongly generalized derivation of order one is automatically continuous under conditions on the four maps and on the star [2023-13, Theorem 2.9, p. 6]. Hosseini later proves the same conclusion from a condition on the products of the two pairs [2025-9, Theorem 2.1, p. 5].

The same tower of classes exists for n-ary algebras, where the maps take $n-1$ arguments [2016-17, p. 15]. For a multiplicative n-Hom Lie superalgebra the classes form a chain, from the centre derivations to the quasiderivations to the generalized derivations [2016-17, Proposition 3.1, p. 6]. Each of these spaces is a subalgebra of the algebra of endomorphisms, and the centre derivations form an ideal of the derivations [2016-17, Proposition 3.1, p. 6]. When the centre of the algebra is zero, every derivation of the larger algebra built from it splits as a quasiderivation plus a centre derivation [2016-17, Proposition 4.3, p. 15].

For a Lie colour algebra the space of generalized derivations splits into two pieces, and Chen, Ma and Ni give the split [2013-3, p. 923]. The quasiderivations are the maps for which the defining identity holds with the same map on both sides [2013-3, Definition 1.5, p. 925]. The quasicentroid is the set of maps that commute with the bracket up to the colour sign [2013-3, Definition 1.7, p. 925].

Theorem 6.13 ([2013-3, Theorem 2.3, p. 928])**.** *Let L be a Lie colour algebra. Then the space of generalized derivations of L is the sum of the quasiderivations and the quasicentroid:* $\mathrm{GDer}(L) = \mathrm{QDer}(L) + \mathrm{QC}(L)$.

The authors also show that the generalized derivations, the quasiderivations and the centroid are Lie colour subalgebras, and that the centre derivations form an ideal of the derivations [2013-3, Proposition 2.1, p. 926]. When the centre of L is zero, the centroid is the intersection of the quasiderivations and the quasicentroid [2013-3, Theorem 2.6, p. 929].

The cohomological side of the same notion is worked out for n-Lie algebras [2021-12, p. 1]. A generalized derivation there is a linear map of $n-1$ arguments that obeys three identities [2021-12, p. 7]. The authors realise it as a one-cocycle in a cohomology theory of the algebra, and the inner generalized derivations are the one-coboundaries [2021-12, p. 3]. The cohomology of the extension by such a derivation then equals the cohomology of the algebra itself [2021-12, Corollary 3.2, p. 27].

On a C^*-algebra the name carried three meanings at once, and Hosseini, Peralta and Su show that they agree [2025-8, p. 1]. The first meaning is the

class of Definition 6.1, with the derivation taking values in the double dual of the bimodule [2025-8, p. 1]. The second reads $F(ab) = F(a)b + aF(b) - a\xi b$, with ξ in the double dual [2025-8, p. 2]. The third allows two further maps, neither of them linear, one on each side of the product [2025-8, Definition 1.1, p. 3].

Theorem 6.14 ([2025-8, Theorem 2.8, p. 10])**.** *Let A be a C^*-algebra and M a Banach A-bimodule. Suppose that $F\colon A \to M$ is a generalized derivation of the third type. Then F is continuous, and it is a generalized derivation of the second type.*

The same paper shows that a generalized Jordan derivation of the third type on a C^*-algebra is continuous [2025-8, Theorem 3.1, p. 13]. Together the two theorems make the three meanings of the name one class for such algebras [2025-8, p. 1].

Further work in the generalized-derivation line, held in the collection: *Higher-order commutators with power central values on rings and algebras involving generalized derivations* [2021-1].

6.2 (σ,τ)-derivations and skew derivations

The generalized derivations of Definition 6.1 keep the right side of the product rule intact. The next line of maps weakens both sides at once, turning one factor by an endomorphism [2018-16, p. 892].

Definition 6.15 ([2018-16, p. 892])**.** Let R be a ring and let α and β be endomorphisms of R. An additive mapping $d\colon R \to R$ is said to be an *(α,β)-derivation* if $d(xy) = d(x)\alpha(y) + \beta(x)d(y)$ holds for all $x, y \in R$.

The ordinary derivations are the case $\alpha = \beta = 1_R$, and the skew derivations of the theory of rings of quotients are the case $\beta = 1_R$ with α an automorphism [2009-5, p. 260].

Definition 6.16 ([2009-5, p. 260])**.** Let σ be an automorphism of R. A *σ-derivation* $\delta\colon R \to R$ is a mapping satisfying $\delta(r+s) = \delta(r) + \delta(s)$ and $\delta(rs) = \sigma(r)\delta(s) + \delta(r)s$ for all $r, s \in R$. For brevity, σ-derivations are generally called *skew derivations*.

Some authors twist the left factor instead, and the two conventions agree after the names of the two maps are exchanged.

Definition 6.17 ([2012-7, p. 885])**.** Let α be an automorphism of a ring R. An additive mapping $D\colon R \to R$ is called an *α-derivation* (or a skew derivation) on R if $D(xy) = D(x)y + \alpha(x)D(y)$ for all $x, y \in R$. In this case α is called the associated automorphism of D.

The simplest σ-derivation is the map $\sigma - 1_R$, and 1_R-derivations are ordinary derivations; the concept generalizes both derivations and automorphisms [2009-5, p. 260]. An element b of the right Martindale quotient ring gives the *inner* α-derivation $D(x) = bx - \alpha(x)b$ [2012-7, p. 885]. Generalized derivations and skew derivations are two natural generalizations of the same notion, and they meet in the expected way [2018-16, p. 892].

Definition 6.18 ([2018-16, p. 892])**.** Let R be a ring and let α and β be endomorphisms of R. An additive mapping $F\colon R \to R$ is called a *generalized (α, β)-derivation* on R if there exists an (α, β)-derivation $d\colon R \to R$ such that $F(xy) = F(x)\alpha(y) + \beta(x)d(y)$ holds for all $x, y \in R$.

A classical theorem re-enters through a twisted version, since Chou and Liu extended the Engel theorem of Lanski from derivations to σ-derivations [2009-5, p. 260].

Theorem 6.19 ([2009-5, Theorem 1.1, p. 260])**.** *Let R be a prime ring and L a noncommutative Lie ideal of R. Suppose that δ is a nonzero σ-derivation of R such that $[\delta(x), x]_k = 0$ for all $x \in L$, where k is a fixed positive integer. Then* $\operatorname{char} R = 2$ *and $R \subseteq M_2(F)$ for some field F.*

A power identity in one variable forces a skew derivation to be ordinary, unless the ring satisfies a small standard identity [2012-7, p. 886].

Theorem 6.20 ([2012-7, Theorem 1.2, p. 886])**.** *Let m, n, r be nonzero fixed positive integers, R a 2-torsion free prime ring, L a non-central Lie ideal of R, and $D\colon R \to R$ a skew derivation of R. If*

$$E(x) = D(x^{m+n+r}) - D(x^m)x^{n+r} - x^m D(x^n)x^r - x^{m+n}D(x^r)$$

vanishes for all $x \in L$, then D is a usual derivation of R or R satisfies $s_4(x_1, \ldots, x_4)$, the standard identity of degree 4.

The theorem of Herstein on nilpotent values has a twisted form in that paper [2018-16, p. 893]. Chang worked with right generalized β-derivations [2018-16, p. 893]. Take a prime ring R with a noncommutative Lie ideal L, and a right generalized β-derivation f of R. If $f(x)^n = 0$ for all $x \in L$, where n is a fixed positive integer, then $f = 0$ [2018-16, p. 893]. The

commutativity theorems of Section 6.1 also survive in a nilpotent form [2018-16, pp. 893–894].

Theorem 6.21 ([2018-16, Theorem 2.4, p. 893]). *Let R be a semiprime ring and n a fixed positive integer. If R admits a generalized derivation F associated with a nonzero derivation d such that $(F(x \circ y) - F(x) \circ y)^n = 0$ for all $x, y \in R$, then R is commutative.*

The anti-commutator can be replaced by the commutator, with the same conclusion [2018-16, p. 894].

Theorem 6.22 ([2018-16, Theorem 2.5, p. 894]). *Let R be a semiprime ring and n a fixed positive integer. If R admits a generalized derivation F associated with a nonzero derivation d such that $(F([x, y]) - [F(x), y])^n = 0$ for all $x, y \in R$, then R is commutative.*

Dhara, Ali and Das put an Engel condition on a generalized derivation itself [2016-10, p. 1]. They allow the exponents in the repeated commutator to differ, one for the value of F and one for each following argument [2016-10, p. 1].

Theorem 6.23 ([2016-10, Theorem 1.1, p. 2]). *Let R be a prime ring with Utumi quotient ring U, let F be a nonzero generalized derivation of R, and let L be a noncentral Lie ideal of R. Suppose that $[F(u^{n_1}), u^{n_2}, \ldots, u^{n_k}] = 0$ for all $u \in L$, where $n_1, n_2, \ldots, n_k \geq 1$ are fixed integers. Then $F(x) = \alpha x$ for some $\alpha \in C$ and all $x \in R$, or R satisfies the standard identity s_4.*

On the whole ring the second alternative disappears, and the map is a scalar multiple of the identity [2016-10, Theorem 1.2, p. 2]. The same paper applies the result to continuous generalized derivations on Banach algebras [2016-10, p. 2].

Chang had asked the Engel question for the twisted map itself [2008-4, p. 1641]. A right generalized β-derivation is an additive map f with $f(xy) = f(x)y + \beta(x)\delta(y)$ for a β-derivation δ, where β is an automorphism [2008-4, p. 1641]. She requires that a fixed element a sends the iterated commutator of $f(x)$ with x to zero [2008-4, p. 1641].

Theorem 6.24 ([2008-4, Main Theorem, p. 1642]). *Let R be a noncommutative prime ring and $a \in R$. Suppose that f is a right generalized β-derivation of R such that $a[f(x), x]_k = 0$ for all $x \in R$, where k is a fixed positive integer. Then $a = 0$, or there exists s in the extended centroid C such that $f(x) = sx$ for all $x \in R$, except when R is the ring of 2×2 matrices over the field with two elements.*

The result extends an earlier one for a β-derivation [2008-4, p. 1641]. The exception is genuine, and the proof separates the X-inner and X-outer cases for the automorphism [2008-4, p. 1647].

Güven put a generalized (σ, τ)-derivation on a pair of ideals and compared its bracket with the twisted bracket [2013-7, p. 1]. Two ideals enter, and the twisted commutator is $[x, y]_{\alpha,\beta} = x\beta(y) - \alpha(y)x$ [2013-7, p. 1]. The main case is an identity between the values of the map and the twisted bracket itself.

Theorem 6.25 ([2013-7, Theorem 4, p. 7])**.** *Let R be a prime ring and let $h\colon R \to R$ be a generalized $(1, \tau)$-derivation associated with a nonzero (λ, γ)-derivation d. Let I and J be nonzero ideals of R. If $h(x, y)_{\alpha,\beta} = (x, y)_{\alpha,\mu}$ for all $x \in I$ and $y \in J$, or the same with a minus sign, then R is commutative.*

The author proves the companion statement for $h[x, y]_{\alpha,\beta}$ in place of $h(x, y)_{\alpha,\beta}$, again with both signs [2013-7, p. 1]. The paper also contains a set of lemmas that move a centralizer of a twisted commutator from an ideal to the whole ring [2013-7, Theorem 1, p. 3].

The twist of Definition 6.15 takes one more parameter, and Bera, Dhara and Kar state it as follows [2023-11, p. 888].

Definition 6.26 ([2023-11, p. 888])**.** Let R be a prime ring, $b \in Q_r$ and α an automorphism of R. A linear map $F\colon R \to Q_r$ is called a *b-generalized skew derivation* of R if there is a skew derivation d of R such that $F(xy) = F(x)y + b\alpha(x)d(y)$ holds for all $x, y \in R$.

Here Q_r is the right Martindale quotient ring of R [2023-11, p. 888]. The map $x \mapsto ax + b\alpha(x)c$ is such a derivation, and the authors call it inner [2023-11, p. 888]. Their main theorem decides an identity in two such maps on the values of a multilinear polynomial [2023-11, Theorem 1.1, p. 889].

Theorem 6.27 ([2023-11, Theorem 1.1, p. 889])**.** *Let R be a prime ring of characteristic different from 2, $Z(R)$ its centre, Q_r its right Martindale quotient ring, C its extended centroid, $f(x_1, \ldots, x_n)$ a non-central multilinear polynomial over C, $b \in Q_r$ and F, G two b-generalized skew derivations of R. If there is $a \in R \setminus Z(R)$ such that*

$$[a, F(f(r_1, \ldots, r_n))f(r_1, \ldots, r_n)] = f(r_1, \ldots, r_n)G\left(f(r_1, \ldots, r_n)\right)$$

for all $r_1, \ldots, r_n \in R$, then $f(x_1, \ldots, x_n)^2$ is central valued on R, and there are $p, q \in Q_r$ with $F(x) = px$ and $G(x) = xq$ for all $x \in R$, where $[a, p] = q$.

So both maps of that identity are one-sided multiplications. The case $G = 0$ leaves one map, and the authors record it as Corollary 1.2 [2023-11, Corollary 1.2, p. 889].

Dhara had proved the one-map case earlier, for a single generalized skew derivation [2019-4, p. 1]. His conclusion is the same pair of alternatives.

Theorem 6.28 ([2019-4, Theorem 1.1, p. 3])**.** *Let R be a noncommutative prime ring of characteristic different from 2, with extended centroid C, and let $f(X_1, \ldots, X_n)$ be a multilinear polynomial over C that is not central valued on R. Suppose that F is a generalized skew derivation of R and that $a \in R \setminus C$ satisfies $[a, [F(f(r)), f(r)]] \in Z(R)$ for all $r \in R^n$. Then $F(x) = \lambda x$ for some $\lambda \in C$ and all $x \in R$, or $F(x) = bx + xb + \lambda x$ for some b in the right Martindale quotient ring and f^2 is central valued on R.*

Dhara reads the second alternative as a centralizing condition on the set of commutators of the values [2019-4, Corollary 1.2, p. 3]. He needs that set to be nonzero, and then its centralizer is the center of the ring [2019-4, Corollary 1.2, p. 3].

A third paper of his puts the centralizing condition on a Lie ideal, where the two alternatives collapse to two explicit forms of the map [2019-5, p. 1]. The characteristic here is not 2 or 3.

Theorem 6.29 ([2019-5, Theorem 1.1, p. 2])**.** *Let R be a prime ring of characteristic different from 2 and 3, with right Martindale quotient ring Q_r and extended centroid C, let F be a nonzero generalized skew derivation of R, and let L be a noncentral Lie ideal of R. If $[[F(u), u], F(u)] \in Z(R)$ for all $u \in L$, then $F(x) = \lambda x$ for some $\lambda \in C$ and all $x \in R$, or R satisfies the standard identity s_4 and $F(x) = ax + xa + \lambda x$ for some $a \in Q_r$ and $\lambda \in C$.*

Dhara shows by example that neither the hypothesis that the ring is prime nor the exclusion of characteristic 3 can be dropped [2019-5, p. 1].

The same class carries a Jordan-type identity in two maps, and Pandey and Pandey settle it [2023-15, p. 3]. Their two maps share one automorphism and act on the values of a noncentral multilinear polynomial [2023-15, Theorem 6, pp. 3–4]. The theorem lists five possible forms for the pair [2023-15, Theorem 6, pp. 3–4]. One form makes the polynomial central valued on the ring, and the others reduce the maps to one-sided multiplications or to a skew derivation [2023-15, Theorem 6, pp. 3–4]. A single map that acts as a 2-Jordan multiplier takes three forms, recorded as Corollary 21 [2023-15, Corollary 21, p. 18].

Over a commutative ring the class has a universal object, in the way the module of differentials serves ordinary derivations [2020-10, p. 3]. Chaudhuri allows two different endomorphisms and a target bimodule [2020-10, p. 2]. Her theorem factors every (σ,τ)-derivation through one universal map [2020-10, Theorem 3.1, p. 3]. The factorization is left linear when τ is invertible and right linear when only σ is [2020-10, Theorem 3.1, p. 3]. Over the ring of integers of a quadratic field the class has a complete description [2020-10, Theorem 4.2, pp. 6–7]. Every $\mathbb{Z}$-linear map with $D(1)=0$ is a (σ,τ)-derivation there, and a condition on the value at $\sqrt{d}$ decides whether it is inner [2020-10, Theorem 4.2, pp. 6–7].

Number rings make the count of these maps explicit [2026-25, p. 9]. For an integral extension $B=A[\theta]$ the (σ,τ)-derivations are the A-linear maps with $D(1)=0$ that satisfy $n-1$ linear equations, one for each power of θ [2026-25, Theorem 3.2, p. 9]. Over a cyclotomic field with two different field automorphisms the resulting space has rank $\varphi(n)$ [2026-25, Corollary 3.5, p. 11]. The same authors count the maps on the rings of integers. For the p-th cyclotomic ring the module has rank $p-1$ [2026-23, Corollary 3.9, p. 17]. For the bi-quadratic ring $\mathbb{Z}[\sqrt{m},\sqrt{n}]$ it has rank 4 [2026-23, Corollary 3.19, p. 24]. On that ring they give a criterion for a map to be inner [2026-23, Theorem 3.22, p. 25]. They also show that non-inner ones exist [2026-23, Corollary 3.24, p. 26].

Argaç and Albaş gave the notion a form in the language of categories, for a map into a bimodule rather than into the ring [2002-1, p. 977]. A generalized (σ,τ)-derivation there is a pair (f,w) with $f(xy)=f(x)\tau(y)+\sigma(x)f(y)+\sigma(x)w\tau(y)$ [2002-1, (2.6), p. 978]. The pair is inner when $f(x)=m\tau(x)+\sigma(x)n$ and $w=-m-n$ [2002-1, (2.9), p. 978].

Theorem 6.30 ([2002-1, Theorem 3.1, p. 979])**.** *Let M be an A/k-bimodule. Suppose that $Am=0$ with $m\in M$ forces $m=0$, and that $\sigma\colon A\to A$ is surjective. Then the following sequence of k-modules is exact and splitting:*

$$0 \longrightarrow M \longrightarrow \mathrm{gDer}_k^{(\sigma,\tau)}(A,M) \longrightarrow \mathrm{Der}_k^{(\sigma,\tau)}(A,M) \longrightarrow 0.$$

The middle term is the module of generalized (σ,τ)-derivations, and the map from M sends m to the pair $(m_l\tau,-m)$ [2002-1, Theorem 3.1, p. 979]. The authors prove the same for the Jordan and the Lie versions [2002-1, Theorems 4.1 and 4.2, pp. 981–982]. They also give examples of a generalized Jordan (σ,τ)-derivation that is not a generalized (σ,τ)-derivation, so the two classes stay apart in general [2002-1, Examples 4.1 and 4.3, pp. 982–983].

The two branches of the chapter now separate. Generalized derivations ask what the weakening does to commutativity and dependence theorems; the (α, β) line asks which maps absorb a twist by an endomorphism. The next section follows a special case of the twist, the σ-derivations built from an endomorphism alone.

6.3 α-derivations and their characterizations

For a fixed automorphism the (α, β)-derivations collapse to a single twist. The theory of these maps was built in the Siberian school, where they were called directional derivatives, for the study of identities with automorphisms [1991-6, p. 1045].

Definition 6.31 ([1991-6, p. 1045])**.** Suppose s is an automorphism of a ring R. The map $\delta\colon R \to R$ will be called an *s-derivative* if $\delta(xy) = \delta(x)y + s(x)\delta(y)$ for all $x, y \in R$. If s is fixed or is not defined, δ is called a directional derivative. The automorphism s is said to be fundamental for δ.

The s-derivatives with a skew structure are the α-derivations of the later literature [2012-7, p. 885].

The s-derivatives of Definition 6.31 are the α-derivations of Definition 6.17, up to the side on which the twist acts [2012-7, p. 885]. An element of the ring supplies one, in the same way that it supplies an inner derivation [1991-6, p. 1045].

Remark 6.32 ([1991-6, Lemma 6, p. 1045])*.* Suppose s is an automorphism of the ring Q and $a \in Q$. Then the map $\mathrm{ad}_s\, a\colon x \mapsto ax - s(x)a$ is an s-derivative. A directional derivative of this form is called inner.

The inner form is one of three basic examples of the twist [2012-7, p. 885].

Remark 6.33 ([2012-7, p. 885])*.* Basic examples of α-derivations are usual derivations and the map $\alpha - 1$, where 1 denotes the identity map. Let R be a ring and fix an element b of the right Martindale quotient ring Q. The map $D\colon R \to R$ defined by $D(x) = bx - \alpha(x)b$ is an α-derivation, called an *inner α-derivation* (an inner skew derivation) defined by b.

The Siberian line tied these maps to the theory of polynomial identities [1991-6, p. 1045]. An essential identity with automorphisms and directional derivatives forces the ring to be a PI-ring:

Theorem 6.34 ([1991-6, Theorem 3, p. 1050])**.** *Suppose a semiprime ring with unity R satisfies an essential polylinear identity possessing automorphisms and directional derivatives. Then R will be a PI-ring if all the automorphisms corresponding to the directional derivatives occur in the notation for this identity and commute with the directional derivatives of this identity.*

Chou and Liu extended the Engel condition of Lanski to a σ-derivation (Theorem 6.19), and De Filippis and Fosner described when a power identity forces one to be ordinary (Theorem 6.20). The generalized form of the twist enters a skew-commutator version [2010-2, p. 843].

Theorem 6.35 ([2010-2, Theorem 1.3, p. 843])**.** *Let R be a prime ring of characteristic different from 2 with extended centroid C, F a non-zero generalized derivation of R, and $f(x_1, \ldots, x_n)$ a polynomial over C. If $F(u) \circ F(v) = 0$ for any $u, v \in f(R)$, then $f(x_1, \ldots, x_n)$ is central valued on R.*

For α the identity, the result reduces to the commutativity theorems of Section 6.1 [1998-5, p. 1148]. The multiplicative special case of the α-derivation is older still. The map $\delta = \alpha - 1$ is an α-derivation, so every endomorphism supplies one [2012-7, p. 885].

The twist has a Lie form, and it enters the enveloping algebra of a Hom-Lie algebra [2026-18, p. 4]. There a map d is an α-derivation when it commutes with the twisting map α and satisfies $d([x, y]) = [d(x), \alpha(y)] + [\alpha(x), d(y)]$ [2026-18, Definition 2.3, p. 4]. The same rule serves Hom-associative algebras, where $D(ab) = D(a)\alpha(b) + \alpha(a)D(b)$ [2026-18, Definition 2.9, p. 5]. A prescribed α-derivation extends to the universal enveloping algebra of the Hom-Lie algebra, and the pair formed by the algebra and the derivation has the expected universal property [2026-18, Theorem 3.11, p. 13]. When the twisting map is bijective, the enveloping algebra is isomorphic, with the derivations, to the twist of the enveloping algebra of the untwisted Lie algebra [2026-18, Theorem 4.8, p. 18]. The Poincaré–Birkhoff–Witt filtration survives the twist [2026-18, Theorem 4.10, p. 19].

The next section builds the skew polynomial rings where σ-derivations are the coefficient maps, the Ore extensions.

6.4 Derivations and Ore extensions

The σ-derivations of the previous sections are the coefficient maps of the skew polynomial ring [2017-6, p. 2838]. The construction goes back to

Ore [1933-1, p. 480]. He built polynomials over a division ring whose multiplication keeps the degree of a product equal to the sum of the degrees of the factors.

Definition 6.36 ([1933-1, p. 481])**.** Let K be a division ring, commutative or not. The product of two polynomials is associative and distributive, and the degree of a product is the sum of the degrees of the factors. It then suffices to define the product xa for $a \in K$, and the degree postulate forces

$$xa \;=\; \bar{a}\, x + a',$$

where $\bar{a}$ and a' lie in K. The element $\bar{a}$ is the conjugate and a' the derivative of a.

The two operations satisfy fixed rules [1933-1, p. 482]. Ore stated them as follows [1933-1, (4)–(5), p. 482].

Remark 6.37 ([1933-1, (4)–(5), p. 482])*.* The conjugate and the derivative obey

$$\overline{a+b} = \bar{a} + \bar{b}, \qquad \overline{ab} = \bar{a}\,\bar{b}, \qquad (a+b)' = a' + b', \qquad (ab)' = a\,b' + a'\,\bar{b}.$$

The conjugation is an automorphism of K, and the derivative is then an operator of derivation type [1933-1, p. 482]. This is the content of the first result of the paper [1933-1, Theorem 1, p. 482].

Theorem 6.38 ([1933-1, Theorem 1, p. 482])**.** *Through the correspondence $a \mapsto \bar{a}$ one obtains a homeomorphism in K; the elements $\bar{a}$ form a subfield $\bar{K}$ of K.*

Two classical theories are the special cases in which the rule is fixed [1933-1, p. 482]. The choice $\bar{a} = a$ gives the differential case, and $a' = 0$ the difference case.

Remark 6.39 ([1933-1, p. 482])*.* In the theory of linear differential equations K is commutative and $\bar{a} = a$, so the derivative has the ordinary properties of a derivative. In the theory of difference equations one puts $a' = 0$, though $\bar{a}$ is usually different from a [1933-1, p. 482].

The elements that commute with the new variable form a subfield [1933-1, p. 483].

Theorem 6.40 ([1933-1, Theorem 3, p. 483])**.** *The elements permutable with x form a field which is the greatest common subfield of the invariant field and the constant field.*

Ore also introduced the division theory of the ring [1933-1, p. 484]. The Euclidean algorithm for right-hand division runs over any K [1933-1, p. 484]. Left-hand division is stronger and needs the conjugate map to be surjective [1933-1, Theorem 6, p. 480].

Theorem 6.41 ([1933-1, Theorem 6, p. 480])**.** *In general, left-hand division can be performed in K only when the conjugate map is an automorphism of K.*

The recent literature gives the earlier construction its current form [2017-6, p. 2838]. The variable is written t and the two maps are named [2017-6, p. 2838].

Definition 6.42 ([2017-6, p. 2838])**.** Let R be a prime ring, σ an automorphism and δ a σ-derivation of R. The Ore extension $R[t;\sigma,\delta]$ is the ring of finite sums $\sum_i a_i t^i$, with componentwise addition and the multiplication subjected to the rule

$$ta \;=\; \sigma(a)\,t + \delta(a) \qquad \text{for } a \in R.$$

It is also called the skew polynomial ring over R twisted by δ.

The two classical cases of Remark 6.39 are the two types named by Chuang [2017-6, p. 2838].

Remark 6.43 ([2017-6, p. 2838])*.* A σ-derivation with σ the identity gives $R[t;\delta]$, the Ore extension of derivation type. A σ-derivation with $\delta(R) = 0$ gives $R[t;\sigma]$, the Ore extension of automorphism type. These two types have been most extensively studied [2017-6, p. 2838].

The skew polynomial ring is the home of the divided powers of differentiation [2010-1, p. 158]. Chuang, Lee, Liu and Tsai built these into higher derivations and used them to describe the endomorphisms of the Ore extension [2010-1, p. 158].

Definition 6.44 ([2010-1, p. 161])**.** By a higher derivation of order $n \geq 1$ of a ring R one means a map $\delta\colon R \to R$ such that, for each $0 < i < n$, there exist higher derivations δ_i and δ'_i of orders i and $n - i$ with

$$\delta(a+b) = \delta(a) + \delta(b), \qquad \delta(ab) = \delta(a)b + \sum_{i=1}^{n-1} \delta_i(a)\,\delta'_i(b) + a\,\delta(b)$$

for $a, b \in R$. Higher derivations of order one are ordinary derivations.

The description of the bimodule endomorphisms is the main application [2010-1, p. 174]. Each one is a series in the divided powers [2010-1, Theorem 12, p. 174].

Theorem 6.45 ([2010-1, Theorem 12, p. 174])**.** *Let R be a prime ring and δ a derivation of R. Any (R,R)-bimodule endomorphism θ of $Q[x;\delta]$ can be written uniquely in the form*

$$\theta(f) = \sum_{n=0}^{\infty} \zeta_n D_n(f) \qquad \text{for } f \in Q[x;\delta],$$

where $\zeta_n \in C_S(R)$ and $D_n = \frac{1}{n!}\frac{d^n}{dx^n}$.

The same framework settles the differential identities of the next section [2010-1, p. 172].

Theorem 6.46 ([2010-1, Theorem 11, p. 172])**.** *For $f_i, g_i \in Q[x;\delta]$, one has $\sum_{i=1}^{n} f_i\, r\, g_i = 0$ for all $r \in R$ if and only if $\sum_{i=1}^{n} f_i \otimes g_i = 0$ in $Q[x;\delta] \otimes_{C_S(R)} Q[x;\delta]$.*

The Ore extension is where the σ-derivation of Definition 6.17 becomes the coefficient map δ in the relation $ta = \sigma(a)t + \delta(a)$ [2017-6, p. 2838]. The next section turns to the differential identities that Theorem 6.46 begins to describe.

6.5 Generalized and skew differential identities

A differential identity is an expression involving unknowns, ring elements and derivations as unary operations [1978-2, p. 155]. Kharchenko built the theory of these identities for prime rings, and Section 3.4 reports it. The present section reports two later extensions [2017-6, p. 2838]. The first has derivations that act through the twist of a generalized derivation [2012-7, p. 885]. The second has a twist that is a skew derivation of the Ore extension type [2017-6, p. 2838].

The first step is to translate an identity about a derivation into an identity about an Ore extension [2010-1, p. 172]. The translation rests on the commutator formula for the variable x [2010-1, p. 172].

Remark 6.47 ([2010-1, p. 172]). In $Q[x;\delta]$ the derivation acts as the commutator with x. One has $\delta(r) = [x, r]$ and, in general,

$$\delta^i(r) = \sum_{s=0}^{i} (-1)^s \binom{i}{s} x^{i-s} r\, x^s.$$

A differential identity $\sum_{ij} a_{ij}\delta^i(r)b_{ij} = 0$ for all $r \in R$ is thereby an equality $\sum_i f_i(x)\, r\, g_i(x) = 0$ with $f_i, g_i \in Q[x;\delta]$.

The translation turns a question about the values of derivations into a question about an Ore extension [2010-1, p. 172]. The answer is then a linear independence statement [2010-1, Theorem 11, p. 172].

Theorem 6.48 ([2010-1, Theorem 11, p. 172])**.** *For $f_i, g_i \in Q[x;\delta]$, one has $\sum_{i=1}^n f_i\, r\, g_i = 0$ for all $r \in R$ if and only if $\sum_{i=1}^n f_i \otimes g_i = 0$ in $Q[x;\delta] \otimes_{C_S(R)} Q[x;\delta]$.*

The theorem is the single-derivation case of a general statement about (R,R)-bimodule endomorphisms [2010-1, p. 164]. Chuang, Lee, Liu and Tsai prove it through these endomorphisms [2010-1, Theorem 3, p. 165]. Every (R,R)-bimodule endomorphism of the Ore extension is a series in the divided powers (Theorem 6.45).

The sources also state the structural consequence that the same authors draw for the module of the extension [2010-1, p. 169].

Theorem 6.49 ([2010-1, Corollary 5, p. 169])**.** *The (R,R)-bimodule $S = Q[X;D]$ is quasi-injective in the sense that any (R,R)-bimodule map from an (R,R)-bisubmodule into S can be extended to an (R,R)-bimodule endomorphism of S.*

The skew case starts one step earlier, at the definition of the identity itself [2017-6, p. 2838]. The twist σ acts on the coefficients, and the Ore extension of Definition 6.42 carries both maps.

The two maps are not independent when one of them is inner [2017-6, p. 2841]. This is the mechanism by which a skew identity reduces to an ordinary one [2017-6, p. 2841].

Remark 6.50 ([2017-6, p. 2841])*.* Let δ be a σ-derivation with $\sigma = I_u$ for a unit u of R. Then $d := u^{-1}\delta$ is an ordinary derivation, and $R[t;d]$ is isomorphic to $R[t;\sigma,\delta]$ by the R-fixing map $t \mapsto u^{-1}t$. If δ is the inner σ-derivation defined by a, so that $\delta(r) = ar - \sigma(r)a$, then $(t-a)r = \sigma(r)(t-a)$, and $R[t;\sigma,\delta]$ is isomorphic to the Ore extension of automorphism type $R[t;\sigma]$ by $t \mapsto t + a$.

The reduction shows where the twisted identities are genuinely new [2017-6, p. 2838]. Some skew derivations are inner, and the reduction above removes their twist [2017-6, p. 2841]. Some are of automorphism type, and a change of variable removes the other part. In the remaining case the skew

derivation keeps a twist [2017-6, p. 2841]. No change of variable removes it; its identities are identities of the twisted polynomial ring [2017-6, p. 2841].

The two sections of this chapter meet at the same phenomenon [2010-1, p. 172]. The Ore extension is the coefficient ring of the identity, and the identities of Theorem 6.48 are the identities of that ring. Section 6.6 turns to the weaker product rules, where the identity is not a twist of the Leibniz rule but a different rule altogether.

6.6 Semiderivations, left derivations, reverse and multiplicative derivations

The rule of Section 6.5 twisted the derivation and kept the derivation value. The four maps collected here change the rule itself [1983-1, p. 267]. Each is specified by a formula that follows the Leibniz rule, and by the place where the formula fails [1957-1, p. 1105].

The first is the most general [1983-1, p. 267]. In 1983 Bergen needed maps that behave like derivations in a polynomial-identity question [1983-1, p. 267].

Definition 6.51 ([1983-1, p. 267])**.** Let R be an associative ring. An additive mapping $f\colon R \to R$ is a *semiderivation* of R if there exists a function $g\colon R \to R$ such that, for all $x, y \in R$,

$$f(xy) = f(x)g(y) + x\,f(y) = f(x)\,y + g(x)f(y), \qquad f(g(x)) = g(f(x)).$$

The two usual cases recover two familiar examples [1990-3, p. 867]. If g is the identity map, a semiderivation is an ordinary derivation [1990-3, p. 867]. If g is any ring endomorphism, the map $f(x) = x - g(x)$ is a semiderivation [1990-3, p. 867].

On a prime ring there is nothing else [1990-3, p. 867]. Chuang proves this structure theorem in 1990, and Brešar proves the same dichotomy in the same issue [1990-2, p. 859].

Theorem 6.52 ([1990-3, p. 867])**.** *Let f be a semiderivation of a prime ring R associated with the endomorphism $g\colon R \to R$. Then either one of the following two cases holds.*

label=(1) *There exists an element λ in the extended centroid of R such that $f(x) = \lambda(x - g(x))$ for all $x \in R$.*

lbbel=(2) *The map g is the identity and f is an ordinary derivation of R.*

The next two maps sit closer to the Leibniz rule [1990-1, p. 7]. A *left derivation* of a ring R into a left R-module X swaps the two factors of the rule.

Definition 6.53 ([1990-1, p. 7])**.** Let R be a ring and X a left R-module. An additive mapping $D\colon R \to X$ is a *left derivation* if $D(ab) = a\,D(b) + b\,D(a)$ for all $a, b \in R$. It is a *Jordan left derivation* if $D(a^2) = 2a\,D(a)$ for all $a \in R$.

A Jordan left derivation is again a strong hypothesis [1990-1, p. 8]. Brešar and Vukman show that over torsion-controlled modules it already forces the base ring to commute [1990-1, p. 8].

Theorem 6.54 ([1990-1, Theorem 1.2, p. 8])**.** *Let R be a ring and X be a 2-torsionfree and 3-torsionfree left R-module. Suppose that $aRx = 0$ with $a \in R$, $x \in X$ implies that either $a = 0$ or $x = 0$. If there exists a nonzero Jordan left derivation $D\colon R \to X$, then R is commutative.*

The next corollary specializes it to a prime ring.

Corollary 6.55 ([1990-1, Corollary 1.3, p. 13])**.** *Let R be a prime ring of characteristic different from 2 and 3. If R admits a nonzero Jordan left derivation $D\colon R \to R$, then R is commutative.*

Reversing the two sides of the rule instead of the two factors gives the oldest of the four notions [1957-1, p. 1105]. Herstein names it in 1957, inside the paper that settles Jordan derivations on prime rings.

Definition 6.56 ([1957-1, p. 1105])**.** Let R be a ring. An additive mapping $D\colon R \to R$ is a *reverse derivation* if $D(xy) = D(y)\,x + y\,D(x)$ for all $x, y \in R$.

Every reverse derivation is a Jordan derivation [1957-1, p. 1105]. On a prime ring such a map collapses the whole theory to the commutative case [1957-1, p. 1106].

Theorem 6.57 ([1957-1, Theorem 2.1, p. 1106])**.** *Let A be a prime ring and suppose that D is a nonzero reverse derivation of A. Then A is a commutative integral domain and D is an ordinary derivation of A.*

The semiprime generalization is the one step the literature reports as already settled [2015-1, p. 200]. Aboubakr and González record the

statement made by Samman and Alyamani in 2007: on a semiprime ring a reverse derivation is a derivation of the ring into its center [2015-1, p. 200].

The reverse rule has a generalized form of its own. An additive map F is an *l-generalized reverse derivation* [2015-1, p. 199] when $F(xy) = F(y)x + yd(x)$ for a fixed reverse derivation d, and an *r-generalized reverse derivation* [2015-1, p. 199] when $F(xy) = d(y)x + yF(x)$. The two are not the same condition. Aboubakr and González separate them: on a 2-torsion free semiprime ring they coincide with each other and with the ordinary generalized derivations [2015-1, pp. 204–205]. Either one, associated with a nonzero reverse derivation, forces a nonzero central ideal [2015-1, Corollary 3.3]. That is the same conclusion the sentence above records for a plain reverse derivation, reached from the weaker hypothesis.

The fourth map removes an axiom rather than the rule [1996-3, p. 190]. One asks only that the Leibniz identity holds, with no additivity and no linearity [1996-3, p. 190].

Definition 6.58 ([1996-3, p. 190])**.** Let $\mathcal{A}$ be an algebra. A *multiplicative derivation* of $\mathcal{A}$ is a mapping $d\colon \mathcal{A} \to \mathcal{A}$ satisfying $d(ab) = a\,d(b) + d(a)\,b$ for all $a, b \in \mathcal{A}$; neither additivity nor continuity is assumed.

On a function algebra the linear derivations all vanish [1996-3, p. 190]. The multiplicative derivations, however, survive, and they are described by a pointwise formula with a logarithm [1996-3, p. 191].

Theorem 6.59 ([1996-3, Theorem 2, p. 191])**.** *Let X be a completely regular Hausdorff space satisfying the first axiom of countability, and assume that X has no isolated points. Let $d\colon C(X) \to C(X)$ be a multiplicative derivation of the algebra $C(X)$ of continuous real-valued functions on X. Then there exists a continuous function $a\colon X \to \mathbb{R}$ such that, for all $f \in C(X)$ and all $x \in X$,*

$$d(f)(x) = \begin{cases} a(x)\, f(x) \log |f(x)| & \text{if } f(x) \neq 0, \\ 0 & \text{if } f(x) = 0. \end{cases}$$

The same paper records the companion question of Martindale, taken up by Daif in 1991 [1991-3, p. 615]. Daif shows that on a ring with an idempotent satisfying his three conditions, every multiplicative derivation is additive [1991-3, p. 615].

A multiplicative skew derivation asks for additivity on a prime ring, and Brox shows that the answer is negative in a sharp form [2020-9, p. 3]. A multiplicative generalized skew derivation, as Ashraf and Mozumder defined it, satisfies two identities at once [2020-9, p. 2]. Brox proves that each

such map on a prime ring is either multiplicative generalized or generalized skew [2020-9, Theorem 2.2, pp. 3–4]. One identity is therefore enough for a genuinely new class [2020-9, p. 1].

The four maps show that the derivation rule is more rigid than its generalizations suggest [1990-3, p. 867]. Relax the rule and the prime ring collapses to the commutative case, the ring becomes commutative, or the map is described by a closed formula [1957-1, pp. 1106, 191].

Two of these notions combine. Aziz, Ghosh and Prakash introduce the skew semiderivation, which generalizes the skew derivation and the semiderivation at once, and study when a multiplicative map of that class becomes additive [2024-8, Abstract]. On generalized gamma rings Sirivoravit and Leerawat define the reverse f-derivation, the reverse condition with the map f in the product rule [2020-22, Abstract].

6.7 f-derivations: three meanings of f in rings, modules and triangular algebras

The previous sections twisted the product rule with a ring map. Here a second map f enters the rule, and the same name, f-derivation, is used for three different objects. We recall each one from the paper that defines it. In the first, f is a function on a lattice. In the second, f is a linear map between modules. In the third, f is a polynomial.

We start with lattices. Xin, Li and Lu introduced the notion of a derivation on a lattice [2008-3, p. 701]. Çeven and Öztürk recall their definition.

Definition 6.60 ([2008-3, Definition 6, p. 702])**.** A function $d : L \to L$ on a lattice L is called a derivation on L if it satisfies the following condition $d(x \wedge y) = (dx \wedge y) \vee (x \wedge dy)$.

Çeven and Öztürk introduce the f-derivation as an extension of this notion [2008-3, p. 701]. Zhan and Liu had earlier introduced left-right f-derivations of BCI-algebras [2008-3, p. 701].

Definition 6.61 ([2008-3, Definition 8, p. 702])**.** Let L be a lattice. A function $d : L \to L$ is called an f-derivation on L if there exists a function $f : L \to L$ such that

$$d(x \wedge y) = (d(x) \wedge f(y)) \vee (f(x) \wedge d(y)) \tag{2.1}$$

for all $x, y \in L$.

They note that d is a derivation when f is the identity function [2008-3, p. 702]. They also note that d can be an f-derivation only when a function f satisfying (2.1) exists [2008-3, p. 702]. The first consequences of the definition are as follows.

Proposition 6.62 ([2008-3, Proposition 1, p. 703])**.** *Let L be a lattice and d be an f-derivation on L. Then the following identities hold for all $x, y \in L$.*

a) $dx \leq fx$,

b) $dx \wedge dy \leq d(x \wedge y) \leq dx \vee dy$,

c) $d(x \wedge y) \leq fx \vee fy$,

d) If L has a least element 0, then $f0 = 0$ implies $d0 = 0$.

Part a) comes from $dx = d(x \wedge x) = dx \wedge fx$ [2008-3, p. 703]. The same paper shows that distributive and modular lattices can be characterized by f-derivations under some conditions [2008-3, p. 701].

We turn to modules. Bland works with an associative ring R with identity and unitary right R-modules [2006-3, p. 379]. Here f is a linear map between modules.

Definition 6.63 ([2006-3, p. 379])**.** An additive mapping $\delta : R \to R$ is a derivation on R if $\delta(ab) = \delta(a)b + a\delta(b)$ for all $a, b \in R$. If δ is a derivation on R and $f : M \to N$ is an R-linear mapping, then an additive mapping $d : M \to N$ is a (δ, f)-derivation if $d(xa) = d(x)a + f(x)\delta(a)$ for all $x \in M$ and all $a \in R$.

Bland fixes δ and then calls a (δ, f)-derivation simply an f-derivation [2006-3, p. 379]. If $f : M \to M$, then $d : M \to M$ is an f-derivation on M, and if $f = \mathrm{id}_M$, then d is a derivation on M [2006-3, p. 379]. He remarks that f-derivations always exist, since $f = 0$ makes d an R-linear mapping [2006-3, p. 379]. His main result extends an f-derivation $d : M \to N$ uniquely to an f_τ-derivation $d_\tau : Q_\tau(M) \to Q_\tau(N)$ when τ is a differential torsion theory [2006-3, p. 380].

Finally we meet triangular algebras. Benkovič lets A be an algebra over a commutative unital ring R and M an A-bimodule [2014-7, Section 1]. The map f is now a polynomial, and the product rule becomes a rule on f.

Definition 6.64 ([2014-7, Section 1, (1.1) and (1.2)])**.** Let

$$f(x_1, x_2, \ldots, x_n) = \sum_{\pi \in S_n} \alpha_\pi x_{\pi(1)} x_{\pi(2)} \cdots x_{\pi(n)}, \quad \alpha_\pi \in R, \tag{1.1}$$

be a fixed nonzero multilinear polynomial in noncommuting indeterminates x_i over R, where S_n denotes the symmetric group of order n. An R-linear map $d : A \to M$ satisfying

$$d(f(x_1, \ldots, x_n)) = \sum_{i=1}^{n} f(x_1, \ldots, x_{i-1}, d(x_i), x_{i+1}, \ldots, x_n) \qquad (1.2)$$

for all $x_1, \ldots, x_n \in A$ is called an f-derivation.

Benkovič observes that every derivation is an f-derivation for $f(x_1, x_2) = x_1x_2$. Every Jordan derivation is one for $f(x_1, x_2) = x_1 \circ x_2 = x_1x_2 + x_2x_1$, and every Lie derivation is one for $f(x_1, x_2) = [x_1, x_2] = x_1x_2 - x_2x_1$ [2014-7, Section 1]. In the same way, a Jordan triple derivation is an f-derivation for $f(x_1, x_2, x_3) = x_1x_2x_3 + x_3x_2x_1$, and a Lie triple derivation is one for $f(x_1, x_2, x_3) = [[x_1, x_2], x_3]$ [2014-7, Section 1]. So the Jordan-type and Lie-type maps of Chapters 4 and 5 are special cases of this one definition.

He writes $\alpha = \sum_{\pi \in S_n} \alpha_\pi$ for the sum of the coefficients of f. An element $r \in R$ is M-regular if $rm = 0$ with $m \in M$ implies $m = 0$ [2014-7, Section 1]. A triangular algebra is a unital algebra with a nontrivial idempotent e such that $A = eAe + eAf + fAf$, where $f = 1 - e$ and the bimodule eAf is faithful on both sides [2014-7, Section 1].

Theorem 6.65 ([2014-7, Theorem 1.1])**.** *Let A be a triangular algebra and let $d : A \to A$ be an f-derivation, where $f \in R\langle x_1, x_2, \ldots \rangle$ is a multilinear polynomial of degree $n \geq 2$ with $\alpha \neq 0$. If A is $(n-1)$-torsion free and α is A-regular, then d is a derivation.*

For $\alpha = 0$ the question stays open. Benkovič asks when such a map has the standard form $d = \delta + \gamma$, with δ a derivation and γ a linear map into the center that vanishes on $f(A, \ldots, A)$ [2014-7, Problem 1.2]. He points to functional identities as the usual tool for prime rings [2014-7, Section 1]; Chapter 9 treats them.

The remaining case is open, and this section states it as a problem.

Problem 6.66 ([2014-7, Problem 1.2])**.** Let A be a triangular algebra over a commutative ring R, and let $d\colon A \to A$ be an f-derivation for a multilinear polynomial f of degree $n \geq 2$ whose coefficients sum to 0. Under which conditions on A and f is d of the standard form $d = \delta + \gamma$, where δ is a derivation and γ is a linear map into the center of A that vanishes on $f(A, \ldots, A)$?

Benkovič proves the statement for the case where the sum of coefficients is not zero [2014-7, Theorem 1.1] and states the remaining case as Problem 1.2 [2014-7, Problem 1.2]. Chapter 10 reads the triangular side in Section 10.2.

The three definitions share the name and one idea: a map d follows the product rule with some other map f in the second slot. They differ in what f is, so a result about one cannot be used for another. We keep the three names apart in the chapters that follow. Lattices continue in Chapter 16, and triangular algebras in Chapter 10.

Yanti and Wijayanti add the Jordan condition to the (δ, f)-derivation and characterize the maps that satisfy it [2025-21, Abstract, p. 1].

The same idea has a two-variable form. Bahmani, Bennis, Ebrahimi Vishki and Fahid define an f-biderivation by replacing the derivation in the biderivation identity with f. They show that every such map is a Jordan biderivation under their conditions. They also prove that every Jordan biderivation of a unital algebra with an idempotent is a sum of a biderivation, an antibiderivation and an extremal biderivation [2021-13, Theorem 2.7].

Chapter 7

Higher, bi- and n-derivations

Why a single derivation is not enough. Chapter 6 changed one map: from a derivation to a σ-derivation, an α-derivation, a semiderivation or a left derivation [2017-6, p. 2841]. This chapter changes the map in four more ways, and each way is read from the paper that carries it.

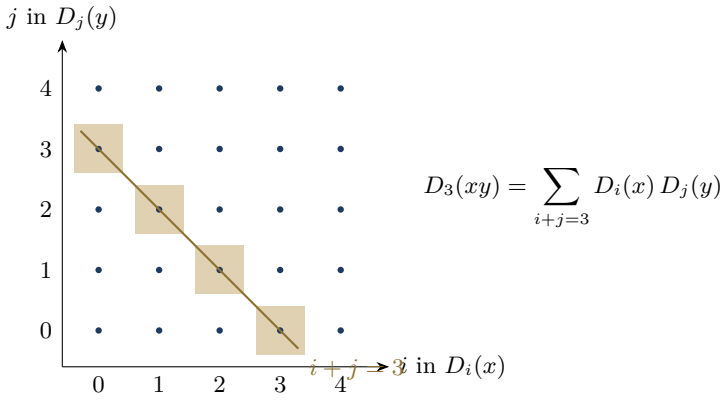


Figure 7.1: The product rule of a higher derivation: $D_n(xy)$ collects the terms $D_i(x)D_j(y)$ on the antidiagonal $i + j = n$, here $n = 3$.

The first way replaces the derivation by a sequence [2002-6, p. 2323]. Hasse and Schmidt built the first sequences in 1937, and their two rules hold for an arbitrary integral domain [1937-1, p. 224]. Section 7.1 recalls those rules and reads the definition from Ferrero and Haetinger [2002-6, p. 2323]. The single-map notion of a q-th order derivation is Nakai [1970-3, p. 1], and the extension theorems are Narváez Macarro [2009-9, pp. 2983–2984].

The second way keeps one map and gives it two arguments [1989-3, p. 245]. Maksa introduced the symmetric biderivation [1989-3, references 5 and 6] and Vukman carried it into prime rings [1989-3, p. 245]; Section 7.3 recalls his Theorem 1 and Theorem 2 [1989-3, pp. 246, 249].

The third way lets the number of arguments grow [2017-15, p. 490]. Section 7.4 introduces the permuting triderivation of Öztürk [1999-1, reference 10] and runs the second theorem of Posner along higher derivations [2021-20, p. 9].

The fourth way studies the trace on its own [1990-6, p. 181] and asks what a biderivation can be on a structured ring [2016-16, p. 1952]. Sections 7.5 and 7.6 close the chapter with the double-commutator theorem [1990-6, p. 187] and Wang decomposition on triangular rings [2016-16, p. 1955].

7.1 Higher derivations and Hasse–Schmidt sequences

Every generalization in Chapter 6 changed one map. The generalization of this chapter replaces one map by a sequence [2002-6, p. 2323]. The first component of the sequence is the identity, and the next is a derivation [2002-6, p. 2323]. The product rule of the whole sequence collects the terms $D_i(x)D_j(y)$ along the *antidiagonal* $i + j = n$, the terms whose two indices sum to n [2002-6, p. 2323].

Definition 7.1 ([2002-6, p. 2323])**.** Let T be a ring. A family of additive mappings $D = (d_i)_{i\in\mathbb{N}}$ of T is a *higher derivation* of T if $d_0 = \mathrm{id}_T$ and, for every $n \in \mathbb{N}$,

$$d_n(ab) = \sum_{i=0}^{n} d_i(a)\, d_{n-i}(b) \qquad \text{for all } a, b \in T.$$

The definition reaches the literature as an exercise [2002-6, p. 2323]. Ferrero and Haetinger record it with page 532, Exercise 4 of the book *Basic Algebra II* by Jacobson [2002-6, p. 2323].

Hasse and Schmidt gave the same sequence for algebraic function fields of one variable in 1937 [1937-1, pp. 215–223]. Their paper states the two rules of the sequence for an arbitrary integral domain [1937-1, pp. 223–224]. The authors set a domain I of characteristic p and a larger domain I^* containing the derivatives [1937-1, p. 224]. They assign to each $y \in I$ a function $y^{(\nu)} \in I^*$ for every natural number ν [1937-1, p. 224]. The

assignment is a *differentiation* when it satisfies the two rules below [1937-1, p. 224].

Definition 7.2 ([1937-1, p. 224])**.** Let $I \subseteq I^*$ be integral domains of characteristic p, and let $y \mapsto y^{(\nu)}$ assign to each $y \in I$ and each $\nu \in \mathbb{N}$ an element $y^{(\nu)} \in I^*$. The assignment is a *differentiation* of I when it satisfies the sum rule

$$(y+z)^{(\nu)} = y^{(\nu)} + z^{(\nu)}$$

and the modified product rule

$$(yz)^{(\nu)} = y^{(\nu)}z + y^{(\nu-1)}z' + \cdots + y'z^{(\nu-1)} + yz^{(\nu)}$$

for all $y, z \in I$ and all ν. The element $y^{(\nu)}$ is the *ν-th derivative* of y.

A function whose derivatives all vanish from some order on is a *differentiation constant* of that order, and the absolute constants are those whose derivatives all vanish [1937-1, p. 224]. The authors note that the constants of at least a given order form a subdomain of I, and that when I is a field they form a subfield [1937-1, p. 224].

The 1937 paper then identifies a sequence with a single map into a ring of power series [1937-1, p. 225]. Over I^* it forms the domain $T^* = I^*\{u\}$ of all power series in u, and assigns to each $y \in I$ the series $Y_u = y + y'u + y''u^2 + \cdots$ [1937-1, p. 225]. The authors call Y_u the general Taylor expansion of y [1937-1, p. 225]. They prove that the assigned series form a subdomain on which the assignment is an isomorphism [1937-1, Satz 1, p. 226]. They prove the converse too [1937-1, Satz 2, p. 226]. Take an isomorphism of I onto a subdomain of T^* that carries each y to a series with first term y. Reading the coefficients as derivatives gives a differentiation [1937-1, Satz 2, p. 226].

Nakai took up the single map rather than the sequence in 1970 [1970-3, p. 1]. He works over commutative rings k and A with A a k-algebra, and he attributes the notion to Osborn [1970-3, p. 1]. His q-th order derivation keeps the product rule on $(q+1)$ arguments, and his Theorem 2 turns that rule into an identity of commutators [1970-3, Theorem 2, p. 3]. The link between the two notions is his Proposition 5: the q-th component of a higher derivation is a q-th order derivation [1970-3, Proposition 5, p. 4]. A q-th order derivation need not be a component of a higher derivation, and Nakai leaves that question open [1970-3, p. 1].

The model is the iterated derivative of a single derivation [2002-6, p. 2322]. If d is a derivation of a rational algebra, the sequence $d_i = d^i/i!$ is a higher derivation of R [2002-6, p. 2322].

The name Hasse–Schmidt marks a second presentation of the same notion [2009-9, p. 2983]. Narváez Macarro states it for algebras, with the 1937 paper as his reference for the definition [2009-9, §1.1, reference 7].

Definition 7.3 ([2009-9, §1.1, p. 2983])**.** Let k be a commutative ring, $f\colon k \to A$ and $g\colon A \to B$ ring homomorphisms, and $m \geq 1$ an integer or $m = \infty$. A *Hasse–Schmidt derivation* over k of length m from A to B is a sequence $D = (D_0, D_1, \dots, D_m)$ or $D = (D_0, D_1, \dots)$ of k-linear maps $D_i\colon A \to B$ with

$$D_0 = g, \qquad D_i(xy) = \sum_{r+s=i} D_r(x)\, D_s(y) \qquad \text{for all } x, y \in A \text{ and all } i.$$

When $A = B$ and $g = \mathrm{id}_A$ one says that D is a Hasse–Schmidt derivation of A over k.

Such a sequence is the same data as one algebra map into a truncated polynomial ring [2009-9, p. 2984]. With $B_m = B[t]/(t^{m+1})$, the sequence D is determined by the k-algebra homomorphism $\Phi\colon A \to B_m$ given by $\Phi(x) = \sum_{i=0}^{m} D_i(x)\, t^i$ [2009-9, p. 2984].

Over a semiprime ring a higher derivation extends to the Martindale ring of quotients in a unique way [2002-6, p. 2323].

Proposition 7.4 ([2002-6, Proposition 1.1, p. 2323])**.** *Let R be a semiprime ring and $D = (d_i)_{i\in\mathbb{N}}$ a higher derivation of R. Then there exists a unique higher derivation $D^* = (d_i^*)_{i\in\mathbb{N}}$ of Q such that $d_n^*|_R = d_n$ for every $n \in \mathbb{N}$.*

The theory of the sequence starts when the terms satisfy a linear relation [2002-6, p. 2323]. A family D satisfies an S-linear relation on T when there exist $a_0, a_1, \dots, a_n \neq 0$ in S with $\sum_{i=0}^{n} a_i d_i(x) = 0$ for all $x \in T$; the minimal n is the length of D. For a single derivation d of a rational algebra the relation is the algebraicity of d, that is, the existence of a polynomial relation among its powers [2002-6, p. 2322]. On a prime ring the relation is rigid, as Corollary 7.5 shows [2002-6, p. 2325].

Corollary 7.5 ([2002-6, Corollary 1.4, p. 2325])**.** *Assume that R is a prime ring and $D = (d_i)_{i\in\mathbb{N}}$ a higher derivation of R that satisfies an R-linear relation of minimal length n on R. Then there exist $q_0 = 1, q_1, \dots, q_{n-1} \in Q$ such that $\sum_{i=1}^{n} q_{n-i} d_i = 0$. In particular, D satisfies a monic Q-linear relation of length n on R. Moreover, $d_1 = d_{q_1}$ and*

$$d_m(x) = d_{q_m}(x) - \sum_{i=1}^{m-1} q_i\, d_{m-i}(x) \qquad \textit{for all } x \in R \textit{ and } 2 \leq m \leq n.$$

The main result compares the possible relations [2002-6, pp. 2327–2328]. It says the ring of coefficients of the relation does not matter, as Theorem 7.6 states [2002-6, p. 2327].

Theorem 7.6 ([2002-6, Theorem 1.8, pp. 2327–2328])**.** *Assume that R is a semiprime ring and D a higher derivation of R. Then the following conditions are equivalent:*

label=(i) *D satisfies an R-linear relation on R;*

lbbel=(ii) *D satisfies a Q-linear relation on R;*

lcbel=(iii) *D^* satisfies a Q-linear relation on Q;*

ldbel=(iv) *D^* satisfies an R-linear relation on Q.*

Moreover, when one of these conditions holds, the minimal lengths of the relations are all the same.

A higher derivation is therefore a derivation whose values are measured in steps, as Definition 7.1 displays. Later sections of this chapter multiply the map in a second direction, two derivation arguments at once in Section 7.3 and three in Section 7.6.

7.2 Coding theory: Hasse–Schmidt derivations in positive characteristic

Error-correcting codes are built over finite fields, where the classical derivative no longer records enough information, and Hasse–Schmidt derivations replace it [2018-15, Introduction]. Narváez Macarro states the multivariate form [2018-15, Definition 2.1.1, p. 15].

Definition 7.7 ([2018-15, Definition 2.1.1, p. 15])**.** Let k be a commutative ring, A a commutative k-algebra, $\mathbf{s} = \{s_1, \ldots, s_p\}$ a set of variables and $\Delta \subset \mathbb{N}^p$ a non-empty co-ideal. A (p, Δ)-*variate Hasse–Schmidt derivation* of A over k is a family $D = (D_\alpha)_{\alpha \in \Delta}$ of k-linear maps $D_\alpha \colon A \to A$ satisfying

$$D_0 = \mathrm{Id}_A, \qquad D_\alpha(xy) = \sum_{\beta+\gamma=\alpha} D_\beta(x) D_\gamma(y) \quad \text{for every } x, y \in A \text{ and every } \alpha \in \Delta.$$

For $p = 1$ and $\Delta = \{q \in \mathbb{N} \mid q \leq m\}$, D is a *Hasse–Schmidt derivation of length m*, also called a higher derivation.

The first component D_1 is a k-derivation of A in the sense of Chapter 2. Every D_i for $i \geq 1$ is a k-linear differential operator of order $\leq i$ vanishing on 1 [2018-15, p. 1]. In characteristic 0, every k-derivation δ of A appears as the 1-component of a Hasse–Schmidt derivation of infinite length, namely the exponential $e^{t\delta} = \sum_{\alpha \geq 0} \frac{\delta^\alpha}{\alpha!} t^\alpha$ [2018-15, p. 2]. In positive characteristic this exponential collapses when the derivation reaches order p, and the higher derivations are the surviving notion [2018-15, p. 2].

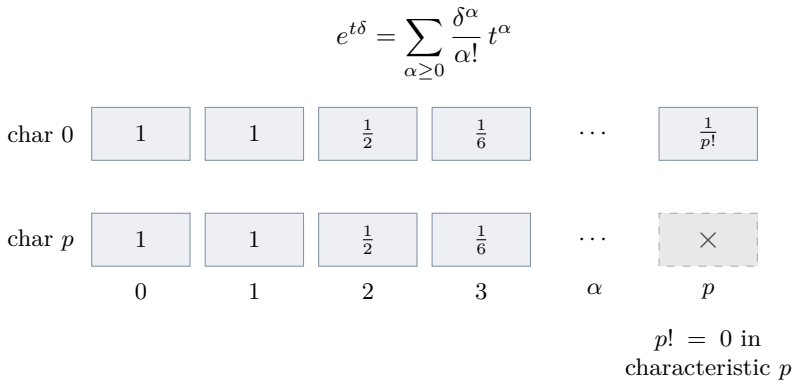


Figure 7.2: Why the exponential $e^{t\delta} = \sum_{\alpha \geq 0} \delta^\alpha t^\alpha / \alpha!$ survives in characteristic 0 and collapses in characteristic p [2018-15, p. 2]. The coefficients $1/\alpha!$ exist for every α in characteristic 0. In characteristic p the factorial $p!$ is a multiple of p, so $1/p!$ does not exist and the series stops at $\alpha = p$; the higher derivations are the notion that survives.

The rule can be weakened by a pair of maps, one for each side, and the sequence keeps its shape [2019-8, Definition 2.1, p. 3]. Hosseini and Rehman call a sequence a higher $\{g_n, h_n\}$-derivation when each term splits the product according to two sequences $\{g_n\}$ and $\{h_n\}$ [2019-8, Definition 2.1, p. 3]. They express every such sequence by a sequence of $\{G, H\}$-derivations [2019-8, Theorem 2.4, p. 7]. A Jordan higher $\{g_n, h_n\}$-derivation on a semiprime algebra is then a higher $\{g_n, h_n\}$-derivation [2019-8, Corollary 2.6, p. 12]. The result extends the semiprime theorem of Brešar for the one-term $\{g, h\}$-derivation [2019-8, p. 1].

The same sequence has a Jacobian form in positive characteristic [2019-13, p. 1]. Nagamine defines the higher derivation of Jacobian type by a tuple $F = (f_1, \ldots, f_{n-1})$ of polynomials [2019-13, Definition 2.4, p. 3]. The tuple gives a sequence of R-linear maps, and the sequence is a higher derivation exactly when the tuple extends to a coordinate system [2019-13,

Definition 2.4, p. 3]. An $(n-1)$-tuple of polynomials is extendable in this sense exactly when the higher derivation exists and has a slice [2019-13, Theorem 2.8, p. 5]. For the polynomial ring in two variables over a field of positive characteristic, the same construction describes the variables [2019-13, Theorem 3.1, p. 6].

A single derivation need not be the first component of a sequence, and the question of which ones are is the question of integrability [2021-21, p. 1]. A k-derivation δ of A is *m-integrable* when there is a Hasse–Schmidt derivation D of length m with $D_1 = \delta$ [2021-21, p. 1]. The set of m-integrable derivations is a module inside the module of derivations, written $\mathrm{IDer}_k(A; m)$ [2021-21, p. 1]. In characteristic zero, or when A is smooth over k, every derivation is integrable [2021-21, p. 1]. In positive characteristic these modules carry more information than the module of all derivations [2021-21, p. 1]. Tirado Hernández computes them for the plane curve $x^n = y^q$, in each of several cases of the exponents [2021-21, Proposition 2.1, p. 4].

Barajas and Duarte tie the sequence to the trivial extension of a ring by a module [2026-6, p. 2]. A derivation from A into M is the same data as a ring homomorphism from A into the trivial extension $A \ltimes M$ that the projection takes back to the identity [2026-6, p. 2]. Anderson, Bennis, Fahid and Shaiea had replaced the one module by n modules [2026-6, p. 2]. Barajas and Duarte replace the derivation by a sequence of maps with the same property, and call it a multi Hasse–Schmidt derivation [2026-6, Definition 3.1, p. 6].

Proposition 7.8 ([2026-6, Proposition 3.5, p. 7])**.** *Let $A, B_1, \ldots, B_n$ be k-algebras, let $g_i \colon A \to B_i$ be k-algebra homomorphisms, and give each B_i the A-module structure induced by g_i. Then the multi Hasse–Schmidt derivations of A in $B_1, \ldots, B_n$ correspond bijectively to the k-algebra homomorphisms from A into the n-trivial extension $A \ltimes_n B_1 \ltimes \cdots \ltimes B_n$ whose components all agree with the g_i.*

When all the algebras B_i are the same, the multi Hasse–Schmidt derivations are the Hasse–Schmidt derivations of Definition 7.3 [2026-6, p. 2]. For $n = 1$ the correspondence is the classical one between derivations and trivial extensions [2026-6, p. 2].

The sequence has a Lie version. A Lie higher derivation of an algebra A is a sequence $L = (L_n)$ of linear maps with L_0 the identity and $L_n([x, y]) = \sum_{i+j=n} [L_i(x), L_j(y)]$ for all x, y [2021-11, p. 1]. The first term is then a Lie derivation [2021-11, p. 1]. The authors ask when such a sequence is proper,

that is, when it is a higher derivation plus a sequence of central maps that vanish on commutators [2021-11, p. 1].

Theorem 7.9 ([2021-11, Theorem 3.1, p. 6])**.** *Let $A = \mathrm{Tri}(A, M, B)$ be a triangular algebra and $L = (L_n)$ a Lie higher derivation of A. Then there is a triangular algebra A_0 containing A with the same identity such that $L_n = \Delta_n + \chi_n$ for every n, where (Δ_n) is a higher derivation of A into A_0 and each χ_n is a linear map of A into the center of A_0 that vanishes on every commutator.*

Moafian proved the corresponding structure theorem for a generalized matrix algebra, and he states conditions under which the sequence is proper [2017-10, Theorem 4.4, p. 12]. Ashraf and Jabeen carry the same splitting to a nonlinear generalized Lie triple higher derivation [2018-5, p. 1]. There each component is the sum of an additive generalized higher derivation and a nonlinear map that vanishes on all Lie triple products [2018-5, Theorem 3.1, p. 5].

Over a domain of positive characteristic the higher derivations of Jacobian type carry the same information as the Jacobian matrix. Nagamine gives a necessary and sufficient condition for an $(n-1)$-tuple of such derivations on $R[x_1, \ldots, x_n]$ [2019-13, Abstract, p. 1].

7.3 Biderivations

Section 7.1 replaced one derivation by a sequence of maps of one argument. The notion of this section keeps one map and gives it two arguments. A *biderivation* is an additive map of two variables that is a derivation in each argument when the other is fixed [1989-3, p. 245].

Maksa introduced the symmetric case [1989-3, references 5 and 6]. Vukman records the notion and gives it its ring-theory start [1989-3, p. 245].

Definition 7.10 ([1989-3, p. 245])**.** Let R be a ring. A *symmetric biderivation* of R is a biadditive symmetric mapping $D\colon R \times R \to R$ such that, for every fixed $y \in R$, the mapping $x \mapsto D(x, y)$ is a derivation of R. The *trace* of D is the map $d\colon R \to R$ given by $d(x) = D(x, x)$.

The question for a prime ring is the now familiar one: when does the new map force the ring to commute [1989-3, p. 245]. The answer uses the trace of Definition 7.10. A map $f\colon R \to R$ is *commuting* on R when $[f(x), x] = 0$ for all $x \in R$ [1989-3, p. 246].

Theorem 7.11 ([1989-3, Theorem 1, p. 246]). *Let R be a noncommutative prime ring of characteristic not two. Let $D\colon R \times R \to R$ be a symmetric biderivation and d its trace. Suppose that d is commuting on R. Then $D = 0$.*

the abstract of Vukman states the same result the other way round [1989-3, p. 245]. A prime ring of characteristic not two with a nonzero symmetric biderivation whose trace satisfies $D(x, x)\, x = x\, D(x, x)$ for all x is commutative [1989-3, p. 245].

The hypothesis can be weakened from commuting to centralizing, at the price of one more excluded characteristic [1989-3, p. 249].

Corollary 7.12 ([1989-3, Theorem 2, p. 249]). *Let R be a noncommutative prime ring of characteristic not two and three. Let $D\colon R \times R \to R$ be a symmetric biderivation and d its trace. Suppose that d is centralizing on R. Then $D = 0$.*

Vukman proves it by reducing it to Theorem 7.11: the centralizing hypothesis gives $[d(x), x] = 0$ for x outside the center, and Theorem 7.11 then applies [1989-3, p. 250]. He asks whether the exclusion of characteristic three is necessary [1989-3, p. 250].

The symmetric case is the whole of the prime case for an arbitrary biderivation, symmetric or not [2000-3, p. 669]. Brešar, Martindale and Miers described the biderivations of a noncommutative prime ring R: each one is of the form $\lambda[x, y]$ with λ in the extended centroid of R [1993-2, reference 1]. In particular, a noncommutative prime ring has no nonzero symmetric biderivation [2000-3, p. 669]. On a commutative ring, by contrast, any sum $\Delta(x, y) = \sum_i d_i(x)\, d_i(y)$ of products of derivations is a symmetric biderivation [2000-3, p. 669].

Hongan and Komatsu then build the universal home of the notion [2000-3, p. 670]. They construct the module of differentials of an algebra extension for left, right and central derivations [2000-3, p. 670]. With it they show that $R \otimes R$ is a universal representing object for the biderivations of $R \times R$ of each kind [2000-3, p. 671]. They also characterize symmetric biderivations of a 2-torsion free semiprime algebra, where every symmetric biderivation is central [2000-3, p. 671].

A biderivation is the first mirror-image of a higher derivation [2002-6, p. 2323]. The higher derivation spreads the Leibniz rule along a sequence of maps of one argument; the biderivation splits the one map into two arguments, each obeying the same rule. Then n-derivations keep the split and let the number of arguments grow.

On a Lie algebra the earlier literature asked the biderivation to be skew-symmetric [2016-15, p. 1]. Tang drops that condition and finds maps that the condition excludes [2016-15, p. 1]. For the Virasoro algebra every biderivation is a scalar multiple of the bracket [2016-15, Theorem 3.1, p. 7]. For the W-algebra $W(2,2)$ a second term appears, built from a linear map ω that sends each generator L_m to H_m [2016-15, Theorem 2.6, p. 7]. The map ω satisfies $[\omega(x), y] = [x, \omega(y)]$ for all x and y [2016-15, p. 6]. So the scalar multiples of the bracket are not the whole space on that algebra.

The deformed Schrödinger–Virasoro algebra shows the same phenomenon [2016-11, p. 1]. Its skew-symmetric biderivations take one of three forms, and the two parameters decide which [2016-11, Theorem 3.2, p. 4]. The commuting maps on the same algebra then take one of four forms, again decided by the parameters [2016-11, Theorem 4.3, p. 11]. Each form is read from the skew-symmetric biderivations [2016-11, p. 1].

For a Lie superalgebra the graded signs enter every definition, and Pandey proves a structure theorem for the super-commuting maps and the super-biderivations of a Hom-Lie superalgebra [2024-16, p. 1]. On a centerless Hom-Lie superalgebra equal to its own derived algebra, every super-biderivation is a twisted bracket with an element of the centroid [2024-16, Theorem 3.8, p. 7]. Every linear super-commuting map of such an algebra lies in the centroid [2024-16, Theorem 4.2, p. 11].

Tang settled the simple case for Lie algebras [2017-13, p. 1]. He drops the skew-symmetric condition here as well, and the biderivations do not grow.

Theorem 7.13 ([2017-13, Theorem 2.4, p. 8])**.** *Let L be a finite-dimensional complex simple Lie algebra. Then a bilinear map $f\colon L \times L \to L$ is a biderivation of L if and only if it is inner, that is, $f(x, y) = \lambda[x, y]$ for all $x, y \in L$ and some complex number λ.*

For the general linear Lie algebra a second term appears, built from the trace [2017-13, Theorem 3.2, p. 8]. It produces a non-inner and non-skew-symmetric biderivation [2017-13, p. 1]. The linear commuting maps of both algebras then take a fixed form [2017-13, Theorems 3.4 and 3.5, p. 9].

On a triangular algebra the biderivations do not have one form either. Liang, Ren and Wei describe the Lie biderivations of a triangular algebra [2020-15, p. 1]. A Lie biderivation is a bilinear map that is a Lie derivation in each argument [2020-15, p. 1].

Theorem 7.14 ([2020-15, Theorem 3.2, p. 7])**.** *Let T be a triangular algebra over a commutative ring R and φ a Lie biderivation of T. Suppose that the*

projections of the center of T are the centers of the two corner algebras, that one corner algebra is noncommutative, that every bimodule endomorphism of the off-diagonal bimodule has the standard form, and that $\alpha a = 0$ with α central and a nonzero forces $\alpha = 0$. Then

$$\varphi(x, y) = \lambda_0[x, y] + [x, [y, \varphi(e, e)]] + \mu(x, y)$$

for all $x, y \in T$, where λ_0 lies in the center of T and μ is a central bilinear map.

So a Lie biderivation is the sum of an inner one, an extremal one and a central one [2020-15, p. 1]. The result applies to block upper triangular matrix algebras and to nest algebras [2020-15, Corollaries 3.8 and 3.9, p. 14].

Di Bartolo and La Rosa put a Lie bracket on the space of right biderivations of a semiprime algebra, so that the biderivations themselves form a Lie algebra [2026-15, Abstract, p. 1].

7.4 n-derivations, permuting maps and multiderivations

Section 7.3 gave the derivation two arguments. This section gives it three, and then n [2017-15, p. 490]. Öztürk introduced the permuting triderivation [1999-1, reference 10], and Yazarlı develops its commutativity theorems [2017-15, p. 489].

A map $D\colon R \times R \times R \to R$ is *permuting* when $D(x, y, z) = D(x, z, y) = D(z, x, y) = D(z, y, x) = D(y, z, x) = D(y, x, z)$ for all $x, y, z \in R$ [2017-15, p. 490]. Its *trace* is the map $d\colon R \to R$ given by $d(x) = D(x, x, x)$ [2017-15, p. 490]. For a triadditive permuting D the trace satisfies the identity $d(x + y) = d(x) + d(y) + 3\,D(x, x, y) + 3\,D(x, y, y)$ [2017-15, p. 490].

Definition 7.15 ([2017-15, p. 490])**.** Let R be a ring. A *permuting triderivation* of R is a permuting triadditive map $D\colon R \times R \times R \to R$ such that $D(xw, y, z) = D(x, y, z)\,w + x\,D(w, y, z)$ for all $x, y, z, w \in R$.

The triderivation lands back on the maps of the two earlier sections [2017-15, p. 490]. For a fixed $a \in R$ the map $D_1(x, y) = D(a, x, y)$ is a symmetric biderivation, so a permuting 2-derivation is a symmetric biderivation [2017-15, p. 490]. The map $d_2(x) = D(a, a, x)$ is a derivation [2017-15, p. 490].

The commutativity theorems follow the pattern of Section 7.3 [2017-15, p. 490].

Theorem 7.16 ([2017-15, Theorem 1, p. 490]). *Let R be a 2, 3-torsion free prime ring and I a nonzero ideal of R. If D is a permuting triderivation with trace d such that $D(d(x), x, x) = 0$ for all $x \in I$, then either $D = 0$ or R is commutative.*

A product of two traces cannot vanish on a noncommutative ring [2017-15, p. 492].

Theorem 7.17 ([2017-15, Theorem 2, p. 492]). *Let R be a 2, 3, 5-torsion free prime ring and I be an ideal of R. If D_1, D_2 are permuting triderivations of R with traces d_1, d_2 such that $d_1(x)\, d_2(x) = 0$ for all $x \in I$, then either $D_1 = 0$ or $D_2 = 0$ unless R is commutative.*

The same triderivation statements hold with a permuting generalized triderivation G in front, and Yazarlı adds the commuting conclusion on the trace [2017-15, p. 494].

Theorem 7.18 ([2017-15, Theorem 3, p. 494]). *Let R be a 2, 3-torsion free prime ring and I a nonzero ideal of R. If G is a permuting generalized triderivation associated with the permuting triderivation D of R with trace d such that $G(d(x), x, x) = 0$ for all $x \in I$, then the following hold:*

1. *$D = 0$,*
2. *R is commutative,*
3. *d is commuting on I.*

The same second theorem of Posner runs along the higher derivations of Section 7.1 [2021-20, p. 4]. Prajapati quotes the result of Posner and Brešar Theorem B [2021-20, p. 4].

Theorem 7.19 ([2021-20, Theorem 1.1 (Theorem B of Brešar), p. 4]). *Let R be a prime ring and I a nonzero left ideal of R. Suppose derivations d and g satisfy $d(x)\, x - x\, g(x) \in Z(R)$ for all $x \in I$. If $d \neq 0$ then R is commutative.*

Prajapati runs the condition along two higher derivations [2021-20, p. 9].

Theorem 7.20 ([2021-20, Theorem 3.3, p. 9]). *Let R be a semiprime ring and I an ideal of R. Let $(d_i)_{i\in\mathbb{N}}$ and $(\delta_j)_{j\in\mathbb{N}}$ be higher derivations of orders m and n on R such that $d_m(x)\, x - x\, \delta_n(x) \in Z(R)$ for all $x \in I$. Then either R is commutative or some linear combination of (δ_j) sends $Z(R)$ to zero.*

The one-derivative case is a corollary [2021-20, p. 9]. A derivation d of a semiprime ring R with $[d(x), x] \in Z(R)$ for all x in an ideal I has either $d(Z(R)) = 0$ or R commutative [2021-20, p. 9].

The number of arguments can grow further, to n of them [2022-2, p. 1]. Ashraf, Ansari and Akhter ask when a multiplicative generalized Jordan n-derivation of a unital ring with a nontrivial idempotent is additive [2022-2, p. 1]. Their hypothesis on the ring is a pair of conditions on the Peirce components of the idempotent [2022-2, p. 3].

Theorem 7.21 ([2022-2, Theorem 3.1, p. 5])**.** *Let A be a 2-torsion free unital ring with a nontrivial idempotent satisfying the stated condition on its Peirce components. Then every multiplicative generalized Jordan n-derivation $\Delta\colon A \to A$ is additive, and $\Delta(t) = \mu t + \delta(t)$ for all $t \in A$, where μ lies in the center of A and δ is a Jordan n-derivation.*

The authors apply the theorem to triangular rings, matrix rings, prime rings, nest algebras, standard operator algebras and von Neumann algebras [2022-2, p. 1].

The permuting triderivation has a stability theory of its own [2020-21, p. 7]. Park states a tri-additive inequality, one that compares a three-variable map with its own averages [2020-21, Theorem 4.3, p. 7]. Every map that satisfies it is close to a unique tri-additive map, with the bound written in the norm of the three arguments [2020-21, Theorem 4.3, p. 7]. If the same map also satisfies the Leibniz rule and the permuting condition within the same bound, then that tri-additive map is a permuting triderivation [2020-21, Theorem 4.3, p. 7]. So the class of Definition 7.15 is stable under small changes of the map [2020-21, p. 1].

The two threads of the chapter join here [2002-6, p. 2323]. A higher derivation is a sequence of maps of one argument [2002-6, p. 2323]. A triderivation is one map of three arguments [2017-15, p. 490]. The same Posner statement holds on both sides [2021-20, p. 4].

The skew form takes an automorphism. For a ring with an automorphism σ, an n-additive map is a skew n-derivation when fixing $n - 1$ arguments leaves a σ-derivation in the remaining one. Xu, Liu and Zhang prove that such a map inherits Bresar's theorem [2013-14, Abstract].

The n-ary form reaches the Hom setting. Mabrouk, Ncib and Silvestrov study generalized derivations and Rota–Baxter operators of n-ary Hom-Nambu superalgebras [2020-19, Abstract].

The ternary condition also appears for conjugate-linear maps. Every continuous conjugate-linear map of a unital C^*-algebra into its dual that is ternary derivable at the unit is a ternary derivation [2020-1, Abstract].

7.5 Symmetric biadditive maps and their traces

Section 7.3 defined the trace of a symmetric biderivation as the diagonal map $f(x) = D(x, x)$ [1989-3, p. 245]. This section studies the trace on its own [1990-6, p. 181]. Maksa introduced the symmetric biderivation and the map that sends x to $D(x, x)$ [1990-6, references 4 and 5]. The two results of Vukman force the ring to commute from the behaviour of the trace alone [1990-6, p. 181].

The first result is the biderivation version of the product theorem of Posner [1990-6, p. 181].

Theorem 7.22 ([1990-6, Theorem 1, p. 185])**.** *Let R be a prime ring of characteristic different from two and three. Suppose there exist symmetric biderivations $D_1 \colon R \times R \to R$ and $D_2 \colon R \times R \to R$, such that $f_1(x)\, f_2(x) = 0$, $x \in R$ holds, where f_1 and f_2 are the traces of D_1 and D_2, respectively. In this case either $D_1 = 0$ or $D_2 = 0$.*

The second weakens the commuting hypothesis of Theorem 7.11 a second time [1990-6, p. 187]. Now even one extra commutator is allowed before the ring is forced to commute [1990-6, p. 187].

Theorem 7.23 ([1990-6, Theorem 2, p. 187])**.** *Let R be a noncommutative prime ring of characteristic different from two and three. Suppose there exists a symmetric biderivation $D \colon R \times R \to R$, such that the mapping $x \mapsto [[f(x), x], x]$, where f stands for the trace of D, is centralizing on R. In this case $D = 0$.*

Vukman records the general statement as a conjecture [1990-6, pp. 188–189]. For some $n \geq 1$ the n-fold commutator of the trace with x lies in the center for all x, and then the symmetric biderivation vanishes [1990-6, pp. 188–189]. He marks the case of arbitrary n as hard [1990-6, p. 189].

The general case is open, and this section states it as a problem.

Problem 7.24 ([1990-6, pp. 188–189])**.** Let R be a noncommutative prime ring of characteristic not 2, let D be a symmetric biderivation of R, and let f be its trace. Suppose that for some $n \geq 1$ the n-fold commutator $[f(x), x]_n$ lies in the center for every $x \in R$. Does D then vanish?

Vukman proves the statement for $n \leq 2$ [1990-6, Theorem 2, p. 187] and states the general case as a conjecture, which he expects to be hard [1990-6, pp. 188–189]. The two sections of the chapter join on the diagonal [2017-15, p. 490]. A vanishing trace of a symmetric biderivation forces the

ring to commute [1990-6, p. 187]. The trace of a permuting triderivation of Section 7.4 is the second.

7.6 Tri-derivations, generalized biderivations and Jordan biderivations

Sections 7.3 to 7.5 studied a biderivation in prime and semiprime rings. Brešar extends the notion to a generalized biderivation and develops its properties [1995-2, reference 5]. This section looks at what a biderivation can be on a structured ring [2016-16, p. 1952].

Two special forms carry the theory [2016-16, p. 1952].

Definition 7.25 ([2016-16, p. 1952])**.** Let R be a noncommutative ring. A biderivation $\phi\colon R \times R \to R$ is *inner* when $\phi(x, y) = \lambda[x, y]$ for all $x, y \in R$, where $\lambda \in Z(R)$. A biderivation $\phi\colon R \times R \to R$ is *extremal* when $\phi(x, y) = [x, [y, a]]$ for all $x, y \in R$, where $a \in R$ with $a \notin Z(R)$ and $[[R, R], a] = 0$.

On a triangular ring every biderivation is built from these two [2016-16, p. 1955]. A unital ring R with a nontrivial idempotent e is *triangular* when $f = 1 - e$, eRf is a faithful (eRe, fRf)-bimodule and $fRe = 0$ [2016-16, p. 1953].

Theorem 7.26 ([2016-16, Theorem 2.1, p. 1955])**.** *Let R be a triangular ring. If the following conditions hold:*

1. $C(fQ_{ml}(R)f, fRf) = C(R)f$,
2. *either $eRC(R)e$ or $fRC(R)f$ does not contain nonzero central ideals,*

then every biderivation on R is a sum of an extremal biderivation and an inner biderivation.

Upper triangular matrices over a unital ring meet the two conditions for rank at least three [2016-16, p. 1956].

Corollary 7.27 ([2016-16, Corollary 2.1, p. 1956])**.** *Let S be a unital ring and let $T_n(S)$ be the ring of all $n \times n$ upper triangular matrices over S (where $n \geq 3$). Then every biderivation of $T_n(S)$ is the sum of an extremal biderivation and an inner biderivation.*

The triangular ring is the right home for the Jordan question [2021-19, p. 1417]. Liu and Liu show that, under some mild conditions, every Jordan biderivation of a triangular ring is a biderivation, and apply it to some upper triangular matrix rings [2021-19, p. 1417]. The full statement sits in their section 2 but the local copy of the paper does not extract its body; only the abstract is quoted [2021-19, p. 1417].

The same two families describe the biderivations of an incidence algebra [2024-14, p. 3]. Guan and Zhang take a commutative ring with unity and a locally finite poset [2024-14, p. 2]. Their inner and extremal biderivations have the forms of Definition 7.25 [2024-14, p. 3]. The decomposition holds once each maximal chain has at least three elements [2024-14, Theorem 4.10, p. 13]. When every maximal chain is infinite, the extremal part disappears and every additive biderivation is a sum of inner ones [2024-14, p. 1]. So the incidence algebra keeps the pattern of the triangular ring, with the shape of the poset in place of the shape of the matrix [2024-14, p. 1].

For a Lie algebra the space of biderivations carries a Lie bracket of its own [2026-15, p. 1]. Di Bartolo and La Rosa call a bilinear map B a biderivation when it is a derivation in each argument [2026-15, Definition 2.1, p. 2]. They also call B a right biderivation when only the first argument obeys the rule, with no linearity asked of the second [2026-15, Definition 3.1, p. 3]. The right biderivations form a Lie algebra under a bracket built from the derivations at each point [2026-15, Theorem 3.5, p. 4]. The left biderivations form one as well [2026-15, Theorem 3.6, p. 4]. For a complete finite-dimensional Lie algebra the right space is a tensor product of functions with the algebra itself [2026-15, Corollary 3.8, p. 5]. The classical biderivations are the maps that are derivations in both arguments, and they form the first cohomology group with coefficients in the algebra [2026-15, p. 1].

The same questions are asked for Leibniz algebras [2025-7, p. 1]. Two definitions of a complete Leibniz algebra appear in the literature, and the authors treat both [2025-7, Definitions 2.6 and 2.7, p. 6]. On such an algebra every biderivation splits into a commutator with a linear map plus a biderivation with values in the Leibniz kernel [2025-7, Theorem 3.1, p. 7]. For a symmetric or a skew-symmetric biderivation the first term lands in the kernel as well [2025-7, Proposition 4.1, p. 10].

Lie superalgebras need a definition that makes the bracket itself a superbiderivation, and the authors give one [2025-6, Definition 2.6, p. 5]. On a complete Lie superalgebra every superbiderivation is a bracket with a linear map of the same degree [2025-6, Theorem 3.6, p. 10]. The authors compare their definition with the earlier ones and record where the two disagree [2025-6, pp. 1–2].

A trivial extension is the triangular matrix ring of a special shape, and Cheraghpour and Jafari describe its Jordan super-biderivations [2024-10, p. 1]. They show that such a map is a Jordan biderivation of the base ring together with two biadditive pieces, and they record the two identities the pieces satisfy [2024-10, Theorem 2.6, p. 9].

Theorem 7.28 ([2024-10, Theorem 2.6, p. 9])**.** *Let B be a Jordan super-biderivation of the trivial extension $T(R, M)$ of a ring R by a bimodule M. Then there are a Jordan biderivation δ of R, a biadditive map $\beta\colon R\times M \to M$ that is a Jordan derivation in the first coordinate with $\beta(r, m\circ r') = \beta(r, m)\circ r' + m\circ\delta(r, r')$, a biadditive map $\eta\colon M\times R \to M$ that is a Jordan derivation in the second coordinate with $\eta(m\circ r', r) = \eta(m, r)\circ r' + m\circ\delta(r', r)$, and a biadditive map $f\colon M\times M \to R$ that is a bimodule Jordan homomorphism in each coordinate, such that*

$$B((r, m), (r', m')) = (\delta(r, r') + f(m, m'),\ \beta(r, m') + \eta(m, r'))$$

for all $r, r' \in R$ and $m, m' \in M$.

So a Jordan super-biderivation of a trivial extension is a Jordan biderivation [2024-10, p. 1]. The same holds for an upper triangular matrix ring under their conditions [2024-10, p. 1].

The chapter ends where it began, through Brešar, Martindale and Miers [2016-16, p. 1952]. A biderivation of a noncommutative prime ring is inner, so it has the form $\lambda[x, y]$ [2016-16, p. 1952]. On a triangular ring it is the sum of an extremal and an inner piece [2016-16, p. 1956]. On the ring $T_n(S)$ over a unital ring S the sum holds for $n \geq 3$ [2016-16, p. 1956]. Liu and Liu add that a Jordan biderivation of such a ring is a biderivation [2021-19, p. 1417].

Chapter 8

Local maps and characterizations through products

A check at each point. The early chapters define a derivation by one law at every pair of points. This chapter weakens the law to a check at a single point or a single pair, and asks how much of the derivation survives. Five threads answer, each read from the papers that carry it.

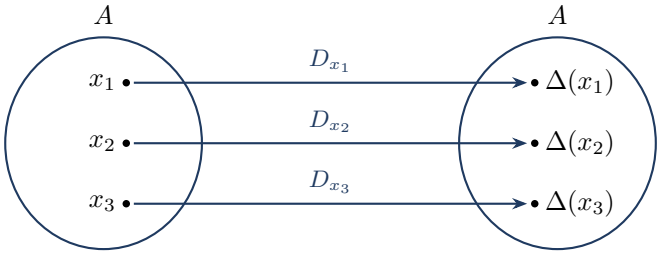


Figure 8.1: A map Δ is a local derivation if for each x there is a derivation D_x with $\Delta(x) = D_x(x)$; the derivation may change from point to point. Is Δ itself a derivation?

The first thread is the local derivation of Kadison [1990-5, p. 494]. Kadison started the line in 1990 with von Neumann algebras, and Section 8.1 reads his Theorem A, the Larson–Sourour companion and the CSL theorems of Crist [1996-1, p. 83]. On every algebra these papers study, a local check already gives a derivation.

The second thread strengthens the check to two points [2012-6, p. 15]. Šemrl asked in 1997 whether a map that agrees with a derivation at each pair of points is a derivation; Section 8.2 recalls his theorem through Ayupov and Kudaybergenov and runs their ladder up to every von Neumann algebra [2015-4, p. 2]. A two-point check again gives a derivation on these algebras.

The third thread checks the law only at zero products [2004-3, p. 218]. Chebotar, Ke and Lee moved the derivation law to the zero products of a prime ring [2004-3, p. 218]. Section 8.3 reads their Theorem 2. It also reads the derivational points of Pan on nest algebras and the zero product determined algebras of Brice and Huang.

The fourth thread comes from preserver problems [2009-2, p. 131]. Alaminos, Brešar, Extremera and Villena started from maps that send zero products to zero products [2009-2, p. 131]. Section 8.4 reads their bilinear tool and their Theorem 2.11 and Theorem 4.6. The answer: a map derivable at zero products is a generalized derivation.

The fifth thread changes the law into the Lie law [2017-8, p. 209]. Section 8.5 reads the local Lie theorem of Liu and Zhang. It then reads the four-condition pattern of He, Li, An and Huang and the structure theorem of Yang and Fang [2019-16, p. 244]. The statement: a linear 2-local Lie derivation on a finite von Neumann algebra without central summands of type I_1 is a Lie derivation [2019-16, p. 250].

8.1 Local derivations: Kadison, Larson–Sourour

Chapter 6 changed the derivation itself: it twisted it, generalized it or gave it more arguments. This chapter changes the form of the map *data*. Kadison names a linear map α of an algebra R into an R-bimodule M a *local derivation* in this case [1990-5, p. 494]. At each $A \in R$ the value $\alpha(A)$ agrees with a true derivation δ_A, whose choice may vary with A [1990-5, p. 494]. His paper opens the study of such maps [1990-5, p. 494]. Brešar and Šemrl give the same definition in their abstract, and they record that Kadison had already studied the question on von Neumann algebras [1993-4, p. 483].

Definition 8.1 ([1990-5, p. 494])**.** Let R be an associative algebra and M an R-bimodule. A linear mapping $\alpha\colon R \to M$ is a *local derivation* when for each $A \in R$ there is a derivation $\delta_A\colon R \to M$ such that $\alpha(A) = \delta_A(A)$.

The question is whether the varying witness can be replaced by one derivation of Definition 8.1. the criterion of Kadison answers it for von

Neumann algebras [1990-5, p. 494]. An R-bimodule M is *dual* when it is a dual space and the module actions are weak* continuous [1990-5, p. 494].

Proposition 8.2 ([1990-5, Theorem A, pp. 495–496])**.** *Let R be a von Neumann algebra and M a dual R-bimodule. If $\delta\colon R \to M$ is a norm-continuous local derivation, then δ is a derivation.*

The proof is organized as five lemmas, and Lemma 4 asks for eleven sublemmas [1990-5, p. 496]. The paragraph after the theorem records the same reading of Proposition 8.2: under these conditions each local derivation is a derivation [1990-5, p. 496].

Proposition 8.3 ([1990-5, Theorem B, p. 507])**.** *Let R be a von Neumann algebra. If $\delta\colon R \to R$ is a norm-continuous linear mapping such that for each $A \in R$ there is a $T_A \in R$ with $\delta(A) = AT_A - T_A A$, then there is a $T \in R$ with $\delta(A) = AT - TA$ for all $A \in R$.*

Kadison reads Theorem B off Proposition 8.2. Every derivation of a von Neumann algebra into itself is inner, so the map is a local inner derivation and one inner derivation then serves all points [1990-5, p. 507]. The statement may surprise even when R is the full complex matrix algebra [1990-5, p. 494].

A Banach-space companion came in the same year, 1990 [1996-1, p. 78]. Larson and Sourour proved the same result for $B(X)$, the bounded operators on a Banach space X [1996-1, p. 78]. Brešar then lowered the hypothesis of Proposition 8.2. A local derivation from a von Neumann algebra into a normed bimodule is still a derivation [1996-1, p. 78].

Crist enlarges the stage from von Neumann algebras to operator algebras with a maximal abelian subalgebra [1996-1, p. 76]. A CSL algebra is an operator algebra whose invariant subspaces form a commutative lattice [1996-1, p. 76].

Theorem 8.4 ([1996-1, Theorem 2.7, p. 83])**.** *Let $\mathcal{A}$ be a finite dimensional CSL algebra. Then any local derivation $\varphi\colon \mathcal{A} \to \mathcal{A}$ must be a derivation.*

The next theorem gives the same conclusion for inner local derivations.

Theorem 8.5 ([1996-1, Theorem 2.8, p. 88])**.** *Let $\mathcal{A}$ be a finite dimensional CSL algebra. If $\varphi\colon \mathcal{A} \to \mathcal{A}$ is an inner local derivation, then φ is an inner derivation.*

In the language of the reflexivity program of Larson this says the derivations, and relatively the inner derivations, are reflexive spaces on these algebras [1996-1, pp. 78–79]. The same conclusion passes to algebras that are direct limits of finite dimensional CSL algebras by $*$-extendable embeddings [1996-1, p. 78]. One step is left behind by Theorem 8.4. Finite dimensional CSL algebras can carry outer derivations, so the results of Kadison and of Larson and Sourour do not carry over verbatim [1996-1, p. 79].

The question of Larson for a general algebra stays open, and this section states it.

Problem 8.6 ([2004-5, p. 703])**.** For which algebras A is the space $\mathrm{der}(A)$ of bounded derivations of A reflexive, that is, equal to its own reflexive closure in the bounded operators on A?

Hadwin and Li answer the question for completely distributive commutative subspace lattices and for J-subspace lattices, where the space of bounded derivations is reflexive [2004-5, Theorem 2.15, p. 708]. They also show that $\mathrm{der}(A, M)$ is algebraically 3-reflexive for every Banach A-bimodule M [2004-5, Theorem 2.19, p. 710]. The local property for automorphisms fails even in finite dimension, since $T \mapsto AT^tA^{-1}$ on $M_n(\mathbb{C})$ is a surjective local automorphism that is not an automorphism [2004-5, p. 713]. The local condition also holds on algebras of measurable operators [2011-4, p. 1]. Let M be a von Neumann algebra with a faithful normal semi-finite trace τ, and let $L^\omega(M, \tau)$ be the Arens algebra of all τ-measurable operators affiliated with M [2011-4, p. 1]. Every local derivation of that algebra is a spatial derivation when M is finite [2011-4, Theorem 2.4, p. 7]. When M is commutative the only local derivation of the Arens algebra is zero [2011-4, Corollary 2.5, p. 7].

Hadwin, Li, Li and Ma answer a question of Kadison with a ring-theoretic argument [2013-8, p. 1]. They prove the following. Let the abelian summand of a von Neumann algebra M be discrete. Then every local derivation of the algebra of measurable operators affiliated with M is a derivation [2013-8, Theorem 1, pp. 2–3]. The proof passes through a lemma on rings that carry a system of matrix units [2013-8, Lemma 1, p. 1].

On the two five-dimensional naturally graded nilpotent associative algebras π_2 and π_3 the answer is negative [2026-5, p. 1]. The matrix form of a derivation and the matrix form of a local derivation are different on each algebra, so a local derivation need not be a derivation [2026-5, p. 1]. The authors write both forms out on π_3 [2026-5, Theorems 8 and 9, pp. 14–15].

The local derivations of each algebra form a vector space, and that space is closed under the commutator [2026-5, Theorems 10 and 11, p. 15]. The matrix form of a local derivation obeys the rules that the matrix form of a derivation obeys, and the authors list them [2026-5, Theorem 5, pp. 10–11].

The vector space of local derivations of an algebra is a Lie algebra under the commutator, with no hypothesis on the algebra [2026-17, Theorem 2.8, p. 4]. Hao and Chen prove it for an arbitrary algebra over a field, and their argument covers every space of operators of one derivation type [2026-17, p. 4]. The result settles half of a question of Ayupov, Elduque and Kudaybergenov [2026-17, p. 1]. The other half concerns the local automorphisms, and it is settled in the same paper [2026-17, p. 1]. The authors also prove that an algebra admits an invertible local derivation exactly when it admits an invertible derivation [2026-17, Theorem 2.18, p. 7].

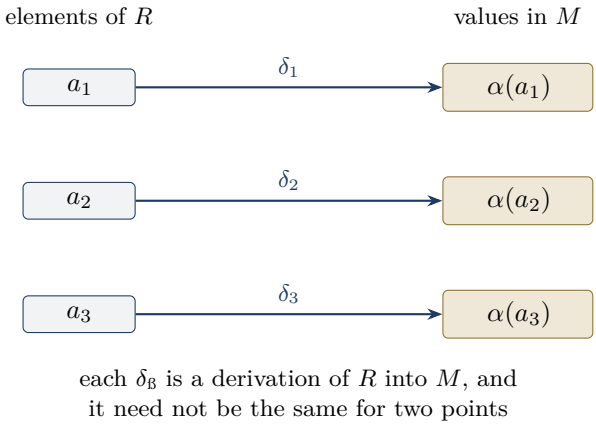


Figure 8.2: A local derivation [1990-5, p. 494]. At each element a of the algebra R the value $\alpha(a)$ equals $\delta_a(a)$ for some derivation δ_a of R, and the derivation may change with a. The question the section answers is when one derivation serves every point.

8.2 2-local derivations: Šemrl and after

Section 8.1 asked for a witness at one point at a time, and Proposition 8.2 replaced it with one derivation. Šemrl asks for a witness only at each pair of points [2012-6, p. 15]. The witness of a pair may still change from pair to pair in Definition 8.7.

Definition 8.7 ([2012-6, p. 15])**.** Let A be an algebra. A map $\Delta\colon A \to A$, not linear in general, is a *2-local derivation* when for every $x, y \in A$ there is a derivation $D_{x,y}\colon A \to A$ such that $\Delta(x) = D_{x,y}(x)$ and $\Delta(y) = D_{x,y}(y)$.

Šemrl introduced the notion in 1997 and described the 2-local derivations and 2-local automorphisms of $B(H)$ for an infinite dimensional separable Hilbert space H [2012-6, p. 15]. Kim and Kim treated the finite dimensional matrix algebra, and Lin and Wong the matrix algebras over finite dimensional division rings [2012-6, p. 15]. A 2-local derivation of Definition 8.7 need not be a derivation at all. Zhang and Li built a 2-local derivation on the upper triangular complex 2×2 matrices that is not a derivation [2012-6, p. 15].

Ayupov and Kudaybergenov drop the separability in Theorem 8.8. Their proof goes through the ideal $F(H)$ of finite rank operators and its trace [2012-6, p. 16].

Theorem 8.8 ([2012-6, Theorem 2.3, p. 16])**.** *Let H be an arbitrary Hilbert space, and let $B(H)$ be the algebra of all bounded linear operators on H. Then every 2-local derivation $\Delta\colon B(H) \to B(H)$ is a derivation.*

Their companion result says every 2-local automorphism of $B(H)$ is an automorphism [2012-6, Theorem 2.4, p. 17]. Ayupov and Kudaybergenov then move the question from $B(H)$ to a general von Neumann algebra [2014-5, p. 9]. Ayupov and Kudaybergenov settled the type I algebras and, with Nurjanov and Alauatdinov, the finite ones; Ayupov and Arzikulov gave a short proof for the semi-finite case [2014-5, p. 9].

Theorem 8.9 ([2014-5, p. 9])**.** *Every 2-local derivation on a semi-finite von Neumann algebra is a derivation.*

The type I case has a proof of its own, through the decomposition of the algebra into summands of type I_n [2011-3, p. 1]. The authors first settle the algebra $C(X) \otimes M_n(\mathbb{C})$ of matrix-valued continuous functions [2011-3, Theorem 1, p. 3]. The summand of type I_n is isomorphic to such an algebra for a hyperstonean compact space X [2011-3, p. 7]. Then every 2-local derivation of a von Neumann algebra of type I is a derivation [2011-3, Theorem 5, p. 7].

The purely infinite piece needs a new tool in Theorem 8.10. Ayupov and Kudaybergenov use the analogue of the Gleason theorem for signed measures on the projection lattice of a von Neumann algebra [2015-4, p. 1].

Theorem 8.10 ([2015-4, Theorem 2.1, p. 2])**.** *Let M be an arbitrary von Neumann algebra. Then any 2-local derivation $\Delta\colon M \to M$ is a derivation.*

On a von Neumann algebra every derivation is inner, so the pair condition of Definition 8.7 reads $\Delta(x) = ax - xa$ and $\Delta(y) = ay - ya$ for one element a [2015-4, p. 2].

A Banach algebra beyond the von Neumann setting calls for another tool, the socle and its trace [2021-18, pp. 322–323]. Huang and Li prove the same conclusion from an ideal-theoretic hypothesis on the socle [2021-18, Theorem 3.6, p. 325].

Theorem 8.11 ([2021-18, Theorem 3.6, p. 325])**.** *Let A be a semisimple Banach algebra. If $\operatorname{soc}(A)$ is an essential ideal of A, then every 2-local derivation $\Delta\colon A \to A$ is a derivation.*

Two applications follow at once [2021-18, Theorem 4.1 and Theorem 4.5, pp. 326 and 328]. A standard operator algebra is a subalgebra of $B(X)$ that contains every finite-rank operator, and it is semisimple and prime [2021-18, p. 326].

Theorem 8.12 ([2021-18, Theorem 4.1, p. 326])**.** *Let A be a semisimple and prime Banach algebra. If $\operatorname{soc}(A) \neq \{0\}$, then every 2-local derivation $\Delta\colon A \to A$ is a derivation.*

Every standard operator algebra meets those hypotheses, so its 2-local derivations are derivations [2021-18, Corollary 4.2, p. 326]. A modular annihilator Banach algebra is one whose socle has zero right and left annihilator, and it gives the second application [2021-18, p. 328].

Theorem 8.13 ([2021-18, Theorem 4.5, p. 328])**.** *Let A be a semisimple modular annihilator Banach algebra. Then every 2-local derivation $\Delta\colon A \to A$ is a derivation.*

Matrix algebras over a semiprime Banach algebra carry the same conclusion from the inner derivation property [2014-4, p. 2]. Take a unital semiprime Banach algebra A in which every derivation is inner, and let $M_{2^n}(A)$ be the algebra of $2^n \times 2^n$ matrices over A with $n \geq 2$ [2014-4, Theorem 2.1, p. 2]. Every 2-local derivation of that matrix algebra is a derivation [2014-4, Theorem 2.1, p. 2]. Every AW*-algebra is semiprime and has the inner derivation property, so every 2-local derivation of an AW*-algebra is a derivation [2014-4, Theorem 3.1, p. 11].

Matrix algebras over a commutative algebra give the same answer for a different reason [2018-11, p. 9]. Here A is a unital commutative algebra over $\mathbb{C}$ that commutes with the bimodule, and every 2-local derivation from $M_n(A)$ to $M_n(A)$ is a derivation [2018-11, Theorem 3.6, p. 10]. The paper also describes every derivation of $M_n(A)$ into $M_n(M)$ as an inner part plus a part induced by a derivation of A [2018-11, Theorem 2.1, p. 4].

The finite von Neumann algebra case has a second route, through the Arens algebra [2011-4, Theorem 3.4, p. 11]. Every 2-local derivation of a finite von Neumann algebra is an inner derivation [2011-4, Theorem 3.4, p. 11]. Every 2-local derivation of the Arens algebra of a von Neumann algebra with a faithful normal semi-finite trace is a derivation [2011-4, Theorem 3.2, p. 9].

For Lie algebras the question has a module-valued form [2026-9, p. 1]. Let $\mathfrak{g} = \mathfrak{s} \oplus \mathfrak{z}$ be a finite-dimensional reductive Lie algebra over an algebraically closed field of characteristic zero, and let V be a finite-dimensional $\mathfrak{g}$-module [2026-9, p. 1]. Every 2-local derivation of $\mathfrak{g}$ into V is a derivation exactly when $\dim \mathfrak{z} \leq 1$ or $V^{\mathfrak{g}} = 0$ [2026-9, Corollary 1.3, p. 2]. The exceptional maps in the remaining case are the homogeneous ones [2026-9, Theorem 1.2, p. 2]. On the infinite-dimensional Block-type Lie algebra the answer is positive again: every 2-local derivation is a derivation [2026-8, Theorem 3.6, p. 9].

The picture of Section 8.1 and of this one now sits at two levels. Local and 2-local maps agree on $B(H)$ and on von Neumann algebras by Theorems 8.8 and 8.10. They part on the triangular matrices, where the example of Zhang and Li lives [2012-6, p. 15].

The problem for a general infinite-dimensional Lie algebra is open, and this section states it.

Problem 8.14 ([2020-7, p. 1])**.** Let L be an infinite-dimensional Lie algebra over a field of characteristic 0. Is every 2-local derivation of L a derivation?

Ayupov and Yusupov answer the question for the Witt algebra and for the positive Witt algebra [2020-7, Theorem 4.1, p. 4, Theorem 4.4, p. 6]. They also build a thin Lie algebra whose 2-local derivations include a map that is not a derivation [2020-7, Theorem 4.8, p. 8]. So the answer depends on the algebra. Chapter 14 reads Lie algebras in Section 14.5.

Arzikulov and Shodiev treat local and 2-local $\frac{1}{2}$-derivations of the n-dimensional complex filiform Leibniz algebra [2026-4, Abstract, p. 1].

A weaker condition tests the witness through one functional at a time. A map Δ is a *weak 2-local derivation* [2017-5, p. 382] when for every x, y and

every bounded functional φ there is a derivation T with $\varphi\Delta(x) = \varphi T(x)$ and $\varphi\Delta(y) = \varphi T(y)$. The derivation may depend on the two points and on the functional, so the condition is weaker than 2-locality. Cabello and Peralta prove that every weak 2-local derivation on $B(H)$ is a linear derivation. They prove the same on a finite-dimensional C^*-algebra [2017-5, Theorems 1.1 and 1.2, p. 383]. The weakening produces no new map on these algebras.

8.3 Derivations through zero products and derivable maps

Sections 8.1 and 8.2 tested the map point by point and pair by pair. Chebotar, Ke and Lee test it on the zero products [2004-3, p. 217]. An additive map $\delta\colon A \to A$ *behaves like a derivation on the zero products* when $\delta(x)y + x\delta(y) = 0$ holds whenever $xy = 0$. Their aim is to get the derivation back, at least almost [2004-3, p. 217].

The property that governs this section has a name. An algebra A is *zero product determined* [2009-4, p. 1487] when for every module X and every bilinear map $\{\cdot,\cdot\}$ into X, the condition $\{x, y\} = 0$ whenever $xy = 0$ forces $\{x, y\} = T(xy)$ for some linear T. Brešar, Grašič and Sánchez Ortega introduce it and prove that the full matrix algebras over a field are zero product determined [2009-4, p. 1487]. A derivable map is the case where $\{\cdot,\cdot\}$ is built from the map and the product, which is why the two notions meet here.

The same question is asked of the Jordan product. A is *zero Jordan product determined* [2015-2, p. 90] when every bilinear φ vanishing on the pairs with $a \circ b = 0$ has the form $\varphi(a, b) = T(a \circ b)$. Write $J(A)$ for the subalgebra generated by the idempotents. An, Ghahramani and Heidari prove that A is zero Jordan product determined whenever $A = J(A)$, by exhibiting the map explicitly: for $a \in A$ and $x \in J(A)$ the value $\varphi(a, x)$ is $\frac{1}{2}\varphi(ax, 1) + \frac{1}{2}\varphi(xa, 1)$ [2015-2, Theorem 2.1, p. 91]. That settles the question left open at the end of the paper cited above.

The condition can be weakened to the idempotents alone. A is *idempotent elements determined* [2011-13, Definition 1.2, p. 2890] when the only pairs tested are those with x idempotent, that is when $\{x, x\} = \{e, x\}$ for every idempotent x already forces $\{x, y\} = f(x \circ y)$. Wang, Li and Chen prove that $M_n(R)$ is idempotent elements determined for a commutative ring with 2 invertible [2011-13, Theorem 2.4]. So on matrix algebras the whole of the Jordan product need not be tested; the idempotents carry it.

Theorem 8.15 ([2004-3, Theorem 2, p. 218])**.** *Let A be a prime ring with center Z, maximal right quotient ring Q and extended centroid C. Let $\delta\colon A \to A$ be an additive map such that $\delta(x)y + x\delta(y) = 0$ for all $x, y \in A$ with $xy = 0$. Suppose that Q contains a nontrivial idempotent e such that $eA \cup Ae \subseteq A$.*

1. *If $1 \in A$, then $\delta(xy) = \delta(x)y + x\delta(y) - \lambda xy$ for all $x, y \in A$, where $\lambda = \delta(1) \in Z$. In particular, if $\delta(1) = 0$, then δ is a derivation on A.*
2. *If $\deg A \geq 3$, there exists $\lambda \in C$ such that $\delta(xy) = \delta(x)y + x\delta(y) - \lambda xy$ for all $x, y \in A$.*

Chebotar, Ke and Lee also treat the multiplicative side [2004-3, p. 217]. A bijective additive map of prime rings that preserves zero products is a scalar multiple of a ring isomorphism [2004-3, Theorem 1, pp. 217–218].

Pan asks the question at one element [2012-13, p. 4251]. A linear map δ of an algebra A into an A-bimodule M is *derivable at $C \in A$* when $\delta(A)B + A\delta(B) = \delta(C)$ for all A, B with $AB = C$ [2012-13, p. 4251].

Definition 8.16 ([2012-13, p. 4251])**.** Let A be a unital algebra and M a unital A-bimodule. An element $C \in A$ is a *derivational point* of $L(A, M)$ when every $\delta \in L(A, M)$ that is derivable at C is a derivation.

Zero products are the point $C = 0$, the Jordan condition is the point $C = I$, and the recent work asks for the points in between [2012-13, p. 4251]. Pan describes the maps derivable at a fixed C through a Peirce decomposition [2012-13, Theorem 2.1, p. 4253] and then decides which elements are derivational points.

Theorem 8.17 ([2012-13, Theorem 4.6, p. 4259])**.** *If $A = \operatorname{alg}\mathcal{N}$ is a nest algebra on a Hilbert space H then every $0 \neq C \in A$ is a derivational point of $L(A, B(H))$.*

For $C = 0$ the condition is exactly the hypothesis of Theorem 8.15; Theorem 8.17 says a nest algebra needs no idempotent hypothesis, since each nonzero element decides the question by itself. Brice and Huang turn the picture around and ask which algebras support it [2015-7, p. 326].

Definition 8.18 ([2015-7, pp. 326–327])**.** Let K be a commutative ring with identity. A K-algebra A, not necessarily associative, is *zero product determined* when each K-bilinear map $\varphi\colon A \times A \to B$ with $\varphi(a_1, a_2) = 0$ whenever $a_1 a_2 = 0$ can be written as $\varphi(a_1, a_2) = \tilde{\varphi}(a_1 a_2)$ for some K-linear $\tilde{\varphi}\colon A^2 \to B$.

The notion is from Brešar, Grašić and Sánchez Ortega, who introduced it for matrix algebras [2009-4, pp. 1486–1498] and state it in the form used here in the later paper of Brice and Huang [2015-7, pp. 326–327]. It was built for zero product preserving linear maps on Banach algebras and on matrix algebras under the standard, Lie and Jordan products [2015-7, p. 327]. Brice and Huang give a tensor criterion for the class of Definition 8.18.

Theorem 8.19 ([2015-7, Theorem 2.3, p. 328])**.** *Let (A, μ) be a K-algebra, with $\mu\colon A \otimes A \to A$ its multiplication map and T_μ the span of all $a_1 \otimes a_2$ with $a_1 a_2 = 0$. Then (A, μ) is zero product determined if and only if $T_\mu = \ker \mu$.*

The criterion of Theorem 8.19 runs the class well. A direct sum of K-algebras belongs to the class exactly when each summand does [2015-7, Theorem 3.1, p. 331]. An algebra tensor product of two zero product determined algebras stays in the class when K is a field or both multiplications are surjective [2015-7, Theorem 4.3, pp. 331–332].

An element can carry the notion instead of the map. Zhu calls Z an all-derivable point of an operator algebra when every continuous derivable map at Z is a derivation, and identifies such points [2007-13, Abstract].

Benkovic and Grasic settle when the zero-product condition determines the map. They show a unital algebra with a nontrivial idempotent is zero product determined exactly when both Peirce subalgebras are, and then characterize the generalized derivations determined by their action on zero products [2014-8, pp. 347–368].

Further work in the operator and zero-product line, held in the collection: *Generalized skew derivations with nilpotent values on left ideals of rings and Banach algebras* [2014-2]. *Aggregate special functions to approximate permuting tri-homomorphisms and permuting tri-derivations associated with a tri-additive ψ -functional inequality in Banach algebras* [2024-1]. *Additive maps on some operator algebras behaving like α β -derivations or generalized α β -derivations at zero-product elements* [2014-1]. *Ring isomorphisms and linear or additive maps preserving zero products on nest algebras* [2004-2].

8.4 Derivations among preserver problems

The zero product condition of Section 8.3 came from a preserver question. A linear map T between algebras *preserves zero products* when $T(a)T(b) = 0$ whenever $ab = 0$ [2009-2, p. 131]. Banach started this line: a linear map of $L_p([0, 1])$ that keeps distances sends two functions whose supports do not meet to two functions whose supports do not meet [2009-2, p. 131].

The standard goal is now to write such a T as a weighted homomorphism [2009-2, p. 132].

Alaminos, Brešar, Extremera and Villena run the same question for Banach algebras [2009-2, p. 131]. Their tool is the bilinear map $\varphi(a,b) = T(a)T(b)$, which vanishes on zero products exactly when T preserves them [2009-2, p. 132].

Theorem 8.20 ([2009-2, Theorem 2.11, p. 143])**.** *Let A be a faithful Banach algebra, let X be a Banach space, and let $\varphi\colon A \times A \to X$ be a continuous bilinear map preserving zero products. Then $\varphi(a\mu, b) = \varphi(a, \mu b)$ for all $a, b \in A$ and $\mu \in \overline{D(A)}^{so}$.*

Here $D(A)$ is the subalgebra of the multiplier algebra $M(A)$ generated by doubly power-bounded elements, and the closure is taken in the strong operator topology [2009-2, p. 143]. The rule of Theorem 8.20 applied to $\varphi(a,b) = T(a)T(b)$ turns a surjective zero product preserving map into a weighted homomorphism on a large class of Banach algebras, group algebras among them [2009-2, Theorem 3.6, p. 152].

The same tool answers the derivational side of Section 8.3 for Banach algebras [2009-2, p. 155]. A continuous linear $D\colon A \to X$ with $D(a)\cdot b + a \cdot D(b) = 0$ whenever $ab = 0$ need not be a derivation: by Theorem 8.21 it lands in a wider class.

Theorem 8.21 ([2009-2, Theorem 4.6, p. 156])**.** *Let A be a Banach algebra with the property of Theorem 8.20 and a bounded approximate identity, let X be a Banach A-bimodule with $\operatorname{Ann}_X(A) = \{0\}$, and let $D\colon A \to X$ be a continuous linear operator satisfying $D(a)\cdot b + a \cdot D(b) = 0$ whenever $ab = 0$. Then D is a generalized derivation: there exists $\xi \in Z_A(X^{**})$ such that $D(ab) = D(a)\cdot b + a \cdot D(b) - a \cdot \xi \cdot b$ for all $a, b \in A$. Accordingly, there exists a continuous derivation $d\colon A \to X^{**}$ such that $D(a) = d(a) + a \cdot \xi$ for all $a \in A$. Moreover ξ can be chosen in $Z_A(X)$ when A has an identity or X is a dual bimodule.*

Theorem 8.21 is the Banach cousin of the prime-ring Theorem 8.15. The correction element λ of Theorem 8.15 becomes a central ξ in the second dual [2009-2, p. 156]. When A is unital, ξ sits in the bimodule itself [2009-2, p. 156].

8.5 Local and 2-local Lie derivations

The local condition of Section 8.1 has a Lie form. A linear map φ of an algebra is a *local Lie derivation* [2017-8, p. 209] when for each A there is a Lie derivation φ_A, depending on A, with $\varphi(A) = \varphi_A(A)$. A Lie derivation here means a linear map φ with $\varphi([A, B]) = [\varphi(A), B] + [A, \varphi(B)]$ [2017-8, p. 209]. Liu and Zhang settled the local Lie question on the classical ground of Section 8.1, von Neumann algebras.

Theorem 8.22 ([2017-8, Theorem 2.1, p. 209])**.** *Let A be a factor von Neumann algebra acting on a complex Hilbert space H with* $\dim(A) \geq 2$*. Then each local Lie derivation φ from A into itself is a Lie derivation.*

A Lie derivation of a von Neumann algebra splits as a derivation plus a linear map into the center that vanishes on each commutator, after Miers 1973 as reported at [2017-8, p. 209]. A Lie derivation with such a splitting is called *standard*, and each Lie derivation on a C^*-algebra is standard, after Mathieu and Villena as reported at [2018-10, p. 725]. He, Li, An and Huang used these two facts to build a transferable pattern [2018-10, p. 727]. Its four hypotheses are simple. The first three ask that each local derivation be a derivation. They also ask that each derivation be inner and that each Lie derivation be standard. The fourth says that no nonzero central element is a sum of commutators [2018-10, p. 727].

Theorem 8.23 ([2018-10, Theorem 3.1, p. 727])**.** *Let B be an algebra over C satisfying the conditions: (1) every Lie derivation on B is standard; (2) every derivation on B is inner; (3) each local derivation on B is a derivation; (4) every nonzero element in $Z(B)$, the center of B, cannot be written as $\sum_{i=1}^{n}[A_i, B_i]$, where $A_i, B_i \in B$. Then each local Lie derivation φ on B is a Lie derivation.*

The four conditions hold widely [2018-10, pp. 727–728]. Each local Lie derivation on a factor von Neumann algebra is a Lie derivation, which recovers Theorem 8.22 with different hypotheses [2018-10, Corollary 3.2, p. 727]. Condition (4) alone follows from the existence of a center-valued trace, so the statement covers every finite von Neumann algebra [2018-10, Theorem 3.4 and Corollary 3.5, p. 728]. The properly infinite case is simpler: every element is already a sum of commutators, so every Lie derivation is a derivation [2018-10, Remark 3.6, p. 728]. The same paper records the conclusion on UHF C^*-algebras and on the Jiang–Su algebra [2018-10, Corollaries 3.7 and 3.8, p. 728].

The 2-local form of the question extends the construction of Section 8.2 to Lie derivations. A map δ on a Banach algebra, not assumed linear, is a *2-local Lie derivation* when δ agrees at each pair of points with a Lie derivation chosen for that pair [2019-16, p. 243]. The condition is that for every A and B there is a Lie derivation $\delta_{A,B}$, depending on A and B, with $\delta(A) = \delta_{A,B}(A)$ and $\delta(B) = \delta_{A,B}(B)$ [2019-16, p. 243].

Theorem 8.24 ([2018-10, Theorem 3.9, pp. 728–729])**.** *Let B be an algebra over C satisfying the conditions: (1) every Lie derivation on B is standard; (2) every derivation on B is inner; (3) each 2-local derivation on B is a derivation; (4) $Z(B) = CI$, where $Z(B)$ is the center of B; (5) every element in B can be written as $\lambda I + \sum_{i=1}^{n}[A_i, B_i]$, where $\lambda \in C$ and $A_i, B_i \in B$. Then each 2-local Lie derivation φ on B is a Lie derivation.*

Factor von Neumann algebras, UHF C^*-algebras and the Jiang–Su algebra meet the conditions of Theorem 8.24, so each 2-local Lie derivation on them is a Lie derivation [2018-10, Corollaries 3.10 to 3.12, p. 729]. Linearity saves the general finite case up to a defect, as Theorem 8.25 will show [2019-16, p. 244]. On a finite von Neumann algebra which is not a factor there is a 2-local Lie derivation, which is not linear, and which is no Lie derivation [2018-10, Example 3.13, p. 729].

Yang and Fang closed the remaining case by describing the defect exactly [2019-16, p. 244].

Theorem 8.25 ([2019-16, Theorem 2.1, p. 244])**.** *Let M be a finite von Neumann algebra without central summands of type I_1. Then every 2-local Lie derivation $\delta\colon M \to M$ can be written in the form $\delta(A) = AE - EA + h(A)$ for all A in M, where E is an element in M and $h\colon M \to Z(M)$ is a homogenous mapping which annihilates each commutator of M.*

Every linear 2-local Lie derivation on a finite von Neumann algebra without central summands of type I_1 is then a Lie derivation [2019-16, Corollary 2.6, p. 250]. On a properly infinite von Neumann algebra, every element is a finite sum of commutators after Sunouchi 1956 as reported at [2019-16, p. 244]. There every 2-local Lie derivation is a 2-local derivation, and a 2-local derivation is a derivation by Theorem 8.10 [2019-16, p. 244].

Chapter 9

Functional identities and commuting maps

The theme of this chapter is a single observation that grew into a method. Posner showed in 1957 that a prime ring admitting a nonzero centralizing derivation must be commutative [1993-1, p. 385]. Mayne proved the parallel statement for centralizing automorphisms [1993-1, p. 385]. Brešar then asked what an arbitrary additive centralizing map looks like [1993-1, p. 385]. He answered it in the prime case: such a map is multiplication by a central element, up to a central remainder [1993-1, Theorem A, p. 385]. Section 9.1 reports this line and its one-sided-ideal refinement of Lee [1997-4, Theorem 2, p. 335].

Out of these commuting-map theorems grew the general theory. Section 9.3 introduces a functional identity as an equation with functions as unknowns, and the d-free set as the concept on which the whole development rests [2007-3, p. 9 and p. x]. Section 9.4 reports the threefold development of that concept from the preface of the monograph [2007-3, p. x]. Section 9.5 carries the theory back to derivations and shows how a two-variable functional identity settles Jordan σ-derivations on prime rings [2016-13, Theorem 2.1, p. 221]. Section 9.6 closes with the twisted form: a map that commutes with an endomorphism still has the central form on semiprime rings [2001-2, Theorem 2, p. 2948].



Figure 9.1: Commuting maps on prime rings (Brešar): the only additive maps commuting with every element are the obvious ones.

9.1 Commuting and centralizing maps

A map F of a ring R into itself is *commuting* on a subset S [1997-4, p. 333] when $[s, F(s)] = 0$ for all s in S. The map is *centralizing* on S [1997-4, p. 333] when $[s, F(s)]$ lies in the center Z of R. The line began with a result of Posner from 1957: the existence of a nonzero centralizing derivation on a prime ring forces the ring to be commutative [1993-1, p. 385]. Mayne proved the analogous statement for centralizing automorphisms [1993-1, p. 385]. The following years brought extensions of these two theorems in several directions [1993-1, p. 385].

Brešar asked for the structure of an arbitrary additive centralizing map, and gave a complete answer in the prime case [1993-1, p. 385].

Theorem 9.1 ([1993-1, Theorem A, p. 385])**.** *Let R be a prime ring. Suppose an additive mapping F of R into itself is centralizing on R. If either R has a characteristic different from 2 or F is commuting on R, then F is of the form $F(x) = \lambda x + \zeta(x)$ for all $x \in R$, where λ is an element from the extended centroid C of R and ζ is an additive mapping of R into C.*

Theorem 9.1 says that a centralizing additive map on a prime ring is, up to a central remainder, multiplication by a central element [1993-1, p. 385]. Brešar later obtained the same conclusion in the semiprime case, and based on these studies he initiated the study of functional identities [1997-4, p. 333]. For a right ideal the same form forces a restriction: every commuting additive map $f\colon \rho \to R$ has the form $f(x) = \lambda x + \mu(x)$ unless $[\rho, \rho]\rho = 0$ [1997-4, p. 333].

The derivations between two centralizing maps are also controlled, at the cost of restricting the ring [1997-4, p. 334]. A ring is *S_4-free* when it does not satisfy the standard identity S_4 of degree 4. Lee proved a result on pairs of derivations [1997-4, Theorem 1, p. 334]. Let δ and d be nonzero derivations of a prime S_4-free ring with $\delta \notin Cd$, and let $[\delta(x), d(x)]$ be central on a right ideal ρ. Then d and δ are inner on ρ through two elements of the Martindale quotient ring [1997-4, Theorem 1, p. 334]. The clean

form of the centralizing theorem itself holds on one-sided ideals [1997-4, Theorem 2, p. 335]. It moves Theorem 9.1 from the whole prime ring to a right ideal [1997-4, p. 335].

Theorem 9.2 ([1997-4, Theorem 2, p. 335])**.** *Let R be a prime ring and ρ a nonzero right ideal of R. Then every centralizing additive mapping $f\colon \rho \to U$ is of the form $f(x) = \lambda x + \mu(x)$ for all $x \in \rho$, where $\lambda \in C$ and $\mu\colon \rho \to C$, unless $[\rho, \rho]\rho = 0$.*

Here U is the right Utumi quotient ring of R [1997-4, p. 334]. Two earlier theorems sit behind Theorem 9.2 [1997-4, p. 334]. An additive map f of a prime ring with $[f(x), x]_n = 0$ for a fixed n is already commuting, provided the characteristic is 0 or greater than n [1997-4, Theorem B, p. 334]. For a derivation d of a semiprime ring with $[d(x^n), x^n]_k = 0$ on a right ideal, the conclusion is $[\rho, R]d(R) = 0$ [1997-4, Theorem L, p. 334]. The general conjecture that a positive-power condition $[f(x), x]_n = 0$ already gives the form $f(x) = \lambda x + \mu(x)$ remained open [1997-4, p. 334].

That conjecture is open, and this section states it as a problem.

Problem 9.3 ([1997-4, p. 334])**.** Let R be a semiprime ring and let $f\colon R \to R$ be an additive map with $[f(x), x]_n = 0$ for one fixed positive power n. Must f be of the form $f(x) = \lambda x + \mu(x)$, with λ in the extended centroid and μ a map into the extended centroid?

Lee proves the statement for $n = 1$ and for several restricted conditions on the ring [1997-4, Theorem L, p. 334], and he records the general power as open [1997-4, p. 334].

So far the sets were a whole ring or a right ideal. Franca and Louza moved the question to subsets of a matrix algebra studied in linear preserver problems, and those subsets are not closed under addition [2024-13, p. 2540].

Throughout their paper F is a field whose characteristic is 0 or at least 5, and M_n is the algebra of all $n \times n$ matrices over F with $n \geq 3$ [2024-13, p. 2541]. Here L is the prime field of F [2024-13, p. 2541]. The set E consists of the idempotent matrices, and V consists of the scalar multiples αE_{ii} of the matrix units [2024-13, p. 2541]. Their sets are the union $\overline{E} = E \cup V$, the set S of matrices whose minimal polynomial has degree 2, and $P = \{X \in M_n \mid \operatorname{tr}(X) \in L\}$ with $\overline{P} = P \cup V$ [2024-13, p. 2541].

An additive map G on M_n has the *standard form* when $G(X) = \lambda X + \mu(X)$ for all $X \in M_n$, where λ is a scalar and μ is an additive map into

the scalar matrices [2024-13, p. 2540]. In the matrix algebra this is the conclusion of Theorem 9.1.

The first of the two answers goes to S [2024-13, p. 2550].

Theorem 9.4 ([2024-13, Theorem 6.6, p. 2550])**.** *Let F be a field whose characteristic is 0 or at least 5, let $n \geq 3$, and let $G\colon M_n \to M_n$ be an additive map which is commuting on S. Then G has the standard form.*

So the condition on the matrices of minimal polynomial of degree 2 is as strong as the condition on the whole algebra. The idempotents behave in another way, and the authors tie them to the trace condition [2024-13, p. 2547].

Theorem 9.5 ([2024-13, Theorem 5.1, p. 2547])**.** *Let F be a field whose characteristic is 0 or at least 5, let $n \geq 3$, and let $G\colon M_n \to M_n$ be an additive map. Then G is commuting on $\overline{P}$ if and only if G is commuting on $\overline{E}$.*

Theorem 9.5 says that the condition on the idempotents already decides it on the larger set $\overline{P}$, which is still not the whole algebra [2024-13, p. 2547]. It does not force the standard form, and the authors build a map that shows this [2024-13, Example 7.1, pp. 2550–2553].

Example 9.6 ([2024-13, Example 7.1, pp. 2550–2553])**.** Let F be a field with $[F : L] \geq 7$ and let $n = 3$. Then there is an additive map $G\colon M_3 \to M_3$ which is commuting on $\overline{E}$ and does not have the standard form. That map is not commuting on the set R_1 of rank-one matrices.

So $\overline{E}$ admits commuting maps outside the standard form, and S does not [2024-13, p. 2553]. The authors compare the three classical sets in their closing commentary: among R_1, E and S only S has all its commuting maps of the standard form [2024-13, p. 2553].

9.2 Functional equations and information measures

A functional equation can carry a derivation inside its solutions [1981-1, p. 195]. The presence of the derivation is what makes those solutions surprising [1981-1, p. 195]. Aczel studies one real function of one variable, written f [1981-1, p. 191]. The variable runs over the open unit interval, and the function comes from a measure of information [1981-1, p. 191]. The

measure of uncertainty behind it satisfies a recursion [1981-1, pp. 192–193]. Once the measure is symmetric, the recursion becomes an equation on f alone

$$f(x) + (1-x)\, f\left(\frac{y}{1-x}\right) = f(y) + (1-y)\, f\left(\frac{x}{1-y}\right),$$

and x and y are positive with $x + y < 1$ [1981-1, pp. 192–193]. Aczel calls the solutions the *symmetric information functions* [1981-1, p. 192].

The general solution of that equation involves a derivation [1981-1, p. 195]. Aczel writes the solution through two auxiliary functions [1981-1, p. 195]. The first one, written e, is any solution of the product equation

$$e(xy) = x\, e(y) + y\, e(x), \qquad x, y \in \,]0, 1[.$$

The second one, written d, is any real derivation [1981-1, pp. 191 and 194]:

$$d(x+y) = d(x) + d(y), \qquad d(xy) = x\, d(y) + y\, d(x).$$

With those two in hand, the general solution reads

$$f(x) = e(x) + e(1-x) + ax$$

for a constant a [1981-1, p. 195].

The point Aczel makes is about a regularity assumption [1981-1, p. 195]. The formula above is correct [1981-1, p. 195]. What fails is the claim that a regular f forces e to be regular in the same class [1981-1, p. 195]. The derivation is the reason [1981-1, p. 195]. If e solves the two equations, then so does $e + d$ [1981-1, p. 195]. A derivation satisfies both by definition, and $d(1-x) = -d(x)$ follows from them [1981-1, p. 195]. So a very regular f can hide a very irregular e [1981-1, p. 195].

Theorem 9.7 ([1981-1, p. 195])**.** *Let e_0 be one particular solution of the two equations above for a given f and a. Then their general solution is $e = e_0 + d$, where d is an arbitrary real derivation, whether zero or not, restricted to* $]0, 1[$.

Aczel gives the converse argument too [1981-1, p. 195]. Take two solutions of the pair of equations [1981-1, p. 195]. Their difference satisfies both halves of the Leibniz rule [1981-1, p. 195]. So the difference is a derivation [1981-1, p. 195].

The same effect appears again one degree higher [1981-1, p. 199]. Aczel also treats information functions of degree α [1981-1, p. 199]. With no regularity assumption their general solution is

$$f(x) = b\,[x^\alpha + (1-x)^\alpha - 1], \qquad \alpha \neq 1,$$

so the case $\alpha = 2$ looked like the rest [1981-1, pp. 198–199]. The expectation was that the same implication holds at $\alpha = 2$ [1981-1, p. 199]. Aczel reports a counterexample instead [1981-1, p. 199].

Example 9.8 ([1981-1, p. 199])**.** Let d be a not identically zero real derivation and b a constant. Then

$$f(x) = b\,[x^2 + (1-x)^2 - 1] + d(x), \qquad x \in \,]0,1[,$$

is a solution of the equation for degree 2. It satisfies neither the symmetry $f(x) = f(1-x)$ nor the equation for a degree other than 2.

The derivation is what breaks the symmetry here [1981-1, p. 199]. A derivation changes sign under $x \mapsto 1 - x$ [1981-1, p. 199]. The square bracket does not change sign under that map [1981-1, p. 199]. So the sum of the two cannot be symmetric. Aczel adds a second observation [1981-1, p. 199]. If an f of this shape is measurable, then the derivation in it is zero [1981-1, p. 199]. That is why the measurable solutions never showed the effect [1981-1, p. 199].

So the derivations of this book are not always constructive [1981-1, p. 191]. Here they are the hidden term. They separate the algebraic solutions of a functional equation from the regular ones [1981-1, p. 199]. That separation is what the two surprises in the title of Aczel count [1981-1, p. 191]. The same question has an answer on infinite-dimensional Lie algebras [2016-15, p. 1]. On the Virasoro algebra every linear commuting map is $x \mapsto \lambda x + \sigma(x)c$, with λ a complex number and σ a linear functional [2016-15, Theorem 4.2, p. 9]. On the W-algebra $W(2,2)$ a second term appears, built from the map ω that replaces each generator L_m by H_m [2016-15, Theorem 4.1, p. 9]. The same two algebras carry no non-trivial commutative post-Lie structure [2016-15, Theorem 4.6, p. 9]. On the deformed Schrödinger–Virasoro algebra the commuting maps have four possible forms, according to the two parameters [2016-11, Theorem 4.3, p. 11]. Each form is read from the skew-symmetric biderivations of the algebra [2016-11, Theorem 3.2, p. 4]. For a Hom-Lie superalgebra the commuting condition carries graded signs, and a linear super-commuting map on a centerless algebra equal to its derived algebra lies in the centroid [2024-16, Theorem 4.2, p. 11]. The same paper describes the super-biderivations of that algebra [2024-16, Theorem 3.8, p. 7]. So on these algebras the commuting maps, the biderivations and the centroid are three views of one object [2024-16, p. 1].

9.3 The theory of functional identities

A functional identity is an equation with functions appearing as unknowns [2007-3, p. 9]. One refers to functions satisfying a certain functional identity as solutions of that identity [2007-3, p. 9]. One calls the identically zero functions the standard solution, and the aim of the theory is to describe exactly when nonstandard solutions exist [2007-3, p. 9]. The first functional identity is the two-term identity $E(x)y + F(y)x = 0$ on a ring A [2007-3, Example 1.1, p. 10]. Its solvability is a structural statement about A [2007-3, Example 1.1, p. 10]. A semiprime ring A has a nonstandard solution exactly when it contains a nonzero central ideal [2007-3, Example 1.1, p. 10]. A prime ring has one exactly when it is commutative [2007-3, Example 1.1, p. 10]. The matrix ring $M_n(C)$ over a commutative unital ring has one exactly when $n = 1$ [2007-3, Example 1.1, p. 10].

The central concept upon which the general theory rests is the *d-free set* [2007-3, p. x]. A ring is d-free when two designated functional identities of degree d have only the standard solution [2007-3, p. 21]. The concept carries the important weight of the whole development [2007-3, p. 21]. The first parts of the book introduce d-free sets and study their basic properties and constructions [2007-3, p. x]. They then prove the existence of important classes of d-free sets inside prime and semiprime rings [2007-3, p. x].

Until this monograph the general functional identity theory existed only in research journals; the book was written because the theory had reached maturity and deserved a comprehensive account [2007-3, p. ix]. The theory grew directly out of the commuting and centralizing maps of Section 9.1: Brešar initiated the study of functional identities based on his characterization of centralizing maps, as reported at [1997-4, p. 333]. The next section reports the central role of d-free sets [2007-3, p. x]. The two following sections report the two families of applications, to derivation problems and to centralizers and multipliers [2007-3, p. x].

9.4 d-free sets and applications

The d-free set of Section 9.3 is the organizing idea of the whole theory: it is the basic concept upon which everything is based [2007-3, p. x]. A ring is d-free when two designated two-variable functional identities of degree d admit only the standard solution [2007-3, p. 21]. The concept is introduced exactly for this purpose, to control when nonstandard solutions can exist [2007-3, p. 21].

The development that the sources report is threefold [2007-3, p. x]. First, the basic properties and constructions of d-free sets are worked out [2007-3, p. x]. Second, functional identities on d-free sets are analyzed in general, through the notions of quasi-polynomials and core functions [2007-3, p. x]. Third, the existence of central classes of d-free sets is proved inside prime and semiprime rings, which are the rings the applications live on [2007-3, p. x].

What the concept buys is a single hypothesis that replaces the separate structural arguments of Section 9.1. The commuting map $f(x) = \lambda x + \mu(x)$ of Theorem 9.2 was obtained ring by ring. On a sufficiently d-free ring the same conclusion follows from the general theory of functional identities on d-free sets [2007-3, p. x]. The applications reported in the sources are of two families: to derivation problems, taken up in Section 9.5, and to centralizers, multipliers and Lie centralizers, taken up in Section 9.6 [2007-3, p. x].

9.5 Applications to derivation problems

The general theory of Sections 9.3 and 9.4 exists largely to settle derivation questions, and its first clean application is to Jordan σ-derivations. Lee used the functional identity method to prove a two-variable identity that carries the weight of every later argument in his paper [2016-13, Section 2, p. 221].

Theorem 9.9 ([2016-13, Theorem 2.1, p. 221])**.** *Let R be a non-commutative prime ring and let $f\colon R \times R \to Q_{mr}(R)$ be a bi-additive map. Suppose that $f(x,y)[x,y] = 0$ for all x, y in R. Then $f = 0$.*

Here $Q_{mr}(R)$ is the maximal right ring of quotients of R [2016-13, p. 221]. Theorem 9.9 turns any bi-additive expression annihilated by the commutator into zero [2016-13, p. 221]. This is exactly the step needed to promote Jordan-type laws to derivation laws [2016-13, p. 221].

Theorem 9.10 ([2016-13, Theorem 2.2, p. 221])**.** *Let R be a non-commutative prime ring with an endomorphism σ. Then every Jordan σ-derivation δ of R satisfying $\delta(xyx) = \delta(x)yx + \sigma(x)\delta(y)x + \sigma(xy)\delta(x)$ is a σ-derivation.*

The proof uses Theorem 9.9: after linearization the defect $\delta(xy) - \delta(x)y - \sigma(x)\delta(y)$ is a bi-additive map whose product with $[x,y]$ vanishes, so it is zero [2016-13, p. 221]. When the characteristic is not 2 the extra identity holds automatically, so every Jordan σ-derivation of a noncommutative prime ring with $\operatorname{char} R \neq 2$ is a σ-derivation [2016-13, Corollary 2.3, p. 221].

This generalizes the earlier characterization of Jordan derivations to the σ-derivation context [2016-13, p. 221].

When σ is not injective the conclusion strengthens: every Jordan σ-derivation is X-inner, meaning $\delta(x) = ax - \sigma(x)a$ for a fixed a in $Q_{mr}(R)$ [2016-13, Theorem 2.4, p. 221]. A Jordan-type law on a prime ring becomes an ordinary derivation law once it is passed through the functional identity Theorem 9.9 [2016-13, p. 221]. The d-free method of Section 9.4 is what lets this happen on rings where no bare commuting-map argument reaches [2007-3, p. x].

9.6 The σ-commuting form

The centralizing maps of Section 9.1 have a twisted version, where the map is asked to commute with an endomorphism instead of with the identity. Lee carried the characterization of Section 9.1 over to this setting and recorded it for semiprime rings [2001-2, p. 2946]. Throughout, R is a semiprime ring and σ is an endomorphism of R [2001-2, p. 2946]. As in Section 9.1, $Z(R)$ is the center, C the extended centroid, U the right Utumi quotient ring and Q the two-sided Utumi quotient ring of R [2001-2, p. 2946].

A map f of a subring S of R into U is *σ-commuting* [2001-2, p. 2946] when $[f(x), \sigma(x)] = 0$ for all x in S. It is *additive modulo C* [2001-2, p. 2946] when $f(x+y) - f(x) - f(y)$ lies in C for all x, y in S. A right ideal of R with zero left annihilator is called *large* [2001-2, p. 2946]. The main result generalizes the commuting-form Theorem 9.2 to the twisted case.

Theorem 9.11 ([2001-2, Theorem 2, p. 2948])**.** *Let R be a semiprime ring, I a subring of R and σ an endomorphism of the ring R such that $\sigma(I)$ contains a large ideal of R. Then every σ-commuting map $f\colon I \to U$, which is additive modulo C, must be of the form $f(x) = \lambda\sigma(x) + \zeta(x)$ for all $x \in I$, where $\lambda \in C$ and $\zeta\colon I \to C$.*

Under the weaker hypothesis that $\sigma(I)$ contains only a large right ideal, the same form holds with values in the two-sided quotient Q [2001-2, Theorem 1, p. 2947]. When σ is an epimorphism the hypothesis is automatic on the whole ring, so every σ-commuting map of R that is additive modulo C has the form $f(x) = \lambda\sigma(x) + \zeta(x)$ [2001-2, Corollary 1, p. 2948]. The proof of Theorem 9.11 rests on a biderivation identity that is itself a functional identity.

Lemma 9.12 ([2001-2, Lemma 1, p. 2947])**.** *Let R be a semiprime ring with an endomorphism σ and let $D\colon S \times S \to U$ be a (σ, σ)-biderivation,*

where S is a subring of R. Then $[\sigma(x),\sigma(y)]D(u,v) = D(x,y)[\sigma(u),\sigma(v)]$ for all x,y,u,v in S.

The σ-commuting form forces a Posner-type conclusion for twisted derivations [2001-2, Theorem 3, p. 2948]. Let σ and τ be endomorphisms of R; a (σ,τ)-derivation is an additive map d with $d(xy) = d(x)\sigma(y)+\tau(x)d(y)$ [2001-2, p. 2948].

Theorem 9.13 ([2001-2, Theorem 3, p. 2948])**.** *Let R be a semiprime ring with d a (σ,τ)-derivation of R. Suppose that I is a subring (right ideal) of R such that both $\sigma(I)$ and $\tau(I)$ contain some large right ideals of R. Suppose that $[d(x),\sigma(x)] = 0$ for all $x \in I$. Then $d(I)$, and likewise $d(R)$, is contained in a central ideal of R.*

For a prime ring the conclusion of Theorem 9.13 becomes commutativity, since a nonzero (σ,τ)-derivation with $[d(x),\sigma(x)] = 0$ on a right ideal forces R to be commutative [2001-2, Corollary 4, p. 2950]. This is the semiprime and prime form of the theorem of Posner for twisted derivations, and it recovers the classical statement of 1957 on specializing $\sigma = \tau = \mathrm{id}$ [2001-2, p. 2946]. On a noncentral Lie ideal the σ-centralizing form still holds, with two low-rank exceptions, as Theorem 9.14 states.

Theorem 9.14 ([2001-2, Theorem 4, p. 2950])**.** *Let R be a prime ring, L a noncentral Lie ideal of R and σ an epimorphism of R with $\sigma(L)$ not contained in $Z(R)$. Then every σ-centralizing map $f\colon L \to U$, which is additive modulo C, must be of the form $f(x) = \lambda\sigma(x) + \zeta(x)$ for all $x \in L$, where $\lambda \in C$ and $\zeta\colon L \to C$ unless $\dim_C RC = 4$ or 9.*

Lee quotes Theorem 9.14 as an immediate consequence of a general functional-identities result of Beidar and Chebotar [2001-2, p. 2950]. This last statement is the point where the twisted commuting theory meets the d-free method of Section 9.4 [2001-2, p. 2950].

Part IV

Concrete classes of associative algebras

Chapter 10

Matrix, triangular and generalized matrix algebras

Chapter 9 settled the commuting-map and functional-identity questions on prime and semiprime rings by means of d-free sets [2007-3, p. 21]. A triangular algebra is a standard example of a ring that is not d-free, so that method does not reach it [2014-7, Section 1]. This chapter takes up the same questions for algebras built from matrices, where the answers come from explicit computations on the matrix blocks.

The literature of the chapter opens with the 1969 paper of Suh on the Zorn vector matrix [1969-3, p. 255]. Suh replaces the three-dimensional vector algebra inside the Zorn matrix by a finite-dimensional Lie algebra carrying a symmetric associative bilinear form [1969-3, p. 255]. He then determines every alternative algebra obtained in this way [1969-3, p. 255]. If the form is nondegenerate the algebra is the split Cayley algebra or a quaternion algebra [1969-3, Theorem 1, p. 257]. The paper shows how a matrix shape turns a structure problem about the whole algebra into a problem about its entries.

The derivation question for matrix-shaped algebras has its own line of papers. For upper triangular matrices over a ring with identity and no further restriction, Coelho and Milies describe every derivation [1993-6, p. 263]. For the subring of $M_n(R)$ cut out by a reflexive and transitive relation on the index set, Nowicki describes every derivation and decides when all of them are inner [1983-6, Theorem 3.7, p. 287]. For triangular algebras in the abstract sense, Cheung introduces the algebra and finds all of its commuting maps [2001-1, p. 119], and later determines all of its Lie derivations [2003-1, Theorem 11, p. 307]. The two questions have the

same shape: a map is described block by block, and the part that is not a derivation lands in the centre.

Section 10.1 reports full matrix algebras and the special subrings of $M_n(R)$, from the Zorn matrix of Suh through Nowicki relation-indexed subrings to the Jordan derivations of Alizadeh [2009-3, Theorem 3.1, p. 575]. Section 10.2 treats triangular algebras, where the derivation, commuting-map and Lie derivation questions are all settled [2003-1, Theorem 11, p. 307]. Section 10.3 passes to generalized matrix algebras, which add a second bimodule and two bilinear maps, and reports the Jordan-derivation theorem of Li, van Wyk and Wei [2013-10, Theorem 3.11, p. 13]. Section 10.4 closes with the algebraic side of nest and reflexive algebras, where a triangular algebra appears again as the compression of a nest algebra at a projection [2003-1, Proposition 19, p. 308].

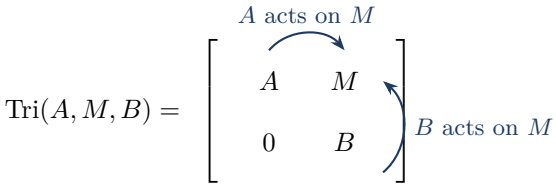


Figure 10.1: A triangular algebra: two algebras joined by a bimodule. Derivations split into parts on A, on B and on M.

10.1 Full matrix algebras

The chapter opens where the literature of the chapter opens, with an algebra that is built from matrices and whose structure was determined by elementary entry-by-entry work [1969-3, p. 255]. Suh takes a finite-dimensional Lie algebra L over a field of characteristic not 2, with an associative symmetric bilinear form (a, b) and the property $[a[bc]] = (a, c)b - (a, b)c$ [1969-3, p. 255]. The set of 2×2 vector matrices

$$\begin{pmatrix} \alpha & a \\ b & \beta \end{pmatrix}, \qquad \alpha, \beta \in \Phi,\ a, b \in L,$$

with the multiplication displayed on the first page of the paper, is a flexible algebra over Φ [1969-3, p. 255]. Suh asks which alternative algebras arise this way and answers completely [1969-3, p. 255].

Theorem 10.1 ([1969-3, Theorem 1, p. 257]). *Let L be a finite dimensional Lie algebra over a field Φ of characteristic not 2, equipped with an associative symmetric bilinear form (a, b) and having the property $[a[bc]] = (a, c)b - (a, b)c$. If (a, b) is nondegenerate, then the algebra $\mathfrak{A}$ built from L is the split Cayley algebra or a quaternion algebra.*

Suh adds that a nondegenerate form forces L to be simple of dimension three or one [1969-3, Corollary, p. 257]. For a form that is degenerate the algebra splits as the direct sum of its radical and a subalgebra that is quaternion or two-dimensional separable [1969-3, p. 255]. The lesson carries over to the derivation question: a matrix shape turns one problem about a large algebra into several problems about its entries [1969-3, p. 255].

The same entry-by-entry method settles derivations of matrix rings [1993-6, p. 263]. Take a ring R with identity and the ring $T_n(R)$ of upper triangular $n \times n$ matrices over R [1993-6, p. 263]. Coelho and Milies describe every derivation of $T_n(R)$ under that one hypothesis [1993-6, p. 263].

Theorem 10.2 ([1993-6, p. 263]). *Let R be a ring with unity and let $d\colon T_n(R) \to T_n(R)$ be a derivation. Then there exists a derivation $\delta\colon R \to R$ and a matrix $A \in T_n(R)$ such that $d = \overline{\delta} + d_A$.*

Here $\overline{\delta}$ sends (x_{ij}) to $(\delta(x_{ij}))$, and d_A is the inner derivation $X \mapsto AX - XA$ [1993-6, p. 263]. The authors compute A from the values of d on the matrix units and then recover δ from d, so the description is effective [1993-6, p. 264].

Proposition 10.3 ([1993-6, p. 266]). *Let $d\colon T_n(R) \to T_n(R)$ be a derivation, and let A, δ be as in Theorem 10.2. If $A' \in T_n(R)$ and $\delta'\colon R \to R$ are such that $d = d_{A'} + \overline{\delta'}$, then there exists an element $a \in R$ such that $\delta' = \delta - \delta_a$ and $A' = A + aI$.*

Jøndrup re-proves the same description for $M_n(A)$ and $T_n(A)$ with a short induction, and adds the corresponding automorphism theorems [1995-4, p. 205].

Theorem 10.4 ([1995-4, Theorem 2, p. 209]). *Let A be a C-algebra. Every C-derivation of $\mathrm{Mat}_n(A)$ or $T_n(A)$ is the sum of an inner derivation and a derivation induced from a C-derivation of A.*

The proof reduces to the case where the derivation vanishes on e_{11} and on every element of $f \,\mathrm{Mat}_n(A) f$, and shows that such a derivation is

inner [1995-4, p. 209]. The reduction uses only the idempotents e_{ii}, so no hypothesis on A beyond the C-algebra structure is needed.

Nowicki asks a finer question [1983-6, p. 281]. Fix a reflexive and transitive relation ρ on $\{1, \ldots, n\}$ and let P be the set of matrices in $M_n(R)$ whose (i, j) entry vanishes whenever $(i, j) \notin \rho$ [1983-6, p. 281]. This P is a subring of $M_n(R)$, called a special subring [1983-6, p. 281]. An R-derivation is a derivation of P that vanishes on the scalar matrices [1983-6, p. 284]. Nowicki describes all of them [1983-6, p. 287].

Theorem 10.5 ([1983-6, Theorem 3.7, p. 287])**.** *Let P be a special subring of $M_n(R)$ with the relation ρ. Every derivation d of P has a unique representation $d = [A, -] + \Delta_f + \Theta_\delta$, where A is a matrix in P with $A_{pp} = 0$ for $p = 1, \ldots, n$, $\delta = \{\delta_t : t \in T_\rho\}$ is a set of derivations of R, and $f : \rho \to R$ is a quasi-trivial transitive mapping.*

The three pieces have different jobs [1983-6, p. 287]. The commutator $[A, -]$ is inner [1983-6, p. 287]. The term Δ_f moves each entry by the value of the transitive mapping f [1983-6, p. 287]. The term Θ_δ applies a derivation of R to the entries, one derivation for each equivalence class of ρ [1983-6, p. 287]. The question of when every derivation is inner then becomes a question about ρ [1983-6, p. 289].

Corollary 10.6 ([1983-6, Corollary 4.4, p. 289])**.** *The following conditions are equivalent: every R-derivation of P is inner, and the relation ρ is regular over $Z(R)$.*

Regularity of ρ over an abelian group G means that every transitive mapping from ρ to G is trivial [1983-6, Definition 2.2, p. 283]. For a relation that is not regular the term Δ_f survives, and a proper derivation exists [1983-6, p. 289]. The special subrings therefore separate the inner derivations from the outer ones by a property of the index relation alone [1983-6, p. 289].

The last question for full matrix algebras is the Jordan one [2009-3, p. 574]. Jacobson and Rickart showed that a full matrix ring over a 2-torsion free unital ring has no proper Jordan derivation, through their theorems on Jordan homomorphisms [2009-3, p. 574]. Alizadeh gives a direct proof for matrices over a ring, with values in a bimodule.

Theorem 10.7 ([2009-3, Theorem 3.1, p. 575])**.** *Let A be a unital associative ring and M a 2-torsion free A-bimodule. Every Jordan derivation from $M_n(A)$ into $M_n(M)$ is a derivation.*

The proof writes D as a family of maps $D_{ijkl}(a) = [D(a \otimes E_{ij})]_{kl}$ from A to M and checks the Jordan law on each family [2009-3, p. 575]. Alizadeh uses no Jordan homomorphism, which is why his proof extends the classical result to values in a module [2009-3, p. 575].

The torsion hypothesis can be dropped for the full matrix algebra over a commutative ring [2018-9, p. 2]. Ghosh and Prakash prove the Jordan statement without it, and add the inner form of the derivations [2018-9, Theorem 3.2, p. 8].

Theorem 10.8 ([2018-9, Theorem 3.1, p. 5])**.** *Let C be a commutative ring with unity and $M_n(C)$ the full matrix algebra over C, with $n \geq 2$. Then every Jordan derivation of $M_n(C)$ into itself is a derivation.*

The same two authors then turn from the Jordan law to the derivations themselves [2018-9, p. 5].

Theorem 10.9 ([2018-9, Theorem 3.2, p. 8])**.** *Let C be a commutative ring with unity and $M_n(C)$ the algebra of all $n \times n$ matrices over C, with $n \geq 2$. Then every derivation of $M_n(C)$ is an inner derivation.*

The upper triangular matrix algebra over the same ring behaves in the same way [2018-9, Theorem 2.1 and Theorem 2.2, pp. 3 and 5]. The proof of the Jordan statement writes the image of a matrix unit in the matrix units and compares coefficients [2018-9, pp. 5–8].

A ladder matrix keeps the block shape of an upper triangular matrix and opens some of the blocks on the diagonal [2015-9, p. 1]. Fix a field of characteristic not 2 and a ladder L whose blocks dominate the diagonal [2015-9, Definition 2.2, p. 2]. The matrices that live on L form a Lie algebra under the commutator [2015-9, p. 1]. Ghimire and Huang describe its derivations as a direct sum of two ideals [2015-9, Theorem 3.1, p. 7]. The first ideal consists of the adjoint action of the block upper triangular matrices [2015-9, p. 7]. The second consists of the linear maps from $M_L/[M_L, M_L]$ to the centre [2015-9, p. 7]. In the simplest case every derivation is an adjoint action plus a sum of traces of the diagonal blocks [2015-9, p. 7]. When the characteristic is not 2 or 3 and the ladder is strongly dominant, every derivation of the commutator algebra extends to the whole algebra [2015-9, Theorem 5.3 and Corollary 5.4, p. 18]. So the entry-by-entry method of this section reaches a family of Lie algebras as well [2015-9, p. 1].

The strictly upper triangular matrices form a nilpotent Lie algebra, and its derivations have a complete description with no hypothesis on the ring [2007-10, p. 378]. Let R be a commutative ring with identity and let

$N(n, R)$ be the Lie algebra of strictly upper triangular $n \times n$ matrices over R [2007-10, p. 378]. Ou, Wang and Yao build five families of derivations: the inner ones, the diagonal ones, the central ones, and two families of extremal ones [2007-10, pp. 379–380]. The diagonal family uses a diagonal matrix, and the central family uses a vector of $n-3$ ring elements [2007-10, pp. 379–380].

Theorem 10.10 ([2007-10, Theorem 3.2, pp. 380–381])**.** *Let R be a commutative ring with identity and let $n > 2$.*

1. *If $n \geq 4$, every derivation φ of $N(n, R)$ is uniquely*

$$\varphi = \operatorname{ad} x + \eta_y + \mu_c + \rho_{r,s}^{(12)} + \rho_{p,q}^{(n-1,n)},$$

   where $x \in N(n, R)$, $y \in D(n, R)$, $c \in R^{n-3}$ and $r, s, p, q \in R$ satisfy $2r = 2p = 0$.

2. *If $n = 3$, every derivation φ of $N(n, R)$ is uniquely $\varphi = \operatorname{ad} x + \eta_y + \theta_r^{(12)} + \theta_s^{(23)}$ with $x \in N(n, R)$, $y \in D(n, R)$ and $r, s \in R$.*

Here η_y is the adjoint action of a diagonal matrix [2007-10, p. 379]. The maps ρ and θ are the extremal derivations, and they exist only in the dimensions shown [2007-10, pp. 379–380].

A left derivation changes the Leibniz rule in a different direction. An additive map μ from a ring R into a left R-module V is a module left derivation when $\mu(xy) = x\mu(y) + y\mu(x)$ [2010-10, p. 753]. A module Jordan left derivation satisfies the weaker law $\mu(x^2) = 2x\mu(x)$ [2010-10, p. 754]. Xu and Zhang settle both questions for full matrix rings [2010-10, pp. 755–757].

Proposition 10.11 ([2010-10, Proposition 2.1, p. 755])**.** *For $n \geq 2$, a module left derivation $\mu\colon M_n(R) \to M_n(V)$ must be zero.*

The Jordan law gives a nonzero answer on the same ring, and the trace controls it.

Theorem 10.12 ([2010-10, Theorem 2.6, pp. 756–757])**.** *For $n \geq 2$, a map $\mu\colon M_n(R) \to M_n(V)$ is a module Jordan left derivation if and only if there are strong module Jordan left derivations $\mu_{ij}\colon R \to V$ with $\mu(A) = (\mu_{ij}(\operatorname{tr} A))$ for every $A \in M_n(R)$. In particular, the only such map is zero when V is 2-torsion free.*

A module Jordan left derivation is strong when $\mu(x^2) = 2\mu(x) = 0$ for every x [2010-10, p. 754]. So the trace gives the argument of each entry map, and the entries only distribute the values [2010-10, pp. 756–757].

The question also has a nonlinear form. A 2-local derivation matches its value at any two points by one derivation, and it need not be additive [2015-3, p. 1]. Ayupov and Arzikulov take inner witnesses on a matrix ring over a commutative ring [2015-3, Theorem 2.1, p. 2].

Theorem 10.13 ([2015-3, Theorem 2.1, p. 2])**.** *Let $\Re$ be a commutative associative ring with identity and let $M_n(\Re)$ be the matrix ring of $n \times n$ matrices over $\Re$, with $n > 1$. Then every inner 2-local derivation on $M_n(\Re)$ is an inner derivation.*

The paper also shows that every derivation of an associative ring extends to a derivation of its matrix ring [2015-3, p. 1].

Full matrix algebras are the case where the two algebras of a triangular algebra coincide in size and the bimodule fills the whole off-diagonal block [2001-1, p. 119]. Section 10.2 keeps the block shape and sets part of it to zero.

Ghimire and Huang compute the derivations of the Lie algebra of dominant upper triangular ladder matrices. That family sits between the strictly upper triangular and the full matrix algebra [2015-9, Abstract, p. 1].

10.2 Triangular algebras

Section 10.1 treated full matrix algebras, where the two algebras A and B of the block picture have the same size and the bimodule M fills the whole off-diagonal block [2001-1, p. 119]. A triangular algebra keeps the block shape and sets one off-diagonal block to zero [2001-1, p. 119]. That is the smallest change which removes the symmetry that the matrix-unit arguments used [2001-1, p. 119].

Definition 10.14 ([2001-1, Definition 2, p. 119])**.** Let A and B be unital algebras over a commutative ring R, and let M be an (A, B)-bimodule that is faithful as a left A-module and as a right B-module. The R-algebra

$$\operatorname{Tri}(A, M, B) = \left\{ \begin{pmatrix} a & m \\ 0 & b \end{pmatrix} : a \in A,\ m \in M,\ b \in B \right\}$$

under the usual matrix addition and formal matrix multiplication is called a triangular algebra. We write $a \oplus b$ for the element with $m = 0$.

Cheung notes that the two natural projections π_A and π_B, which read off a and b, are the only maps needed to state the results [2001-1, Definition 3, p. 119]. Cheung gives the centre a short description, and that description carries the whole theory [2001-1, p. 119].

Proposition 10.15 ([2001-1, Proposition 3, p. 119])**.** *The centre of* $\Omega = \mathrm{Tri}(A, M, B)$ *is*

$$Z(\Omega) = \{a \oplus b : a \in A,\ b \in B,\ am = mb \text{ for every } m \in M\}.$$

Moreover $\pi_A(Z(\Omega)) \subseteq Z(A)$ *and* $\pi_B(Z(\Omega)) \subseteq Z(B)$*, and there is a unique algebra isomorphism* τ *from* $\pi_A(Z(\Omega))$ *to* $\pi_B(Z(\Omega))$ *such that* $am = m\tau(a)$ *for every* $m \in M$*.*

The faithfulness of M on both sides enters here. A central element $a \oplus b$ forces $a \in Z(A)$ and $b \in Z(B)$, since M separates A from B [2001-1, p. 119].

Derivations of a triangular algebra split in an explicit way. Cheung records the split as a lemma, and the rest of the paper uses it [2003-1, Lemma 5, p. 303].

Lemma 10.16 ([2003-1, Lemma 5, p. 303])**.** *A derivation* δ *on* $\Omega = \mathrm{Tri}(A, M, B)$ *is of the form*

$$\delta\begin{pmatrix} a & m \\ 0 & b \end{pmatrix} = \begin{pmatrix} p_A(a) & an - nb + f(m) \\ 0 & p_B(b) \end{pmatrix},$$

where $n \in M$ *and*

1. p_A *is a derivation on* A *and* $f(am) = p_A(a)m + af(m)$*;*
2. p_B *is a derivation of* B *and* $f(mb) = mp_B(b) + f(m)b$*.*

The element n gives the inner part of the derivation. The map f is the part that acts on the bimodule [2003-1, p. 303]. Suppose M is faithful on the left. Then the first condition alone forces p_A to be a derivation, so the two conditions are not independent [2003-1, p. 303].

One more name is needed, for the maps that the derivation lemma does not cover [2003-1, p. 299].

Definition 10.17 ([2003-1, p. 299])**.** A proper Lie derivation is a Lie derivation of an associative algebra that can be written as the sum of a derivation and a linear map with image in the centre of the algebra. A Lie derivation that is not a proper Lie derivation is called an improper Lie derivation.

Cheung first describes the shape of an arbitrary Lie derivation [2003-1, Proposition 4, p. 302]. The description has one element $n \in M$ and six linear maps [2003-1, Proposition 4, p. 302]. A map is proper exactly when two centre-valued maps h_A and h_B pass a test [2003-1, Theorem 6, p. 303].

The main theorem asks when every Lie derivation is proper. Its conditions involve only A and B [2003-1, Theorem 11, p. 307]. For an algebra X, let $W(X)$ be the smallest subalgebra that meets conditions (i) to (vii) of Proposition 10 of that paper [2003-1, Proposition 10, p. 305].

Theorem 10.18 ([2003-1, Theorem 11, p. 307])**.** *Every Lie derivation of* $\Omega = \mathrm{Tri}(A, M, B)$ *is proper if the following two conditions hold:*

1. *either* M *is a faithful left* A*-module and* $Z(B) = \pi_B(Z(\Omega))$*; or* M *is a faithful left* A*-module and* $A = W(A)$*; or every Lie derivation of* A *is proper and* $A = W(A)$*;*

2. *either* M *is a faithful right* B*-module and* $Z(A) = \pi_A(Z(\Omega))$*; or* M *is a faithful right* B*-module and* $B = W(B)$*; or every Lie derivation of* B *is proper and* $B = W(B)$*.*

Cheung gives three simple ways to meet $A = W(A)$. One is that A is generated by commutators and idempotents. Another is that two elements of A commute to an invertible element [2003-1, p. 307]. The upper triangular matrix algebra meets the first one, which gives the next corollary.

Corollary 10.19 ([2003-1, Corollary 17, p. 308])**.** *Every Lie derivation of* $T_n(R)$*, the* $n \times n$ *upper triangular matrix algebra over* R*, has the form* $a \mapsto ax - xa + h(a)1$*, where* $x \in T_n(R)$ *is fixed and* h *is a linear functional defined on the span of* $\{E_{jj} : j = 1, \ldots, n\}$*.*

Over a semiring the upper triangular matrix algebra still has derivations [2018-17, p. 5]. Take an additively idempotent semiring, where the sum of two equal terms adds nothing [2018-17, p. 2]. Set one principal dense submatrix to zero and keep every other entry. That map is a derivation [2018-17, Theorem 1, p. 5]. The partial identity matrices produce these maps, and the theorem counts them [2018-17, p. 5].

The commuting-map question has a parallel answer on the same class of algebras [2001-1, Theorem 2, p. 124]. Cheung proves that every commuting map is proper under three conditions [2001-1, Theorem 2, p. 124]. Each centre must be the image of the centre of Ω. One element $m_0 \in M$ must separate the central elements [2001-1, Theorem 2, p. 124]. The proof first describes an arbitrary commuting map. Its data are two commuting maps,

one on A and one on B, and four maps into centres [2001-1, Proposition 4, p. 121].

The conditions are needed. A triangular algebra can carry an improper commuting map even when A and B are both commutative [2001-1, Example 1, p. 123]. That example is why the chapter keeps the commuting-map and the derivation theorems apart [2001-1, Example 1, p. 123].

The Jordan law holds on triangular algebras without the unital hypothesis. One condition on the two diagonal algebras is enough [2007-5, Theorem 2.10, p. 5]. Cheng and Jing say that A has property (P) when $xay + yax = 0$ for all $x, y \in A$ forces $a = 0$ [2007-5, p. 2].

Theorem 10.20 ([2007-5, Theorem 2.10, p. 5])**.** *Let A and B be two algebras over a 2-torsion free commutative ring R, both with property (P), and let M be a faithful (A, B)-bimodule. Then every Jordan derivation of* $\mathrm{Tri}(A, M, B)$ *into itself is a derivation.*

Neither A nor B need have an identity, and the proof does not use the two idempotents P and Q of the block picture [2007-5, p. 1]. The authors prove the three component statements separately: δ is a derivation on each of T_{11}, T_{12} and T_{22} [2007-5, Lemmas 2.7–2.9, pp. 4–5].

The same law reaches a bimodule when the map is generalized by its value at the identity [2007-9, Theorem 2.4, p. 362]. A generalized Jordan derivation Φ satisfies $\Phi(ab + ba) = \Phi(a)b + a\Phi(b) + \Phi(b)a + b\Phi(a) - a\Phi(I)b - b\Phi(I)a$ [2007-9, p. 355]. A generalized derivation Ψ satisfies $\Psi(ab) = \Psi(a)b + a\Psi(b) - a\Psi(I)b$ [2007-9, p. 355].

Theorem 10.21 ([2007-9, Theorem 2.4, p. 362])**.** *Let Φ be a generalized Jordan derivation from $T_n(R)$ into a $T_n(R)$-bimodule M. Then there are a unique generalized derivation Ψ and an antiderivation Λ from $T_n(R)$ to M, with $\Lambda(D_n(R)) = 0$, such that $\Phi = \Psi + \Lambda$.*

Here $D_n(R)$ is the subalgebra of diagonal matrices [2007-9, p. 356]. The condition on $D_n(R)$ makes the factorization unique. Without it an antiderivation can appear twice [2007-9, p. 362].

The Jordan question also has a nonlinear form. A nonlinear Jordan derivation is a map that satisfies the Jordan law with no additivity assumed [2012-15, p. 1]. Xiao shows that on a triangular algebra the nonlinear law forces additivity and the derivation rule together [2012-15, Theorem 2.1, p. 2].

Theorem 10.22 ([2012-15, Theorem 2.1, p. 2]). *Let* $T = \begin{pmatrix} A & M \\ 0 & B \end{pmatrix}$ *be a triangular algebra and let* d *be a nonlinear Jordan derivation of* T*. Then* d *is an additive derivation.*

The proof first finds the image of each corner. The image of T_{12} lies in T_{12}, and the image of T_{11} lies in $T_{11} + T_{12}$ [2012-15, Lemmas 2.2–2.3, p. 2]. The nonlinear hypothesis then gives additivity, and the derivation rule follows [2012-15, pp. 2–7]. Take a nest algebra on an infinite dimensional Hilbert space. There every nonlinear Jordan derivation is inner. It is an additive derivation, and every derivation of a nest algebra is inner [2012-15, Proposition 2.11, p. 7].

Lie derivations of a triangular algebra carry a generalization of their own. A generalized Lie derivation f comes with an associated linear map d and satisfies $f([x, y]) = [f(x), y] + [x, d(y)]$ [2011-5, p. 1535]. Benkovic reduces the question to a condition on the single element $d(1)$ [2011-5, p. 1539].

Theorem 10.23 ([2011-5, Theorem 3.6, p. 1541]). *Let* $\mathcal{A} = \mathrm{Tri}(A, M, B)$ *be a triangular algebra and let* $f \colon \mathcal{A} \to \mathcal{A}$ *be a generalized Lie derivation with associated linear map* d*. Suppose that*

1. $\pi_A(Z(\mathcal{A})) = Z(A)$ *or* $[B, B] = B$*, and*
2. $\pi_B(Z(\mathcal{A})) = Z(B)$ *or* $[A, A] = A$*.*

Then $f = \Delta + \tau$*, where* Δ *is a generalized derivation of* $\mathcal{A}$ *and* $\tau \colon \mathcal{A} \to Z(\mathcal{A})$ *is a linear map that vanishes on all commutators of* $\mathcal{A}$*.*

The two conditions let the theorem of Cheung on proper Lie derivations reach the Lie part [2011-5, p. 1541]. Take a Hilbert space nest algebra. Its centre is one-dimensional. There every generalized Lie derivation is a generalized inner derivation plus a centre-valued linear map [2011-5, Corollary 3.8, p. 1542]. The same paper settles the generalized Jordan question on a 2-torsion free triangular algebra [2011-5, Theorem 4.2, p. 1542].

The bilinear form of the question has the same shape. A biderivation of an algebra A is a bilinear map that is a derivation in each argument [2020-15, p. 2]. Its Lie form is a Lie derivation in each argument [2020-15, p. 2].

Theorem 10.24 ([2020-15, Theorem 3.2, p. 7]). *Let* $T = \begin{pmatrix} A & M \\ 0 & B \end{pmatrix}$ *be a triangular algebra over a commutative ring* R *and let* $\varphi \colon T \times T \to T$ *be a Lie biderivation. Suppose that*

1. *$\pi_A(Z(T)) = Z(A)$ and $\pi_B(Z(T)) = Z(B)$;*

2. *at least one of A and B is noncommutative;*

3. *every bimodule homomorphism $f\colon M \to M$ has the standard form;*

4. *$\alpha a = 0$ with $\alpha \in Z(A)$ and $0 \neq a \in A$ forces $\alpha = 0$.*

Then $\varphi(x, y) = \lambda_0[x, y] + [x, [y, \varphi(e, e)]] + \mu(x, y)$ for some $\lambda_0 \in Z(T)$ and some central bilinear map μ.

A bimodule homomorphism has the standard form when $f(m) = a_0 m + mb_0$ for central a_0 and b_0 [2020-15, equation (3.3), p. 7]. On a block upper triangular matrix algebra over a domain the sum reduces to one extremal biderivation plus one inner biderivation [2020-15, Corollary 3.8, p. 8].

A superalgebra version of the triangular ring has the same block shape with a sign in the product [2024-10, p. 1]. A Jordan superderivation of degree 0 satisfies the Jordan law, and one of degree 1 satisfies it with the super sign [2024-10, p. 2]. The upper triangular matrix ring here is the trivial extension of $R \times S$ by the bimodule M [2024-10, p. 3].

Theorem 10.25 ([2024-10, Theorem 2.3, pp. 4–6])**.** *Let d_0 and d_1 be Jordan superderivations of degree 0 and 1 of the upper triangular matrix ring $T = \begin{pmatrix} R & M \\ 0 & S \end{pmatrix}$, with R and S 2-torsion free. Then there are Jordan derivations δ_1 of R and δ_2 of S, an additive map g on M with $g(rm+ms) = rg(m) + \delta_1(r)m + g(m)s + m\delta_2(s)$, and an element $m^* \in M$ such that*

$$d_0 \begin{pmatrix} r & m \\ 0 & s \end{pmatrix} = \begin{pmatrix} \delta_1(r) & g(m) \\ 0 & \delta_2(s) \end{pmatrix}, \qquad d_1 \begin{pmatrix} r & m \\ 0 & s \end{pmatrix} = \begin{pmatrix} 0 & rm^* - m^*s \\ 0 & 0 \end{pmatrix}.$$

So the even part is a triangular Jordan derivation. The odd part is the inner superderivation of one element [2024-10, p. 6]. The corollary is that every Jordan superderivation of the ring is a Jordan derivation [2024-10, p. 6].

Two results come from other chapters and are not repeated here. The f-derivation theorem of Benkovič concerns triangular algebras. It says that an f-derivation is a derivation when the coefficients of f have a nonzero sum that is regular [2014-7, Theorem 1.1], and it is stated as Theorem 6.65. The triangular-ring form is a unital ring with a nontrivial idempotent e such that $f = 1 - e$ and $fRe = 0$. It appears in Section 7.6 [2016-16, p. 1953].

Recall that a triangular algebra has two square blocks of possibly different sizes. The generalized matrix algebra of Section 10.3 keeps both off-diagonal blocks, and the two bilinear maps that multiply them are what remains of the matrix product.

The block upper triangular case has the same shape. Ghahramani, Ghosseiri and Heidarizadeh prove that a linear map on $T(n_1, \ldots, n_k)$ satisfying the Jordan zero-product condition has the form $D + \alpha + \cdot\Delta(I)$, with D a derivation [2020-13, Abstract].

10.3 Generalized matrix algebras

Section 10.2 set one off-diagonal block to zero and kept a single bimodule [2001-1, p. 119]. The next step puts a bimodule in each off-diagonal position, and then the matrix product needs two extra maps which multiply the two bimodules into the diagonal algebras [2012-9, p. 2720]. The result is a generalized matrix algebra, and it contains both earlier shapes as special cases [2012-9, p. 2720].

Definition 10.26 ([2012-9, p. 2720])**.** Let A and B be unital algebras over a commutative ring R, let ${}_AM_B$ and ${}_BN_A$ be bimodules, and let $\Phi_{MN}\colon M \otimes_B N \to A$ and $\Psi_{NM}\colon N \otimes_A M \to B$ be bimodule maps that satisfy the two compatibility diagrams of a Morita context [2012-9, pp. 2719–2720]. The set

$$G = \begin{pmatrix} A & M \\ N & B \end{pmatrix} = \left\{ \begin{pmatrix} a & m \\ n & b \end{pmatrix} : a \in A,\ m \in M,\ n \in N,\ b \in B \right\}$$

with matrix-like addition and multiplication is called a generalized matrix algebra [2012-9, p. 2720]. We write $a \oplus b$ for the element with $m = n = 0$ [2012-9, p. 2720].

Du and Wang assume that M is faithful as an (A, B)-bimodule, and they record that G is a triangular algebra when $N = 0$ [2012-9, p. 2720]. The centre has the same shape as before, with one extra condition coming from N [2010-9, p. 2185].

Proposition 10.27 ([2010-9, Lemma 3.1, p. 2184])**.** *The centre of G is*

$$Z(G) = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} : am = mb,\ na = bn \text{ for every } m \in M,\ n \in N \right\}.$$

The two projections π_A and π_B read off a and b, and $\pi_A(Z(G))$ and $\pi_B(Z(G))$ are subalgebras of $Z(A)$ and $Z(B)$ [2010-9, p. 2185]. Xiao and Wei show that a unique algebra isomorphism carries the first onto the second [2010-9, Lemma 3.2, p. 2185]. When M is faithful on both sides, the conditions $a \in Z(A)$ and $b \in Z(B)$ follow from the two commuting conditions, so they drop out of the formula [2010-9, p. 2185].

The Lie question has a local form on G, where the Lie condition is imposed only on pairs whose product is zero [2012-9, Theorem 1, pp. 2720–2721]. The conclusion has the usual shape: a derivation plus a map into the centre that vanishes on the commutators of such pairs [2012-9, p. 2721].

Theorem 10.28 ([2012-9, Theorem 1, pp. 2720–2721])**.** *Let G be a generalized matrix algebra. Suppose that $Z(A) = \pi_A(Z(G))$ and $Z(B) = \pi_B(Z(G))$, and that either A or B contains no nonzero central ideal. If an R-linear map δ of G satisfies $\delta([x,y]) = [\delta(x), y] + [x, \delta(y)]$ for all $x, y \in G$ with $xy = 0$, then there is a derivation d of G and an R-linear map $\tau\colon G \to Z(G)$ that vanishes at those commutators, such that $\delta(x) = d(x) + \tau(x)$ for every $x \in G$.*

Du and Wang replace the central-ideal condition by a hypothesis on the bimodule [2012-9, p. 2725]. In the proof of Theorem 2 they show that a noncommutative A with such a bimodule has no nonzero central ideal, so the previous theorem applies [2012-9, p. 2725]. They call M loyal when $aMb = 0$ forces $a = 0$ or $b = 0$ [2012-9, p. 2725].

Theorem 10.29 ([2012-9, Theorem 2, p. 2725])**.** *Let G be a generalized matrix algebra. Suppose that $Z(A) = \pi_A(Z(G))$ and $Z(B) = \pi_B(Z(G))$, that M is loyal as an (A, B)-bimodule, and that at least one of A and B is noncommutative. Then every R-linear map δ of G with $\delta([x,y]) = [\delta(x), y] + [x, \delta(y)]$ on all pairs x, y with $xy = 0$ has the form $\delta = d + \tau$ of Theorem 10.28.*

Full matrix algebras give the first corollary, and they need one lemma: $M_n(A)$ has no nonzero central ideal [2012-9, Lemma 1, p. 2726].

Corollary 10.30 ([2012-9, Corollary 1, p. 2726])**.** *Let $M_n(A)$ be a full matrix algebra with $n \geq 3$. If an R-linear map $\delta\colon M_n(A) \to M_n(A)$ satisfies $\delta([x,y]) = [\delta(x), y] + [x, \delta(y)]$ for all x, y with $xy = 0$, then $\delta = d + \tau$, where d is a derivation and τ is a map into $Z(A)I_n$ that vanishes at those commutators.*

The triple bracket asks for one more step, and Ashraf and Akhtar answer it on the same algebra [2020-6, p. 3651]. A Lie triple derivation of G is an R-linear map L with $L([[x,y],z]) = [[L(x),y],z]+[[x,L(y)],z]+[[x,y],L(z)]$ for all $x,y,z \in G$ [2020-6, p. 3651]. Such a map is proper when it is a derivation plus a map into the centre that vanishes on second commutators [2020-6, p. 3651]. Their Theorem 2.1 describes every Lie triple derivation of G through eight component maps and their side conditions [2020-6, Theorem 2.1, pp. 3653–3654]. The question of this section is whether such a map must be proper, and their Theorem 2.2 answers it in five conditions [2020-6, Theorem 2.2, p. 3657].

Theorem 10.31 ([2020-6, Theorem 2.2, p. 3657])**.** *Let G be a generalized matrix algebra and L a Lie triple derivation of G of the form given in Theorem 2.1 of the same paper. Then L is proper if and only if $B_{11}(B) \subseteq \pi_A(Z(G))$, $A_{22}(A) \subseteq \pi_B(Z(G))$, $B_{11}(nm) \oplus A_{22}(mn) \in Z(G)$, $D_{12} = 0$ and $C_{21} = 0$.*

The hypothesis set shrinks in the triangular case, and the authors record that case as their Corollary 2.2, which is Theorem 4.3 of Li and Shen [2020-6, Corollary 2.2, p. 3659]. They also exhibit a generalized matrix algebra that carries an improper Lie triple derivation which is not a Lie derivation [2020-6, Example 2.1, pp. 3658–3659].

The commuting-map question runs on the same algebra and in the same order [2010-9, p. 2178]. Xiao and Wei first describe an arbitrary commuting map of G [2010-9, Proposition 3.3, p. 2185]. The description has a fixed element of G and eight linear maps, four into the centre of A and four into the centre of B [2010-9, p. 2185]. They then decide when such a map is proper [2010-9, Proposition 3.5, p. 2189]. A commuting map of G is proper when it has the form $X \mapsto XC + \xi(X)$ with $C \in Z(G)$ and ξ mapping G into $Z(G)$ [2010-9, p. 2184].

Theorem 10.32 ([2010-9, Theorem 3.6, p. 2192])**.** *Let Θ be a commuting map of G. Then Θ is proper if the following three conditions hold:*

1. *$Z(B) = \pi_B(Z(G))$, or $A = [A,A]$;*
2. *$Z(A) = \pi_A(Z(G))$, or $B = [B,B]$;*
3. *there are $m_0 \in M$ and $n_0 \in N$ such that $Z(G)$ consists of the diagonal elements $a \oplus b$ with $am_0 = m_0 b$ and $n_0 a = b n_0$.*

The third condition asks one pair of elements to separate the central elements [2010-9, p. 2192]. It is the same kind of hypothesis as the one element m_0 in the triangular case of Section 10.2 [2001-1, Theorem 2, p. 124]. Xiao and Wei check the three conditions on the standard examples: full matrix algebras, upper and lower triangular matrix algebras, triangular Banach algebras and nest algebras [2010-9, Section 2, pp. 2182–2184].

Jordan derivations of generalized matrix algebras split into two classes, and the pairings decide which class applies [2013-10, p. 13]. Li, van Wyk and Wei call a pairing nondegenerate when $\Phi_{MN}(m, N) \neq 0$ for every nonzero m and $\Phi_{MN}(M, n) \neq 0$ for every nonzero n, and the same holds for Ψ_{NM} [2013-10, Definition 3.7, p. 12]. For the first class they prove a vanishing theorem [2013-10, Proposition 3.10, p. 13]. Here an antiderivation is a map Θ with $\Theta(xy) = \Theta(y)x + y\Theta(x)$ for all x, y [2013-10, p. 2].

Proposition 10.33 ([2013-10, Proposition 3.10, p. 13])**.** *Let G be a generalized matrix algebra over the commutative ring R, and let $\Theta\colon G \to G$ be R-linear. If one of the pairings $\Phi_{MN}\colon M \otimes N \to A$ and $\Psi_{NM}\colon N \otimes M \to B$ is nondegenerate, then Θ is an antiderivation if and only if $\Theta = 0$.*

The second class has both pairings zero [2013-10, p. 13]. There every Jordan derivation separates into a derivation and an antiderivation [2013-10, p. 13].

Theorem 10.34 ([2013-10, Theorem 3.11, p. 13])**.** *Let G be a 2-torsion free generalized matrix algebra over a commutative ring R. If both pairings Φ_{MN} and Ψ_{NM} are zero, then every Jordan derivation of G is the sum of a derivation and an antiderivation.*

Upper triangular matrices are the model case: $T_n(R)$ is a generalized matrix algebra with zero pairings, and on it the antiderivation part vanishes [2013-10, Example 3.13, p. 14]. So every Jordan derivation of $T_n(R)$ is a derivation [2013-10, Example 3.13, p. 14]. That is the theorem of Zhang and Yu for triangular algebras, recalled as Theorem 4.31 in Section 4.4 [2007-5, p. 1].

The Lie higher derivations of the same algebra ask for the same split one level up [2017-10, p. 8]. A Lie higher derivation is a sequence of linear maps, with the first term the identity, whose Leibniz rule holds for the commutator [2017-10, p. 1]. Such a sequence is proper when it is a higher derivation plus a sequence of centre-valued maps that vanish on commutators [2017-10, p. 1]. Moafian describes the proper sequences of a generalized matrix algebra by three conditions on two sequences of maps into the centres [2017-10,

Theorem 3.1, p. 8]. On a weakly faithful algebra the property holds under two conditions [2017-10, Theorem 4.3, p. 12]. The centres of G must project onto the centres of A and of B. One of the two bimodules must be strongly faithful [2017-10, Theorem 4.3, p. 12].

The Lie triple law carries the same shape, and it also survives without linearity [2019-10, Theorem 2.1, p. 3]. A multiplicative generalized Lie triple derivation is a map, not assumed additive, with $G_L([[a,b],c]) = [[G_L(a),b],c] + [[a,L(b)],c] + [[a,b],L(c)]$ for an associated multiplicative Lie triple derivation L [2019-10, p. 1].

Theorem 10.35 ([2019-10, Theorem 2.1, p. 3])**.** *Let G be a 2-torsion free generalized matrix algebra such that*

1. *$Z(A) = \pi_A(Z(G))$ and $Z(B) = \pi_B(Z(G))$;*
2. *A or B contains no nonzero central ideal;*
3. *if $N \neq 0$, then $mN = 0 = Nm$ for $m \in M$ forces $m = 0$;*
4. *if $M \neq 0$, then $nM = 0 = Mn$ for $n \in N$ forces $n = 0$.*

If $G_L \colon G \to G$ is a multiplicative generalized Lie triple derivation, then there is an additive generalized derivation δ of G and a map $\phi \colon G \to Z(G)$ vanishing at $[[x_1,x_2],x_3]$ such that $G_L = \delta + \phi$.

The same conclusion holds for the multiplicative generalized Lie derivations, where ϕ vanishes at the commutators [2019-10, Corollary 3.1, p. 13]. It also holds on a full matrix algebra of size at least three [2019-10, Corollary 3.2, p. 13].

The local form of the Lie question asks only for a witness at each point. A linear map δ of G is a local Lie derivation when for every X there is a Lie derivation δ_X with $\delta(X) = \delta_X(X)$ [2020-17, p. 2]. Liu shows that the local law forces the global one under three conditions [2020-17, Theorem 2.1, p. 3].

Theorem 10.36 ([2020-17, Theorem 2.1, p. 3])**.** *Let G be a generalized matrix algebra. Suppose that*

1. *$A = J(A)$ and $B = J(B)$, where J is the subalgebra generated by the idempotents;*
2. *$Z(A) = \pi_A(Z(G))$ and $Z(B) = \pi_B(Z(G))$;*
3. *either A or B contains no nonzero central ideal.*

If δ is a local Lie derivation of G, then there are a derivation d of G and a linear map $h\colon G \to Z(G)$ vanishing on the commutators with $\delta(X) = d(X) + h(X)$ for every $X \in G$.

So every local Lie derivation of such an algebra is a Lie derivation [2020-17, p. 3]. The full matrix case follows from the same theorem [2020-17, Theorem 3.1, p. 18].

The multilinear form of the question has an answer in the same shape. An n-Lie derivation is an n-linear map that is a Lie derivation in each argument [2026-21, p. 1]. Liang, Wang and Wei separate every such map into an extremal piece and a central-valued piece [2026-21, Theorem 4.1, p. 25].

Theorem 10.37 ([2026-21, Theorem 4.1, p. 25])**.** *Let $G = \begin{pmatrix} A & M \\ N & B \end{pmatrix}$ be a generalized matrix algebra over a commutative ring R and let $\varphi\colon G \times \cdots \times G \to G$ be an n-Lie derivation with $n \geq 3$. Suppose that*

1. *$\pi_A(Z(G)) = Z(A)$ and $\pi_B(Z(G)) = Z(B)$;*
2. *either A or B contains no nonzero central ideal;*
3. *$\alpha a = 0$ with $\alpha \in Z(G)$ and $0 \neq a \in G$ forces $\alpha = 0$;*
4. *$MN = 0 = NM$ forces at least one of A and B to be noncommutative;*
5. *every special pair of bimodule homomorphisms has the standard form.*

Then $\varphi = \kappa + \psi$, where $\kappa(x_1, \ldots, x_n) = [x_1, [x_2, [\cdots, [x_n, X_0] \cdots]]]$ for all $x_i \in G$ and ψ is an n-linear central-valued map that vanishes on commutators in each component.

The element X_0 is fixed by the values of φ at the unit, $X_0 = e\varphi(e, \ldots, e)f + (-1)^n f\varphi(e, \ldots, e)e$ [2026-21, p. 2]. A second theorem reaches the same conclusion from a condition on the bimodule instead of the commutativity of A or B [2026-21, Theorem 4.3, p. 28].

The three-block form asks one more question, and the multiplicative law has an answer there too [2020-11, Theorem 2.2, p. 5]. A triangular 3-matrix ring $T = T_3(R_i; M_{ij})$ has three diagonal rings and three bimodules, one for each pair [2020-11, p. 2]. The hypothesis is that the three diagonal idempotents cut the centre correctly, $Q_i Z(T) Q_i = Z(T_i)$ for $i = 1, 2, 3$ [2020-11, Theorem 2.2, p. 5].

Theorem 10.38 ([2020-11, Theorem 2.2, p. 5]). *Let $T = T_3(R_i; M_{ij})$ be a triangular 3-matrix ring with $Q_i Z(T) Q_i = Z(T_i)$ for $i = 1, 2, 3$. Then every multiplicative Lie derivation $\phi\colon T \to T$ has the form $\phi = \delta + \gamma$, where δ is a derivation of T and $\gamma\colon T \to Z(T)$ is a central map with $\gamma([T, T]) = 0$.*

The proof separates the three diagonal blocks and the three bimodules, as in the two-block case [2020-11, pp. 5–6]. The theorem therefore carries the standard form from triangular algebras to triangular 3-matrix rings [2020-11, p. 5].

A nest algebra is the last and largest member of the family, because its two diagonal algebras are again nest algebras [2003-1, Proposition 19, p. 308]. The next section states the results which this description makes available [2003-1, p. 308].

A triangular ring has its own zero-product characterization. Qi and Hou show that an additive map on $\mathrm{Tri}(A, M, B)$ is centralized at zero products exactly when it is a centralizer, and characterize the generalized derivations the same way [2013-12, Abstract].

Further work in the matrix and triangular line, held in the collection: *On the finite dual of a cocommutative Hopf algebroid Application to linear differential matrix equations and Picard-Vessiot theory* [2021-4].

10.4 Nest and reflexive algebras, the algebraic side

Section 10.3 kept two diagonal algebras and put a bimodule in each off-diagonal block [2012-9, p. 2720]. A nest algebra has exactly that shape, with one diagonal algebra at the top and one at the bottom, so the theorems of the chapter apply to it [2003-1, Proposition 19, p. 308].

A nest $\mathcal{N}$ is a totally ordered set of closed subspaces of a Hilbert space H [2003-1, p. 308]. It contains $\{0\}$ and H, and it is closed under intersections and closed linear spans [2003-1, p. 308]. The nest algebra of $\mathcal{N}$ consists of the bounded operators T with $TN \subseteq N$ for every $N \in \mathcal{N}$ [2003-1, p. 308]. A nest algebra is a reflexive algebra with a commutative subspace lattice [1977-1, p. 156]. For an orthonormal basis the nest of finite initial spans gives the operators whose matrix is upper triangular [2003-1, p. 308].

Proposition 10.39 ([2003-1, Proposition 19, p. 308]). *Let $N \in \mathcal{N} \setminus \{0, H\}$ and let E be the projection of H onto N. Then $E\mathcal{N}E$ and $(1-E)\mathcal{N}(1-E)$ are nests, $\mathcal{T}(E\mathcal{N}E) = E\mathcal{T}(\mathcal{N})E$ and $\mathcal{T}((1-E)\mathcal{N}(1-E)) = (1-E)\mathcal{T}(\mathcal{N})(1-$*

E). Furthermore

$$\mathcal{T}(\mathcal{N}) = \begin{pmatrix} \mathcal{T}(E\mathcal{N}E) & E\mathcal{T}(\mathcal{N})(1-E) \\ 0 & \mathcal{T}((1-E)\mathcal{N}(1-E)) \end{pmatrix}.$$

Both diagonal algebras are again nest algebras, so the same description applies to them [2003-1, p. 308]. The centre is one-dimensional, and this is what makes the Lie and commuting theorems of the chapter available [2003-1, Proposition 20, p. 309].

Proposition 10.40 ([2003-1, Proposition 20, p. 309])**.** *The centre of a nest algebra is $\mathbb{C}I$.*

Every derivation of a nest algebra is inner [2003-1, Proposition 21, p. 309]. Christensen proved this earlier, together with a quantitative statement [1977-1, Theorem 3.9, p. 160]. He introduces the automorphism implementation property: an algebra has AIP(r, s) when every automorphism within distance r of the identity is implemented by an element x with $\|I - x\| \leq s\|\alpha - \mathrm{id}\|$ [1977-1, Definition 3.1, p. 158]. The next theorem turns that property into a vanishing statement for cohomology [1977-1, Theorem 3.2, p. 158].

Theorem 10.41 ([1977-1, Theorem 3.2, p. 158])**.** *Let $\mathcal{A}$ be a weakly closed unital algebra of operators on a Hilbert space H. If $\mathcal{A}$ has AIP(r, s) for some pair of positive reals, then $H^1(\mathcal{A}, \mathcal{A}) = 0$.*

Christensen shows that every nest algebra has AIP$(1/24, 26)$ [1977-1, Corollary 3.10, p. 161]. With Theorem 10.41 this gives the vanishing theorem for every ultraweakly closed algebra of operators containing it [1977-1, Corollary 3.11, p. 161]. He also proves that two cohomology groups of a reflexive algebra with a commutative lattice agree [1977-1, Proposition 3.4, p. 158]. Every derivation of such an algebra into a dual normal module is continuous [1977-1, Corollary 2.3, p. 157].

Cheung records the derivation theorem in the shape needed for the chapter, and then reads the Lie statement off it [2003-1, Proposition 21, p. 309].

Proposition 10.42 ([2003-1, Proposition 21, p. 309])**.** *Every derivation of a nest algebra is inner, that is, of the form $a \mapsto ax - xa$ for some fixed element x of the nest algebra.*

The Lie case follows from this proposition.

Corollary 10.43 ([2003-1, Corollary 22, p. 309])**.** *Every Lie derivation of a nest algebra has the form $a \mapsto ax - xa + h(a)1$, where x is a fixed element of the nest algebra and h is a linear functional that annihilates the commutators.*

The proof uses Theorem 10.18 on the triangular form of Proposition 10.39, because both diagonal algebras have centre $\mathbb{C}I$ [2003-1, p. 309]. A nest is continuous when every member is the intersection of the members strictly above it [2003-1, p. 309]. On the nest algebra of a continuous nest every element is a sum of two commutators, so the functional h in Corollary 10.43 can be absorbed [2003-1, p. 309].

Corollary 10.44 ([2003-1, Corollary 23, p. 309])**.** *Every Lie derivation on the nest algebra of a continuous nest is an inner derivation.*

Cheung adds two corollaries for triangular algebras whose top block is a nest algebra of a continuous nest [2003-1, Corollaries 24–25, p. 309]. In both, every Lie derivation of $\mathrm{Tri}(A, M, B)$ is proper [2003-1, p. 309].

The commuting-map question has the same answer [2001-1, Corollary 7, p. 126]. Cheung gives the algebraic form of Proposition 10.39 and the one-dimensional centre first [2001-1, Propositions 5–6, p. 126], then applies his triangular theorem [2001-1, Corollary 1, p. 124].

Corollary 10.45 ([2001-1, Corollary 7, p. 126])**.** *Every commuting map of a nest algebra has the form $a \mapsto ta + h(a)1$, where $t \in \mathbb{C}$ and h is a linear functional.*

For a continuous nest and a bimodule generated by one element the same form holds on the triangular algebra built from it [2001-1, Corollary 8, p. 127]. The Lie triple statement for nest algebras is the one recalled in Section 5.5 [2015-6, Corollary 5.12, p. 163]. On a nest algebra of a Banach space the Lie n-derivations have the standard form as well [2017-17, Theorem 3.1, p. 6]. That theorem is stated in Section 5.5 as Theorem 5.26 [2017-17, p. 6].

The operator-algebra side of these results belongs to Chapter 19, where the continuity of derivations and the cohomology of nest algebras are treated with analytic tools [1977-1, Theorem 2.1, p. 156]. Banach algebras with the same triangular shape are a separate source of examples, and their derivations have an explicit description in operator terms [1996-2, Proposition 2.1, p. 443]. Forrest and Marcoux describe a continuous derivation by two derivations of the corner algebras, an element of the bimodule and

a linear map of the bimodule [1996-2, Proposition 2.1, p. 443]. They then state which linear maps occur [1996-2, Proposition 2.2, p. 444], and prove a vanishing theorem under hypotheses on the two corner algebras [1996-2, Theorem 2.8, p. 448].

We close here. The matrix, triangular, generalized matrix and nest algebras form one family [2012-9, p. 2720]. On each member the same three questions receive the same kind of answer: about derivations, about commuting maps and about Lie derivations [2003-1, Theorem 11, p. 307].

Chapter 11

Path, incidence, trivial extension and Leavitt path algebras

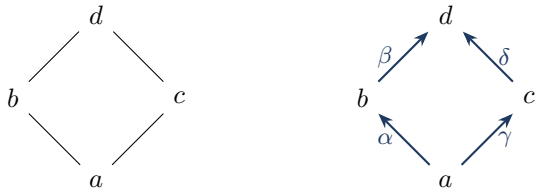
Chapter 10 read the derivations of algebras whose elements are matrices, and its answers come from the shape of the matrix. An incidence algebra keeps the shape of a matrix and replaces the entries by the intervals of a poset [2015-13, p. 1358]. Ward first considered these algebras as generalized algebras of arithmetic functions [2015-13, p. 1358]. Rota and Stanley then made them the base of counting and of the theory of arithmetic functions [2015-13, p. 1358]. This chapter asks for their derivations, and it follows the same shape through a quiver, a trivial extension and a Leavitt path algebra [2019-12, p. 59].

Baclawski answered the derivation question for a locally finite ordered set [1972-1, p. 352]. Every derivation of the incidence algebra is an inner derivation plus one attached to an additive function [1972-1, Theorem 2, p. 352]. When the poset has a smallest or a largest element, the second part is inner as well [1972-1, p. 354]. Xiao states the same split for a pre-ordered set over a commutative ring, and he shows that no Jordan derivation is new there [2015-13, Theorem 3.3, p. 1366].

A path algebra gives a second reading of the same object [2018-14, p. 81]. Guo and Li describe its derivations by conditions on the coefficients of an operator on paths [2014-12, Theorem 2.3, p. 548]. Their operators $D_{r,s}$ and the inner derivations form a basis of the Lie algebra of derivations [2014-12, Theorem 3.4, p. 555]. Ortega shows that an incidence algebra is a

path algebra modulo the differences of parallel paths [2006-7, p. 232]. A derivation of the incidence algebra is then a derivation of a path algebra that kills the differences of parallel paths [2006-7, p. 232].

A trivial extension adds a module with one crossed rule for the product [2017-4, p. 4]. Bennis and Fahid describe its derivations in four parts and decide which of them are inner [2017-4, Theorem 3.12, p. 14]. A Leavitt path algebra adds a reverse edge for every arrow [2019-12, p. 61]. Lopatkin describes its derivations, and he shows that the outer derivations are infinite exactly when the graph has an infinite path [2019-12, Corollary 3.8, p. 79].



Hasse diagram quiver, relation $\beta\alpha = \delta\gamma$

Figure 11.1: A poset and its quiver. The incidence algebra of the poset is the path algebra of the quiver modulo the relation $\beta\alpha = \delta\gamma$.

Section 11.1 gives the path algebra of a quiver. It states the conditions of Guo and Li on a derivation, and the basis of the Lie algebra of derivations [2014-12, Theorem 3.4, p. 555]. It closes with the Jordan derivations of a quiver with relations [2018-14, Theorem 3.4, p. 85]. Section 11.2 gives the incidence algebra, the two theorems of Baclawski [1972-1, pp. 352, 354] and the split of Xiao [2015-13, Proposition 2.5, p. 1362]. Section 11.3 gives the trivial extension and the innerness criterion of Bennis and Fahid [2017-4, Theorem 3.12, p. 14]. Section 11.4 closes the chapter with the Leavitt path algebra, its generators and the dimension of its outer derivations [2019-12, Corollary 3.8, p. 79].

11.1 Path algebras of quivers

The triangular algebra of Chapter 10 is a path algebra. Take a quiver that is a line of arrows, and the algebra of its paths is the algebra of upper triangular matrices [2018-14, Example 2.1, p. 82]. A quiver keeps the order of the matrix algebra and allows branching [2018-14, p. 81]. This section

gives the derivations of a path algebra, first for a plain quiver and then for a quiver with relations.

A quiver is a directed graph with a set of vertices and a set of arrows between them [2014-12, p. 547]. Its paths form a basis of an algebra. The product of two paths is their composition when the head of the first is the tail of the second, and zero otherwise [2014-12, p. 547].

Definition 11.1 ([2014-12, p. 547])**.** Let $\Gamma = (V, E, t, h)$ be a quiver with vertex set V, arrow set E, and maps $t, h\colon E \to V$ giving the tail and the head of an arrow. Let P be the set of paths, a path being a sequence of arrows in which the head of each arrow is the tail of the next. The *path algebra* is

$$k\Gamma = \bigoplus_{p \in P} kp, \qquad p \cdot q = \delta_{h(p),t(q)}\, pq,$$

where $\delta_{h(p),t(q)}$ is the delta function and $pq = 0$ when $h(p) \neq t(q)$.

Guo and Li characterize the derivations of $k\Gamma$ by conditions on the coefficients [2014-12, p. 548]. Write $D(p) = \sum_{q \in P} c_q^p q$ for a linear operator D, with $c_q^p \in k$ [2014-12, p. 548]. Call two paths p and q parallel, written $p \parallel q$, when they have the same tail and the same head [2014-12, p. 548]. A path is acyclic when its head differs from its tail, and P_A denotes the set of acyclic paths [2014-12, p. 548].

Theorem 11.2 ([2014-12, Theorem 2.3, pp. 548–549])**.** *Let Γ be a quiver and $D\colon k\Gamma \to k\Gamma$ a linear operator. Then D is a derivation if and only if it satisfies the following conditions.*

label=(a) For $v \in V$,

$$D(v) = \sum_{q \in P_A,\, t(q) = v} c_q^v q + \sum_{q \in P_A,\, h(q) = v} c_q^v q = \sum_{q \in P_A,\, t(q) = v \text{ or } h(q) = v} c_q^v q.$$

lbbel=(b) For $p \in P \setminus V$,

$$D(p) = \sum_{q \in P_A,\, h(q) = t(p)} c_q^{t(p)} qp + \sum_{q \parallel p} c_q^p q + \sum_{q \in P_A,\, t(q) = h(p)} c_q^{h(p)} pq,$$

where the coefficients obey two rules.

label=(i) For every $q \in P_A$, $c_q^{h(q)} + c_q^{t(q)} = 0$.

lbbel=(ii) For every $p = p_1p_2$ with $p_1, p_2 \in P \setminus V$ and every $q \parallel p$,

$$c_q^p = c_q^{p_1p_2} = \begin{cases} c_{qp_1}^{p_1} + c_{q_2}^{p_2}, & p_2 \leq_h q \text{ with } q = q_1p_2 \text{ and } p_1 \leq_t q \text{ with} \\ c_{q_1}^{p_1}, & p_2 \leq_h q \text{ with } q = q_1p_2 \text{ and } q_1 \not\leq_t p_1, \\ c_{q_2}^{p_2}, & p_1 \leq_t q \text{ with } q = p_1q_2 \text{ and } q_2 \not\leq_h p_2, \\ 0, & p_2 \not\leq_h q \text{ and } p_1 \not\leq_t q, \end{cases}$$

where $p \leq_t q$ means that $q = pr$ and $p \leq_h q$ means that $q = rp$ for some path r.

The conditions of Theorem 11.2 come in two families. One family fixes the value on a vertex by values on acyclic paths at that vertex [2014-12, p. 549]. The other fixes the value on a longer path by the values on its head, its tail and its parallel paths [2014-12, p. 549]. A derivation is therefore built from one arrow to a parallel path [2014-12, pp. 553–554].

Fix an arrow $r \in E$ and a path s parallel to r. Guo and Li define an operator $D_{r,s}$ by $D_{r,s}(p) = 0$ for $\ell(p) = 0$, where ℓ is the length, and on a path $p = p_1\tilde{p}$ by

$$D_{r,s}(p) = \begin{cases} s\tilde{p} + p_1 D_{r,s}(\tilde{p}), & p_1 = r, \\ p_1 D_{r,s}(\tilde{p}), & p_1 \neq r, \end{cases}$$

[2014-12, pp. 553–554]. This operator is a derivation [2014-12, p. 554].

Theorem 11.3 ([2014-12, Theorem 3.2, p. 554])**.** *For $r \in E$ and $s \parallel r$, the linear operator $D_{r,s} \colon k\Gamma \to k\Gamma$ defined as above is a derivation.*

A quiver without arrows has no derivation but zero [2014-12, p. 554], so the condition on the quiver cannot be dropped.

Corollary 11.4 ([2014-12, Corollary 3.3, p. 554])**.** *There is a nonzero derivation on the path algebra $k\Gamma$ of a quiver Γ if and only if Γ is a non-trivial quiver, that is, Γ has at least one arrow. Equivalently,* $\mathrm{Der}(k\Gamma)$ *is a non-zero Lie algebra if and only if Γ is a non-trivial quiver.*

Inside the Lie algebra of derivations, the operators of Theorem 11.3 and the inner derivations form a basis. For a path s, write $D_s(q) = sq - qs$ for the inner derivation with respect to s [2014-12, p. 555].

Theorem 11.5 ([2014-12, Theorem 3.4, p. 555])**.** *Let Γ be a quiver. A basis of the Lie algebra* $\mathrm{Der}(k\Gamma)$ *is given by the set*

$$\mathcal{B} := \mathcal{B}_1 \cup \mathcal{B}_2, \qquad \mathcal{B}_1 := \{D_s \mid s \in P_A\}, \quad \mathcal{B}_2 := \{D_{r,s} \mid r \in E,\ s \parallel r\}.$$

Thus $\mathrm{Der}(k\Gamma) = D_1 \oplus D_2$*, where* D_i *is the* k*-linear space with basis* $\mathcal{B}_i$ *for* $i = 1, 2$*.*

A relation on a quiver is a linear combination of parallel paths [2018-14, p. 81]. With a set ρ of relations, the algebra $K(\Gamma, \rho) = K\Gamma/\langle\rho\rangle$ is a quiver with relations [2018-14, p. 81]. Every basic finite-dimensional algebra over an algebraically closed field has this form [2018-14, p. 81].

For a quiver without oriented cycles, Li and Wei remove a source, and the algebra becomes a triangular algebra over the part that remains [2018-14, p. 83]. Repeating the step reaches the field, and the step is called a one-point extension [2018-14, p. 83]. The step carries a statement about the field to every finite-dimensional path algebra without oriented cycles [2018-14, p. 85].

Theorem 11.6 ([2018-14, Theorem 3.4, p. 85])**.** *Let* K *be a field with* $\mathrm{Char}\, K \neq 2$ *and* $\Lambda = K(\Gamma, \rho)$ *a finite-dimensional path algebra over* K *of the quiver* (Γ, ρ) *with relations. Then every Jordan derivation on* Λ *is a derivation.*

The theorem carries over to one step of that construction [2018-14, p. 86].

Corollary 11.7 ([2018-14, Corollary 3.6, p. 86])**.** *Let* K *be a field and* A *a finite-dimensional* K*-algebra with every Jordan derivation being a derivation. Let* Λ *be a one-point extension of* A *by the bimodule* ${}_AM_K$*. Then every Jordan derivation of* Λ *is a derivation.*

A local derivation of a path algebra turns out to need no such freedom. Adrabi, Bennis and Fahid prove that every local derivation of a path algebra is a derivation, and the same for local generalized derivations [2023-6, Theorem 3, p. 391]. Their theorem settles the Kadison question of Section 8.1 for this whole family at once. The Lie generalized derivations of a bound quiver algebra are treated in [2021-8, p. 1950], and the generalized σ-derivations of that setting have a categorical description in [2023-5, p. 4754].

Ortega presents the incidence algebra of a poset as a path algebra with relations [2006-7, p. 232]. Take the poset as the vertex set and put one arrow from v to w when $v < w$ and no element lies between them [2006-7, p. 232]. The path algebra modulo the differences of parallel paths is the incidence algebra of the poset [2006-7, p. 232]. The ideal of differences vanishes exactly when the poset is a tree, and then the incidence algebra is hereditary [2006-7, p. 232]. Incidence algebras deserve their own treatment, and Section 11.2 gives it.

11.2 Incidence algebras of posets

The path algebra of Section 11.1 counts paths. An incidence algebra counts the intervals of a poset, and it is the same algebra written with less data [2006-7, p. 232]. Rota and Stanley made it the base of counting and of the theory of arithmetic functions, and Baclawski asked for its derivations [2015-13, p. 1358]. This section gives his answer, then the answer for a pre-ordered set over an arbitrary ring.

The elements of the algebra are functions on the intervals of a pre-ordered set [2006-7, p. 230]. The product adds over the points between the two ends [2006-7, p. 230].

Definition 11.8 ([2006-7, Definition 1.10, p. 230])**.** Let X be a finite pre-ordered set and F a commutative ring with unit. The *incidence algebra* of X over F is

$$I(X,F) = \{\, f\colon X \times X \to F \mid f(x,y) = 0 \text{ if } x \not\leq y \,\},$$

with

$$(f{+}g)(x,y) = f(x,y){+}g(x,y), \qquad (f{\cdot}g)(x,y) = \sum_{x\leq z\leq y} f(x,z)\, g(z,y), \qquad (\lambda f)(x,y) =$$

for all $f, g \in I(X,F)$, $\lambda \in F$ and $x, y, z \in X$.

Baclawski works over a locally finite ordered set and a field [1972-1, p. 351]. He calls an element a of $I(P)$ *additive* when $a(x,z) = a(x,y){+}a(y,z)$ for $x \leq y \leq z$, and attaches to it the derivation $L_a(f) = af$ [1972-1, p. 352]. The inner derivations are the maps $d_f(g) = fg - gf$ [1972-1, p. 352]. His first result measures how many connected components the poset has [1972-1, p. 352].

Theorem 11.9 ([1972-1, Theorem 1, p. 352])**.** *Let P be a locally finite ordered set having c connected components: $P_1, \ldots, P_c$. Then there is a short exact sequence of K-homomorphisms*

$$0 \longrightarrow K^c \longrightarrow I(P) \longrightarrow \mathcal{D}(I(P)) \longrightarrow 0,$$

where $\mathcal{D}(I(P))$ is the space of inner derivations of $I(P)$.

The second result gives the decomposition [1972-1, p. 352]. Every derivation is an inner derivation plus one that comes from an additive function [1972-1, p. 352].

Theorem 11.10 ([1972-1, Theorem 2, p. 352])**.** *If P is a locally finite ordered set, then*

$$\mathrm{Der}(I(P)) = \mathrm{Inn}(I(P)) + \mathcal{M}(I(P)),$$

where $\mathrm{Inn}(I(P))$ is the space of inner derivations and $\mathcal{M}(I(P))$ is the space of derivations L_a attached to additive elements a.

When the poset has a smallest or a largest element, the additive part is inner as well [1972-1, p. 354]. Then every derivation of the incidence algebra is inner [1972-1, p. 354].

Xiao states the same decomposition for a pre-ordered set over a commutative ring [2015-13, p. 1362]. The additive rule becomes the *transitive* rule, and the derivation attached to it sends each matrix unit to a scalar multiple of itself [2015-13, p. 1362]. Xiao records that the definition of a transitive mapping first appeared in the paper of Nowicki on finite incidence algebras [2015-13, Remark 2.4, p. 1362].

Definition 11.11 ([2015-13, p. 1362])**.** A mapping $f\colon \leq \to R$ is *transitive* if $f(i,j) + f(j,k) = f(i,k)$ for all $i,j,k \in X$ with $i \leq j$ and $j \leq k$. Every mapping $\sigma\colon X \to R$ determines a transitive mapping $(i,j) \mapsto \sigma(i) - \sigma(j)$; such a mapping is called trivial. For a transitive f, the map Δ_f is defined on the matrix units by $\Delta_f(e_{ij}) = f(i,j)e_{ij}$.

A transitive map of this kind gives a derivation, and the derivation is inner only in the trivial case [2015-13, p. 1362].

Lemma 11.12 ([2015-13, Lemma 2.3, p. 1362])**.** *With the notation above, Δ_f is a derivation. Moreover, Δ_f is inner if and only if f is trivial.*

Every derivation of the algebra splits into these two pieces [2015-13, p. 1362].

Proposition 11.13 ([2015-13, Proposition 2.5, p. 1362])**.** *Let D be a derivation of $I(X,R)$. Then there exist an element $g \in I(X,R)$ and a transitive mapping f such that*

$$D = \mathrm{Inn}_g + \Delta_f.$$

The decomposition decides innerness for the whole algebra [2015-13, p. 1363].

Corollary 11.14 ([2015-13, Corollary 2.6, p. 1363])**.** *Every derivation of $I(X,R)$ is inner if and only if every transitive mapping is trivial.*

The same description settles Jordan derivations [2015-13, p. 1364]. A Jordan derivation satisfies $D(x^2) = D(x)x + xD(x)$ [2015-13, p. 1357]. Xiao proves that on a 2-torsion free ring no such map is new [2015-13, p. 1366].

Theorem 11.15 ([2015-13, Theorem 3.3, p. 1366])**.** *Let R be a 2-torsion free commutative ring with identity. Then every Jordan derivation of the incidence algebra $I(X, R)$ is a derivation.*

The generalized Jordan form of the same question has the same answer [2018-8, p. 6]. A generalized Jordan derivation is a map d with $d(a^2) = d(a)a + a\tau(a)$ for a Jordan derivation τ [2018-8, p. 1]. On an incidence algebra over a 2-torsion free commutative ring each such map is a generalized derivation [2018-8, Theorem 2.1, p. 6]. The theorem is a direct consequence of the Jordan result of Theorem 11.15 [2018-8, p. 1].

The biderivations of the same algebra split in the same two ways [2024-14, p. 13]. The decomposition into inner and extremal pieces, which Section 7.6 meets on a triangular ring, holds here under a condition on the maximal chains [2024-14, Theorem 4.10, p. 13].

By Proposition 11.13, the derivations of an incidence algebra split into an inner part and a part read from an additive function. A trivial extension is a second construction that puts two algebras together, and Section 11.3 treats it.

11.3 Trivial extensions

The triangular algebra of Chapter 10 is a trivial extension [2017-4, p. 5]. A trivial extension puts a ring and a bimodule together with one rule for the product [2017-4, p. 4]. The triangular algebra is the special case where the module has the form $M = eSf$ [2017-4, p. 5]. This section gives the derivations of the construction and the test for innerness.

Let A be a ring and M an A-bimodule. The product of two pairs keeps the first coordinate and adds a crossed term to the second [2017-4, p. 4].

Definition 11.16 ([2017-4, p. 4])**.** Let A be a unital R-algebra and M a unital A-bimodule. The direct product $A \times M$, with pairwise addition and the product

$$(a, m)(b, n) = (ab,\ an + mb) \qquad (a, b \in A,\ m, n \in M),$$

is a unital algebra, called the *trivial extension* of A by M and written $A \ltimes M$.

Every linear map on the extension splits into four parts [2017-4, p. 8]. Two parts act on A and M separately, and two carry M to A and A to M [2017-4, p. 8].

Lemma 11.17 ([2017-4, Lemma 3.1, p. 8])**.** *Every linear map $D\colon A \ltimes M \to A \ltimes M$ has the form*

$$D(a,m) = (D_A(a) + T(m),\ D_M(a) + S(m)) \qquad (a \in A,\ m \in M)$$

for some linear maps $D_A\colon A \to A$, $D_M\colon A \to M$, $T\colon M \to A$ and $S\colon M \to M$. We write $D = (D_A + T, D_M + S)$. Moreover, D is a derivation if and only if D_A and D_M are derivations, $T \in E(M)$ and S is a module generalized D_A-derivation.

The notion of $E(M)$ in Lemma 11.17 needs its own definition [2017-4, p. 7]. It is the set of bimodule maps whose sum over two arguments vanishes [2017-4, p. 7]. A module generalized derivation satisfies the product rule on both sides, with the same derivation d of A on each side [2017-4, p. 7].

Definition 11.18 ([2017-4, p. 7])**.** Let $E(M)$ be the R-submodule of $\mathrm{Hom}_{A-A}(M,A)$ consisting of all bimodule homomorphisms f such that $f(m)n + mf(n) = 0$ for all $m, n \in M$. A linear map $S\colon M \to M$ is a *module generalized d-derivation* for some $d \in \mathrm{Der}(A)$ if

$$S(am) = aS(m) + d(a)m, \qquad S(ma) = S(m)a + md(a)$$

for all $m \in M$ and $a \in A$.

Innerness of the whole map is decided by its four parts [2017-4, p. 14]. One part must vanish, and the two parts on the module must be inner [2017-4, p. 14].

Theorem 11.19 ([2017-4, Theorem 3.12, p. 14])**.** *Consider a derivation $D\colon A \ltimes M \to A \ltimes M$ of the form $D = (D_A + T, D_M + S)$, where D_A, T, D_M and S are as in Lemma 11.17. Then the following assertions hold.*

label=(1) If D is inner, then $T = 0$ and both S and D_M are inner.

lbbel=(2) The converse holds if $\mathrm{l.Ann}_A(M) \cap \mathrm{r.Ann}_A(M) = 0$.

When the module is the ring itself, the criterion takes an exact form [2017-4, p. 14]. The two module parts then agree [2017-4, p. 14].

Corollary 11.20 ([2017-4, Corollary 3.13, p. 14])**.** *Assume $M = A$ and consider a derivation $D\colon A \ltimes M \to A \ltimes M$ of the form $D = (D_A + T, D_M + S)$ with the parts as in Lemma 11.17. Then D is inner if and only if $T = 0$, D_M is inner, $S = D_A$ and S is inner. Precisely, the inner derivations on $A \ltimes A$ are only of the form $(d_1, d_0 + d_1)$, where d_0 and d_1 are inner derivations on A.*

Boudine and Zerra give the same description over a general ring with two projections [2024-9, p. 461]. Write $d = (d_1, d_2)$ for the two components, and write $s_1(m) = d_2((0, m))$ and $s_2(r) = d_2((r, 0))$ [2024-9, p. 461]. Their conditions are the three product rules of Bennis and Fahid, written out.

Theorem 11.21 ([2024-9, Theorem 2.1, p. 461])**.** *With the notation above, the map d is a derivation on $R \propto M$ if and only if the following statements hold:*

label=(1) d_1 *is a (π_1, π_1)-derivation;*

lbbel=(2) s_2 *is a derivation;*

lcbel=(3) $s_1(rm) = r\, s_1(m) + d_1(r, 0)m$ *and* $s_1(mr) = s_1(m)r + m\, d_1(r, 0)$ *for every $r \in R$ and $m \in M$.*

Their innerness test lists four conditions, one for each part [2024-9, p. 463]. The list matches Corollary 11.20 in the case $M = R$.

Theorem 11.22 ([2024-9, Theorem 2.3, p. 463])**.** *With the notation above, the derivation d is the inner derivation on $R \propto M$ induced by $(r, m) \in R \propto M$ if and only if the following statements hold:*

label=(1) $d_1 \circ i_2 = 0$*;*

lbbel=(2) $d_1 \circ i_1$ *is the inner derivation on R induced by r;*

lcbel=(3) s_1 *is the inner derivation on M induced by r;*

ldbel=(4) s_2 *is the inner derivation on R into M induced by m.*

Over a prime ring the classical theorems of Herstein, Posner and Bell carry over to the extension [2024-9, p. 467]. The first one kills the part of the derivation that stays on R [2024-9, p. 467].

Theorem 11.23 ([2024-9, Theorem 3.5, p. 467])**.** *Let R be a noncommutative prime ring with* $\operatorname{char}(R) \neq 2$*, M an R-bimodule, and d a derivation on $R \propto M$. If $d((r,m))d((r',m')) = d((r',m'))d((r,m))$ for every $(r,m),(r',m') \in R \propto M$, then $d_1 \circ i_1 = 0$ and $s_1 \in \operatorname{End}_R(M)$. Furthermore, for every $m, m' \in M$, $[s_2(m), d_1((0,m'))] = [d_1((0,m)), s_2(m')]$.*

The centralizing form gives the corresponding statement for the theorem of Posner [2024-9, p. 468].

Proposition 11.24 ([2024-9, Proposition 3.6, p. 468])**.** *Let R be a noncommutative prime ring, M an R-bimodule such that $Z_M(R) = M$, and d a derivation on $R \propto M$. Then d is centralizing if and only if, for every $(r,m) \in R \propto M$, the following statements hold:*

label=(1) $d_1 \circ i_1 = 0$*;*

lbbel=(2) $d_1((0,m))r - rd_1((0,m)) \in Z_R(R)$*.*

By Theorem 11.21, the extension keeps the derivations of its first factor and adds a crossed part. That pattern returns once more with a graph in place of the module, and Section 11.4 gives the graph case.

The construction repeats for a finite family of modules, and the derivations become a sequence of maps [2026-6, p. 2]. Anderson, Bennis, Fahid and Shaiea defined the n-trivial extension of a ring by modules $M_1, \ldots, M_n$ [2026-6, p. 2]. Barajas and Duarte ask for the maps that stand to it as derivations stand to the one-module case [2026-6, p. 2]. When all the modules are the same algebra, the answer is the Hasse–Schmidt sequence of Section 7.1, which is the higher derivation under its other name [2026-6, p. 2]. For different algebras they define a multi Hasse–Schmidt derivation, a sequence of additive maps whose Leibniz rule carries the maps φ_{ij} between the modules [2026-6, Definition 3.1, p. 6]. The n-trivial extension is then a quotient of a polynomial ring, exactly as the one-module extension is the ring of dual numbers [2026-6, Theorem 4.2, p. 13]. The multi Hasse–Schmidt derivations with first term the identity correspond to the algebra maps into that quotient [2026-6, Corollary 4.3, p. 15].

11.4 Leavitt path algebras

The trivial extension of Section 11.3 adds one crossed term to an algebra. The Leavitt path algebra of this section adds a reverse edge for every arrow, and the presentation lists the reverse edges as new generators [2019-12,

p. 61]. Lopatkin describes every derivation of it on a row-finite graph [2019-12, p. 59]. This section gives his description, the generating set, and the dimension of the outer derivations.

A row-finite graph has finitely many edges with each source [2019-12, p. 60]. The algebra has a vertex for each point of the graph, an edge for each arrow, and a reverse edge for each arrow [2019-12, p. 61]. Four sets of relations present it, the last of which splits a vertex among the edges that leave it [2019-12, p. 61].

Definition 11.25 ([2019-12, Definition 1.1, p. 61])**.** Let K be a commutative ring and $\Gamma = (V, E)$ a row-finite graph. The *Leavitt path algebra* $L_K(\Gamma)$ of Γ with coefficients in K is the K-algebra presented by the generators $\{v,\ v \in V\}$ and $\{e, e^* \colon e \in E\}$ and the relations:

label=(1) $vu = \delta_{v,u} v$ for all $v, u \in V$;

lbbel=(2) $s(e)e = er(e) = e$ and $r(e)e^* = e^* s(e) = e^*$ for all $e \in E$;

lcbel=(3) $e^* f = \delta_{e,f} r(e)$ for all $e, f \in E$;

ldbel=(4) $v = \sum_{s(e)=v} ee^*$ for an arbitrary vertex $v \in V \setminus \{\text{sinks}\}$.

Two familiar algebras appear as examples [2019-12, p. 61]. A line of n vertices gives the full matrix algebra [2019-12, p. 61]. A single loop gives the Laurent polynomial algebra [2019-12, p. 61].

Example 11.26 ([2019-12, Example 1.2, p. 61])**.** Let A_n denote the graph

$$\bullet_{v_1} \xrightarrow{e_1} \bullet_{v_2} \xrightarrow{e_2} \cdots \xrightarrow{e_{n-1}} \bullet_{v_n}.$$

Then $L_K(A_n) \cong M_n(K)$, the full $n \times n$ matrix algebra. The isomorphism sends v_i to $E_{i,i}$, e_i to $E_{i,i+1}$ and e_i^* to $E_{i+1,i}$ for $1 \leq i \leq n-1$.

The second example has one vertex and one loop, and there the reverse edge acts as the inverse variable [2019-12, p. 61].

Example 11.27 ([2019-12, Example 1.3, p. 61])**.** Let R_1 denote the graph with one vertex v and one loop e. Then $L_K(R_1) \cong K[t, t^{-1}]$, the Laurent polynomial algebra. The isomorphism is $v \mapsto 1$, $e \mapsto t$ and $e^* \mapsto t^{-1}$.

The algebra has a basis in normal form [2019-12, p. 63]. A word in the basis is a path, a reverse path, or a pair of paths of equal length that end at the same vertex [2019-12, p. 63]. The last pair must not both end in the same special edge [2019-12, p. 63].

Theorem 11.28 ([2019-12, Theorem 1.6, p. 63]). *Let Γ be a row-finite graph and K a commutative ring. Then the following elements form a K-basis $\mathcal{B}$ of the Leavitt path algebra $L(\Gamma)$: (1) the set of all vertices V; (2) the set of all paths P; (3) the set $P^* = \{p^* \colon p \in P\}$; (4) a set M of words of the form wh^*, where $w = w_0 \cdots w_z \in P$ and $h^* = [h_0 \cdots h_z]^* = h_z^* \cdots h_0^* \in P^*$ are paths that end at the same vertex $r(w) = r(h)$, with the condition that the edges w_z and h_z are either distinct or equal but not special.*

Every derivation is fixed by its values on the generators [2019-12, pp. 63–64]. Those values are sums over the basis, and the coefficients satisfy three linear relations [2019-12, pp. 63–64].

Theorem 11.29 ([2019-12, Theorem 2.1, pp. 63–64]). *Let Γ be a row-finite graph and K a commutative ring. Every derivation D of the Leavitt path algebra $L_K(\Gamma)$ can be described as follows:*

$$D(v) = \sum_{wh^* \in \mathcal{B}} F_{wh^*}(v) \cdot \mathrm{ad}_{wh^*}(v),$$

$$D(e) = \sum_{wh^* \in \mathcal{B}} F_{wh^*}(s(e))\mathrm{ad}_{wh^*}(e) + \sum_{\substack{wh^* \in \mathcal{B} \\ s(w)=s(e),\ s(h)=r(e)}} F_{wh^*}(e)wh^*,$$

$$D(e^*) = \sum_{wh^* \in \mathcal{B}} F_{wh^*}(s(e))\mathrm{ad}_{wh^*}(e^*) + \sum_{\substack{wh^* \in \mathcal{B} \\ s(w)=r(e),\ s(h)=s(e)}} F_{wh^*}(e^*)wh^*,$$

for every $v \in V$ and $e \in E$. Further, almost all coefficients are zero, and they satisfy the relations

$$F_p(e^*) + F_{pff^*}(e^*) + F_{epf}(f) = 0, \qquad epf \neq 0,$$

$$F_{[fpe]^*}(e^*) + F_{p^*}(f) + F_{e[pe]^*}(f) = 0, \qquad fpe \neq 0,$$

$$F_{w[fh]^*}(e^*) + F_{ewh^*}(f) = 0, \qquad w[fh]^* = pff^*,\ ewh = e[pe]^*,$$

where $p, w, h \in P \cup V$ and $e, f \in E$ with $s(e) = s(f)$.

The derivations of Theorem 11.29 are generated by four families [2019-12, p. 64]. Each family is attached to a path or a pair of paths [2019-12, p. 64].

Corollary 11.30 ([2019-12, Corollary 2.1, p. 64]). *The K-module* $\mathrm{Der}(L_K(\Gamma))$ *is generated by the following set of derivations:*

label=(1) $\bigcup_{pq^* \in \mathcal{B}} \{\mathrm{ad}_{pq^*}\}$;

lbbel=(2) $\bigcup_{wh^*\in M,\ e\in E,\ s(w)=s(h)}\{D_{wh^*}, D_{ee^*}\}$, *where* $D_{wh^*}(h_0) = w[h/h_0]^*$, $D_{wh^*}(w_0^*) = -[w/w_0]h^*$, *and* $D_{wh^*} = 0$ *otherwise;*

lcbel=(3) $\bigcup_{c\in P,\ s(c)=r(c)}\{D_c\}$, *where* $D_c(e) = ce$ *for* $e \in E$, $D_c(c_0^*) = -[c/c_0]^*$, *and* $D_c = 0$ *otherwise;*

ldbel=(4) $\bigcup_{c\in P,\ s(c)=r(c)}\{D_{c^*}\}$, *where* $D_{c^*}(e^*) = -[ce]^*$ *for* $e \in E$, $D_{c^*}(c_0^*) = [c/c_0]^*$, *and* $D_{c^*} = 0$ *otherwise.*

Modulo the inner derivations, the same families generate the outer derivations, with explicit relations among them [2019-12, p. 78]. The relations record when a generator dies or splits [2019-12, p. 78].

Theorem 11.31 ([2019-12, Theorem 3.2, p. 78])**.** *Let* $\Gamma = (V, E)$ *be a row-finite graph and* $L_K(\Gamma)$ *the Leavitt path algebra over a commutative ring* K*. The* K*-module* $HH^1(L_K(\Gamma))$ *of outer derivations of* $L_K(\Gamma)$ *can be presented by the following sets of generators:*

label=(1) $\bigcup_{wh^*\in M,\ e\in E,\ s(w)=s(h)}\{D_{wh^*}, D_{ee^*}\}$, *where* $D_{ee^*}(e) = e$, $D_{ee^*}(e^*) = -e^*$, *and* $D_{ee^*} = 0$ *otherwise, and* $D_{wh^*}(h_0) = w[h/h_0]^*$, $D_{wh^*}(w_0^*) = -[w/w_0]h^*$, *and* $D_{wh^*} = 0$ *otherwise;*

lbbel=(2) $\bigcup_{c\in P,\ s(c)=r(c)}\{D_c\}$, *where* $D_c(e) = ce$ *for every* $e \in E$, $D_c(c_0^*) = -[c/c_0]^*$, *and* $D_c = 0$ *otherwise;*

lcbel=(3) $\bigcup_{c\in P,\ s(c)=r(c)}\{D_{c^*}\}$, *where* $D_{c^*}(e^*) = -[ce]^*$ *for every* $e \in E$, $D_{c^*}(c_0^*) = [c/c_0]^*$, *and* $D_{c^*} = 0$ *otherwise;*

and the following sets of relations among those generators:

label=(1) $D_{wh^*} = 0$ *whenever* $|\{p \in P\colon r(p) = s(w)\}| < \infty$*;*

lbbel=(2) $D_{wh^*} = \sum_{e\in E} D_{ew[eh]^*}$ *whenever* $|\{e \in E\colon r(e) = s(w)\}| < \infty$*;*

lcbel=(3) if the edge e *is special,* $s(e) = r(e)$, $\operatorname{supp}_e(s(e)) \cap \operatorname{supp}_e(r(e)) = \varnothing$*, and both supports are finite, then*

$$D_{ee^*} = \alpha \sum_{\substack{f,g\in E\setminus\vartheta(V)\\ r(g)\in\operatorname{supp}_e(s(e)),\ s(f)\in\operatorname{supp}_e(s(e))}} (D_{gg^*} - D_{ff^*}) + \beta \sum_{\substack{f,g\in E\setminus\vartheta(V)\\ r(g)\in\operatorname{supp}_e(r(e)),\ s(f)\in\operatorname{supp}_e}}$$

with $\alpha, \beta \in K$ *and* $\alpha - \beta = 1$*;*

ldbel=(4) $D_c = D_{\zeta^k c} - \sum_{e\in E} D_{\zeta^k cee^*}$ *whenever* $\delta_{c_0,\vartheta}\cdots\delta_{c_{k-1},\vartheta} = 0$, $\delta_{c_k,\vartheta} = 1$ *and* $|\{p \in P\colon r(p) = s(c_0),\ p_z = c_k\}| < \infty$.

Over a field the dimension of the space of outer derivations has two possible values [2019-12, p. 79]. It is infinite exactly when the graph has an infinite path [2019-12, p. 79].

Corollary 11.32 ([2019-12, Corollary 3.8, p. 79])**.** *Let K be a field. For the vector space $HH^1(L_K(\Gamma))$ of outer derivations of the Leavitt path algebra $L_K(\Gamma)$ we have*

$$\dim HH^1(L_K(\Gamma)) = \begin{cases} \infty, & \text{if there exists a path in } \Gamma \text{ with infinite length,} \\ 0, & \text{otherwise.} \end{cases}$$

By Corollary 11.32, a Leavitt path algebra carries as many outer derivations as its graph has infinite paths. The record of the chapter is in Sections 11.1 to this one, and Chapter 12 turns to algebras built from a polynomial ring.

Further work in the Leavitt and graded-ring line, held in the collection: *Prime group graded rings with applications to partial crossed products and Leavitt path algebras* [2021-6].

Chapter 12

Polynomial, Weyl and differential operator algebras

The derivative returns to its first home. Chapter 1 opened the book with Leibniz and the derivative of a polynomial. This chapter comes back to that home and asks what the Leibniz rule becomes on the polynomial ring $k[x_1, \dots, x_n]$ and on the noncommutative rings that imitate it. Four threads answer, each read from the papers that carry it.

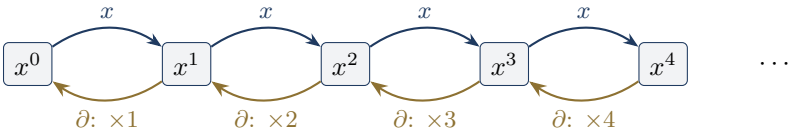


Figure 12.1: The Weyl algebra acting on $k[x]$: multiplication by x raises the degree; $\partial = d/dx$ lowers it, with $\partial(x^n) = nx^{n-1}$; and $\partial x - x\partial = 1$.

The first thread asks what a derivation of a polynomial ring can be [2017-7, p. 75]. the fourteenth problem of Hilbert and the Jacobian conjecture made the question urgent [2017-7, p. xii]. Weitzenböck gave the first answer in 1932: for a linear locally nilpotent derivation the kernel is finitely generated [2009-6, p. 41]. Section 12.1 defines the named families: locally nilpotent, triangular, elementary, monomial, Weitzenböck [2017-7, p. 76], Shamsuddin [2016-9, p. 2033] and Lotka–Volterra [2025-17, p. 2]. It then reads the kernel theorems. These are the Weitzenböck theorem and the confirmed Nowicki conjecture [2009-6, Theorem 3, p. 44]. The section closes with the fixed-group theorems of Mendes and Pan [2017-9, Theorem 1, p. 2]

and of Rewri and Kour [2025-17, Theorem 4.1, p. 21]. The automorphisms that commute with a derivation d form the fixed group of d there. Bedratyuk gives an algorithm for the kernel of a Weitzenböck derivation, built from Casimir elements of the derivation [2005-1, p. 1]. Mishra, Mondal and Shukla compute the fixed group of a monomial derivation whose exponent matrix is invertible [2026-27, Theorem 3.1, p. 5]. On a simple derivation that takes no unit value, the fixed group reduces to the translations [2025-16, Theorem 2.4, p. 5]. A derivation of a polynomial ring is a first-order step of a Hasse–Schmidt system, and the four named families live on the polynomial ring as such steps. Locally nilpotent derivations are the nil case: each polynomial dies under some power of the derivation [2017-7, p. 3]. The general definitions live in Section 7.1 and Section 2.1.

The second thread carries the derivative into a noncommutative algebra [1968-1, p. 209]. Dixmier named the Weyl algebra in 1968, after a suggestion of Segal, and listed six problems at the end of his paper [1968-1, p. 241]. Section 12.2 reads his definition, his structure theorem for A_n and his first problem, now the Dixmier conjecture. Belov-Kanel and Kontsevich prove that this conjecture agrees, in a stable sense, with the Jacobian conjecture [2007-2, Theorem 1, p. 2]; the two threads of the chapter close on the same question.

The third thread counts the order of an operator [2013-9, p. 4]. Grothendieck defined the differential operators of a commutative algebra by commuting them with multiplications; Section 12.3 recalls the recursive characterization as Hazewinkel states it [2013-9, Theorem 6.2, p. 4]. Sweedler carried the same commutator to noncommutative algebras and named the right derivation [1980-2, p. 29], the first kind of the chapter with the right hand rule of Section 2.1.

The fourth thread builds polynomials over a derivation [1933-1, p. 480]. Section 12.4 takes the differential form $R[t;\delta]$ of the Ore extension of Section 6.4 to its concrete ground. The Weyl algebra is the first example, and the d-simple rings make $R[t;\delta]$ simple [2008-9, p. 801]. On $k[x][t;\delta]$ every local derivation is already a derivation [2004-8, Theorem 4.1, p. 4569]. Chapter 17 follows the derivations of polynomial rings into geometry.

12.1 Derivations of polynomial algebras

The chapter returns to the polynomial ring, the first home of the derivative. The fourteenth problem of Hilbert asked whether certain rings of invariants are finitely generated. The Jacobian conjecture asked whether a polynomial

map with a unit Jacobian determinant is invertible [2017-7, p. xii]. Derivations answer on both fronts, and the answers organize the section: kernels, the named families, and the kernel theorems.

The named families live on $k[x_1, \ldots, x_n]$ with k a field of characteristic zero. Section 2.5 met the locally nilpotent derivation in general; Freudenburg makes it concrete on polynomial rings [2017-7, p. 76]. Recall: a derivation D is locally nilpotent when every element is killed by a high enough power of D [2017-7, p. 21].

Definition 12.1 ([2017-7, p. 76])**.** Let $B = k[x_1, \ldots, x_n]$ and $D \in \mathrm{Der}_k(B)$. The derivation D is a *triangular derivation* of B if $D(x_1) \in k$ and $D(x_i) \in k[x_1, \ldots, x_{i-1}]$ for $i = 2, \ldots, n$. The derivation D is a *monomial derivation* if each image $D(x_i)$ is a monomial in $x_1, \ldots, x_n$. The derivation D is an *elementary derivation* if for some j with $1 \leq j \leq n$, $D(x_i) = 0$ for $1 \leq i \leq j$ and $D(x_i) \in k[x_1, \ldots, x_j]$ for $j + 1 \leq i \leq n$.

Triangularity depends on the choice of coordinates, and every triangular, elementary or nice derivation of $k[x_1, \ldots, x_n]$ is locally nilpotent [2017-7, p. 76]. Every elementary derivation is both triangular and nice [2017-7, p. 76]. In one direction the converse holds after a permutation of the variables, as Proposition 12.2 states [2017-7, p. 76].

Proposition 12.2 ([2017-7, Proposition 3.4, p. 76])**.** *For the polynomial ring $B = k[x_1, \ldots, x_n]$, every locally nilpotent monomial derivation of B is triangular relative to some ordering of $x_1, \ldots, x_n$.*

Monomial derivations have a second home: the body of constants. Definition 12.1 states them through the images of single variables, and Moulin Ollagnier and Nowicki study them for their rings of constants [2011-10, p. 3138]. Their setting adds the field of rational constants, and a classical example of the family sits inside it [2011-10, p. 3139].

Example 12.3 ([2011-10, p. 3139])**.** The *Jouanolou derivation* with integer parameters $n \geq 2$ and $s \geq 1$ is the derivation $J(n, s)$ of $k[x_1, \ldots, x_n]$ given by $d(x_1) = x_2^s$, $d(x_2) = x_3^s$, $\ldots$, $d(x_{n-1}) = x_n^s$ and $d(x_n) = x_1^s$. Jouanolou proved in 1979 that $J(3, s)$ has no nontrivial Darboux polynomial for every $s \geq 2$ [2011-10, p. 3139].

Li carries the question of Darboux polynomials to a monomial derivation of four variables [2015-10, p. 1]. A polynomial F is a Darboux polynomial of d when $d(F) = \Lambda F$ for some polynomial Λ [2015-10, p. 1]. For a normal monomial derivation with $\det(\beta_{ij} - I) \neq 0$, the absence of Darboux

polynomials is the same as a trivial field of constants [2015-10, p. 2]. Li treats the case where that determinant vanishes, for the derivation $d(x) = t^2$, $d(y) = zt$, $d(z) = y^2$, $d(t) = xy$ [2015-10, p. 2].

Theorem 12.4 ([2015-10, Theorem 2.3, p. 3])**.** *Let d be the monomial derivation of $k[x, y, z, t]$ with $d(x) = t^2$, $d(y) = zt$, $d(z) = y^2$ and $d(t) = xy$. Then d has no Darboux polynomial if and only if its field of constants is trivial.*

So the equivalence survives the vanishing of the determinant, for this one case. The proof restricts a possible Darboux polynomial to its homogeneous parts and finds a contradiction from the degrees [2015-10, p. 3].

The nicest monomial way into the kernel is linear. A derivation of $k[x_1, \ldots, x_n]$ is linear when it maps the variables into their own linear span [2005-4, p. 394], and a Weitzenböck derivation is the linear locally nilpotent case. Weitzenböck studied it in 1932, and the name comes with his theorem [2009-6, p. 41]. His paper could not be downloaded for this book; the definition and the theorem are quoted here from Drensky and Makar-Limanov.

Definition 12.5 ([2009-6, p. 41])**.** A derivation Δ of the polynomial algebra $K[Z] = K[z_1, \ldots, z_m]$ over a field K of characteristic zero is a *Weitzenböck derivation* if it is linear and locally nilpotent, that is, Δ is a nilpotent linear operator on the vector space $Kz_1 \oplus \cdots \oplus Kz_m$ extended to a derivation of $K[Z]$.

The next theorem is the finiteness result of Weitzenböck.

Theorem 12.6 ([2009-6, Weitzenböck 1932, p. 42])**.** *If Δ is a Weitzenböck derivation of $K[Z]$, then the algebra of constants $K[Z]^\Delta$ is a finitely generated algebra.*

A Weitzenböck derivation is determined up to a change of basis by its Jordan normal form, whose cells are upper triangular with zero diagonal [2009-6, p. 42]. For two-by-two cells only, Nowicki made a conjecture on the generators of the kernel, and Drensky and Makar-Limanov proved it [2009-6, pp. 42–45].

Theorem 12.7 ([2009-6, Theorem 3, p. 44])**.** *Let $X = \{x_1, \ldots, x_n\}$, $Y = \{y_1, \ldots, y_n\}$, $n \geq 1$, and let Δ be the Weitzenböck derivation of $K[X, Y]$ defined by $\Delta = \sum_{i=1}^{n} x_i \, \partial/\partial y_i$. The algebra of constants $K[X, Y]^\Delta$ is generated by x_i, $i = 1, \ldots, n$, and $u_{ij} = x_i y_j - x_j y_i$, $1 \leq i < j \leq n$.*

The generators of such a kernel can also be built one at a time. Bedratyuk copies the Casimir elements of a Lie algebra over to a derivation [2005-1, p. 2]. For a derivation D of $k[X]$ he takes two dual D-modules U and U^*, with bases u_i and u_i^*, and forms the element $\Delta(U,U^*) = \sum_i u_i u_i^*$ [2005-1, Definition 2.4, p. 2]. That element lies in the kernel of D [2005-1, Theorem 2.2, p. 2].

Theorem 12.8 ([2005-1, Theorem 2.2, p. 2])**.** *Let D be a derivation of $k[X]$ and let U and U^* be two dual D-modules in $k[X]$. Then the Casimir element $\Delta(U,U^*)$ belongs to the kernel $k[X]^D$.*

For a linear derivation the converse holds, and every homogeneous element of the kernel is such an element [2005-1, Theorem 2.3, pp. 2–3]. The Weitzenböck derivation of Definition 12.5 is linear, so its kernel is generated by Casimir elements [2005-1, pp. 3–4]. That gives an algorithm, and Bedratyuk runs it to list the kernel for at most six variables [2005-1, p. 1].

The kernel theorems of Weitzenböck and Nowicki describe the constants of a linear derivation. Nonlinear families raise a second question: which derivations of $k[x,y]$ generate no invariant ideals at all [2016-9, p. 2031]? A derivation d of a ring is simple when the only d-stable ideals are 0 and the ring itself [2016-9, p. 2031]; The class is named after Shamsuddin, and Lequain characterizes which members of it are simple derivations [2008-8, pp. 801–807].

Definition 12.9 ([2016-9, p. 2033])**.** A derivation d of $k[x,y]$ is a *Shamsuddin derivation* if it has the form

$$d = \partial_x + (a(x)\,y + b(x))\,\partial_y,$$

where $a(x), b(x) \in k[x]$.

Among all derivations of $k[x,y]$, the simple ones turn out to be lonely: Mendes and Pan prove that no nontrivial automorphism of $k[x,y]$ commutes with them. Write $\mathrm{Aut}(D)$ for the subgroup of $\mathrm{Aut}_K(K[x,y])$ whose elements commute with D; this is the *isotropy group* of D [2017-9, p. 2]. Their theorem reads as follows [2017-9, p. 2].

Theorem 12.10 ([2017-9, Theorem 1, p. 2])**.** *Let K be an algebraically closed field of characteristic zero and let D be a simple derivation of $K[x,y]$. Then the isotropy group $\mathrm{Aut}(D)$ of D in $\mathrm{Aut}_K(K[x,y])$ is trivial.*

For Shamsuddin derivations the converse holds under a hypothesis, as the companion theorem states [2017-9, p. 2].

Theorem 12.11 ([2017-9, Theorem 2, p. 2])**.** *Let D be a Shamsuddin derivation $D = \partial_x + (ay + b)\partial_y$ of $K[x, y]$ with $a \neq 0$. Then D is simple if and only if* $\mathrm{Aut}(D) = \{\mathrm{id}\}$.

A third named family comes from ecology [2025-17, p. 2]. It looks quadratic rather than linear, and it models the predator and prey populations through a system of differential equations [2025-17, p. 2].

Definition 12.12 ([2025-17, (1), p. 2])**.** Let K be a field of characteristic zero, $n \geq 3$, and let the indices of the variables $x_1, \ldots, x_n$ of $K[X] = K[x_1, \ldots, x_n]$ be regarded as elements of $\mathbb{Z}_n$. A K-derivation $d\colon K[X] \to K[X]$ of the form

$$d(x_i) = x_i \left(x_{i-1} - C_i\, x_{i+1}\right)$$

where $C_i \in K$ for all $i \in \mathbb{Z}_n$, is a *Lotka–Volterra derivation.*

Lotka–Volterra derivations are not simple [2025-17, p. 2], so Theorem 12.10 does not speak for them. Rewri and Kour compute their isotropy groups directly [2025-17, pp. 21–23].

Theorem 12.13 ([2025-17, Theorem 4.1, p. 21])**.** *Let d be the Lotka–Volterra derivation of $K[X]$ as in Definition 12.12. If $C_i \in K^*$ for all i, then the isotropy group of d is finite.*

When all the coefficients agree and lie in $\{\pm 1\}$, the group is dihedral.

Corollary 12.14 ([2025-17, Corollary 4.3, p. 23])**.** *Let d be the Lotka–Volterra derivation of $K[X]$ as in Definition 12.12. If $C_1 = \cdots = C_n$ and $C_1 \in \{\pm 1\}$, then the isotropy group of d is isomorphic to the dihedral group of order $2n$.*

The monomial family of Definition 12.1 also has a known isotropy group. Mishra, Mondal and Shukla treat a monomial derivation whose exponent matrix has nonzero determinant [2026-27, Theorem 3.1, p. 5]. Such an automorphism acts on the variables one at a time, up to a diagonal change [2026-27, Theorem 3.1, p. 5].

Theorem 12.15 ([2026-27, Theorem 3.1, p. 5])**.** *Let k be a field of characteristic zero and let d be the monomial derivation of $k[x_1, \ldots, x_n]$ with $d(x_i) = \alpha_i x_1^{\beta_{i1}} \cdots x_n^{\beta_{in}}$ for $\alpha_i \in k^*$ and non-negative integers β_{ij}. Put*

$A = (\beta_{ij} - \delta_{ij})$ and suppose that $\beta_{ii} \geq 1$ for all i, that $\det A \neq 0$, and that the field of rational constants of d is k. Let $\rho \in \mathrm{Aut}(k[X])$. Then ρ commutes with d if and only if there are a permutation σ and constants $c_1, \ldots, c_n \in k^*$ with $\rho(x_i) = c_i x_{\sigma(i)}$, $P_\sigma A = A P_\sigma$ and $\prod_t c_t^{a_{ti}} = \alpha_{\sigma(i)}/\alpha_i$ for all i.

For the Jouanolou-type derivation $J = \sum_i x_i^r \partial_{x_{\sigma(i)}}$ the commuting automorphisms are again monomial [2026-27, Theorem 4.2, p. 11]. Each one is fixed by a permutation that commutes with σ and by constants that satisfy $\lambda_{\sigma(i)} = \lambda_i^r$ [2026-27, Theorem 4.2, p. 11]. The full group is then a semidirect product written out cycle by cycle [2026-27, Theorem 4.6, p. 14].

A simple derivation that takes no unit value has a small group too. Mishra, Mondal and Shukla extend a non-invertible simple derivation of an initial block of variables to more variables [2025-16, p. 1]. Under their conditions the commuting group of the extension lies in the translations [2025-16, Theorem 2.4, p. 5].

Theorem 12.16 ([2025-16, Theorem 2.4, p. 5])**.** *Let i, n be positive integers with $i \geq 2$ and $n > i$. Let d_i be a simple derivation of $k[x_1, \ldots, x_i]$ that satisfies the three conditions of the cited theorem, and let d_n extend it to $k[x_1, \ldots, x_n]$ with $d_n(x_j) \in k[x_{j-1}] \setminus k$ for $i + 1 \leq j \leq n$. Then the commuting group of d_n is $\{(x_1, \ldots, x_{n-1}, x_n + c) \mid c \in k\}$.*

So the isotropy group is as small as it can be, a one-parameter group of translations [2025-16, p. 1]. This continues the pattern of Theorem 12.10: simplicity leaves almost no symmetry.

The named families of the section return again in the geometry of polynomial maps. Rentschler classified the planar case in 1968: every nonzero locally nilpotent derivation of $k[x, y]$ is conjugate, by a tame automorphism, to a triangular one [2017-7, Theorem 4.1, p. 113].

Theorem 12.17 ([2017-7, Theorem 4.1, p. 113])**.** *Let $k[x, y] = k^{[2]}$ and let $D \in \mathrm{LND}(k[x, y])$ be nonzero. There exists $p(x) \in k[x]$ and a tame automorphism $\alpha \in \mathrm{GA}_2(k)$ such that $\alpha D \alpha^{-1} = p(x)\, \partial_y$.*

the theorem of Rentschler also gives the theorem of Jung on planar automorphisms as a corollary [2017-7, p. 113]. The Jacobian conjecture closes the section: it asks whether a polynomial endomorphism F of $\mathbb{A}_k^n$ with $\det J(F)$ a nonzero constant is an automorphism [2007-2, p. 1]. The conjecture has a formulation in derivations: if the Jacobian derivation Δ_f of $f = (f_1, \ldots, f_{n-1})$ has a slice s, then $k[f_1, \ldots, f_{n-1}, s]$ is the whole

polynomial ring [2017-7, p. 81]. The section that follows moves the same question to the Weyl algebra, where it is the Dixmier conjecture [1968-1, p. 241].

12.2 The Weyl algebra and the problems of Dixmier

Section 12.1 closed with the Jacobian conjecture. This section follows it into the noncommutative world: the polynomial ring becomes the algebra of Definition 12.18 below. The conjecture becomes the first problem of Dixmier [1968-1, p. 241]. Dixmier named the algebra in 1968, after a suggestion of Segal [1968-1, p. 209].

Definition 12.18 ([1968-1, p. 209])**.** Let k be a commutative field of characteristic 0 and $n \geq 1$ an integer. The *n-th Weyl algebra* $A_n(k)$, or A_n, is the unital associative k-algebra with $2n$ generators $p_1, q_1, \ldots, p_n, q_n$ and the relations

$$[p_i, q_i] = 1, \qquad [p_i, p_j] = [q_i, q_j] = [p_i, q_j] = 0 \quad (i \neq j),$$

and $A_0(k) = k$.

The ordered products $p_1^{i_1} \cdots p_n^{i_n} q_1^{j_1} \cdots q_n^{j_n}$ form a basis of A_n over k, and A_n is a simple domain with center k, left and right Noetherian [1968-1, p. 210]. The algebra is a model of the derivative: A_n is the algebra of polynomial differential operators on affine n-space, with q_i acting as multiplication by x_i and p_i as $\partial/\partial x_i$ [2007-2, p. 1]. The derivations of the algebra are all inner [1968-1, p. 210].

Proposition 12.19 ([1968-1, p. 210])**.** *Every derivation of the Weyl algebra A_n is inner.*

The automorphisms are harder: the exponential of the locally nilpotent derivation $\mathrm{ad}(\lambda q^{n+1})$ gives a well-defined automorphism of A_1 [1968-1, p. 210]. Dixmier posed six problems at the end of the paper [1968-1, p. 241], and the first is still open.

Remark 12.20 ([1968-1, Problème 11.1, p. 241]). Dixmier asked whether every endomorphism of A_1 is an automorphism: if P and Q in A_1 satisfy $[P, Q] = 1$, do they generate A_1? The question in rank n is now called the Dixmier conjecture [2007-2, p. 1].

Belov-Kanel and Kontsevich tied the conjecture back to the polynomial thread of Section 12.1. Their theorem makes the two conjectures one question [2007-2, p. 2].

Theorem 12.21 ([2007-2, Theorem 1, p. 2])**.** *The Jacobian conjecture in dimension $2n$ implies the Dixmier conjecture in rank n. In particular the stable conjectures JC_∞ and DC_∞ are equivalent.*

The implication $DC_n \Rightarrow JC_n$ was already known [2007-2, p. 1]. Theorem 12.21 is the reverse, so the two threads of the chapter close on the same question.

The conjecture is open, and this section states it as a problem.

Problem 12.22 ([2007-2, p. 1])**.** Let $A_{n,k}$ be the n-th Weyl algebra over a field k of characteristic 0. Is every algebra endomorphism of $A_{n,k}$ an automorphism?

Bavula notes that the Dixmier conjecture has been open since 1968 for every $n \geq 1$ [2017-3, pp. 1–2]. Belov-Kanel and Kontsevich prove that the Jacobian conjecture in dimension $2n$ implies the Dixmier conjecture in dimension n, so the two stable forms are equivalent [2007-2, Theorem 1, p. 3].

The next section builds the same algebra operator by operator, as the differential operators of the polynomial ring itself [2013-9, Theorem 6.2, p. 4].

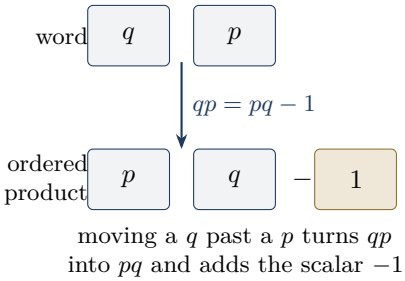


Figure 12.2: The relation $[p, q] = 1$ of Definition 12.18 [1968-1, p. 209], written out for one pair as $qp = pq - 1$. The ordered products $p^i q^j$ form a basis [1968-1, p. 210], so a word is brought to that order by moving each q past the copies of p, and every crossing adds the scalar -1.

12.3 Rings of differential operators

Section 12.2 gave the Weyl algebra by generators and relations. This section builds the same algebra by operators. The order of an operator counts how many commutators kill it, and the recursive characterization of Grothendieck makes that count a definition [2013-9, Theorem 6.2, p. 4]. The characterization is the affine case of EGA IV, propositions 16.8.8 and 16.8.9 [2013-9, p. 5].

Theorem 12.23 ([2013-9, Theorem 6.2, p. 4])**.** *Let D be a k-module endomorphism of the commutative k-algebra A. Then D is a differential operator of order $\leq n$ if and only if for every $(n+1)$-tuple of elements $x_0, x_1, \ldots, x_n \in A$ the iterated commutator*

$$\operatorname{ad}(x_0)\operatorname{ad}(x_1)\cdots\operatorname{ad}(x_n)D \;=\; 0.$$

The differential operators of a commutative algebra form an algebra under composition: the product of an operator of order $\leq n$ and an operator of order $\leq n'$ has order $\leq n+n'$ [2013-9, 6.4, p. 5]. The filtered algebra they form is denoted $\operatorname{Diff}(A)$ [2013-9, 6.4, p. 5]. On $\mathbb{Q}[X,Y]$ this is exactly the naïve algebra of finite sums of the monomial operators $f_{i,j}\partial_X^i\partial_Y^j$; on $\mathbb{Z}[X,Y]$ it is larger, since the divided powers $\frac{1}{i!}\partial_X^i$ enter [2013-9, 6.4, p. 5].

Sweedler carried the same commutator rule to noncommutative algebras and named the right derivation [1980-2, p. 29].

Remark 12.24 ([1980-2, p. 29]). A *right derivation* is a linear map $d\colon A \to N$ with $d(ab) = d(a)b + d(b)a$. Right differential operators of order n are defined inductively: $D_A^0(M,N) = \operatorname{Hom}_A(M,N)$, and $D_A^n(M,N)$ is the set of $f \in \operatorname{Hom}(M,N)$ with $[f,a] \in D_A^{n-1}(M,N)$ for all $a \in A$, where $[f,a](m) = f(ma) - f(m)a$ [1980-2, p. 29].

The module of order-one operators splits into the multiplications and the right derivations [1980-2, p. 29], which places the right derivation beside the derivation of Section 2.1 in the family tree of Section 2.8.

The polynomial case closes the circle: the algebra of polynomial differential operators on affine n-space is the Weyl algebra A_n of Definition 12.18 [2007-2, p. 1]. The generators of the last section are the multiplications and the partial derivatives of this one. On the Weyl algebra every derivation is inner by Proposition 12.19. The next section varies the ground ring instead of the operators, and builds polynomials over a derivation itself [1933-1, p. 480].

Further work in the differential-operator line, held in the collection: *Representations of finite-dimensional Lie algebras by first-order differential*

operators Some local results in the transitive case [1995-1]. *Derivation based differential calculi for noncommutative algebras deforming a class of three dimensional spaces* [2018-2]. *Infinite-order differential operators in the module of formal generalized functions and in a ring of formal power series* [2022-1].

12.4 Differential polynomial rings

Section 12.3 built operators on a fixed ring. This section moves the derivation into the multiplication: a polynomial variable acts on the coefficient ring through a derivation. Ore built the general skew polynomial ring, which the book also calls the Ore extension, in 1933 [1933-1, p. 480], and Section 6.4 recalls his construction. The differential twist of his Section 2 is the case of this section [1933-1, p. 482]. The coefficient ring is commutative, the conjugation is the identity, and the twist of a by x is the derivative a' [1933-1, p. 482].

Remark 12.25 ([1933-1, p. 482]). Ore states the differential case as follows. In the theory of linear differential equations the coefficient field K is commutative, and the conjugate $\bar{a}$ equals a. The derivative then satisfies $(a+b)' = a' + b'$ and $(ab)' = ab' + a'b$, and it has the ordinary properties of a derivative [1933-1, p. 482]. In the theory of difference equations the derivative is 0 and the conjugate is not [1933-1, p. 482].

The differential Ore extension of a ring R by a derivation d is denoted $R[X; d]$. Its elements are polynomials in X with coefficients in R. The multiplication rule is $Xa = aX + d(a)$ [1933-1, p. 481, (3)]. The Weyl algebra A_1 of Section 12.2 is the first example, since A_1 is the algebra of polynomial differential operators in one variable [2007-2, p. 1].

A ring R with a derivation d is *d-simple* when every d-ideal of R is trivial [2008-9, p. 801]. Goodearl and Warfield call the same extension a formal differential operator ring $R[x; \delta]$ [2004-4, p. 28].

Proposition 12.26 ([2004-4, Proposition 2.1, p. 31]). *Let R be a $\mathbb{Q}$-algebra and δ a derivation on R. Then $R[x; \delta]$ is a simple ring if and only if R is δ-simple and δ is outer.*

Lequain gives the criterion in a form ready for computation, and recalls the criterion of Shamsuddin for one step further, where the derivation extends to $R[Y]$ with $d(Y) = aY + b$ [2008-9, p. 802].

Theorem 12.27 (Shamsuddin, [2008-9, Theorem 2.1, p. 802])**.** *Let R be a ring, Y an indeterminate over R and d a derivation of $R[Y]$ such that $d(R) \subseteq R$, R is d-simple and $d(Y) = aY + b$ with $a, b \in R$. Then the following are equivalent:*

1. *$R[Y]$ is d-simple;*

2. *the equation $d(Z) = aZ + b$ has no solution in R.*

The concrete ground for this chapter is the Ore extension $A = k[x][t; d]$, with d a nonzero derivation of $k[x]$. Over a field of characteristic zero, every local derivation of this ring is a derivation, by Theorem 12.28 below.

Theorem 12.28 ([2004-8, Theorem 4.1, p. 4569])**.** *Let $k[x]$ be the polynomial ring in one variable over a field k of characteristic zero, and let d be a nonzero derivation of $k[x]$. Let $A = k[x][t; d]$ be the Ore extension of $k[x]$ with respect to d. Every local derivation of A is a derivation.*

The local derivations of Section 2.1 therefore give nothing new on this ring. Chapter 17 follows polynomial derivations into the geometry of the Jacobian question that Section 12.1 left open [2007-2, p. 1].

Chapter 13

Group algebras, Hopf algebras and quantum groups

Group rings came with a question about derivations from the start. Derivations of group algebras of torsion-free nilpotent groups were studied by 1978 [1995-3, p. 434]. From then on the field asks one question in many settings: which derivations of a group ring or a group algebra are inner. Spiegel answered it for the integral group ring of a finite group in 1994 [1994-5, p. 2955]. Johnson asked it for the Banach algebra $L^1(G)$ of a locally compact group [2008-10, p. 221]. Each answer says where a new, outer derivation can live.

This chapter follows the question through four settings. Section 13.1 treats the group ring itself. The integral group ring of a finite group has only inner derivations there. An abelian group splits off a free part, and the free part measures the outer derivations. An infinite direct sum of finite p-groups has an outer derivation over every field. The derivation algebra is graded by the quotient of the group by a normal subgroup with abelian quotient [2023-7, Theorem 1, pp. 1, 8]. Over a finite group the question becomes a block question, and the first Hochschild cohomology of a block with nontrivial defect group is often nonzero [2026-3, Theorem A, p. 2]. Section 13.3 moves to a Hopf algebra, where a derivation meets the coproduct. The twisted derivations of Davydov are shown to be symmetries of the category of representations, taken up to first order. They are computed for a universal enveloping algebra and for the four-dimensional Hopf algebra. Section 13.4 turns to quantum groups, where Woronowicz builds the differential calculus. A derivation into a bimodule is the basic object, and the calculi on a compact matrix pseudogroup are classified by right ideals. The book defines

each technical term where it first appears. Section 13.5 ends with crossed homomorphisms, the group-map form of a skew derivation, and with the theorem of Losert that every derivation of $L^1(G)$ is inner. The thread through all four sections is the same map with a new symmetry attached at each step.

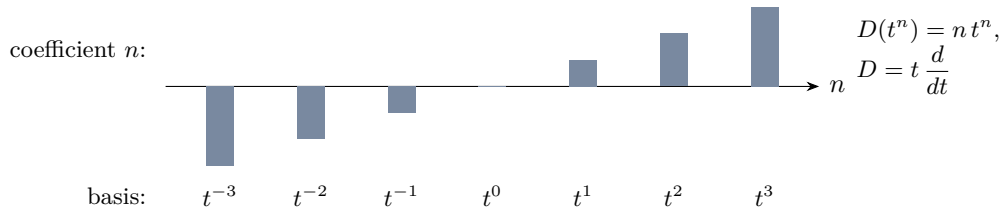


Figure 13.1: The group algebra $k[\mathbb{Z}] = k[t, t^{-1}]$. The homomorphism $n \mapsto n$ from $\mathbb{Z}$ to $(k, +)$ gives the derivation $D(t^n) = n\,t^n$, that is $D = t\,d/dt$.

13.1 Group algebras and group rings

Group rings join a group to a ring, and the elements of the group now act on the product rule. The group ring RG of a group G over a commutative ring R with 1 is the free R-module on G, with the product extended from G [1994-5, p. 2956]. Write D for a derivation of RG. On a group element the Leibniz rule reads $D(xy) = D(x)y + xD(y)$, so the values of D on G decide its values on whole of RG by Lemma 13.1.

Lemma 13.1 ([1994-5, Lemma 1, p. 2956])**.** *Let R be a commutative ring with 1, G a group, and $D\colon G \to RG$ a map with $D(xy) = D(x)y + xD(y)$ for all $x, y \in G$. Then D extends to a derivation of RG.*

Spiegel proves the extension formula by linearity: write an element of RG as a finite sum $\sum r_g g$ and send it to $\sum r_g D(g)$ [1994-5, p. 2956]. He then asks which derivations of the integral group ring $\mathbb{Z}G$ occur, first for a finite group [1994-5, p. 2955].

Theorem 13.2 ([1994-5, Theorem 1, p. 2956])**.** *Let $\mathbb{Z}G$ be the integral group ring of the finite group G. Every derivation of $\mathbb{Z}G$ is inner.*

The proof extends a derivation D of $\mathbb{Z}G$ to the rational group algebra $\mathbb{Q}G$ by Lemma 13.1. The theorem of Maschke makes $\mathbb{Q}G$ semisimple [1994-5, p. 2956]. On each simple component the derivation is linear over the centre, so a consequence of the Noether–Skolem theorem makes it inner [1994-5,

pp. 2956–2957]. Spiegel then clears the fractions [1994-5, pp. 2957–2958]: $D = D_f$ with $f \in \mathbb{Q}G$ and $D(h) \in \mathbb{Z}G$ for $h \in G$ give $(r_g - r_{h^{-1}gh}) \in \mathbb{Z}$. The fractional parts of the coefficients of f form a central element, so f may be taken inside $\mathbb{Z}G$ [1994-5, pp. 2957–2958]. For an abelian group G there is more to say: $L(x) = D(x)x^{-1}$ is a homomorphism from G to the additive group of $\mathbb{Z}G$ [1994-5, p. 2958]. The image $L(G)$ is free abelian [1994-5, p. 2958]. Take a basis of $L(G)$ and choose one element of G above each basis element; these elements and $\ker L$ give the direct product [1994-5, p. 2958].

Theorem 13.3 ([1994-5, Theorem 2, pp. 2958–2959])**.** *Let G be an abelian group.*

1. *If D is a derivation of $\mathbb{Z}G$, there exist subgroups A, B of G, with A a free abelian group having a basis $\{a_i \mid i \in I\}$, and a linearly independent set $\{y_i \mid i \in I\}$ of the additive group of $\mathbb{Z}G$ such that G is the direct product of A and B, and if $g \in G$ with $g = ab$, $a \in A$, $b \in B$ and $a = a_{i_1}^{e_1} \cdots a_{i_n}^{e_n}$, then $D(g) = (e_1 y_{i_1} + \cdots + e_n y_{i_n})g$.*

2. *Conversely, if G is the direct product of groups A, B with A a free abelian group with basis $\{a_i \mid i \in I\}$, and $\{y_i \mid i \in I\}$ is a set of linearly independent elements of the additive group of $\mathbb{Z}G$, then the mapping $D\colon G \to \mathbb{Z}G$ given by $D(g) = (e_1 y_{i_1} + \cdots + e_n y_{i_n})g$ extends to a derivation of $\mathbb{Z}G$. Here $g = ab$, $a \in A$, $b \in B$ and $a = a_{i_1}^{e_1} \cdots a_{i_n}^{e_n}$.*

So a derivation of $\mathbb{Z}G$ for G abelian is a choice of a free factor of G and of a free family in $\mathbb{Z}G$. Torsion-free factors give outer derivations [1994-5, Theorem 2, pp. 2958–2959].

So the story now moves from $\mathbb{Z}$ to a semiprime coefficient ring. Ferrero, Giambruno and Polcino Milies take a ring with unity R and call a derivation $d\colon RG \to RG$ an R-derivation when $d(R) = 0$ [1995-3, p. 434]. They also record a decomposition [1995-3, p. 434]: a derivation d of RG with $d(R) \subseteq R$ writes as $d = d_1 + d_2$. Here d_1 is a derivation of R extended by $d_1(G) = 0$, and d_2 is an R-derivation of RG. Their main theorem puts a finiteness condition on the centre.

Theorem 13.4 ([1995-3, Theorem 1.1, p. 434])**.** *Let R be a semiprime ring and G a torsion group such that $[G : Z(G)] < \infty$, where $Z(G)$ denotes the center of G. Suppose that either* $\operatorname{char} R = 0$ *or for every characteristic p of R, p does not divide $o(g)$, for all $g \in G$. Then every R-derivation of RG is inner.*

The proof first treats a finite group G [1995-3, pp. 434–436]. The ring R embeds into a direct sum of algebraically closed fields, where the theorem of Maschke applies [1995-3, p. 435]. The general case reduces to a finite subgroup generated by the supports of the values of d [1995-3, pp. 434–436]. Section 3 of the paper shows the hypothesis on G is needed.

Example 13.5 ([1995-3, §3, p. 437])**.** Let H be a finite non-abelian p-group and let $G = \bigoplus_i H_i$ be the direct sum of infinitely many copies of H. Then G is a locally finite nilpotent group. Let F be any field, and choose a non-central element x_i of H_i for each i. The map $d\colon FG \to FG$ with $d(a) = \sum_{i \geq 1} [x_i, a]$ is a derivation of FG: on FG_n it is the inner derivation induced by $x_1 + \cdots + x_n$. It is not an inner derivation of FG, since $d(H_i) \neq 0$ for all indices i.

So an infinite group can carry a derivation that is inner on every finitely generated subgroup but outer on the whole group.

Manju and Sharma count the inner derivations over any field, and they read the count from the class sums [2026-24, p. 5]. The class sums form a basis of the centre of FG [2026-24, p. 5]. The proof extends that basis to a basis of FG [2026-24, p. 5]. The inner derivations indexed by the added elements then form a basis of the inner ones [2026-24, p. 5].

Theorem 13.6 ([2026-24, Theorem 2.15, p. 5])**.** *Let F be an arbitrary field of any characteristic and G a finite group of order n having r conjugacy classes. Then the dimension of the space of inner derivations of FG is $n - r$. If $C_1, \ldots, C_r$ are the distinct conjugacy classes of G with representatives $x_1, \ldots, x_r$ respectively and s is a positive integer such that $|C_i| = 1$ for all $i \in \{1, \ldots, s\}$ and $|C_i| \geq 2$ for all $i \in \{s+1, \ldots, r\}$, then the set $B_0 = \{d_g \mid g \in \bigcup_{i=s+1}^{r} (C_i \setminus \{x_i\})\}$ is an F-basis of the space of inner derivations of FG.*

Here d_g is the inner derivation $d_g(\alpha) = \alpha g - g\alpha$ [2026-24, Definition 2.2, p. 3]. So the count of Theorem 13.6 is fixed by the class number alone, for every finite group [2026-24, p. 5]. The paper then works the count out for three families and asks when every derivation is inner [2026-24, Corollary 3.6, p. 10]. The dihedral group is $D_{2n} = \langle a, b \mid a^n = 1,\ b^2 = 1,\ (ab)^2 = 1 \rangle$ of order $2n$, with $n \geq 3$ [2026-24, p. 6].

Corollary 13.7 ([2026-24, Corollary 3.6, p. 10])**.** *Let F be an algebraic extension of a prime field and p an odd rational prime.*

1. *If* $\operatorname{char} F$ *is either* 0 *or* p *with* $\gcd(n,p)=1$*, then every derivation of* FD_{2n} *is inner.*

2. *If* $\operatorname{char} F = p$ *with* $\gcd(n,p)\neq 1$*, then the inner derivations form a proper subspace of the space of all derivations of* FD_{2n}*.*

The authors prove the two cases by counting both spaces [2026-24, Theorem 3.5, p. 10]. Their count of the inner derivations of FD_{2n} is $3\lfloor (n-1)/2 \rfloor$ over any field [2026-24, Theorem 3.5, p. 10]. Their count of all derivations of FD_{2n} equals it exactly when $\operatorname{char} F$ is 0 or a prime not dividing n [2026-24, Theorem 3.3, p. 7]. A prime dividing the group order therefore leaves room for outer derivations on a finite group algebra [2026-24, Corollary 3.6, p. 10].

The derivation algebra itself carries a grading. Arutyunov and Zhiltsov identify a derivation with a character of the groupoid of the adjoint action [2023-7, p. 1]. Fix a normal subgroup N of G with G/N abelian. Then the derivations split into pieces indexed by the cosets of N [2023-7, p. 1].

Theorem 13.8 ([2023-7, Theorem 1, pp. 1, 8])**.** *Let* G *be a group and let* N *be a normal subgroup of* G *with* G/N *abelian and* $|G/N|>1$*. Then the derivation algebra of the group algebra is graded by* G/N*:*

$$\operatorname{Der} = \bigoplus_{k\in G/N} \operatorname{Der}_k, \qquad [\operatorname{Der}_k, \operatorname{Der}_l] \subset \operatorname{Der}_{kl} \quad \text{for all } k,l \in G/N.$$

The piece Der_k collects the derivations whose character is supported on the coset k [2023-7, p. 1]. Every group that is not perfect has a nontrivial grading, because such a group has a proper normal subgroup with abelian quotient [2023-7, p. 1].

Arutyunov and Naianzin carry the derivation problem to a family of groups that they call BC-groups [2023-8, p. 2]. The defining condition bounds the diameter of the conjugate of any finite set [2023-8, p. 2]. For such a group every bounded derivation from the group ring into an ℓ^q space is inner [2023-8, Theorem 2, p. 2]. This extends the case of Johnson, who settled the derivations of $\ell^1(G)$ for a discrete group [2023-8, p. 2].

The question has a block form for finite groups. Let kG be a group algebra over a field k of characteristic p, and let B be a block of kG with defect group P [2026-3, p. 2]. Linckelmann made the conjecture that the first Hochschild cohomology of B is nonzero whenever P is nontrivial [2026-3, Conjecture, p. 2]. Antrobus, Briggs and Rubio y Degrassi prove many cases of it [2026-3, Theorem B, p. 2].

Theorem 13.9 ([2026-3, Theorem A, p. 2])**.** *Let B be a block of kG with defect group P. Then the dimension of $HH^1(B)$ is at least the number of conjugacy classes of extra-strong non-Schur elements of G relative to P.*

The cases include the principal blocks with abelian defect group, and all blocks of the symmetric and alternating groups. They include the blocks of finite groups of Lie type in defining characteristic, and the blocks of general linear groups in any characteristic [2026-3, Theorem B, p. 2]. When $p > 5$ the conjecture holds for all blocks of maximal defect [2026-3, Theorem C, p. 3].

13.2 Harmonic analysis: derivations on group rings

Fox built the harmonic analysis of a group ring in 1953 [1953-1]. The classical theory begins with a definition of derivation that is the Leibniz rule on the group itself [1953-1, equations (1.1)–(1.2), p. 549].

Definition 13.10 ([1953-1, equations (1.1)–(1.2), p. 549])**.** Let JG be the group ring of a group G over a commutative ring J with unit. By a *derivation* in JG is meant any mapping D of JG into itself which satisfies

$$D(u+v) = Du + Dv, \qquad D(u * v) = Du \cdot v^0 + u * Dv \quad \text{for } u, v \in JG,$$

where $v^0 \in J$ is the coefficient sum of v.

For elements of the group itself, the second law takes the simpler form $D(gh) = Dg + g\,Dh$ [1953-1, (1.2)', p. 549], which is the Leibniz rule for the group multiplication. Fox shows the set of derivations in JG forms a right JG-module, with $(D \cdot v)(u) = Du\,v$ [1953-1, p. 549]. On the free group ring, the structure is fully described by [1953-1, Theorem, equations (2.1) and (2.2), p. 550].

Theorem 13.11 ([1953-1, Theorem, equations (2.1) and (2.2), p. 550])**.** *Let X be a free group with generators $(x) = (x_1, x_2, \dots)$ and let JX be its group ring. To each generator x_j of X there corresponds a derivation $f(x) \mapsto D_j f(x) = f_{x_j}(x) = \partial f(x)/\partial x_j$, called the* derivative with respect to x_j *[1953-1, p. 550], which has the property $\partial x_k/\partial x_j = \delta_{jk}$. Furthermore, there is one and only one derivation $f(x) \mapsto f'(x)$ mapping $x_1, x_2, \dots$ into prescribed elements $h_1(x), h_2(x), \dots$ of JX; it is given by $f'(x) = \sum_j f_{x_j}(x) \cdot h_j(x)$.*

On the continuous side, the same rule governs the operator algebras of Chapter 19, where each derivation of a von Neumann algebra is inner [1966-1, Theorem 15, p. 290].

Section 13.3 moves from the group ring to a Hopf algebra, where the coproduct itself asks a derivation to respect a second operation.

13.3 Twisted derivations of Hopf algebras

The group ring of Section 13.1 is cocommutative: every group element g satisfies $\Delta(g) = g \otimes g$. In a bialgebra H a derivation $d\colon H \to H$ deals only with the product, and the element $\Delta(d(x))$ need not equal $(I \otimes d + d \otimes I)(\Delta(x))$: Davydov lets the difference be the commutator $[\varphi, \Delta(x)]$ for a tensor φ [2013-4, p. 572]. He calls a pair (d, φ) a twisted derivation [2013-4, p. 572].

Definition 13.12 ([2013-4, p. 572])**.** A *twisted derivation* of a bialgebra H is a pair (d, φ), where $d\colon H \to H$ is an algebra derivation and φ is an element of $H \otimes H$, such that

$$\begin{aligned}(I \otimes d + d \otimes I)(\Delta(x)) - \Delta(d(x)) &= [\varphi, \Delta(x)] \quad \text{for all } x \in H,\\ 1 \otimes \varphi + (I \otimes \Delta)(\varphi) &= \varphi \otimes 1 + (\Delta \otimes I)(\varphi),\\ \varepsilon d &= 0,\\ (\varepsilon \otimes I)(\varphi) &= (I \otimes \varepsilon)(\varphi) = 0.\end{aligned}$$

The set of twisted derivations of H is denoted by $\mathrm{Dert}_w(H)$.

Twisted derivations appear as infinitesimal symmetries of the category of representations [2013-4, p. 567]. They are the infinitesimal versions of the twisted automorphisms of the bialgebra [2013-4, p. 567]. They also form a Lie algebra, the tangent algebra of the group of twisted automorphisms [2013-4, p. 567]. There is a gauge action [2013-4, pp. 567–568]: an element $a \in H$ sends (d, φ) to $(d + [a, -],\ \varphi + a \otimes 1 + 1 \otimes a - \Delta(a))$. The quotient of $\mathrm{Dert}_w(H)$ by this action is the Lie algebra of outer twisted derivations $\mathrm{OutDert}_w(H)$ [2013-4, pp. 567–568]. For a universal enveloping algebra the computation is complete over a field of characteristic zero.

Theorem 13.13 ([2013-4, Theorem 2.1, p. 578])**.** *Let $\mathfrak{g}$ be a Lie algebra over a field of characteristic zero. Then the Lie algebra of outer twisted derivations of $U(\mathfrak{g})$ has a form:*

$$\mathrm{OutDert}_w(U(\mathfrak{g})) \simeq \mathrm{OutDer}(\mathfrak{g}) \ltimes (\Lambda^2 \mathfrak{g})^{\mathfrak{g}},$$

where the crossed product is taken with respect to the natural action of $\mathrm{OutDer}(\mathfrak{g})$ *on the Abelian Lie algebra* $(\Lambda^2\mathfrak{g})^{\mathfrak{g}}$.

The proof gauges φ to a skew-symmetric tensor and then uses cocommutativity [2013-4, pp. 577–578]. In the first equation of Definition 13.12 one side is symmetric and the other is skew-symmetric, so both vanish. The twisted derivation then separates into a bialgebra derivation and an invariant twist [2013-4, pp. 577–578]. His second example is small and not cocommutative: the four-dimensional Hopf algebra [2013-4, p. 580].

Example 13.14 ([2013-4, p. 580])**.** the Hopf algebra H_4 is the algebra

$$H_4 = k\langle g, x \mid g^2 = 1,\ x^2 = 0,\ gx + xg = 0\rangle.$$

It is a Hopf algebra with the comultiplication $\Delta(g) = g \otimes g$, $\Delta(x) = 1 \otimes x + x \otimes g$, the counit $\varepsilon(g) = 1$, $\varepsilon(x) = 0$, and the antipode $S(g) = g$, $S(x) = gx$.

On this Hopf algebra the twisted derivations form a Lie algebra.

Theorem 13.15 ([2013-4, Theorem 2.14, p. 580])**.** *The algebra* $\mathrm{OutDert}_w(H_4)$ *of twisted derivations of the Hopf algebra is a two-dimensional non-Abelian Lie algebra.*

The two generators are the bialgebra derivation d with $d(g) = 0$ and $d(x) = x$, and the invariant infinitesimal twist $\psi = x \otimes gx$; their bracket is $[(d, 0), (0, \psi)] = (0, 2\psi)$ [2013-4, p. 580]. So a non-cocommutative coproduct ties the algebra derivations and the twists together. Section 13.4 follows the same idea into a quantum group, where the derivation is asked to respect the left and right shifts of the group.

13.4 Quantum groups and the differential calculus of Woronowicz

the algebra of Sweedler of Section 13.3 is the simplest quantum semigroup. A quantum group in the sense of Woronowicz is a compact matrix pseudogroup: an algebra A of smooth functions on the group with a comultiplication [1989-4, p. 126]. On a Lie group the exterior derivative satisfies the Leibniz rule, and the left and right shifts of the group act on the differential forms [1989-4, pp. 126–127]. Woronowicz takes these two properties as axioms and lets the derivative range over a bimodule: for a given quantum group there exist many non-equivalent differential calculi [1989-4, p. 126].

Definition 13.16 ([1989-4, Definition 1.1, p. 128])**.** Let A be an algebra with unity, Γ be a bimodule over A and $d\colon A \to \Gamma$ be a linear map. We say that (Γ, d) is a *first order differential calculus* over A if

1. for any $a, b \in A$, $d(ab) = (da)b + a\,db$;
2. any element $\rho \in \Gamma$ is of the form $\rho = \sum_{k=1}^{K} a_k\,db_k$, where $a_k, b_k \in A$ and K is a natural number.

So a first order differential calculus is a derivation into a bimodule, with the bimodule generated by the image. Let A^2 be the kernel of the multiplication map $m\colon A \otimes A \to A$, made an A-bimodule through the first factor, and set $Db = 1 \otimes b - b \otimes 1$. Then (A^2, D) is a first order differential calculus over A, and it is universal [1989-4, p. 128].

Proposition 13.17 ([1989-4, Proposition 1.1, pp. 128–129])**.** *Let $\mathcal{N}$ be a sub-bimodule of A^2, $\Gamma = A^2/\mathcal{N}$, π be the canonical epimorphism $A^2 \to \Gamma$ and $d = \pi \circ D$. Then (Γ, d) is a first order differential calculus over A. Any first order differential calculus over A can be obtained in this way.*

So the calculi over a fixed algebra A are the quotients of the universal calculus (A^2, D). On a quantum group the calculus is asked to be left-covariant: the left shifts of the group lift to Γ [1989-4, p. 132]. Woronowicz classifies such calculi by right ideals of A contained in the kernel of the counit ε [1989-4, p. 132].

Theorem 13.18 ([1989-4, Theorem 1.5, p. 132])**.** *Let $\mathcal{N}$ be a right ideal of A contained in $\ker\varepsilon$ and $\mathcal{N}' = r^{-1}(A \otimes \mathcal{N})$. Then $\mathcal{N}'$ is a sub-bimodule of A^2. Moreover, let $\Gamma = A^2/\mathcal{N}'$, π be the canonical epimorphism $A^2 \to \Gamma$ and $d = \pi \circ D$. Then the first order differential calculus (Γ, d) is left-covariant. Any left-covariant first order differential calculus on G can be obtained in this way.*

For a Lie group the choice $\mathcal{N} = (\ker\varepsilon)^2$ recovers the usual differential calculus [1989-4, p. 132]. For covariance under both shifts the ideal $\mathcal{N}$ has to be ad-invariant [1989-4, p. 134]. A linear subset $T \subset A$ is ad-invariant when $\mathrm{ad}(T) \subset T \otimes A$, where ad is the adjoint coaction [1989-4, p. 133].

Theorem 13.19 ([1989-4, Theorem 1.8, p. 134])**.** *Let $\mathcal{N}$ be a right ideal of A contained in $\ker\varepsilon$ and (Γ, d) be the left-covariant first order differential calculus described in Theorem 13.18. Then (Γ, d) is bicovariant if and only if $\mathcal{N}$ is ad-invariant.*

The next example is a calculus that is not bicovariant.

Example 13.20 ([1989-4, p. 132, Example 2, p. 134, remark])**.** The 3D-calculus on the twisted $SU(2)$ group $SU_\mu(2)$ is left-covariant; its right ideal $\mathcal{N}$ is generated by six elements, and $\dim(\ker\varepsilon/\mathcal{N}) = 3$. This calculus is not bicovariant, while the standard differential calculus on a Lie group is bicovariant.

A bicovariant calculus then lifts to higher order forms, so the exterior derivative is well defined on the whole external algebra [1989-4, p. 125]. The classifications of this section all pass through quotient bimodules of $A \otimes A$: Proposition 13.17 for a bare algebra, and Theorem 13.18 and Theorem 13.19 for a quantum group. Section 13.5 returns to the group itself: a skew derivation of a group ring is a crossed homomorphism on G, and the theorem of Losert makes each of them principal [2008-10, p. 222].

13.5 Crossed products and crossed homomorphisms

Section 13.4 ended with quotients of the kernel of the multiplication map. The same tensor $1 \otimes b - b \otimes 1$ has a group form, the difference $\mu - x \circ \mu$ of Definition 13.22 below. When G acts on the ground ring, the Leibniz rule restricted to group elements reads $D(xy) = D(x)y + xD(y)$, and the map $x \mapsto D(x)x^{-1}$ of Section 13.1 becomes a crossed homomorphism. We first recall the cocycle that builds a crossed product.

Definition 13.21 ([2013-4, second condition, p. 567])**.** An element φ of $H \otimes H$ satisfies the *2-cocycle condition* for the comultiplication Δ of a bialgebra H when

$$1 \otimes \varphi + (I \otimes \Delta)(\varphi) = \varphi \otimes 1 + (\Delta \otimes I)(\varphi).$$

The cocycle condition is the second equation of a twisted derivation, Definition 13.12; it says that the first components of φ in a coassociative expansion agree. Its group version is older: Losert calls it a crossed homomorphism, with a left Banach G-module and measures in place of tensors [2008-10, p. 222].

Definition 13.22 ([2008-10, p. 222])**.** Let Ω be a locally compact space and G a discrete group acting on Ω by homeomorphisms. A mapping $\Phi\colon G \to M(\Omega)$ (or more generally, $\Phi\colon G \to X$, where X is a left Banach

G-module) is called a *crossed homomorphism* if $\Phi(xy) = \Phi(x) + x \circ \Phi(y)$ for all $x, y \in G$. The map Φ is called *bounded* if $\sup_{x \in G} \|\Phi(x)\| < \infty$. For $\mu \in M(\Omega)$, the special example $\Phi_\mu(x) = \mu - x \circ \mu$ is called a *principal crossed homomorphism*. It is the principal derivation of Section 2.4 under its third name, and the three are one object: an inner derivation of a ring is the same map as a principal crossed homomorphism of a group.

In $\Phi_\mu = \mu - x \circ \mu$ the measure μ plays the role of the inducing element of an inner derivation, and Losert proves the converse for every group: Theorem 13.23 below makes every bounded crossed homomorphism principal. The derivation problem of Johnson asks whether every derivation $D\colon L^1(G) \to L^1(G)$ is inner, induced by a bounded measure [2008-10, p. 221]. The question goes back to Williamson around 1965, and Johnson settled the discrete groups, the amenable groups and the connected groups [2008-10, p. 221]. Losert proves the full statement through crossed homomorphisms [2008-10, p. 222].

Theorem 13.23 ([2008-10, Theorem 1.1, p. 222])**.** *Let Ω be a locally compact space, G a discrete group with a left action of G on Ω by homeomorphisms. Then any bounded crossed homomorphism Φ from G to $M(\Omega)$ is principal. There exists $\mu \in M(\Omega)$ with $\|\mu\| \leq 2\|\Phi\|$ such that $\Phi = \Phi_\mu$.*

So a derivation $D\colon L^1(G) \to M(G)$, having values in $L^1(G)$ and being bounded, defines a bounded crossed homomorphism from G to $M(G)$ for the conjugation action of G on G; Theorem 13.23 makes it principal, and the inducing measure gives the inner derivation [2008-10, p. 222].

Corollary 13.24 ([2008-10, Corollary 1.2, p. 222])**.** *Let G denote a locally compact group. Then any derivation $D\colon L^1(G) \to M(G)$ is inner.*

So the chapter closes where it began: a derivation of a group algebra is again the bracket with a single element, once the group and the norm allow it. From the integral group ring of a finite group to the measure algebra of a locally compact group, only three ways for a derivation to fail to be inner appeared. So they are the free factors of Theorem 13.3, the infinite sum of Example 13.5, and the twists of Section 13.3.

Part V

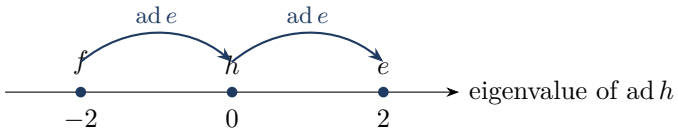
Nonassociative algebras

Chapter 14

Derivations of Lie algebras

Jacobson began the study of derivations as a piece of Lie theory in 1937. His paper *Abstract derivation and Lie algebras* [1937-2] showed that the derivations of any algebra form a Lie algebra under the commutator bracket. It also asked how $\operatorname{Der} L$ reflects the shape of L. The first answers came at the two extremes. A semisimple Lie algebra in characteristic zero has only inner derivations [1972-3, Theorem in §5.3, p. 23]; a nilpotent Lie algebra always has an outer one [1972-3, Exercise 10, p. 14]. The theory since then has walked between these poles, and the chapter follows that walk.

Section 14.1 reports the two extremal answers. Section 14.2 moves one step toward the nilpotent side: the characteristically nilpotent Lie algebras of Dixmier and Lister, where every derivation is nilpotent. On the other side, modular simple Lie algebras of Cartan and of Albert–Zassenhaus type grow outer derivations that no characteristic-zero analogue has; Section 14.3 recounts this with the classification theorem of Wilson and the computation of Winter of $\operatorname{Der} L / \operatorname{ad} L$. Section 14.4 turns to the infinite Lie algebras of vector fields, where derivations become concrete operators on polynomial functions. Section 14.5 closes with affine Kac–Moody algebras, where the loop construction builds derivations into the algebra itself. It also records the 2-local derivations of a reductive Lie algebra, which fail to be derivations only when the centre is large [2026-9, Corollary 1.3, p. 2].



$\mathfrak{sl}_2$: $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$;
in characteristic 0 every derivation is ad of an element

Figure 14.1: The adjoint action on $\mathfrak{sl}_2$. In characteristic 0 every derivation of a semisimple Lie algebra is inner.

14.1 The theorem of Jacobson and nilpotent Lie algebras

This section reports the two extremal answers. Semisimple Lie algebras in characteristic zero have only inner derivations [1972-3, Theorem in §5.3, p. 23]. Nilpotent Lie algebras always have an outer one [1972-3, Exercise 10, p. 14]. The first answer starts with the Lie algebra formed by the derivations themselves [1937-2, Theorem 1, p. 206].

Theorem 14.1 ([1937-2, Theorem 1, pp. 206–207])**.** *Let $\mathfrak{A}$ be an arbitrary algebra over a commutative field F, not necessarily associative or finite-dimensional. Then the derivations of $\mathfrak{A}$ form a Lie algebra $\mathfrak{D}$ of linear transformations of $\mathfrak{A}$ under the commutator bracket; over a field of characteristic $p > 0$ it is a restricted Lie algebra.*

Jacobson called $\mathfrak{D}$ the *derivation algebra* or *d-algebra* of $\mathfrak{A}$ [1937-2, p. 207]. His paper then turned to associative algebras and asked how far derivations recover inner structure [1937-2, pp. 209–215]. For a normal simple algebra over its center his answer is the statement that owes his name.

Theorem 14.2 ([1937-2, Theorem 8, p. 215])**.** *Let $\mathfrak{A}$ be a finite-dimensional normal simple associative algebra over a field F. Every derivation of $\mathfrak{A}$ is inner: there is $a \in \mathfrak{A}$ with $D(x) = [x, a]$ for every $x \in \mathfrak{A}$.*

For Lie algebras the corresponding result is Theorem 5.3 of Humphreys, proved by the same Killing-form argument.

Theorem 14.3 ([1972-3, Theorem in §5.3, p. 23])**.** *Let L be a finite-dimensional semisimple Lie algebra over a field of characteristic zero. Then every derivation of L is inner, that is,* $\operatorname{ad} L = \operatorname{Der} L$.

The proof rests on two observations [1972-3, p. 23]. The Killing form of a semisimple L is nondegenerate. Also $[\delta, \operatorname{ad} x] = \operatorname{ad}(\delta x)$ for every $x \in L$ and $\delta \in \operatorname{Der} L$, so $\operatorname{ad} L$ is an ideal of $\operatorname{Der} L$ [1972-3, p. 23]. Its complement under the Killing form is zero, because an ideal perpendicular to $\operatorname{ad} L$ kills L [1972-3, p. 23].

Nilpotent Lie algebras sit at the opposite extreme [1972-3, Exercise 10, p. 14]; the construction in [1972-3, p. 14] is left as an exercise with a hint that already contains the argument.

Theorem 14.4 ([1972-3, Exercise 10, p. 14])**.** *Let L be a finite-dimensional nilpotent Lie algebra over a field F. Then L has an outer derivation.*

Humphreys constructs the derivation explicitly [1972-3, p. 14]. Write $L = K \oplus Fx$ for an ideal K of codimension one. Pick n with $C_L(K) \subset L^n$ but $C_L(K) \not\subset L^{n+1}$. Take $z \in C_L(K) \setminus L^{n+1}$, and let δ be the map with $\delta(K) = 0$ and $\delta(x) = z$. Because z lies in $C_L(K)$ one checks that δ is a derivation, and because z was chosen outside L^{n+1} the derivation is not inner.

Together Theorem 14.3 and Theorem 14.4 give the two answers the introduction promised. Between them sits a question that drives Section 14.2: what happens when every derivation of L is nilpotent as a linear transformation?

14.2 Characteristically nilpotent Lie algebras

Section 14.1 ended with two extremes: semisimple Lie algebras have only inner derivations (Theorem 14.3), and nilpotent ones always have outer derivations (Theorem 14.4). A stronger way for a Lie algebra to sit near the nilpotent pole is to require that every derivation be nilpotent as a linear map; Togo works with the following definition [1969-4, p. 209], attributed there to Dixmier–Lister and to Leger–Togo.

Definition 14.5 ([1969-4, p. 209])**.** Let L be a Lie algebra over a field F and let $\mathcal{D}(L) = \operatorname{Der} L$ be its algebra of derivations. Put $L^{(1)} = L\mathcal{D}(L)$, the F-subspace spanned by all $D(x)$ with $x \in L$ and $D \in \mathcal{D}(L)$, and inductively $L^{(n)} = L^{(n-1)}\mathcal{D}(L)$ for $n \geq 2$. Then L is called *characteristically nilpotent* provided there exists an integer k such that $L^{(k)} = (0)$.

Every characteristically nilpotent Lie algebra is nilpotent, since the inner derivations alone drive the lower central series to zero. The converse is false: examples of nilpotent Lie algebras that admit a non-nilpotent

derivation already appear in the papers of Leger and Togo [1969-4, p. 209]. Togo describes the class by a nilpotent characteristic ideal and its quotient [1969-4, Theorem, p. 210].

Theorem 14.6 ([1969-4, Theorem, p. 210])**.** *Let A be a nonassociative algebra over a field F and let N be a nilpotent characteristic ideal of A such that N^k is an ideal of A for every $k \geq 2$. Then A is characteristically nilpotent if and only if A/N^n is characteristically nilpotent for some integer n.*

For Lie and alternative algebras the assumption on N^k is automatic [1969-4, p. 211], so the statement specializes without it.

Corollary 14.7 ([1969-4, Corollary, p. 211])**.** *Let A be a Lie or alternative (associative) algebra over a field F and let N be a nilpotent characteristic ideal of A. Then A is characteristically nilpotent if and only if A/N^n is characteristically nilpotent for some integer n.*

Togo closes the paper with a three-dimensional nonassociative algebra. It is characteristically nilpotent without being a Lie algebra [1969-4, p. 211]. The basis x_1, x_2, x_3 has $x_1 x_2 = x_3$ and all other products zero, so $A^{(2)} = (0)$ [1969-4, p. 211].

Derivations also serve as a construction tool. The method of adjoining a derivation turns a nilpotent Lie algebra $\mathfrak{g}$ and a nilpotent derivation δ of it into the semidirect product $\widetilde{\mathfrak{g}}_\delta = \mathbb{C}\delta \oplus \mathfrak{g}$, with $\mathbb{C}$ acting through δ [2010-6, p. 724]. Magnin applies the method to every complex nilpotent Lie algebra of dimension six [2010-6, p. 724].

The result is a new determination of all seven-dimensional complex nilpotent Lie algebras, with a cross table between the six- and seven-dimensional lists [2010-6, p. 724]. Among the seven-dimensional algebras the rank-zero family is exactly the characteristically nilpotent one [2010-6, p. 726]. The check corrected earlier lists, and the paper records the corrections [2010-6, p. 724].

Adjoining a derivation once is a special case of a general construction. A derivation of a subalgebra can always be extended to one of a larger algebra that contains it [1998-7, Corollary 5.2, p. 84]. Wasserman builds the extension as an HNN extension of Lie algebras, in the pattern of group theory [1998-7, p. 79]. Given a Lie algebra L, a subalgebra A and a derivation $d\colon A \to L$, the extension H is presented by adding one generator t with the relations $[t, a] = d(a)$ for $a \in A$ [1998-7, p. 83].

Theorem 14.8 ([1998-7, Corollary 5.2, p. 84])**.** *Let L be a Lie algebra over a field k, let A be a subalgebra and let $d\colon A \to L$ be a derivation. Then L is embedded in the HNN extension H defined by L, A and d, and the map* $\operatorname{ad} t$ *extends d to a derivation of H.*

The proof gives a normal form for the elements of H through a Lyndon basis [1998-7, Theorem 5.1, p. 84]. The extension has two extreme cases. When $A = L$ it is the semidirect product with a one-dimensional Lie algebra acting through d. When $A = 0$ it is the free product with a one-dimensional Lie algebra [1998-7, p. 83]. The construction also embeds every Lie algebra of finite or countable dimension in one generated by two elements [1998-7, Theorem 7.2, p. 88].

The contrast with Section 14.1 is sharp. Theorem 14.4 says a nilpotent Lie algebra always has an outer derivation. A characteristically nilpotent Lie algebra, by contrast, admits no semisimple derivation at all: its derivation algebra is nilpotent in the sense of the theorem of Engel. Section 14.3 returns to the other pole, where positive characteristic gives simple Lie algebras an outer-derivation behavior with no counterpart in characteristic zero.

14.3 Derivations of simple and semisimple Lie algebras

Section 14.2 closed with the nilpotent side of the theory. We now return to simple Lie algebras, this time over fields of positive characteristic, where the characteristic-zero answer "all derivations are inner" breaks down.

The first work is to identify the simple modular Lie algebras that can carry outer derivations; the classification theorem of Wilson [1976-3, Theorem 1, pp. 418–419] reduces the candidates.

Theorem 14.9 ([1976-3, Theorem 1, pp. 418–419])**.** *Let L be a finite-dimensional simple Lie algebra over an algebraically closed field F of characteristic $p > 3$. Suppose L contains a maximal subalgebra L_0 such that the pair (L, L_0) satisfies the following condition: either* $\dim L/L_0 = 1$*, or else there exists Y in the image of the natural representation of L_0 on L/L_0 with* $\operatorname{rank} Y = 1$ *and* $(\operatorname{ad} Y)^2 \neq 0$*. Then L is either classical of type A or C, or is of generalized Cartan type.*

The generalized Cartan type Lie algebras go back to Kostrikin and Shafarevich [1976-3, p. 418]: they are built from truncated polynomial

rings and their derivations. The classical algebras of Seligman and of Mills–Seligman sit inside this family with rank-one root spaces [1984-1, p. 962]. For all of them Winter computes the full derivation algebra [1984-1, Theorems 3.4 and 3.6, pp. 970–971]. Winter works with a Lie algebra L that is CAZ (*classical–Albert–Zassenhaus*) [1984-1, p. 962]. Its Cartan subalgebra H splits, and each nonzero root space L_α is one-dimensional. Also $[L_\alpha, L_\beta] = L_{\alpha+\beta}$ whenever the root sum is again a root [1984-1, p. 962]. The irreducible components R_{cl} of the root system are the classical ones; R_{AZ} are the Albert–Zassenhaus ones, meaning additive subgroups of $k\alpha$ for some nonzero root α [1984-1, p. 962].

Theorem 14.10 ([1984-1, Theorem 3.4, p. 970])**.** *Let L be an extended Lie algebra with extended CAZ Cartan subalgebra H. Then*

$$\operatorname{Der} L \;=\; T(H) \;+\; D(L,H) \;+\; \operatorname{ad} L,$$

where $\dim T(H) = \operatorname{rank} R$ *and* $D(L,H) = D(L/L^\infty, C_H(L^\infty))$.

Here $L^\infty = \bigcap_{i\geq 1} L^i$ is the intersection of the lower central series. $T(H)$ is a maximal torus of the Fitting zero component of $\operatorname{Der} L$ under $\operatorname{ad} H$. $D(L/L^\infty, C_H(L^\infty))$ is the space of derivations of the quotient L/L^∞ into the centraliser $C_H(L^\infty)$ [1984-1, p. 961]. The toral piece $T(H)$ measures outer derivations coming from the dual space of the root system; the middle piece $D(L,H)$ captures central-valued ones [1984-1, Theorem 3.4, p. 970].

For a CAZ Lie algebra in the strict sense ($L = L^2$ and center zero), the count of Winter is a formula in the geometry of the root system.

Theorem 14.11 ([1984-1, Theorem 3.6, p. 971])**.** *Let L be a CAZ Lie algebra with CAZ Cartan subalgebra H. Then*

$$\dim \operatorname{Der} L / \operatorname{ad} L \;=\; a \;-\; b \;+\; \operatorname{rank} R_{AZ},$$

where a is the number of irreducible components of R_{cl} of type A_r with $p \mid r+1$, and b is the number of irreducible components of R_{cl}.

The cases match the two families in Theorem 14.9. Classical components of type A_r with $p \mid r+1$ contribute outer derivations, and the Albert–Zassenhaus ones contribute through the dual rank. The introduction of Winter records the consequence for classical Lie algebras. All derivations of a classical Lie algebra L are inner if and only if L has no component of type A_r with $p \mid r+1$ [1984-1, p. 961].

The same innerness holds for the bilinear form of the question. A biderivation of a Lie algebra L is a bilinear map that is a derivation in

each argument [2017-13, Definition 1.1, p. 2]. The map $(x, y) \mapsto \lambda[x, y]$ is a biderivation for every scalar λ, and it is called inner [2017-13, p. 2]. Tang shows that on a simple complex Lie algebra no other map occurs, with no skew-symmetry assumed [2017-13, Theorem 2.4, p. 7].

Theorem 14.12 ([2017-13, Theorem 2.4, p. 7])**.** *Let L be a finite-dimensional complex simple Lie algebra. A bilinear map $f\colon L \times L \to L$ is a biderivation of L if and only if it is inner, that is, $f(x, y) = \lambda[x, y]$ for some $\lambda \in \mathbb{C}$ and all $x, y \in L$.*

The proof reduces both arguments to a common scalar on each root space [2017-13, pp. 6–7]. On the general linear Lie algebra the answer grows: a biderivation of $\mathfrak{gl}_n(\mathbb{C})$ is $\mu \operatorname{tr}(x) \operatorname{tr}(y) I_n + \lambda[x, y]$ for two complex numbers μ and λ [2017-13, Theorem 3.2, p. 7].

Positive characteristic gives a second family in which the derivations are understood. Let H be the finite-dimensional Hamiltonian superalgebra over a field of prime characteristic, and let H_0 be its even part [2005-6, p. 1]. Liu, Su and Zhang describe the derivations of H_0 into the even part W_0 of the generalized Witt superalgebra [2005-6, p. 1]. They introduce an ideal N of H_0 and show that the derivations of H_0 are controlled by those of N [2005-6, Theorem 2.4, p. 6]. A set of generators of N makes that reduction effective [2005-6, Theorem 2.7, p. 7]. Three series of exceptional derivations remain after the reduction [2005-6, p. 2].

The contrast with Theorem 14.3 could not be sharper. In characteristic zero a simple finite-dimensional Lie algebra has $\operatorname{Der} L = \operatorname{ad} L$; in characteristic $p > 0$ simple Lie algebras exhibit $\operatorname{Der} L$ strictly larger than $\operatorname{ad} L$ in ways tied to the root data. Section 14.4 lifts the same characteristic-zero-versus-outer tension to infinite-dimensional Lie algebras of vector fields.

14.4 Lie algebras of vector fields

Section 14.3 showed that simple modular Lie algebras can grow outer derivations. Over fields of characteristic zero some infinite-dimensional simple Lie algebras do the same [1969-5, 1990-4, 2019-11]. The examples here are the polynomials under the Poisson bracket and the algebras of vector fields studied in [1969-5, 1990-4, 2019-11, 2022-6].

Definition 14.13 ([1969-5, p. 315])**.** Let E denote the collection of all C^∞ complex-valued functions of $2n$ real variables $x = (q_1, \ldots, q_n, p_1, \ldots, p_n)$,

and define the *Poisson bracket* of two elements $f, g \in E$ by

$$\{f, g\} = \sum_{j=1}^{n} \Big(\frac{\partial f}{\partial q_j} \frac{\partial g}{\partial p_j} - \frac{\partial f}{\partial p_j} \frac{\partial g}{\partial q_j} \Big).$$

The collection of all complex polynomials in the variables x forms a Lie subalgebra of E under this bracket, denoted P.

The first observation of Wollenberg is that some derivations of P are not inner [1969-5, Lemma 1, p. 316].

Lemma 14.14 ([1969-5, Lemma 1, p. 316])**.** *For any set of complex numbers $\alpha = (\alpha_1, \ldots, \alpha_n)$, the map*

$$D_\alpha \colon f \longmapsto f \; - \; \sum_{i=1}^{n} \Big(\alpha_i p_i \frac{\partial f}{\partial p_i} + (1 - \alpha_i) q_i \frac{\partial f}{\partial q_i} \Big)$$

is an outer derivation of P.

Together with the inner ones these give all derivations of P [1969-5, Theorem 1, p. 318].

Theorem 14.15 ([1969-5, Theorem 1, p. 318])**.** *Every derivation D of P is a linear combination of a derivation of the form D_α of Lemma 14.14 and an inner derivation. That is, one can find a polynomial A and complex numbers c, α_i such that $D(f) = c\, D_\alpha(f) + \{A, f\}$ for all $f \in P$.*

The decomposition is not unique [1969-5, Remark, p. 318]: shifting the α_i and changing A give the same derivation. Wollenberg also pins down the inner ones [1969-5, Lemma 3, p. 317].

Lemma 14.16 ([1969-5, Lemma 3, p. 317])**.** *If D is a derivation of P with $D(1) = 0$, then D is inner.*

A second family is the generalized Witt family of Ikeda and Kawamoto [1990-4, p. 47]. Its index set G is closed under addition.

Definition 14.17 ([1990-4, p. 47])**.** Let f be a field of characteristic zero, I a non-empty index set, and G an additive submonoid of $\bigoplus_{i \in I} f t_i$ with t_i copies of the additive group f^+. The *generalized Witt algebra* $W(G, I)$ is the Lie algebra over f with basis $\{w(a, i) \mid a \in G, i \in I\}$ and bracket $[w(a, i), w(b, j)] = a_j w(a + b, i) - b_i w(a + b, j)$.

For this algebra Ikeda and Kawamoto determine the full derivation algebra [1990-4, Theorem 2, p. 54].

Theorem 14.18 ([1990-4, Theorem 2, p. 54])**.** *Let $G = \bigoplus_{i\in I} \mathbb{Z}_i$ and let $W = W(G, I)$. Then W is simple and $\operatorname{Der} W = \operatorname{Lin} W$. In particular, if $|I|$ is finite and $G = \mathbb{Z}^I$, then $\operatorname{Der} W = \operatorname{Inn} W$.*

Here $\operatorname{Lin} W$ denotes the *locally inner* derivations of W [1990-4, p. 47]: a derivation δ is locally inner when every finite subset $F \subset W$ lies in a finite-dimensional subspace V with $\delta|_V = \operatorname{ad} x|_V$ [1990-4, p. 47]. In the finite-rank case every locally inner derivation is inner; in infinite rank this fails in general [1990-4, Remark, p. 55], giving the first outer derivations of the section.

Li settles a parallel case [2019-11, Theorem 3.6]: the ring $\mathbb{C}[z_1^{\pm 1}, \dots, z_n^{\pm 1}]$ of Laurent polynomials, its divergence-zero derivations, and once more an answer close to inner.

Theorem 14.19 ([2019-11, Theorem 3.6])**.** *Let $A = \mathbb{C}[z_1^{\pm 1}, \dots, z_n^{\pm 1}]$ and let S_n be the Lie algebra of divergence-zero derivations of A. Let $L = [S_n \cdot A, S_n \cdot A]$ be the derived subalgebra of the semi-direct product. Then $\operatorname{Der} L = \operatorname{Inn} L \oplus \operatorname{Der}(L)_0$, where $\operatorname{Der}(L)_0$ has dimension $2n + 1$ and is spanned by explicit degree derivations.*

In this decomposition the extra summand $\operatorname{Der}(L)_0$ counts the degree derivations and one extra direction [2019-11, Theorem 3.6]. On the remaining weight spaces every derivation is inner [2019-11, Theorem 3.6]. Once more the outer derivations come from the grading data, as in the Poisson case of Theorem 14.15. Randriambololondrantomalala and Asif give an explicit description of this quotient $\operatorname{Der}(P)/\operatorname{ad}_P$ [2022-6, Theorem 2.4, p. 52].

Theorem 14.20 ([2022-6, Theorem 2.4, p. 52])**.** *Let P be a Lie subalgebra of the Lie algebra $\chi(\mathbb{R}^n)$ of vector fields on $\mathbb{R}^n$, containing all constant vector fields and the Euler vector field, and write $H^1(P) := \operatorname{Der}(P)/\operatorname{ad}_P$. Then $H^1(P)$ is isomorphic to N/P, where N is the normaliser of P in $\chi(\mathbb{R}^n)$; if all linear diagonal fields lie in P, then $H^1(P) = 0$.*

Together Theorem 14.20 identifies all outer derivations of such a P with an explicit quotient, and Theorems 14.15–14.19 compute it in the classical examples.

The Hamiltonian Lie algebras on the torus give a case where the answer is inner. Let N be an even positive integer, and let H_N be the Hamiltonian Lie

algebra on the torus, with derived subalgebra H'_N [2026-7, p. 1]. Bisht, Rani and Tantubay compute both derivation algebras [2026-7, Theorems 17–18, p. 18].

Theorem 14.21 ([2026-7, Theorems 17–18, p. 18])**.** *Let N be an even positive integer. Then* $\mathrm{Der}(H'_N) = H'_N \rtimes \mathfrak{h} \cong H_N$ *and* $\mathrm{Der}(H_N) \cong H_N$.

So every derivation of the Hamiltonian Lie algebra is inner, and the derived subalgebra has one more family, indexed by the diagonal fields $\mathfrak{h}$ [2026-7, p. 18]. The authors also compute the second cohomology of H_N [2026-7, Corollary 24, p. 23].

The same question has a geometric answer on the plane. Let k be an algebraically closed field of characteristic zero and let W_2 be the Lie algebra of all k-derivations of $k[x, y]$ [2009-7, p. 1]. For a fixed rational function $u \in k(x, y) \setminus k$, the annihilator $A_{W_2}(u)$ is the set of derivations $D \in W_2$ with $D(u) = 0$ [2009-7, p. 2]. It is a Lie subalgebra of W_2 and a $k[x, y]$-submodule of the module W_2 [2009-7, p. 2].

Theorem 14.22 ([2009-7, Theorem 6, p. 4])**.** *For an arbitrary rational function $\phi \in k(x, y) \setminus k$, its annihilator $A_{W_2}(\phi)$ is a free submodule of rank one of the $k[x, y]$-module $W_2(k)$. As a free generator one may take the derivation δ_ϕ.*

The generator δ_ϕ is reduced, meaning its two polynomial coefficients are coprime [2009-7, p. 2]. The paper also describes the centralisers of elements of $A_{W_2}(\phi)$ and its maximal abelian subalgebras [2009-7, p. 2]. The annihilator itself has a different structure for a polynomial and for a quotient of two algebraically independent polynomials [2009-7, p. 2].

The polynomial ring in infinitely many variables gives an algebra whose derivations are all inner. Let F be a field of characteristic zero and let $W_X(F)$ be the Lie algebra of all derivations of $F[X]$ in infinitely many variables [2025-2, p. 1]. Two subalgebras sit inside it, $W_\infty(F)$ and $W_{\mathrm{fin}}(F)$ [2025-2, p. 2].

Theorem 14.23 ([2025-2, Theorems 1–3, p. 2])**.** *Let F be a field of characteristic zero. Every derivation of $W_\infty(F)$ is the restriction of an adjoint operator* $\mathrm{ad}(a)$ *with $a \in W_{\mathrm{fin}}(F)$. Every derivation of $W_{\mathrm{fin}}(F)$ is inner, and every derivation of $W_X(F)$ is inner.*

So $W_\infty(F)$ has derivations that are not inner, and its inner derivations are dense in its derivations for the Tychonoff topology [2025-2, p. 2]. The algebra $W_{\mathrm{fin}}(F)$ and the full algebra $W_X(F)$ have no outer derivation at all

[2025-2, p. 2]. This differs from the Poisson case of Theorem 14.15, where the degree derivation gave an outer one [1969-5, Lemma 1, p. 316].

Section 14.5 pushes the same question into Kac–Moody theory, where the algebra is no longer a vector-field algebra but an affine extension of a finite semisimple one.

Further work in the Lie-algebra line, held in the collection: *The groups of automorphisms of the Lie algebras of polynomial vector fields with zero or constant divergence* [2017-2]. *Space of second-order linear differential operators as a module over the Lie algebra of vector fields* [1997-1]. *Classification of finite-dimensional Lie superalgebras whose even part is a three-dimensional simple Lie algebra over a field of characteristic not two or three* [2019-1]. *The groups of automorphisms of the Lie algebras of formally analytic vector fields with constant divergence* [2014-3].

14.5 Infinite-dimensional Lie algebras

Section 14.4 described derivations of vector-field Lie algebras. The center of gravity now shifts to the *Kac–Moody* algebras [1983-3, pp. 1–2]. In the affine case one of these algebras is a central extension of a loop algebra with one derivation added [1983-3, pp. 74–75].

Kac builds a Kac–Moody algebra $\mathfrak{g}(A)$ from a generalized Cartan matrix A and derives its root space decomposition [1983-3, pp. 1–11]. The first derivation result appears as one of the exercises of Chapter 1 [1983-3, Exercise 1.9, p. 11].

Proposition 14.24 ([1983-3, Exercise 1.9, p. 11])**.** *Let $\mathfrak{g}$ be a Lie algebra, $\mathfrak{h} \subset \mathfrak{g}$ a finite-dimensional diagonalizable subalgebra, and $\mathfrak{g} = \bigoplus_{\alpha \in Q} \mathfrak{g}_\alpha$ the root space decomposition such that $\mathfrak{g}_0 = \mathfrak{h}$. Then $\operatorname{der}\mathfrak{g} = (\operatorname{der}\mathfrak{g})_0 + \operatorname{ad}\mathfrak{g}$, where $(\operatorname{der}\mathfrak{g})_0$ consists of endomorphisms preserving the root space decomposition.*

The deduction in the hint of Kac is to add an inner derivation so that the image of $\mathfrak{h}$ stays in $\mathfrak{h}$ [1983-3, p. 11]. The next exercise applies Proposition 14.24 to Kac–Moody algebras [1983-3, Exercise 1.10, p. 11].

Proposition 14.25 ([1983-3, Exercise 1.10, p. 11])**.** *All derivations of the Lie algebra $\mathfrak{g}(A)/\mathfrak{c}$ are inner, provided that the generalized Cartan matrix A has no zero rows.*

Here $\mathfrak{c}$ is the centre of $\mathfrak{g}(A)$. The hypothesis on A rules out the rows whose quotient would carry a large centre [1983-3, Exercise 1.10, p. 11].

For the affine case Kac constructs the algebra as a semi-direct product with a derivation. Let $\mathfrak{g}$ be a simple finite-dimensional Lie algebra over $\mathbb{C}$, and let $L(\mathfrak{g}) = \mathbb{C}[t, t^{-1}] \otimes \mathfrak{g}$ be its loop algebra [1983-3, p. 74]. Kac shows that $d_s \colon t^{s+1}\frac{d}{dt}$ extends to a derivation of the central extension of $L(\mathfrak{g})$ [1983-3, Proposition 7.3, p. 74], and uses the case $s = 0$ to define the affine algebra:

Theorem 14.26 ([1983-3, §7.2, pp. 74–75])**.** *Let $\mathfrak{g}$ be a simple finite-dimensional Lie algebra over $\mathbb{C}$ and $\tilde{L}(\mathfrak{g})$ the one-dimensional central extension of $L(\mathfrak{g})$ by the cocycle $\psi(a, b) = \operatorname{Res}\left(\frac{da}{dt}|b\right)_t$. Let $\hat{L}(\mathfrak{g})$ be the Lie algebra $\mathbb{C}[t, t^{-1}] \otimes \mathfrak{g} + \mathbb{C}c + \mathbb{C}d$ with bracket*

$$[t^k \otimes x + \lambda c + \mu d,\ t^{k_1} \otimes y + \lambda_1 c + \mu_1 d] = (t^{k+k_1} \otimes [x, y] + \mu k_1 t^{k_1} \otimes y - \mu_1 k t^k \otimes x) + k\delta_{k,-k_1}(x \mid y)c.$$

Then $\hat{L}(\mathfrak{g})$ is an affine Kac–Moody algebra associated to the affine matrix A of type $X^{(1)}$.

The derivation $d = -t\frac{d}{dt}$ is, by construction, an outer one [1983-3, p. 74]: it measures the t-degree of loop elements. Kac shows that $d_s|_{L(\mathfrak{g})} = -t^{s+1}\frac{d}{dt}$ and $d_s(c) = 0$ defines a Lie algebra derivation of $\hat{L}(\mathfrak{g})$ for every integer s [1983-3, Proposition 7.3, p. 74].

Together Proposition 14.25 and Theorem 14.26 sum up the affine picture: every derivation of $\mathfrak{g}(A)/\mathfrak{c}$ is inner, while the loop extension to $\hat{L}(\mathfrak{g})$ is built by adding one outer degree derivation d. This matches the pattern of Section 14.4: outer derivations of infinite-dimensional Lie algebras carry the grading, and inner ones recover the bracket.

The same two questions, which maps are derivations and when they are inner, were later asked without linearity. A 2-local derivation of a Lie algebra matches its value at any two points by one derivation [2026-9, p. 1]. Chen, Luo and Xu classify them on a reductive Lie algebra, with values in a module [2026-9, p. 1].

Theorem 14.27 ([2026-9, Corollary 1.3, p. 2])**.** *Let F be an algebraically closed field of characteristic zero, let $\mathfrak{g} = \mathfrak{s} \oplus \mathfrak{z}$ be a finite-dimensional reductive Lie algebra over F, and let V be a finite-dimensional $\mathfrak{g}$-module. Then every 2-local derivation of $\mathfrak{g}$ into V is a derivation if and only if $\dim \mathfrak{z} \leq 1$ or $V^{\mathfrak{g}} = 0$.*

So the exceptional 2-local derivations appear exactly when the centre is at least two-dimensional and the module has an invariant vector [2026-9, p. 2]. In that case the nonlinear homogeneous maps give all of them [2026-9,

p. 2]. For a semisimple Lie algebra the centre is zero, so the condition holds and every 2-local derivation is a derivation [2026-9, p. 1]. That special case is due to Ayupov, Kudaybergenov and Rakhimov, as the paper records [2026-9, p. 1]. The same question on the infinite-dimensional Block-type Lie algebra has the answer that every 2-local derivation is a derivation [2026-8, Theorem 3.6, p. 9].

The superconformal algebras give a second family with a complete answer. The twisted $N = 2$ superconformal algebra has a basis of Virasoro modes L_n, a family of modes T_r, a family of odd modes G_p and a central element c [2013-6, p. 1]. Fa computes its derivation algebra and its automorphism group [2013-6, p. 2].

Theorem 14.28 ([2013-6, Theorem 1.1, p. 2])**.** *For the twisted $N = 2$ superconformal algebra L, every derivation is inner:* $\operatorname{Der}(L) = \operatorname{ad}(L)$.

The group of inner automorphisms is isomorphic to $\mathbb{C}^*$, and the full automorphism group is $\mathbb{C}^* \rtimes \mathbb{Z}_4$ [2013-6, Theorem 1.5 and Corollary 1.6, p. 4]. For the generalized twisted algebra the answer has one more summand, and the derivation algebra is $\operatorname{ad}(L[\mathrm{s}\Gamma]) \oplus \operatorname{Hom}_{\mathbb{Z}}(\Gamma, F)$ [2013-6, Theorem 4.1, p. 18].

The Schrödinger–Virasoro algebra carries a biderivation that is neither inner nor skew-symmetric. Let $\mathrm{SV}(\varepsilon)$ be the Schrödinger–Virasoro algebra with its parameter ε [2016-14, p. 1]. A biderivation of a Lie algebra is a bilinear map that is a derivation in each argument [2016-14, p. 2]. Tang computes all of them.

Theorem 14.29 ([2016-14, Theorem 3.6, p. 9])**.** *A bilinear map f is a biderivation of* $\mathrm{SV}(\varepsilon)$ *if and only if there are $\lambda \in \mathbb{C}$ and a finite set Ω of complex numbers with $f(x, y) = \lambda[x, y] + \chi_\Omega(x, y)$ for all $x, y \in \mathrm{SV}(\varepsilon)$, where χ_Ω is the map of Definition 3.5 of the same paper.*

The map χ_Ω is a sum of terms indexed by the finite set Ω, and it need not be skew-symmetric [2016-14, p. 9]. Every skew-symmetric biderivation of $\mathrm{SV}(\varepsilon)$ is inner, however [2016-14, Corollary 3.7, p. 10]. The biderivations also classify the commutative post-Lie structures on the same algebra [2016-14, Theorem 4.3, p. 12].

The deformed version of the same algebra behaves the same way. Fan, Su and Xu compute the skew-symmetric biderivations of the deformed Schrödinger–Virasoro algebras, which contain the twisted and the original algebras [2016-11, p. 1]. They obtain the form of each linear commuting map as a consequence [2016-11, Theorem 4.3, p. 11]. For some values of

the deformation parameter the algebra has non-inner biderivations and non-standard commuting maps [2016-11, p. 1].

We close the chapter here. The affine case of Theorem 14.26 already displays the full picture: the grading forces the outer derivations, and the inner ones give the rest, as in Sections 14.3 and 14.4.

The Schrödinger algebra is another infinite-dimensional case worked out in full. Wu and Tang determine every derivation of the Schrödinger algebra in $(n+1)$-dimensional space-time, and then every biderivation [2022-7, Abstract]. They do the same for the n-th Schrödinger algebra $\mathfrak{sch}_n = \mathfrak{sl}_2 \ltimes \mathfrak{h}_n$, the semidirect product with the n-th Heisenberg algebra [2023-16, Abstract].

Chapter 15

Jordan, alternative, Leibniz algebras and superalgebras

The chapters so far stayed inside associative rings, Lie algebras, and operator algebras. The derivation algebra of a Lie algebra is again a Lie algebra, and many classical results about derivations use associativity of the underlying product. In this chapter we ask what survives when the product is no longer associative nor antisymmetric. The classes include the Jordan product of hermitian matrices, the octonionic product, the Leibniz identity that Loday isolated in 1993, and the brackets of superalgebras. They also include the twisted Jacobi rules of Hom-Lie algebras and Lie conformal algebras.

Some of these questions are older than the pages below may suggest. Jacobson realised in the 1950s that the exceptional Lie algebras of types G_2 and F_4 arise as derivation algebras of the octonions and of the Albert algebra [2004-7, p. 52]. Loday introduced Leibniz algebras in 1993 to explain why a certain chain complex in cyclic homology is closed [1993-10, p. 139]. Hartwig, Larsson and Silvestrov isolated a twisted Jacobi identity in 2003 to describe deformations of the Witt algebra [2006-5, p. 1]. Each line has its own reason for existing and its own derivative-like linear maps.

What we collect below is the literature on derivations in these wider classes, with one result from each. We ask in every section: what is a derivation here, does the analogue of "every derivation of a simple finite-dimensional algebra is inner" hold, and what does the derivation algebra compute?

Section 15.1 works with Jordan algebras proper. The inner derivations of a Jordan algebra are the maps $D_{x,y} = 4[L_x, L_y] = [V_x, V_y]$. They span a Lie ideal of $\mathrm{Der}(J)$ [2004-7, Problem 3.6, p. 185]. The 27-dimensional Albert

algebra of hermitian octonionic 3×3 matrices has derivation algebra F_4 [2004-7, p. 52]. Section 15.2 records the definition of Artin of an alternative algebra and the theorem of Davenport that derivations of a semisimple Malcev algebra in characteristic $\neq 2, 3$ are inner [2002-2, p. 148], [1972-2, p. 565]. Section 15.3 opens the definition of Loday and gives the derivation structure of complex semisimple Leibniz algebras [1993-10, Definition (1.1), p. 141], [2020-8, Theorem 4.5], [2021-14, Proposition 3.5, p. 2193]. At dimension three the description of the derivation algebras is complete [2026-20, Theorem 1.1, p. 2]. Section 15.4 states the definition of Musson of a superderivation and ties it to the classification of Kac of simple Lie superalgebras [2012-11, Lemma 1.1.2, p. 2], [1977-2, p. 1375]. It records that a two-step nilpotent Lie superalgebra carries local superderivations that are not superderivations [2026-10, Theorems 3.2 and 3.5, pp. 7, 12]. Section 15.5 treats the δ-derivations of alternative and Malcev algebras (Filippov 2000) and of simple Jordan superalgebras (Kaygorodov 2007). It also treats the related $\frac{1}{2}$-derivations on Lie algebras that encode right multiplication in transposed Poisson algebras [2000-2, Theorems 2 and 3, pp. 356–358], [2007-6, Theorems 2.5 and 4.4], [2021-17, Definition 5, p. 4]. Section 15.6 gives the definition of Li of a Poisson derivation and the theorem that every Poisson derivation of the symplectic Poisson algebra S_n is inner [2018-13, Definitions, p. 658], [2018-13, Theorem 2.1, pp. 658–659]. Section 15.7 compares two results: the Hom-Lie algebras of Hartwig, Larsson and Silvestrov, and the triviality of such structures on simple Lie algebras proved by Jin and Li [2006-5, Definition 4, p. 18], [2008-7, Proposition 2.2, p. 1400]. It closes with the definition of a conformal derivation of a Lie conformal algebra of Fan, Hong and Su [2019-6, Definitions 2.1 and 2.7, pp. 2–3]. Section 15.8 closes the chapter with the derivations of the ternary Malcev algebra M_8 and with the $(n+1)$-ary derivations of simple Mal'tsev algebras [2006-8, Theorem 2.3, p. 3598], [2014-14, Theorem 5, p. 582].

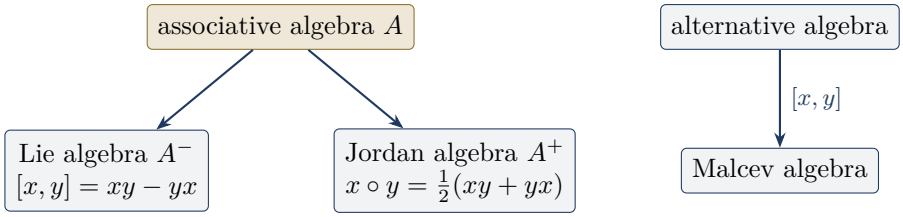


Figure 15.1: From associative to Lie and Jordan algebras, and from alternative to Malcev algebras. A derivation of A is also a derivation of A^- and of A^+.

15.1 Derivations of Jordan algebras

Chapter 15 opens with Jordan algebras, the historical root of this chapter. Recall that a *Jordan algebra* [2004-7, Section 2.4, p. 57] is a commutative nonassociative algebra J over a field k of characteristic $\neq 2$ that satisfies the Jordan identity $(x \circ x) \circ (y \circ x) = ((x \circ x) \circ y) \circ x$. Albert showed that $H_3(\mathbb{K})$ is a 27-dimensional exceptional Jordan algebra [2004-7, p. 971]. Zel'manov later proved that no other simple exceptional Jordan algebra exists, even in infinite dimensions [1979-2, pp. 103–111].

A *derivation* of a Jordan algebra J has the same defining property as in Chapter 2: a k-linear map $\delta\colon J \to J$ with $\delta(xy) = \delta(x)y + x\delta(y)$. For an associative algebra A the inner derivations are the adjoint maps $\mathrm{ad}(x)$, $x \in A$, and they exhaust the derivations in nice cases. For a Jordan algebra a similar mechanism exists, but it needs two indices: the commutators of two multiplication operators.

Proposition 15.1 ([2004-7, Problem 3.6 (3), p. 185])**.** *Let J be a Jordan algebra over a ring in which $\frac{1}{2}$ exists, and write L_x for left multiplication by x. For each pair $x, y \in J$ the map*

$$D_{x,y} \;=\; 4\,[L_x, L_y] \;=\; [V_x, V_y]$$

is a derivation of J, where $[L_x, L_y] = L_xL_y - L_yL_x$ is the usual commutator of multiplication operators.

Question 3.1 of McCrimmon 3.1 records that the derivations $\mathrm{Der}(J)$ form a Lie algebra of endomorphisms under the bracket $[D, E] = DE - ED$, with the inner derivations as an ideal [2004-7, Question 3.1, p. 185]. For special Jordan algebras $J \subseteq A^+$ derivations come from the associative side: any associative derivation of A that leaves J invariant is a Jordan derivation of J [2004-7, Problem 3.6 (2), p. 185].

The exceptional case gives the famous answer.

Theorem 15.2 ([2004-7, Historical remarks, p. 52])**.** *Let $\mathcal{A} = H_3(\mathbb{K})$ be the Albert algebra over an algebraically closed field of characteristic 0, where $\mathbb{K}$ is the Cayley algebra. Then $\mathrm{Der}(\mathcal{A})$ is a 52-dimensional simple Lie algebra of type F_4, and $\mathrm{Aut}(\mathcal{A})$ is the exceptional simple group of type F_4.*

McCrimmon records this as a discovery by Chevalley, Schafer, Freudenthal and others in the early 1950s, with G_2 playing the analogous role for the Cayley algebra itself [2004-7, p. 52]. Zel'manov proved that there is no simple exceptional Jordan algebra other than the Albert algebras [1979-2,

pp. 103–111]. Together with Theorem 15.2 it singles out F_4 as the only derivation algebra of an exceptional simple Jordan algebra.

We close with the Jordan-variant built from a symmetric bilinear form.

Example 15.3 ([2004-7, Spin factor example, p. 360 and Problem 3.7 (1), p. 185])**.** Let M be a k-module with a symmetric bilinear form σ. The *spin factor* $JSpin(M,\sigma) = k\,\mathbf{1} \dot{+} M$ carries the Jordan product

$$(\alpha\,\mathbf{1} \dot{+} x) \bullet (\beta\,\mathbf{1} \dot{+} y) \;=\; \big(\alpha\beta + \sigma(x,y)\big)\,\mathbf{1} \dot{+} (\beta x + \alpha y).$$

Its derivations δ satisfy $\delta(\mathbf{1}) = 0$ and $\sigma(\delta(x), x) = 0$ for all $x \in M$; thus $\delta|_M$ is skew-symmetric for σ.

The Cayley algebra meets the Jordan side again in Section 15.2: it is an alternative algebra [2002-2, p. 148], and the next section studies its derivations.

15.2 Alternative and Malcev algebras

The previous section built the Albert algebra from the Cayley algebra $\mathbb{K}$. An algebra A is *alternative* if the subalgebra generated by any two elements is associative [2002-2, p. 148]. This is the same as asking all three identities $(aa)b = a(ab)$, $(ab)a = a(ba)$, $(ba)a = b(aa)$ for all $a, b \in A$ [2002-2, p. 148]. Any two of the three identities imply the third, and the first two are usually taken as the definition [2002-2, p. 148].

For instance, the four real division algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ are alternative; conversely, they are the only alternative division algebras over $\mathbb{R}$ [2002-2, Theorem, p. 148].

Definition 15.4 ([1972-2, Introduction, p. 565])**.** A *Malcev algebra* M over a field F is a nonassociative algebra satisfying

(i) $x^2 = 0$ for every $x \in M$;

(ii) $(xy)(xz) = ((xy)z)x + ((yz)x)x + ((zx)x)y$ for every $x, y, z \in M$.

Malcev introduced this product as the commutator $xy - yx$ on an alternative algebra in 1955; Sagle named the class [1972-2, p. 565]. Every Lie algebra satisfies conditions (i) and (ii), because condition (ii) then becomes the Jacobi identity. On the trace-zero elements of the Cayley division algebra, the commutator algebra is semisimple and seven-dimensional [2002-2, p. 148], [2004-7, p. 52].

Keeping the product rule of Chapter 2 on the nonassociative side, we ask the question posed and answered by the following theorem.

Theorem 15.5 ([1972-2, Introduction and Corollary 4, p. 565])**.** *Let A be a finite-dimensional Malcev algebra over a field F of characteristic $\neq 2, 3$, and suppose that the Killing form on A and the Lie multiplication algebra $L(A)$ are nondegenerate. Then every derivation of A is inner. In particular, over a field of characteristic* 0 *every derivation of a semisimple Malcev algebra is inner.*

This is the main result of [1972-2], proved by reducing to the Lie algebra $L(A)$ and using the Killing-form hypothesis [1972-2, p. 565].

For the Cayley algebra over an algebraically closed field of characteristic 0, the derivation algebra is the exceptional simple Lie algebra of type G_2 [2004-7, p. 52]. At the group level one has $\mathrm{Aut}(\mathbb{O}) \cong G_2$ [2002-2, p. 159].

We can also ask what happens when we weaken the Leibniz rule itself. Filippov studied the following class of maps for Lie algebras in earlier work and here for alternative and Malcev algebras [2000-2, Introduction, p. 354].

Definition 15.6 ([2000-2, Introduction, equation (1), p. 354])**.** Let Φ be a commutative associative ring with unity containing $\frac{1}{2}$, let A be a Φ-algebra, and let $\delta \in \Phi$ be fixed. A *δ-derivation* of A is a Φ-linear map $d\colon A \to A$ with

$$d(xy) \;=\; \delta\big(d(x)\,y + x\,d(y)\big) \qquad \text{for every } x, y \in A.$$

Filippov calls the cases $\delta \in \{0, \frac{1}{2}, 1\}$ trivial: a 1-derivation is a usual derivation, a 0-derivation is an endomorphism vanishing on products, and every element of the centroid of A is a $\frac{1}{2}$-derivation [2000-2, Introduction, p. 354].

Filippov described δ-derivations of prime alternative Φ-algebras with $\frac{1}{2} \in \Phi$ [2000-2, Theorem 2, p. 356].

Theorem 15.7 ([2000-2, Theorem 2, p. 356])**.** *A prime alternative Φ-algebra with $\frac{1}{2} \in \Phi$ has no non-trivial δ-derivations: every δ-derivation is either a derivation, a suitable multiple of the identity, or an antiderivation in the special cases $\delta \in \{0, \frac{1}{2}, 1, -1\}$.*

The content is that no other maps appear.

Theorem 15.8 ([2000-2, Theorem 3, p. 358])**.** *A prime non-Lie Malcev Φ-algebra with $\frac{1}{2} \in \Phi$ has no non-trivial δ-derivations.*

The Cayley algebra has a companion in dimension eight whose derivations are known in every characteristic, the Okubo algebra. Let F be a field and let $(\mathcal{O}, *)$ be the split Okubo algebra over F, a symmetric composition algebra of dimension eight [2013-5, p. 4]. The derivations of $(\mathcal{O}, *)$ lie in the Lie algebra that preserves its norm [2013-5, p. 5].

Theorem 15.9 ([2013-5, Theorem 6, pp. 5–6])**.** *Let F be a field of characteristic not three. Then* $\mathrm{Der}(\mathcal{O}, *)$ *is isomorphic to* $\mathfrak{sl}_3(F)$*. More precisely,* $\mathrm{Der}(\mathcal{O}, *) = \{\mathrm{ad}^*_x \colon y \mapsto x * y - y * x : x \in \mathcal{O}\}$.

So in characteristic not three every derivation is inner, and the underlying vector space of $\mathcal{O}$ is $\mathfrak{sl}_3(F)$ [2013-5, p. 5]. In characteristic three the answer changes. The inner derivations span a proper ideal of the derivation algebra, and the full derivation algebra is ten-dimensional and semisimple [2013-5, Theorem 9, p. 6]. Its derived algebra is simple and coincides with the span of the inner derivations [2013-5, Theorem 9, p. 6]. The derivation algebra is then isomorphic to the derivation algebra of a simple nodal noncommutative Jordan algebra [2013-5, Theorem 9, p. 6].

Section 15.5 below records the analogous δ-derivation results for simple Jordan algebras and superalgebras.

15.3 Leibniz algebras

The previous sections asked for the Jordan or Malcev identities. This section keeps only one Leibniz rule on the bracket. Loday introduced the class in 1993, while he studied a noncommutative form of the Chevalley–Eilenberg complex [1993-10, Introduction, p. 139].

Definition 15.10 ([1993-10, Definition (1.1), p. 141])**.** A *Leibniz algebra* $\mathfrak{g}$ over a commutative ring k is a k-module with a bilinear bracket $[-,-] \colon \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ satisfying the Leibniz identity

$$[x,[y,z]] \;=\; [[x,y],z] - [[x,z],y] \qquad \text{for all } x,y,z \in \mathfrak{g}.$$

This is the right-Leibniz convention, in which identity (1.1) controls the right factor. The left-Leibniz convention swaps the two factors [1993-10, Definition (1.1), p. 141]. A Leibniz algebra with $[x,x] = 0$ for every x is a Lie algebra, since the Leibniz identity then becomes the Jacobi identity [1993-10, Definition (1.1), p. 141]. Ayupov and Omirov note that the right multiplications of a Leibniz algebra act like the inner derivations of a Lie algebra [1998-1, p. 2].

Proposition 15.11 ([1998-1, p. 2])**.** *Let L be a Leibniz algebra over a field F, and for $x \in L$ write $R_x(z) = [z, x]$ for right multiplication by x. Then the set $R(L) = \{R_x : x \in L\}$ with the commutator product is a Lie algebra, and $R_x R_y - R_y R_x = R_{[y,x]}$ holds in $R(L)$.*

Biyogmam and Tcheka measure the gap from the Lie case by one ideal [2021-14, p. 2191].

Definition 15.12 ([2021-14, p. 2191])**.** For a Leibniz algebra $\mathfrak{g}$, Biyogmam and Tcheka write $\mathrm{Leib}(\mathfrak{g})$ for the subspace spanned by all elements $[x, x]$ with $x \in \mathfrak{g}$. It is a two-sided ideal of $\mathfrak{g}$, called the *Leibniz kernel*, and the quotient $\mathfrak{g}_{\mathrm{Lie}} = \mathfrak{g}/\mathrm{Leib}(\mathfrak{g})$ is a Lie algebra.

This kernel brings innerness and completeness to Leibniz algebras [2021-14, Definition 2.4 and Definition 3.3, p. 2192].

Definition 15.13 ([2021-14, Definition 2.4, p. 2192])**.** A derivation $d\colon \mathfrak{g} \to \mathfrak{g}$ of a Leibniz algebra $\mathfrak{g}$ is an *inner derivation* if $d(\mathfrak{g}) \subseteq R_x(\mathfrak{g}) + \mathrm{Leib}(\mathfrak{g})$ for some $x \in \mathfrak{g}$.

A derivation that is not inner is an *outer derivation* [2021-14, Definition 3.3, p. 2192]. Biyogmam and Tcheka call $\mathfrak{g}$ *complete* when $Z(\mathfrak{g}/\mathrm{Leib}(\mathfrak{g})) = 0$ and $\mathfrak{g}$ has no outer derivation [2021-14, Definition 3.3, p. 2192]. Every semisimple Leibniz algebra in characteristic zero is complete. The derivations themselves can be written down. On a simple complex Leibniz algebra $L = G \dotplus I$, every derivation is $R_a + \alpha + \varphi$. Here $a \in G$, the map α on the ideal respects the bracket, and φ carries G into that ideal [2017-12, Theorem 3.2]. Rakhimov, Masutova and Omirov give the same description for a subclass of the semisimple algebras [2017-12, Abstract, p. 1].

Proposition 15.14 ([2021-14, Proposition 3.5, p. 2193])**.** *Let $\mathfrak{g}$ be a semisimple Leibniz algebra over a field of characteristic zero. Then $\mathfrak{g}$ admits no outer derivations. More precisely, each derivation d of $\mathfrak{g}$ can be written as $d = R_z + a$ for some $z \in \mathfrak{g}$ and some derivation $a\colon \mathfrak{g} \to \mathrm{Leib}(\mathfrak{g})$ that vanishes on the Levi subalgebra of $\mathfrak{g}$.*

At dimension three the description is complete. Kurdachenko, Pypka and Semko take the non-Lie left Leibniz algebras of dimension three over any field [2026-20, p. 1]. Earlier work split them into sixteen types, and the authors list the derivations of each type by direct calculation [2026-20, p. 1].

Theorem 15.15 ([2026-20, Theorem 1.1, p. 2])**.** *Let L be a three-dimensional non-Lie left Leibniz algebra over an arbitrary field F. Then its derivation*

algebra $\mathrm{Der}(L)$ *is the subspace of* $M_3(F)$ *given by the formulas proved or quoted in the cited paper and collected in its Table 2. The table covers every parameter family and every case of characteristic two in the sixteen-type list.*

The table shows that one derivation algebra grows in characteristic two. In that characteristic several types gain derivations [2026-20, p. 1]. So the dimension can change along a family. The authors also record how the resulting Lie algebras split into ideals and subalgebras [2026-20, p. 1].

Nilpotent Leibniz algebras of class three also have an explicit answer at dimension three. Kurdachenko, Semko and Yashchuk take the algebras of dimension three whose class is three, and those whose centre has dimension two [2024-15, p. 819]. Take the first algebra. Its derivation algebra is a semidirect sum. The ideal is abelian. The complement is generated by one derivation, which scales the three basis vectors by 1, 2 and 3 [2024-15, Theorem 3.1, pp. 821–822].

The derivation algebra also has a matrix shape. It is isomorphic to the Lie algebra of matrices

$$\begin{pmatrix} \alpha_1 & 0 & 0 \\ \alpha_2 & 2\alpha_1 & 0 \\ \alpha_3 & \alpha_2 & 3\alpha_1 \end{pmatrix}, \qquad \alpha_1, \alpha_2, \alpha_3 \in F,$$

so it is three-dimensional [2024-15, Theorem 3.1, pp. 821–822]. The second algebra of the paper has a derivation algebra of another shape [2024-15, Theorem 3.2, p. 825].

The filiform families have a dimension count that depends on the structure constants. Rakhimov and Al-Hossain take three naturally graded complex filiform algebras. They also take two further families that come from them [2012-14, Theorem 1, p. 3]. They compute the dimension of the derivation algebra in each family [2012-14, Theorems 3–5, pp. 4, 8 and 12].

Theorem 15.16 ([2012-14, Theorems 3–5, pp. 4, 8 and 12])**.** *Let* $n \geq 2$*. The derivation algebras of* NGF_1*,* NGF_2 *and* NGF_3 *have dimensions* $n+1$*,* $n+2$ *and* $2n-1$*. For* L *in* FL_b^n *the dimension lies between* $n-1$ *and* $n+1$*, and for* L *in* SL_b^n *it lies between* $n-1$ *and* $n+2$*.*

So the dimension is not fixed by the filiform class. The structure constants of each family decide it [2012-14, pp. 8–12].

Almost inner derivations ask for less than innerness. A derivation D of a Leibniz algebra L is almost inner when every $x \in L$ has some $a_x \in L$ with

$D(x) = [x, a_x]$ [2025-11, Definition 2.4, p. 1527]. Every inner derivation is almost inner. The question is whether the two spaces agree [2025-11, p. 1527]. On a simple Leibniz algebra the answer depends on the shape of the algebra.

Theorem 15.17 ([2025-11, Theorems 4.1–4.2, pp. 1530–1531])**.** *Let $L = \mathfrak{g} \ltimes I$ be a simple Leibniz algebra over $\mathbb{C}$.*

1. *If $\mathfrak{g}$ and I are isomorphic as $\mathfrak{g}$-modules, then every almost inner derivation of L is inner.*
2. *If they are not isomorphic, then every almost inner derivation of L is inner when $\mathfrak{g} \cong \mathfrak{sl}_2$, and there is an almost inner derivation that is not inner otherwise.*

The proof tells the two cases apart by one strongly regular element of a Cartan subalgebra [2025-11, p. 1530]. So the exceptional case belongs to the smallest simple Lie algebra.

Complete Leibniz algebras settle the bilinear form of the same question. The literature has two definitions of completeness, and they differ [2025-7, Definitions 2.6–2.7, p. 6]. Under the first, L is complete when $Z(L/\mathrm{Leib}(L)) = 0$ and every derivation is inner up to an element of $\mathrm{Leib}(L)$ [2025-7, Definition 2.6, p. 6]. Under the second, L is complete when $Z(L) = 0$ and every derivation of L is inner [2025-7, Definition 2.7, p. 6].

Theorem 15.18 ([2025-7, Theorems 3.1–3.2, pp. 7–8])**.** *Let L be a complete Leibniz algebra. Then a bilinear map $f\colon L \times L \to L$ is a biderivation of L if and only if there are linear maps $\varphi, \psi\colon L \to L$ with $f(x, y) = [\varphi(x), y] = [\psi(y), x]$ for all $x, y \in L$, in the sense of Definition 2.7. In the sense of Definition 2.6 the same holds up to a right and a left biderivation with values in* $\mathrm{Leib}(L)$*.*

The two versions differ exactly in the Leibniz kernel [2025-7, pp. 7–8]. One example is complete in one sense and not in the other [2025-7, Example 2.7.1, p. 6]. So the two statements are not interchangeable.

The non-nilpotent side has a calculation of its own. Kurdachenko, Semko and Subbotin take some non-nilpotent Leibniz algebras of low dimension and list the derivations of each [2025-12, Theorems 1–3, pp. 120–126]. A second paper does the same for further families [2025-13, Theorems 1–2, pp. 209–211].

The next section separates the even and odd parts of an algebra: it treats derivations that respect a grading by $\mathbb{Z}/2\mathbb{Z}$.

Leibniz algebras have a dialgebra companion, and the same maps live there. Restrepo and Sanchez carry derivations and biderivations to dialgebras [2026-28, Abstract].

On the Lie side, biderivations reach the algebras of maximal class: of Fialowski's three such algebras with one-dimensional homogeneous components, the biderivations, commuting maps and 2-local derivations are computed in [2024-2, Abstract].

15.4 Lie and Jordan superalgebras

The Leibniz algebras of the previous section keep one product rule. A superalgebra separates the even part from the odd part, and the product rule picks up a sign [2012-11, pp. 1–2].

Definition 15.19 ([2012-11, pp. 1–2])**.** Let A be a $\mathbb{Z}_2$-graded algebra over a field K that need not be associative. A K-linear map $\partial\colon A \to A$ of degree $\alpha \in \mathbb{Z}_2$ is a *left superderivation* if

$$\partial(bc) \;=\; \partial(b)c + (-1)^{\alpha\bar{b}}\, b\,\partial(c) \qquad \text{for all } b, c \in A,$$

where $\bar{b}$ is the degree of the homogeneous element b. A *superderivation* is a left superderivation. A superderivation of degree 0 is called *even*, and one of degree 1 is called *odd*.

The graded bracket of the next definition supplies the maps on which superderivations act.

Definition 15.20 ([2012-11, p. 1])**.** A *Lie superalgebra* is a $\mathbb{Z}_2$-graded vector space $\mathfrak{g} = \mathfrak{g}_0 \oplus \mathfrak{g}_1$ over K with a bilinear map $[-,-]\colon \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ that is graded skew-symmetric and satisfies the graded Jacobi identity.

Musson proves the following equivalence [2012-11, Lemma 1.1.2, pp. 2–3].

Lemma 15.21 ([2012-11, Lemma 1.1.2, pp. 2–3])**.** *Let A be a $\mathbb{Z}_2$-graded algebra whose product $(a,b) \mapsto [a,b]$ is graded skew-symmetric. The following conditions are equivalent.*

1. *A satisfies the graded Jacobi identity.*
2. *For every $a \in A$ the map $\operatorname{ad} a : b \mapsto [a,b]$ is a left superderivation of degree $\bar{a}$.*

3. *For every homogeneous $a \in A_\alpha$ the map $b \mapsto [b, a]$ is a right superderivation of degree α.*

Musson quotes the following theorem of Scheunert, where a nondegenerate Killing form stands in place of simplicity [2012-11, Theorem 2.6.2, pp. 19–20].

Theorem 15.22 ([2012-11, Theorem 2.6.2, pp. 19–20])**.** *Let $\mathfrak{g}$ be a Lie superalgebra whose Killing form is nondegenerate. Then every superderivation of $\mathfrak{g}$ is inner, that is, of the form $\operatorname{ad} a$ for some $a \in \mathfrak{g}$.*

Musson points out that the superalgebras with a nondegenerate Killing form behave like semisimple Lie algebras [2012-11, p. 20]. Kac classified the simple finite-dimensional Lie superalgebras as follows [1977-2, Theorem 1, pp. 1385–1386].

Theorem 15.23 ([1977-2, Theorem 1, pp. 1385–1386])**.** *For a simple finite-dimensional Lie superalgebra over an algebraically closed field of characteristic 0, all $\mathbb{Z}$-gradings are known explicitly: the list in [1977-2] includes every $\mathbb{Z}$-grading (up to automorphism) of every such superalgebra.*

A local superderivation matches its value at one point at a time. Chi, Zhang, Meng and Tang work over a field of characteristic not two with a finite-dimensional nilpotent Lie superalgebra [2026-10, p. 1]. A local superderivation is called pure when it is not a superderivation [2026-10, p. 1].

Theorem 15.24 ([2026-10, Theorems 3.2 and 3.5, pp. 7, 12])**.** *Let L be a finite-dimensional two-step nilpotent Lie superalgebra over a field of characteristic not two. Then L admits a pure local superderivation and a pure local anti-superderivation.*

So on the nilpotent side local maps are genuinely wider than global ones. The authors give sufficient conditions for the same failure at higher nilpotency index [2026-10, p. 1]. Every three-step nilpotent Lie superalgebra admits a pure local superderivation [2026-10, Theorem 4.2, p. 15].

A superderivation of a prime superalgebra obeys a polynomial identity only in three ways. A superderivation of a superalgebra keeps the grading, so it is a sum $D = D_0 + D_1$ of an even and an odd part [2014-10, p. 304]. Fošner and Fošner impose two commutator identities on each part.

Theorem 15.25 ([2014-10, Theorem 1, p. 305])**.** *Let A be a prime associative superalgebra and let $D = D_0 + D_1$ be a nonzero superderivation of A. Suppose that $D_i(x)[D_i(x), x]_s = 0$ and $[D_i(x), x]_s D_i(x) = 0$ for every homogeneous element x and $i = 0, 1$. Then one of the following holds:*

1. *$A_1 = 0$ and $D(A_0) \subseteq Z(A)$;*
2. *$D(A_0) = 0$ and A is commutative;*
3. *$D^2 = 0$.*

Here $[a, b]_s = ab - (-1)^{\bar{a}\bar{b}} ba$ is the super commutator [2014-10, p. 304]. The paper proves the analogous statements for each degree separately, and the theorem is their union [2014-10, Propositions 1–2, pp. 305–306].

The bilinear form of the question has its own definition for superalgebras. A super-biderivation is a bilinear map that is a superderivation in each argument, with the super sign carried through [2024-16, p. 2]. On a perfect Hom-Lie superalgebra it satisfies the relation $\varphi(\omega, [\zeta, \xi]) = (-1)^{|\varphi||\omega|}[\omega, \varphi(\zeta, \xi)]$ [2024-16, Lemma 3.7, p. 7]. On a centerless algebra it has the centroid form.

Theorem 15.26 ([2024-16, Theorem 3.8, p. 7])**.** *Let H be a centerless Hom-Lie superalgebra with $H = H'$. Then every super-biderivation $\varphi\colon H \times H \to H$ has the form $\varphi(\zeta, \xi) = \alpha^{-1}\delta([\zeta, \xi])$ for a map δ in the centroid of H.*

So the Hom-Lie superalgebra case has the shape of the Lie algebra case in Theorem 14.12, with the twist α^{-1} and a centroid element in place of a scalar [2024-16, Theorem 3.8, p. 7]. A special super-biderivation of a perfect centerless algebra is zero [2024-16, Theorem 3.10, p. 9].

The definition of a superbiderivation itself has more than one form in the literature. Di Bartolo, Di Fatta and La Rosa propose one that requires the bracket of the superalgebra to be a superbiderivation, a condition the earlier definitions do not meet [2025-6, p. 2]. Their two equivalent forms are Definitions 2.5 and 2.6 [2025-6, Definitions 2.5–2.6, p. 5]. A nonzero bilinear map of odd degree cannot be a superbiderivation under both the new and the earlier definitions at once [2025-6, p. 7]. On a complete Lie superalgebra every superbiderivation has the standard shape [2025-6, Theorem 3.6, p. 10].

With the classification in place we return to a map that appears both in the alternative picture and in the Jordan superalgebra picture: the δ-derivation of Section 15.5.

Han proves that the derivations of a rational vertex operator algebra are inner, the vertex-algebra analogue of the same question for algebras [2026-16, Abstract, p. 1].

A graded ring gives both an ordinary and a graded answer. Ghahramani, Ghosseiri and Safari characterize the derivations and biderivations of the quaternion ring $H(S)$ over a ring S in which 2 is invertible, and, reading $H(S)$ as a $\mathbb{Z}_2$-graded ring, its superbiderivations [2020-14, Abstract].

15.5 δ-derivations and related maps

Section 15.2 introduced the δ-derivation maps and the alternative and Malcev cases (Definition 15.6). Kaygorodov treats the Jordan algebras and superalgebras [2007-6]. For a simple finite-dimensional Jordan algebra there is exactly one non-trivial value of δ.

Theorem 15.27 ([2007-6, Theorem 2.5])**.** *Let φ be a non-trivial δ-derivation of a simple finite-dimensional Jordan algebra A over an algebraically closed field F of characteristic distinct from 2. Then $\delta = \frac{1}{2}$ and $\varphi(x) = \alpha x$ for some $\alpha \in F$.*

Kaygorodov notes that, for Lie algebras, a $\frac{1}{2}$-derivation is an element of the centroid of the algebra [2007-6, Introduction]. The Jordan superalgebra case follows from the same classification method [2007-6, Theorem 4.4].

Theorem 15.28 ([2007-6, Theorem 4.4])**.** *Let A be a simple finite-dimensional Jordan superalgebra over an algebraically closed field of characteristic 0, and let φ be a non-trivial δ-derivation of A. Then $\delta = \frac{1}{2}$ and $\varphi(x) = \alpha x$ for some $\alpha \in F$ and all $x \in A$.*

The two operations of the next class exchange the roles of multiplication and bracket in the Leibniz rule [2018-13, p. 657].

Definition 15.29 ([2018-13, p. 657])**.** A *Poisson algebra* is a vector space B over a field k with a commutative associative multiplication $x \cdot y$ and a Lie bracket $\{x, y\}$ satisfying

$$\{x \cdot y,\, z\} \;=\; x \cdot \{y, z\} + \{x, z\} \cdot y, \qquad x, y, z \in B.$$

The next definition states the derivations of a Poisson algebra.

Definition 15.30 ([2018-13, p. 657])**.** A *Poisson derivation* of a Poisson algebra B is a linear map D on B with $D(x \cdot y) = D(x) \cdot y + x \cdot D(y)$ and $D(\{x, y\}) = \{D(x), y\} + \{x, D(y)\}$ for all $x, y \in B$.

The next example is the symplectic Poisson algebra.

Example 15.31 ([2018-13, p. 657])**.** The *symplectic Poisson algebra* S_n is the polynomial algebra $k[x_1, y_1, \ldots, x_n, y_n]$ with the bracket determined by $\{x_i, y_j\} = \delta_{ij}$, $\{x_i, x_j\} = 0$ and $\{y_i, y_j\} = 0$.

On this algebra every derivation is inner.

Theorem 15.32 ([2018-13, Theorem 2.1, pp. 658–659])**.** *Let D be a derivation of the symplectic Poisson algebra $S_n = k[x_1, y_1, \ldots, x_n, y_n]$. Then $D^* = 0$, and there exists $f \in S_n$ such that $D = \mathrm{ad}_f$, that is, $D(g) = \{f, g\}$ for every $g \in S_n$. Here D^* denotes the divergence of D [2018-13, p. 657].*

Section 15.6 develops the free side of the same picture.

15.6 Poisson algebras

The δ-derivations of the previous section relax the product rule by a scalar. A Poisson algebra instead carries two products, and a derivation must respect both [2018-13, p. 657].

Definition 15.33 ([2018-13, p. 657])**.** A vector space B over a field k is a *Poisson algebra* if B carries two bilinear operations, a multiplication $x \cdot y$ and a bracket $\{x, y\}$, such that B is a commutative associative algebra under the multiplication, B is a Lie algebra under the bracket, and

$$\{x \cdot y, z\} \;=\; x \cdot \{y, z\} + \{x, z\} \cdot y \qquad \text{for all } x, y, z \in B.$$

The last identity says that the map $x \mapsto \{x, z\}$, for a fixed z, is a derivation of the associative product. The parallel requirement defines the derivations of the structure.

Definition 15.34 ([2018-13, p. 657])**.** Let B be a Poisson algebra. A linear map D on B is a *Poisson derivation* if

$$D(xy) \;=\; D(x)y + xD(y) \quad \text{and} \quad D(\{x, y\}) \;=\; \{D(x), y\} + \{x, D(y)\} \qquad \text{for all } x, y \in B.$$

The basic example is built on a polynomial algebra.

Example 15.35 ([2018-13, p. 657])**.** The *symplectic Poisson algebra* S_n is the polynomial algebra $k[x_1, y_1, \ldots, x_n, y_n]$ with the bracket given by

$$\{x_i, y_j\} = \delta_{ij}, \qquad \{x_i, x_j\} = 0, \qquad \{y_i, y_j\} = 0,$$

where δ_{ij} is the Kronecker symbol and $1 \le i, j \le n$.

A derivation D of a polynomial algebra in n generators can be written as $D = \sum_{i=1}^{n} D(x_i)\,\partial/\partial x_i$ [2018-13, p. 657]. Li denotes by

$$D^* \;=\; \sum_{i=1}^{n} \frac{\partial}{\partial x_i} D(x_i)$$

the *divergence* of D [2018-13, p. 658]. For a Poisson derivation of the symplectic algebra the divergence vanishes, and more holds.

Theorem 15.36 ([2018-13, Theorem 2.1, pp. 658–659])**.** *If D is a derivation of the symplectic Poisson algebra $S_n = k[x_1, y_1, \ldots, x_n, y_n]$, then $D^* = 0$ and there exists $f \in S_n$ such that $D = \mathrm{ad}_f$, that is, $D(g) = \{f, g\}$ for any $g \in S_n$.*

Every Poisson derivation of S_n is therefore inner. A transposed Poisson algebra trades the Leibniz rule of the bracket for a factor one half. Let P carry a commutative associative product and a Lie bracket such that $2x \cdot [y, z] = [x \cdot y, z] + [y, x \cdot z]$ for all x, y, z [2025-1, equation (12), p. 489]. A transposed Poisson superalgebra carries the graded form of the same identity [2025-1, Definition 2.1, p. 490]. Abramov and Sovetnikov use an even derivation to build new structures from it [2025-1, p. 487].

Theorem 15.37 ([2025-1, Theorem 4, p. 495])**.** *Let $(P, \cdot, [\,,\,])$ be a transposed Poisson superalgebra and let D be its even derivation. Define $[x, y, z] = D(x) \cdot [y, z] + (-1)^{|x||yz|} D(y) \cdot [z, x] + (-1)^{|xy||z|} D(z) \cdot [x, y]$. Then $(P, [\,,\,,\,])$ is a 3-Lie superalgebra.*

So an even derivation of a transposed Poisson superalgebra produces a ternary bracket that satisfies the Filippov–Jacobi identity [2025-1, p. 495]. The paper also shows that the even derivations of such an algebra form a Jordan algebra. The odd ones close on the same bracket exactly when the derivation squares to zero [2025-1, Theorem 2, p. 492].

The next section keeps the Lie product and twists the Jacobi identity by an algebra endomorphism: Hom-Lie algebras and Lie conformal algebras.

15.7 Hom-Lie algebras and Lie conformal algebras

The Poisson algebras of the previous section carry a Lie bracket as one of their two operations. The two classes that follow deform that bracket itself.

Definition 15.38 ([2006-5, Definition 4, p. 18])**.** A *hom-Lie algebra* (L, ς) is a non-associative algebra L together with an algebra homomorphism $\varsigma\colon L \to L$ such that

$$[x, y]_L = -[y, x]_L$$

and

$$[(\mathrm{id} + \varsigma)(x), [y, z]_L]_L + [(\mathrm{id} + \varsigma)(y), [z, x]_L]_L + [(\mathrm{id} + \varsigma)(z), [x, y]_L]_L = 0$$

for all $x, y, z \in L$, where $[\cdot, \cdot]_L$ denotes the product in L.

If L is a Lie algebra, it is a hom-Lie algebra with ς the identity map [2006-5, p. 18]. A hom-Lie algebra structure on a Lie algebra amounts to a Lie algebra automorphism [2008-7, Section 2, pp. 1399–1400], and on a simple Lie algebra only the identity one works.

Proposition 15.39 ([2008-7, Proposition 2.2, p. 1400])**.** *Let L be a finite-dimensional simple Lie algebra. If (L, σ) is a hom-Lie algebra, then $\sigma = \mathrm{id}$. In other words, hom-Lie algebra structures on simple Lie algebras are trivial.*

The second class replaces the one bracket by a family of brackets indexed by a formal parameter.

Definition 15.40 ([2019-6, Definition 2.1, p. 2])**.** A *Lie conformal algebra* R is a $\mathbb{C}[\partial]$-module endowed with a $\mathbb{C}$-bilinear map

$$R \otimes R \to R[\lambda], \qquad a \otimes b \mapsto [a_\lambda b],$$

that satisfies, for all $a, b, c \in R$, the three axioms of conformal sesquilinearity

$$[\partial a_\lambda b] = -\lambda [a_\lambda b], \qquad [a_\lambda \partial b] = (\partial + \lambda)[a_\lambda b],$$

skew-symmetry

$$[a_\lambda b] = -[b_{-\lambda - \partial} a],$$

and the Jacobi identity

$$[a_\lambda [b_\mu c]] = [[a_\lambda b]_{\lambda+\mu} c] + [b_\mu [a_\lambda c]].$$

Fan, Hong and Su define the derivations of such an algebra among the conformal linear maps.

Definition 15.41 ([2019-6, Definition 2.7, pp. 2–3])**.** Let R be a Lie conformal algebra. A conformal linear map $D_\lambda\colon R \to R[\lambda]$ is a *conformal derivation* if

$$D_\lambda([a_\mu b]) \;=\; [(D_\lambda a)_{\lambda+\mu} b] + [a_\mu (D_\lambda b)] \qquad \text{for all } a, b \in R.$$

For any $a \in R$ the map $(\operatorname{ad} a)_\lambda$ given by $(\operatorname{ad} a)_\lambda b = [a_\lambda b]$ is a conformal derivation. Those of this form are the inner conformal derivations. They form a conformal subalgebra of the algebra of all conformal derivations [2019-6, Definition 2.7, p. 3].

The generalized families of derivations carry over to the twisted case. A Hom-Lie conformal algebra (R, α) is multiplicative when α is an algebra endomorphism [2018-18, p. 3]. Zhao, Yuan and Chen define the α^k-derivations, quasiderivations, centroids and quasicentroids of R by the same identities as in the untwisted case, with α^k inserted in each argument [2018-18, p. 16].

Theorem 15.42 ([2018-18, Theorem 5.4, p. 18])**.** *Let (R, α) be a multiplicative Hom-Lie conformal algebra. Then* $\mathrm{GDer}(R) = \mathrm{QDer}(R) + \mathrm{QC}(R)$*, the sum of the quasiderivations and the quasicentroids of R.*

The same paper shows that $\mathrm{ZDer}(R)$ is an ideal of $\mathrm{Der}(R)$, and that the generalized derivations, quasiderivations and centroids are subalgebras [2018-18, Proposition 5.2, p. 17]. The quasicentroid is again a Hom-Lie conformal algebra exactly when it coincides with the centroid [2018-18, Proposition 5.6, p. 19].

The conformal version of a biderivation has the centroid shape as well. A conformal super-biderivation of a Lie conformal superalgebra is a $\mathbb{C}$-bilinear map of two arguments that is a conformal derivation in each [2023-9, Definition 2.2, p. 2]. Asif and Yao treat the current algebra $L \otimes A$ built from a Lie conformal superalgebra L and a commutative associative algebra A [2023-9, p. 1].

Theorem 15.43 ([2023-9, Theorem 2.9, p. 3])**.** *Every skew-symmetric biderivation of a centerless perfect Lie conformal superalgebra L has the form $\phi_\lambda(x, y) = \alpha_\mu([x_\lambda y])$ for a conformal linear map α_μ of L.*

The same shape passes to the current algebra: if every super-biderivation of L has the centroid form, then every one of $L \otimes A$ does [2023-9, Theorem 3.3, p. 7]. The conformal super-commuting maps of $L \otimes A$ belong to the centroid under the same hypothesis [2023-9, Theorem 4.4, p. 8].

The last section of the chapter moves from binary brackets to n-ary ones.

15.8 n-Lie algebras and ternary algebras

The Hom-Lie algebras of the previous section keep a binary bracket.

The classes of this section replace the bracket itself by an n-ary operation.

Example 15.44 ([2006-8, Section 1, pp. 3595–3596])**.** Let K be a field of characteristic different from 2 and 3. Let A be a composition algebra over K with involution $a \mapsto \bar{a}$ and unit 1, whose symmetric bilinear form $\langle x, y\rangle = \frac{1}{2}(x\bar{y} + y\bar{x})$ is nonsingular. With the ternary multiplication

$$\langle x, y, z\rangle \;=\; (x\bar{y})z - \langle y, z\rangle x + \langle x, z\rangle y - \langle x, y\rangle z,$$

A becomes a ternary Malcev algebra, denoted $M(A)$. If $\dim A = 8$, then $M(A)$ is not a 3-Lie algebra and is denoted by M_8.

Pojidaev and Saraiva call a derivation of M_8 inner if it belongs to the Lie algebra generated by the right multiplications of M_8 [2006-8, Section 2, p. 3596].

Theorem 15.45 ([2006-8, Theorem 2.3, p. 3598])**.** *All derivations of M_8 are inner.*

The derivation algebra of M_8 is 21-dimensional and isomorphic to the simple Lie algebra B_3 [2006-8, Section 2, p. 3598]. The next notion extends the derivation concept itself to n-ary algebras.

Definition 15.46 ([2014-14, p. 576])**.** Let A be an n-ary algebra. An $(n+1)$-*ary derivation* of A is a collection $(f_0, f_1, \ldots, f_n)$ of linear endomorphisms of A such that

$$f_0[x_1, \ldots, x_n] \;=\; \sum_{i=1}^{n} [x_1, \ldots, f_i(x_i), \ldots, x_n]$$

for arbitrary $x_1, \ldots, x_n \in A$.

An $(n+1)$-ary derivation $(f_0, f_1, \ldots, f_n)$ with $f_1 = \cdots = f_n$ is an $(n+1)$-ary quasiderivation, and the collections arising from the centroid or from an ordinary derivation $(D, D, \ldots, D)$ are called trivial [2014-14, pp. 576–577]. For the simple non-Lie n-ary Mal'tsev algebras, nothing nontrivial occurs.

Theorem 15.47 ([2014-14, Theorem 5, p. 582])**.** *The simple ternary Mal'tsev algebra M_8 is free of nontrivial 4-ary derivations.*

Kaygorodov also proves the same for the simple 7-dimensional Mal'tsev algebra M_7: it is free of nontrivial 3-ary derivations [2014-14, Theorem 1, p. 579]. Writing $\Delta(A)$ for the Lie algebra of generalized derivations of A modulo its annihilator, the two results combine as follows.

Corollary 15.48 ([2014-14, Corollary 6, p. 583])**.** $\Delta(M_7) \cong \Delta(M_8)$.

The generalized derivations of an n-Lie algebra have a cohomological home. A generalized derivation of an n-Lie algebra L is a linear map $D\colon L^{n-1} \to L$ that satisfies the two identities of Ateşli, Esen and Sütlü [2021-12, equations (2.9)–(2.10), p. 12]. A derivation of L is the case of a map $D\colon L \to L$ with the n-ary Leibniz rule [2021-12, p. 12].

Theorem 15.49 ([2021-12, Proposition 2.13 and Corollary 3.2, pp. 23 and 27])**.** *Let L be an n-Lie algebra and let $D\colon L^{n-1} \to L$ be a generalized derivation. Then the associated map $D^\sharp$ is a cocycle of the cohomology of L with coefficients in $\mathfrak{gl}(L)$, and the cohomology of the generalized derivation extension $L \oplus k$ coincides with that of L.*

The cohomology that classifies generalized derivations is not the usual cohomology that classifies abelian extensions, and the two agree only for $n = 3$ [2021-12, p. 2]. The paper also lifts the Hochschild–Serre spectral sequence to n-Lie algebras [2021-12, pp. 2–3].

A derivation can also be carried along as part of the structure. A 3-LieDer pair is a 3-Lie algebra together with a derivation of it, and Xu and Liu define a cohomology for such pairs with coefficients in a representation [2021-23, p. 1]. They use it to describe the deformations and the abelian extensions of the pair [2021-23, p. 1]. The same paper raises the notion one level, to a 3-Lie$_2$Der pair, and shows that the skeletal objects are classified by a triple [2021-23, p. 1].

This closes the tour of nonassociative classes. The next chapter returns to associative and commutative algebras.

Chapter 16

Derivations of near-rings, semirings, Gamma-rings and lattices

The earlier chapters studied derivations on rings, algebras, algebras of operators and Lie type structures. This chapter asks how the product rule behaves when the ambient system loses part of the ring structure. A near-ring drops one distributive law. A semiring drops the additive negatives. A Γ-ring replaces the single product by a family of products. A lattice keeps only the order. A BCI, MV or subtraction algebra keeps only an implication or a difference.

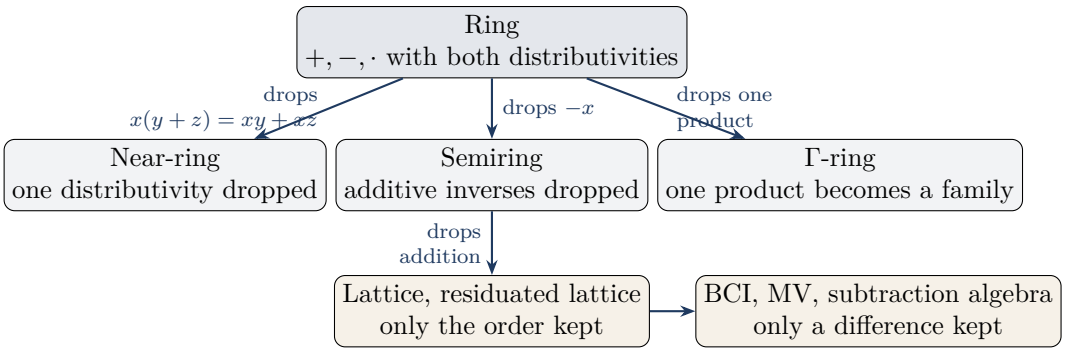


Figure 16.1: What each system in this chapter keeps of the ring axioms, and which axiom it gives up.

Kuratowski asked the question first, but in the language of topology. His closure and derivation operators on parts of a space were maps on a Boolean algebra. Goldberg recalls them in this form in 1975 [1975-3, p. 365]. The derived set of a set A is the set of its limit points. The operator $A \mapsto A^d$ is then a derivation on the Boolean algebra of all parts of the space. This is the oldest derivation on a system with no ring addition.

Bell and Mason moved the question to near-rings in 1987. A near-ring has the product rule but only one distributive law. The proofs from prime rings do not carry over. Their note gave the first commutativity theorems for near-rings with a derivation [1987-2, pp. 31–33]. Section 16.1 opens with these.

Between rings and lattices sits another family. Nobusawa Γ-rings replace the product xy by a product $x \alpha y$ indexed by a set Γ of scalars. Dey, Paul and Rakhimov treat derivations and generalized derivations in this setting [2012-8, pp. 2–3]. Section 16.2 follows their definitions.

Semirings drop the negatives, so the proofs that use subtraction break. Ahmed and Dudek show how left Jordan derivations force commutativity on a class of semirings with an additive-inverse structure [2021-9, pp. 624–631]. Section 16.3 takes this up.

Lattices come next. Xin, Li and Lu defined a derivation on a lattice by the rule $d(x \wedge y) = (dx \wedge y) \vee (x \wedge dy)$ [2008-11, Definition 3.1, p. 309]. Çeven and Öztürk then replaced the second argument by a second map f and obtained the f-derivations [2008-3, Definition 8, p. 702]. A residuated lattice carries a product and its residuum, so He, Xin and Zhan could define a multiplicative derivation there [2016-12, Definition 3.1, p. 100]. Section 16.4 tells this part.

The last two sections treat algebras of logic. Jun and Xin applied the derivation idea to BCI-algebras [2004-6, Definition 3.5, p. 170]. Ciungu defined two types of implicative derivations on pseudo-BCK algebras [2021-15, p. 178]. Wang, She and Qian studied implicative derivations on MV-algebras [2019-15, Definition 3.1, p. 262]. the subtraction of Schein algebras, studied by Öztürk and Çeven [2009-1, Definition 3.1, p. 511], give the last family. Sections 16.5 and 16.6 close the chapter.

16.1 Near-rings and prime near-rings with derivation

The previous section listed the systems of this chapter, and the first of them is the near-ring [1987-2, p. 31]. A ring has two distributive laws; a

near-ring keeps only one [1987-2, p. 31]. The product rule of a derivation still makes sense, but the proofs from prime rings that use the missing law break [1987-2, p. 31]. Bell and Mason asked in 1987 what survives [1987-2, p. 31]. Their setting follows.

Definition 16.1 ([1987-2, p. 31])**.** Let N be a zero-symmetric left near-ring with multiplicative center Z. A *derivation* on N is an additive endomorphism D of $(N,+)$ satisfying the product rule $D(xy) = x\,D(y) + D(x)\,y$ for all $x, y \in N$. An element $x \in N$ with $D(x) = 0$ is called a *constant*. The derivation D is *commuting* if $[x, D(x)] = 0$ for all $x \in N$.

The commutativity theorems below need the near-ring analogue of a prime ring [1987-2, p. 31].

Definition 16.2 ([1987-2, p. 31])**.** The near-ring N is *prime* if $a\,N\,b = \{0\}$ implies $a = 0$ or $b = 0$ for all $a, b \in N$.

Bell and Mason note that this condition implies the usual definition of a prime near-ring [1987-2, p. 31].

The first surprise is that the product rule alone gives a weak form of the missing distributive law.

Lemma 16.3 ([1987-2, Lemma 1, p. 32])**.** *Let D be a derivation on a near-ring N. Then N satisfies the partial distributive law*

$$\big(a\,D(b) + D(a)\,b\big)\,c = a\,D(b)\,c + D(a)\,b\,c \qquad (a, b, c \in N).$$

Bell and Mason proved the first commutativity theorem for a near-ring with a derivation [1987-2, Theorem 1, p. 32].

Theorem 16.4 ([1987-2, Theorem 1, p. 32])**.** *Let N have no nonzero divisors of zero. If N admits a nontrivial commuting derivation D, then $(N,+)$ is abelian.*

They then moved to prime near-rings, where the analogue of the first Posner theorem on rings holds [1987-2, Theorem 2, p. 33].

Theorem 16.5 ([1987-2, Theorem 2, p. 33])**.** *If a prime near-ring N admits a nontrivial derivation D for which $D(N) \subseteq Z$, then $(N,+)$ is abelian. Moreover, if N is 2-torsion-free, then N is a commutative ring.*

The same conclusion follows when the derivations commute with each other [1987-2, Theorem 3, p. 33].

Theorem 16.6 ([1987-2, Theorem 3, p. 33]). *Let N be a prime near-ring admitting a nontrivial derivation D such that $[D(x), D(y)] = 0$ for all $x, y \in N$. Then $(N, +)$ is abelian. If N is 2-torsion-free as well, then N is a commutative ring.*

Bell continued alone in 1997 [1997-3, p. 191]. His condition now holds only on a subset [1997-3, p. 191]. He called a nonempty subset U of N a semigroup right ideal if $UN \subseteq U$, a semigroup left ideal if $NU \subseteq U$, and a semigroup ideal if both hold [1997-3, p. 191].

Proposition 16.7 ([1997-3, Abstract, p. 191]). *Let N be a 3-prime zero-symmetric left near-ring, and let d be a nonzero derivation on N. Let U be a nonzero subset of N. Then N is a commutative ring if one of the following holds:* (a) *U is a semigroup right or left ideal of N and $d(U)$ is multiplicatively central;* (b) *U is a semigroup ideal of N, $d^2 \neq 0$ and $[d(U), d(U)] = \{0\}$.*

Argaç asked in 1997 when a derivation on a near-ring is a homomorphism on a subset [1997-2, p. 737]. For rings, Bell and Kappe had shown that such a derivation vanishes [1997-2, p. 737]. The same holds for near-rings [1997-2, p. 737].

Theorem 16.8 ([1997-2, Theorem, p. 737]). *Let N be a semiprime right near-ring, and let d be a derivation on N. Let A be a subset of N such that $0 \in A$ and $AN \subseteq A$. If d acts as a homomorphism on A or as an anti-homomorphism on A, then $d(A) = \{0\}$.*

In a near-ring the prime ideals behind these theorems come in four strengths, listed by Groenewald in 1991 [1991-4, Definition 1, p. 2669].

Definition 16.9 ([1991-4, Definition 1, p. 2669]). Let N be a near-ring and P an ideal of N. Then P is 0-*prime* if $AB \subseteq P$ implies $A \subseteq P$ or $B \subseteq P$ for all ideals A, B of N; 1-*prime* if the same holds for all left ideals A, B; 2-*prime* if the same holds for all N-subgroups A, B; and 3-*prime* if $aNb \subseteq P$ implies $a \in P$ or $b \in P$ for all $a, b \in N$. The near-ring N is ν-*prime* ($\nu = 0, 1, 2, 3$) if the zero ideal is ν-prime.

The 3-prime near-rings are the ones named 3-prime in the 1997 paper of Bell above [1997-3, p. 191]. The next section moves to Γ-rings, where the product itself becomes a family [2012-8, p. 2].

16.2 Gamma-rings

Recall that a near-ring keeps only one distributive law. A Γ-ring goes further: it replaces the product by a family of products [2012-8, p. 2]. There is one product for each element of a set Γ [2012-8, p. 2]. Nobusawa introduced the notion in 1964; Dey, Paul and Rakhimov state it in the current form [2012-8, p. 2].

Definition 16.10 ([2012-8, p. 2])**.** Let M and Γ be additive abelian groups. Then M is a Γ-*ring* if there is a mapping $M \times \Gamma \times M \to M$, written $(a, \alpha, b) \mapsto a\,\alpha\,b$, such that for all $a, b, c \in M$ and $\alpha, \beta \in \Gamma$: (i) $a\,\alpha\,b \in M$; (ii) $(a+b)\,\alpha\,c = a\,\alpha\,c + b\,\alpha\,c$, $a\,(\alpha+\beta)\,b = a\,\alpha\,b + a\,\beta\,b$, $a\,\alpha\,(b+c) = a\,\alpha\,b + a\,\alpha\,c$, and $(a\,\alpha\,b)\,\beta\,c = a\,\alpha\,(b\,\beta\,c)$.

The next definition states the derivations on these rings.

Definition 16.11 ([2012-8, pp. 2–3])**.** Let M be a Γ-ring. An additive mapping $D\colon M \to M$ is a *derivation* of M if

$$D(x\,\alpha\,y) = D(x)\,\alpha\,y + x\,\alpha\,D(y) \qquad (x, y \in M,\ \alpha \in \Gamma).$$

It is a *Jordan derivation* if $D(x\,\alpha\,x) = D(x)\,\alpha\,x + x\,\alpha\,D(x)$ for all $x \in M$ and $\alpha \in \Gamma$.

The next definition allows a second derivation on one side.

Definition 16.12 ([2012-8, p. 3])**.** An additive mapping $D\colon M \to M$ is a *generalized derivation* of M if there is a derivation d of M such that

$$D(x\,\alpha\,y) = D(x)\,\alpha\,y + x\,\alpha\,d(y) \qquad (x, y \in M,\ \alpha \in \Gamma).$$

It is a *Jordan generalized derivation* if $D(x\,\alpha\,x) = D(x)\,\alpha\,x + x\,\alpha\,d(x)$ for all $x \in M$ and $\alpha \in \Gamma$.

The main theorem of Dey, Paul and Rakhimov says that every Jordan generalized derivation is generalized, under a restriction on M [2012-8, Abstract, p. 1]. The restriction is the assumption

$$a\,\alpha\,b\,\beta\,c = a\,\beta\,b\,\alpha\,c \qquad (a, b, c \in M,\ \alpha, \beta \in \Gamma), \tag{16.1}$$

which the authors impose on the Γ-ring M [2012-8, pp. 1–2].

Theorem 16.13 ([2012-8, Abstract, p. 1])**.** *Let M be a 2-torsion-free semiprime Γ-ring satisfying* (16.1)*. Let $D\colon M \to M$ be an additive mapping such that, for some derivation d of M, $D(x\,\alpha\,x) = D(x)\,\alpha\,x + x\,\alpha\,d(x)$ for all $x \in M$ and $\alpha \in \Gamma$. Then D is a generalized derivation of M.*

The theorem lifts the same statement for rings, due to Hvala, to the Γ setting [2012-8, p. 1]. The next section moves to semirings, where a different obstacle appears: addition has no negatives at all [2021-9, p. 625].

16.3 Semirings

The Γ-rings of the previous section still had additive negatives, and a semiring drops them: proofs that use subtraction then break, and the methods change [2021-9, p. 624]. Ahmed and Dudek describe the setting as follows [2021-9, p. 625].

Definition 16.14 ([2021-9, p. 625])**.** A *semiring* $(S, +, \cdot)$ is a nonempty set S with two binary operations $+$ and $\cdot$ such that $\cdot$ is left and right distributive over $+$, $(S, +)$ is a semigroup with neutral element 0, and $(S, \cdot)$ is a semigroup with zero 0, that is $0\,a = a\,0 = 0$ for all $a \in S$. If $(S, \cdot)$ is commutative, the semiring S is called *commutative*.

The next definition states the additively inverse semirings.

Definition 16.15 ([2021-9, p. 625])**.** A semiring S is *additively inverse* if for every $a \in S$ there is a unique $a' \in S$ such that $a + a' + a = a$ and $a' + a + a' = a'$. An additively inverse semiring S with commutative addition such that $a + a'$ lies in the center $Z(S)$ for every $a \in S$ is called an *M A-semiring*.

The next definition states the derivations on an additively inverse semiring.

Definition 16.16 ([2021-9, p. 626])**.** Let S be an additively inverse semiring. An additive mapping $d\colon S \to S$ is a *derivation* if $d(xy) = d(x)\,y + x\,d(y)$ for all $x, y \in S$, and a *Jordan derivation* if $d(x^2) = d(x)\,x + x\,d(x)$ for all $x \in S$. It is a *left derivation* if $f(xy) = x\,f(y) + y\,f(x)$, and a *left Jordan derivation* if

$$f(x^2) + 2\,x'\,f(x) = 0 \qquad (x \in S).$$

Ahmed and Dudek classify the prime M A-semirings on which a nonzero left Jordan derivation exists: they are commutative [2021-9, pp. 624–631]. Their main theorem is the analogue of the result of Brešar and Vukman on prime rings.

Theorem 16.17 ([2021-9, Theorem 4.3, p. 631])**.** *If on a prime,* 2*- and* 3*-torsion free M A-semiring S there exists a non-zero left Jordan derivation, then S is commutative.*

The proof replaces subtraction by the additive-inverse map $a \mapsto a'$ and keeps the torsion conditions to make the commutator work [2021-9, pp. 631–632]. The next section keeps the order but drops the product and moves to derivations on lattices [2008-11, Definition 3.1, p. 309].

Vladeva studies the Jordan derivations of the finite endomorphism semiring, where the additive structure is a semigroup rather than a group [2017-14, Abstract, p. 1].

16.4 Lattices and residuated lattices

Here lattices stand further from a ring than any system above. A lattice has only an order and two idempotent operations [2008-11, p. 309]. So a product rule in the old sense has no product to use [2008-11, p. 309]. For the way in, look to topology: the derived set of a subset is a map on a Boolean algebra [1975-3, p. 365].

Definition 16.18 ([1975-3, p. 365])**.** Let R be a Boolean algebra. A map $A \mapsto A^d$ of R into itself is a *d-operator* if, for all $A, B \in R$, (I_d) $(A \cup B)^d = A^d \cup B^d$, ($\mathrm{II}_d$) $A^{dd} \subseteq A^d$, (III_d) $0^d = 0$, and (IV_d) $1^d = 1$. A map satisfying (I_d)–(III_d) is a *δ-operator*. A Boolean algebra with a d-operator (or a δ-operator) is called a Boolean algebra with a d-derivation (or a δ-derivation).

The names come from topology. Kuratowski introduced the operator, and if R is the algebra of all parts of a space without isolated points, then A^d is the derived set of A [1975-3, p. 365].

Xin, Li and Lu brought the idea back to lattices in 2008: their definition reads like the product rule, but it is a rule for the meet [2008-11, Definition 3.1, p. 309].

Definition 16.19 ([2008-11, Definition 3.1, p. 309])**.** Let L be a lattice. A function $d\colon L \to L$ is a *derivation* on L if

$$d(x \wedge y) = (dx \wedge y) \vee (x \wedge dy) \qquad (x, y \in L).$$

A derivation d on L is *isotone* if $x \leq y$ implies $dx \leq dy$, *monomorphic* if it is one-to-one, and *epic* if it is onto [2008-11, Definition 3.7, p. 310].

Every derivation on a lattice is decreasing: $dx \leq x$ for all x [2008-11, Proposition 3.6, p. 310]. The identity is then forced by any of the usual extra conditions [2008-11, Theorem 3.17, p. 312].

Theorem 16.20 ([2008-11, Theorem 3.17, p. 312])**.** *Let d be a derivation on a lattice L. Then the following are equivalent:* (1) *d is the identity derivation;* (2) *$d(x \vee y) = (x \vee dy) \wedge (dx \vee y)$ for all $x, y \in L$;* (3) *d is a monomorphic derivation;* (4) *d is an epic derivation.*

Isotone derivations mark out modular and distributive lattices [2008-11, Theorem 3.19, p. 313].

Theorem 16.21 ([2008-11, Theorem 3.19, p. 313])**.** *Let L be a modular lattice and d a derivation on L. Then the following are equivalent:* (1) *d is isotone;* (2) *$d(x \wedge y) = dx \wedge dy$ for all $x, y \in L$;* (3) *$dx = x$ implies $d(x \vee y) = dx \vee dy$.*

The next theorem gives the corresponding criterion for a distributive lattice.

Theorem 16.22 ([2008-11, Theorem 3.21, p. 313])**.** *Let L be a distributive lattice and d a derivation on L. Then the following are equivalent:* (1) *d is isotone;* (2) *$d(x \wedge y) = dx \wedge dy$ for all $x, y \in L$;* (3) *$d(x \vee y) = dx \vee dy$ for all $x, y \in L$.*

Çeven and Öztürk replaced the second argument by a second map and opened a wider family of maps (Section 16.6 of this volume returns to them).

Definition 16.23 ([2008-3, Definition 8, p. 702])**.** Let L be a lattice. A function $d\colon L \to L$ is an *f-derivation* on L if there is a function $f\colon L \to L$ such that

$$d(x \wedge y) = \big(d(x) \wedge f(y)\big) \vee \big(f(x) \wedge d(y)\big) \qquad (x, y \in L).$$

If f is the identity function, then d is a derivation on L [2008-3, p. 702].

The next proposition records two bounds that every f-derivation obeys.

Proposition 16.24 ([2008-3, Proposition 1, pp. 703–704])**.** *Let L be a lattice and d an f-derivation on L. Then, for all $x, y \in L$,* (a) *$dx \leq fx$, and* (b) *$dx \wedge dy \leq d(x \wedge y) \leq dx \vee dy$.*

A residuated lattice carries a product and its residuum, so it has room for a multiplicative rule.

Definition 16.25 ([2016-12, Definition 2.1, p. 99])**.** A *residuated lattice* is an algebra $(L, \wedge, \vee, \cdot, \to, 0, 1)$ of type $(2, 2, 2, 2, 0, 0)$ equipped with an order $\leq$ such that (1) $(L, \wedge, \vee, 0, 1)$ is a bounded lattice relative to $\leq$, (2) $(L, \cdot, 1)$ is a commutative monoid, and (3) $x \cdot y \leq z$ if and only if $x \leq y \to z$ for all $x, y, z \in L$.

The next definition states the multiplicative derivations on a residuated lattice.

Definition 16.26 ([2016-12, Definition 3.1, p. 100])**.** Let $(L, \wedge, \vee, \cdot, \to, 0, 1)$ be a residuated lattice. A map $d\colon L \to L$ is a *multiplicative derivation* on L if

$$d(x \cdot y) = (d(x) \cdot y) \vee (x \cdot d(y)) \qquad (x, y \in L).$$

The next definition introduces the ideal derivations.

Definition 16.27 ([2016-12, Definition 3.6, p. 101])**.** A derivation d on a residuated lattice L is *isotone* if $x \leq y$ implies $d(x) \leq d(y)$, *contractive* if $d(x) \leq x$ for all $x \in L$, and an *ideal derivation* if it is both.

The next theorem describes the ideal derivations that are idempotent.

Theorem 16.28 ([2016-12, Theorem 3.10, p. 102])**.** *Let $(L, \wedge, \vee, \cdot, \to, 0, 1)$ be a residuated lattice and d a derivation on L. Then the following are equivalent:* (1) *d is an ideal derivation on L and $d^2 = d$;* (2) *d satisfies $d(x) \to d(y) = d(x) \to y$ for all $x, y \in L$.*

The next section moves to the algebras of logic, where the same rule appears on a BCI, a BCK or an MV-algebra [2004-6, p. 167].

16.5 BCK, BCI and MV-algebras

Here a BCI or BCK algebra keeps only an operation. This operation behaves like the difference $x - y$ of a group or the implication $y \to x$ of a logic [2004-6, p. 167]. Jun and Xin introduced derivations on BCI-algebras in 2004 [2004-6, p. 167].

Definition 16.29 ([2004-6, p. 168])**.** A nonempty set X with a constant 0 and a binary operation, written by juxtaposition, is a *BCI-algebra* if, for all $x, y, z \in X$, (I) $((xy)(xz))(zy) = 0$, (II) $(x(xy))y = 0$, (III) $xx = 0$, and (IV) $xy = 0$ and $yx = 0$ imply $x = y$. A BCI-algebra X with $0 \leq x$ for all $x \in X$, where $x \leq y$ means $xy = 0$, is a *BCK-algebra* [2004-6, p. 168].

The next definition states the derivations on a BCI-algebra.

Definition 16.30 ([2004-6, Definition 3.5, p. 170])**.** Let X be a BCI-algebra and write $x \wedge y = y(yx)$. A self-map d of X is a *left-right derivation* (an *(l, r)-derivation*) of X if

$$d(xy) = d(x)y \wedge x\, d(y) \qquad (x, y \in X),$$

and a *right-left derivation* (an (r,l)*-derivation*) if $d(xy) = x\,d(y) \wedge d(x)y$ for all $x, y \in X$.

A derivation d on a BCI-algebra is *regular* if $d(0) = 0$ [2004-6, Definition 3.12, p. 173]. Regular derivations detect the p-semisimple BCI-algebras [2004-6, Theorem 3.17, p. 174].

Theorem 16.31 ([2004-6, Theorem 3.17, p. 174])**.** *Let X be a BCI-algebra. Then the following are equivalent:* (i) *X is p-semisimple;* (ii) *if X has a regular derivation, then it is trivial;* (iii) *$d^{-1}(0) = \{0\}$ for every regular derivation d of X.*

The next theorem gives an ideal-theoretic test for regularity.

Theorem 16.32 ([2004-6, Theorem 3.22, p. 176])**.** *Let d be a derivation of a BCI-algebra X. Then d is regular if and only if every ideal of X is d-invariant.*

Ciungu widened the class to the pseudo-BCK algebras, where the single arrow splits into two [2021-15, Definition 2.1, p. 177].

Definition 16.33 ([2021-15, Definition 2.1, p. 177])**.** A *pseudo-BCK algebra* is a structure $(A, \leq, \to, \rightsquigarrow, 1)$ where $\leq$ is a binary relation on A, $\to$ and $\rightsquigarrow$ are binary operations on A and $1 \in A$, satisfying, for all $x, y, z \in A$: $x \to y \leq (y \to z) \rightsquigarrow (x \to z)$ and $x \rightsquigarrow y \leq (y \rightsquigarrow z) \to (x \rightsquigarrow z)$; $x \leq (x \to y) \rightsquigarrow y$ and $x \leq (x \rightsquigarrow y) \to y$; $x \leq x$; $x \leq 1$; if $x \leq y$ and $y \leq x$ then $x = y$; and $x \leq y$ if and only if $x \to y = 1$ if and only if $x \rightsquigarrow y = 1$.

The next definition states the implicative derivations on this algebra.

Definition 16.34 ([2021-15, pp. 178–179])**.** Let $(A, \to, \rightsquigarrow, 1)$ be a pseudo-BCK algebra. A map $d\colon A \to A$ is a *type I implicative derivation* on A if, for all $x, y \in A$,

$$d(x \to y) = (x \to d(y)) \vee_1 (d(x) \to y), \qquad d(x \rightsquigarrow y) = (x \rightsquigarrow d(y)) \vee_2 (d(x) \rightsquigarrow y),$$

and a *type II implicative derivation* if the two identities hold with the summands on the right interchanged, where $\vee_1$ and $\vee_2$ are the two joins of the algebra.

An MV-algebra is the model of Łukasiewicz logic that carries both an addition and a logical not [2019-15, p. 261].

Definition 16.35 ([2019-15, p. 261])**.** An *MV-algebra* is an algebra $L = (L, \oplus, {}^*, 0)$ satisfying, for all $x, y \in L$: (MV1) $(L, \oplus, 0)$ is a commutative monoid; (MV2) $(x^*)^* = x$; (MV3) $0^* \oplus x = 0^*$; (MV4) $(x^* \oplus y)^* \oplus y = (y^* \oplus x)^* \oplus x$. Write $1 = 0^*$ and $x \to y = x^* \oplus y$.

The next definition states the derivations that respect the implication.

Definition 16.36 ([2019-15, Definition 3.1, p. 262])**.** Let L be an MV-algebra. A mapping $g\colon L \to L$ is an *implicative derivation* on L if, for all $x, y \in L$,

$$g(x \to y) = (g(x) \to y) \vee (x \to g(y)).$$

The set of all implicative derivations of L is denoted $I(L)$.

On a Boolean algebra, the set $I(L)$ is itself a Boolean algebra, and the original algebra appears as its algebra of implicative derivations [2019-15, Theorem 3.10, p. 267].

Theorem 16.37 ([2019-15, Theorem 3.10, p. 267])**.** *Let $(L, \wedge, \vee, {}^*, 0, 1)$ be a Boolean algebra. Then $(I(L), \cap, \cup, \neg, g_1, 1_g)$, with $g_1(x) = x$ and $1_g(x) = 1$, is a Boolean algebra.*

The next theorem recovers the original algebra from its derivations.

Theorem 16.38 ([2019-15, Theorem 3.11, p. 268])**.** *Let $(L, \wedge, \vee, {}^*, 0, 1)$ be a Boolean algebra. Then L is isomorphic to the Boolean algebra $(I(L), \cap, \cup, \neg, g_1, 1_g)$ of all implicative derivations on L, by the map $a \mapsto \varphi(a)$, where $\varphi(a)(x) = a \vee x$.*

The next section takes up the last family of the chapter, the subtraction algebras, where the meet rule takes the place of the product rule once more [2009-1, Definition 3.1, p. 511].

16.6 Subtraction algebras

Here the last section kept only an implication. Schein went one step further and kept only a difference [2009-1, p. 509]. He studied systems of functions closed under composition and set difference [2009-1, p. 509]. He called the second part by itself a subtraction algebra [2009-1, p. 509]. Its axioms are three short identities [2009-1, p. 510].

Definition 16.39 ([2009-1, p. 510])**.** An algebra $(X, -)$ with a single binary operation $-$ is a *subtraction algebra* if, for all $x, y, z \in X$, (S1) $x-(y-x) = x$, (S2) $x - (x - y) = y - (y - x)$, and (S3) $(x - y) - z = (x - z) - y$. The subtraction determines an order on X by $a \leq b$ if and only if $a - b = 0$, where $0 = a - a$ does not depend on the choice of a. The ordered set $(X, \leq)$ is then a semi-Boolean algebra: a meet semilattice with zero 0 in which every interval $[0, a]$ is a Boolean algebra, with $a \wedge b = a - (a - b)$ [2009-1, p. 510].

Öztürk and Çeven worked on a wider class. For $a, b \in X$ write $G(a, b) = \{x \in X : x - a \leq b\}$. The algebra X is a *complicated subtraction algebra* (a *c-subtraction algebra*) when every $G(a, b)$ has a greatest element [2009-1, Definition 2.3, p. 511]. This greatest element is written $a + b$ [2009-1, Definition 2.3, p. 511]. It is the least upper bound of a and b [2009-1, Definition 2.3, p. 511]. Their derivation uses the meet and this addition.

Definition 16.40 ([2009-1, Definition 3.1, p. 511])**.** Let X be a c-subtraction algebra. A function $d\colon X \to X$ is a *derivation* on X if

$$d(x \wedge y) = (dx \wedge y) + (x \wedge dy) \qquad (x, y \in X).$$

As on a lattice, every derivation is decreasing: $dx \leq x$ for all x [2009-1, Proposition 3.4, p. 512]. The two-sided bound of Xin, Li and Lu has an exact analogue [2009-1, Theorem 3.6, p. 512].

Theorem 16.41 ([2009-1, Theorem 3.6, p. 512])**.** *Let X be a c-subtraction algebra and d a derivation on X. Then*

$$dx \wedge dy \leq d(x \wedge y) \leq dx + dy \qquad (x, y \in X).$$

The identity criterion of lattices also carries over [2009-1, Theorem 3.10, p. 513].

Theorem 16.42 ([2009-1, Theorem 3.10, p. 513])**.** *Let X be a c-subtraction algebra and d a derivation on X. Then the following are equivalent:* (i) *d is the identity derivation;* (ii) *$d(x + y) = (dx + y) \wedge (x + dy)$ for all $x, y \in X$;* (iii) *d is one-to-one;* (iv) *d is onto.*

The same paper defines the first generalized derivation on a subtraction algebra [2009-1, Definition 3.11, p. 513].

Definition 16.43 ([2009-1, Definition 3.11, p. 513])**.** Let X be a c-subtraction algebra and $d\colon X \to X$ a derivation. A function $D\colon X \to X$ is a *generalized derivation determined by* d if

$$D(x \wedge y) = Dx \wedge y + x \wedge dy \qquad (x, y \in X).$$

So this last family completes the list of systems in the title of the chapter. The open questions they leave are collected in Appendix B.

Arzanesh, Shirazi and Hosseini carry the notion of a generalized shift to any algebra that has derivations, and read the shift through them [2021-10, Abstract, p. 1].

Part VI

Commutative and differential algebra

Chapter 17

Derivations in commutative algebra and geometry

The first chapters of this book treated a derivation as an operator on any ring. When the ring is commutative, two new facts appear.

The first fact is that the derivations of a commutative algebra A into an A-module M are themselves an A-module. The second fact is that they are all shadows of one universal derivation. That is the map d from A to the module of Kähler differentials of A (Figure 17.1).

The module of Kähler differentials makes calculus inside algebraic geometry. Later authors asked how much of the geometry of A the derivations alone can see. This chapter tells that story in five steps. It answers one question in each step: which geometric facts about a commutative algebra are encoded in its derivations?

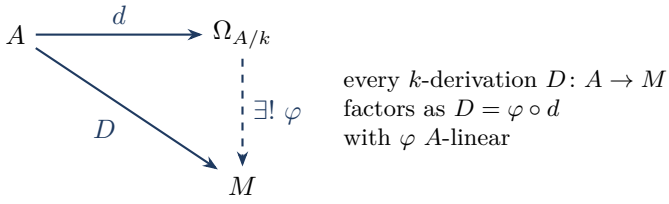


Figure 17.1: The universal derivation $d\colon A \to \Omega_{A/k}$: every derivation into an A-module factors through it.

The first step is the definition itself. Section 17.1 recalls the module of Kähler differentials. It explains that derivations out of A and linear maps

out of $\Omega_{A/k}$ are the same thing. Johnson states this universal property again in 1969 [1969-1, p. 92]. This gives the first cohomological reading of derivations of a commutative algebra.

The second step is a regularity question. If the module of derivations of a local ring is free, must the ring be regular? Lipman raised the question in 1965, guided by a remark of Zariski. He proved that freeness forces normality.

Section 17.2 states the Zariski–Lipman conjecture as it is quoted today [2014-9, Conjecture 1.1, p. 270]. It reports the cases that are settled: graded domains, hypersurfaces, complete intersections, and most surfaces. It also reports the counterexample of Lipman in prime characteristic [2022-4, p. 116]. There the module of derivations of a singular surface is free. A second question about a singularity is qualitative, and it asks for a grading with no derivation of negative degree. Wahl made the conjecture that one always exists [2014-11, Conjecture, p. 2]. Granger and Schulze prove it for complete intersections of order at least three, and they give counterexamples above embedding dimension five [2014-11, Theorems 1.1 and 1.3, p. 2].

The third step restricts to a special class. A derivation D is locally nilpotent when each element is killed by some power of D [1998-3, p. 354]. On a polynomial ring over a field of characteristic zero, these derivations are exactly the infinitesimal generators of algebraic actions of the additive group [1998-3, p. 354].

Section 17.3 asks when a combination of two commuting locally nilpotent derivations is again locally nilpotent [1992-3, Theorem, p. 47]. It then recalls the description of Rentschler for the locally nilpotent derivations of the plane [1998-3, pp. 356–357]. It ends with the Makar-Limanov invariant. Makar-Limanov used this invariant to prove that the Russell cubic threefold is not affine space [2021-22, p. 66]. Figure 17.2 places the locally nilpotent case in the plane.

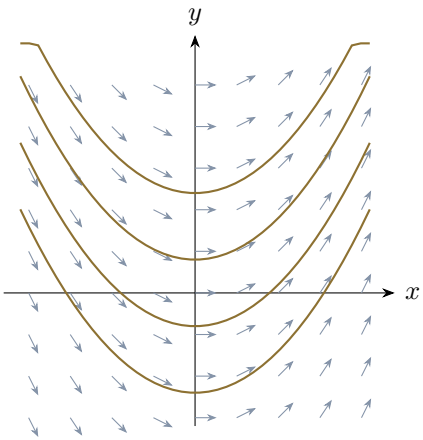


Figure 17.2: The locally nilpotent derivation $D = \partial_x + x\,\partial_y$ of $k[x, y]$ as a vector field. Its exponential is the polynomial automorphism $(x, y) \mapsto (x + t,\ y + tx + t^2/2)$, whose orbits are the parabolas shown.

The section also reports two recent lines. The invariant is the tool of the cancellation problem, and a 2025 survey follows that use in commutative algebras, noncommutative algebras and skew extensions [2025-19, pp. 1–3]. In dimension two the invariant gives a dichotomy, and the affine plane is cancellative [2025-19, Theorems 2.2–2.3, p. 7]. Separately, Cid and Veloso compute the automorphisms that commute with a locally finite derivation of the plane [2026-13, p. 1]. Every nonzero such derivation has a nontrivial commuting group [2026-13, Proposition 3.1, p. 3]. Chitayat, Daigle and Regeta show that a solvable Lie algebra of locally finite derivations is weakly locally finite [2026-11, Corollary 4, p. 2]. On a module over the algebra, the kernel of a locally nilpotent module derivation is finitely generated in dimension at most three [2010-8, Corollary 4.7, p. 10]. A locally nilpotent Lie algebra of derivations of an integral domain has a series of ideals with abelian quotients, one for each unit of rank [2017-11, Theorem 1, p. 5].

The fourth step is the Jacobian conjecture. Keller asked in 1939 whether a polynomial map with a nowhere-vanishing Jacobian determinant must be invertible. The question is still open. It has striking forms in the language of derivations.

Section 17.4 reports two of them. The first is a criterion in terms of derivations of the plane with zero divergence [2021-22, Example 5.2.14, pp. 130–131]. The second is a formulation inside the Lie algebra of formal vector fields [1998-4, Proposition 1, p. 211]. The section also records the

equivalence of the Jacobian, Dixmier and Poisson conjectures [2021-22, Theorem 4.4.1, p. 101]. Chapter 12 of this book develops the polynomial side of the story.

The fifth step leaves algebra for differential geometry. On a smooth manifold M, a derivation of the ring $C^\infty(M)$ is the same thing as a smooth vector field. Section 17.5 recalls the definition of a vector field as a derivation [2020-20, p. 122]. It then proves that the Lie algebra of vector fields with compact support determines the manifold itself [1954-1, Theorem 3, p. 471]. Here the dictionary of the chapter closes. Commutative algebra and geometry describe the same object in two languages.

Section 17.6 carries the dictionary one step further. It recalls the definition of a tangent vector at a point as a derivation of the smooth functions at that point [2023-14, p. 70]. The coordinate derivations then form a basis of the tangent space there [2023-14, equation (2.19), p. 70], and a vector field is a smooth choice of one such derivation at every point.

17.1 Kähler differentials and the universal derivation

Let all rings here be commutative with identity. We look now for the single derivation that carries all the others, as the introduction promised [1969-1, p. 93].

This section recalls that construction. It also fixes the notation used in the rest of the chapter, so that one symbol serves for one object throughout.

Let A be a ring and B an A-algebra. Johnson constructs the module of Kähler differentials from the multiplication map of B [1969-1, p. 93]. Let $\mu\colon B \otimes_A B \to B$ be the A-algebra map with $\mu(b \otimes b') = bb'$. Let I be its kernel.

The quotient I/I^2 is a module over $B = (B \otimes_A B)/I$. There is an A-derivation of B into that module, given by

$$b \longmapsto (b \otimes 1 - 1 \otimes b) + I^2$$

[1969-1, p. 93]. So the kernel of the multiplication already carries a derivation, and the derivation measures the failure of the two factors to commute. The next definition names both objects [1969-1, p. 93].

Definition 17.1 ([1969-1, p. 93])**.** Let A be a ring and B an A-algebra. With I the kernel of the multiplication $\mu\colon B \otimes_A B \to B$, the *module of Kähler differentials* of B over A is the B-module $\Omega_{B/A} = I/I^2$. The *universal*

derivation is the A-linear map $d_{B/A}\colon B \to \Omega_{B/A}$ given by $d_{B/A}(b) = (b \otimes 1 - 1 \otimes b) + I^2$.

The name *universal derivation* comes from a universal property that Johnson states. Every derivation into a B-module factors through $d_{B/A}$ [1969-1, p. 92]. The factor is unique [1969-1, p. 92]. The next theorem records that property in full [1969-1, p. 92].

It is the form the rest of this chapter uses, since every other derivation is read through it.

Theorem 17.2 ([1969-1, p. 92])**.** *Let A be a ring and B an A-algebra. Let M be a B-module and let $D\colon B \to M$ be an A-derivation. Then there is a unique B-linear map $\varphi\colon \Omega_{B/A} \to M$ with $D = \varphi \circ d_{B/A}$.*

Take $M = B$ in Theorem 17.2. This identifies the derivations of B with the dual of $\Omega_{B/A}$ [1969-1, p. 92]. Composition with $d_{B/A}$ maps $\mathrm{Hom}_B(\Omega_{B/A}, B)$ bijectively to $\mathrm{Der}_A(B)$ [1969-1, p. 92].

So questions about the derivations of a commutative algebra are questions about one module [1969-1, p. 92]. When $\mathrm{Der}_A(B)$ is free, one asks whether the geometry of B is smooth [2014-9, p. 270]. That is the Zariski–Lipman problem of Section 17.2.

Graf takes the same duality as the definition of the tangent sheaf [2022-4, p. 115]. The tangent sheaf of a variety X is the dual of the sheaf of Kähler differentials [2022-4, p. 115].

Section 17.2 starts from this duality. It asks what one learns about the singularities when the module of derivations is free [2014-9, p. 270].

17.2 The Zariski–Lipman conjecture

Section 17.1 identified the derivations of a commutative algebra with the dual of one module. Lipman asked in 1965 how much this module says about the singularities of the algebra [2014-9, p. 270]. The paper is *Free derivation modules on algebraic varieties* [2014-9, p. 270].

The question is this [2014-9, p. 270]. Suppose that the module of derivations of a local ring is free. Must the ring then come from a smooth point [2014-9, p. 270]? The question is now named after Zariski and Lipman [2014-9, p. 270].

Theorem 17.3 ([2014-9, Conjecture 1.1, p. 270])**.** *Let V be an algebraic variety over a field k of characteristic 0, let p be a closed point of V and let*

R be the local ring of V at p. If the module of k-derivations $\mathrm{Der}_k(R)$ *is a free R-module, then V is smooth at p.*

The statement of Theorem 17.3 is a conjecture [2014-9, p. 270], quoted as it stands in the literature, and no proof is claimed here. Its hypotheses carry weight [2014-9, p. 270]. Lipman proved that if $\mathrm{Der}_k(R)$ is R-free, then R is normal [2014-9, p. 270]. So the question concerns only normal varieties. If the ring is not normal, the answer can be negative already in dimension one [2014-9, p. 270].

Several cases are settled [2014-9, p. 270]. Biswas, Gurjar and Kolte collect the main ones [2014-9, p. 270]. Hochster proved the conjecture for positively graded domains, and Scheja and Storch proved it for hypersurfaces [2014-9, p. 270]. Becker reduced it to the case of isolated singularities [2014-9, p. 270], and van Straten and Steenbrink settled that case in dimension at least three [2014-9, p. 270].

Flenner settled non-isolated singularities of dimension at least three in codimension at least three [2014-9, p. 270]. Källström proved the conjecture for complete intersections [2014-9, p. 270]. Druel and, independently, Graf proved it for local rings with log canonical singularities [2014-9, p. 270].

Part of this list is recalled in the cited paper as a single statement [2014-9, Theorem 2.2, pp. 272–273]. That statement is Theorem 17.4 below. It is the form later work quotes, because it gathers the settled cases into one hypothesis.

Theorem 17.4 ([2014-9, Theorem 2.2, pp. 272–273])**.** *Let V be an affine algebraic variety such that the algebraic tangent bundle of the smooth locus V^0 is trivial. Then V is smooth in each of the following cases: the coordinate ring of V is a positively graded domain; V is a complete-intersection singularity; or V is a surface with at most rational singularities.*

For surfaces the conjecture is nearly closed. Biswas, Gurjar and Kolte prove it for a wide class of affine surfaces: those whose smooth locus has logarithmic Kodaira dimension at most one [2014-9, Theorem 1.2, p. 270].

Three authors worked on the rest: Lipman, Becker and Flenner [2022-4, p. 115]. Graf reports their work, and says that the case of a normal surface is essentially the only open case left [2022-4, p. 115].

In prime characteristic the picture changes [2022-4, p. 115]. Lipman himself found counterexamples, and Graf recalls them [2022-4, p. 116].

Example 17.5 ([2022-4, p. 116])**.** Let k be a perfect field of characteristic p, and let $X \subset \mathbb{A}^3_k$ be the surface defined by $xy - z^n = 0$, where p divides

n. Then the tangent sheaf of X is freely generated by the two derivations $\delta_1 = x\partial_x - y\partial_y$ and $\delta_2 = \partial_z$, although X is singular.

So Graf studies in the cited paper how far the failure in prime characteristic goes [2022-4, p. 115]. He proves that for rational double points the conjecture holds except for a short finite list of cases. Among rational double points in characteristic $p \geq 7$, he adds, the counterexample of Lipman is the only one [2022-4, p. 115].

A second question about the derivations of a singularity is qualitative. Give a quasihomogeneous singularity its positive grading. Does some grading exist for which no derivation lowers the degree [2014-11, p. 1]? Wahl made the conjecture that it does, for every quasihomogeneous isolated normal singularity [2014-11, Conjecture, p. 2]. Granger and Schulze settle the conjecture for isolated complete intersections of order at least three [2014-11, Theorem 1.1, p. 2].

Theorem 17.6 ([2014-11, Theorem 1.1, p. 2])**.** *Let A be a quasihomogeneous normal isolated complete intersection singularity of order at least three. Then A has no derivation of negative weighted degree with respect to any positive grading.*

They also settle every isolated complete intersection of embedding dimension at most five [2014-11, Theorem 1.3, p. 2]. Above embedding dimension five the conjecture fails, and they give a family of counterexamples [2014-11, Example 1.2, pp. 1–2]. The first has embedding dimension six, and on it the derivation η of degree -1 is written out explicitly [2014-11, Example 1.2, p. 2]. So the sign of the degree of a derivation separates the cases, and the separation is exact at embedding dimension five [2014-11, Theorem 1.3, p. 2].

So Theorem 17.3 compares an algebraic property with a geometric one [2014-9, p. 270]: the module of derivations is free, and the variety is smooth.

The conjecture of Theorem 17.3 is open, and this section records it as the first problem of the chapter.

Problem 17.7 ([2014-9, Conjecture 1.1, p. 270])**.** Let V be an algebraic variety over a field k of characteristic 0. Let p be a closed point of V, and let R be the local ring of V at p. If the module of k-derivations $\mathrm{Der}_k(R)$ is a free R-module, then V is smooth at p.

The conjecture is settled for positively graded domains, for hypersurfaces, for complete intersections and for log canonical singularities [2014-9, p. 270].

Biswas, Gurjar and Kolte settle it for every affine surface whose smooth locus has logarithmic Kodaira dimension at most one [2014-9, Theorem 1.2, p. 270]. They leave one case of the projective problem open.

Problem 17.8 ([2014-9, Remark 1.3, p. 271])**.** Let V be a projective surface whose smooth locus has a trivial tangent bundle and logarithmic Kodaira dimension $-\infty$. Suppose the base of the $\mathbb{P}^1$-fibration of a resolution of V is an elliptic curve, and the exceptional divisor has only reduced feathers. Is V then smooth?

Problem 17.7 is a statement about characteristic 0 only. The conjecture fails in prime characteristic [2022-4, p. 116], and the example of Lipman above is the smallest failure recorded in this section.

The next section keeps the algebraic side of the conjecture, and asks for derivations that are locally nilpotent [1992-3, Theorem, p. 47]. From those it reads polynomial coordinates off.

17.3 Locally nilpotent derivations

Section 17.2 asked what the algebra feels when its derivations form a free module. This section asks a sharper question [1998-3, p. 354]: what happens when a derivation behaves like a nilpotent operator on every element?

Chapter 2 defined the class [2017-7, Section 1.1.9, p. 12]. A derivation D is *locally nilpotent* when every element b is killed by some power of D [2017-7, Section 1.1.9, p. 12]. For such a derivation the map $\exp D$ is well defined [2017-7, Section 1.1.9, p. 12].

On a polynomial ring these derivations have a geometric meaning [1998-3, p. 354]. Algebraic actions of the additive group $\mathbb{G}_a$ on affine space $\mathbb{A}^n$ correspond to locally nilpotent derivations [1998-3, p. 354].

So the question of the section is this [1992-3, Theorem, p. 47]. When does a derivation built from locally nilpotent pieces stay locally nilpotent [1992-3, Theorem, p. 47]? And what do these derivations look like [1992-3, Theorem, p. 47]?

Ferrero, Lequain and Nowicki answer the first part for a combination of two commuting derivations [1992-3, Theorem, p. 47].

Theorem 17.9 ([1992-3, Theorem, p. 47])**.** *Let R be a commutative, reduced, $\mathbb{Z}$-torsion free ring, let d and δ be two locally nilpotent derivations of R which commute, and let $b \in R$ be such that $\delta(b) = d(b) = 0$. Let $\Delta = a\,d + b\,\delta$ with $a \in R$. Then Δ is locally nilpotent if and only if $d(a) = 0$.*

For the polynomial ring in two variables the criterion becomes a clean description. It lists all locally nilpotent derivations of a triangular form [1992-3, Corollary 6, p. 49].

Corollary 17.10 ([1992-3, Corollary 6, p. 49])**.** *Let $K[x, y]$ be the polynomial ring in two indeterminates over a reduced, $\mathbb{Z}$-torsion free ring K. Let $f \in K[x, y]$ and $b \in K$, and let Δ be the K-derivation of $K[x, y]$ defined by $\Delta(x) = f$ and $\Delta(y) = b$. Then Δ is locally nilpotent if and only if $f \in K[y]$.*

In dimension two much more is true [1998-3, pp. 356–357]. The theorem of Rentschler from 1968 describes every locally nilpotent derivation of the plane: its kernel is generated by one coordinate [1998-3, pp. 356–357]. Daigle and Freudenburg recall it in Jacobian form, where for $P \in L[X, Y]$ they write D_P for the derivation $h \mapsto P_X h_Y - P_Y h_X$ [1998-3, pp. 356–357].

Theorem 17.11 (Rentschler; [1998-3, pp. 356–357])**.** *Let L be a field of characteristic zero and let D be a nonzero locally nilpotent derivation of $L[X, Y]$. Then there exist P, Q such that $L[X, Y] = L[P, Q]$ and $\ker D = L[P]$. Moreover, there exists $a \in L[P]$ such that $Dh = a\,(P_X h_Y - P_Y h_X)$ for all $h \in L[X, Y]$.*

Then Daigle and Freudenburg extend the theorem of Rentschler. Their version works over a factorial base ring in place of a field [1998-3, Theorem 2.4, pp. 358–359].

Theorem 17.12 ([1998-3, Theorem 2.4, pp. 358–359])**.** *Let R be a unique factorisation domain containing $\mathbb{Q}$, let $B = R[X, Y]$ and let K be the field of fractions of R. For a nonzero R-derivation D of B, the following are equivalent: D is locally nilpotent; and $D = a\,D_P$ for some $P \in B$ which is a variable of $K[X, Y]$ with $\gcd_B(P_X, P_Y) = 1$ and some $a \in R[P] \setminus \{0\}$. In this case $\ker D = R[P]$.*

Beyond dimension two, coordinates no longer classify the kernels [2021-22, p. 65]. One measures them instead, and van den Essen, Kuroda and Koenig explain how the measure arose [2021-22, p. 65]. Their work produced a list of hypersurfaces that might be counterexamples [2021-22, p. 65].

The best known is a hypersurface V in $\mathbb{C}^4$, given by $x + x^2y + z^2 + t^3 = 0$ [2021-22, p. 65]. It is smooth, factorial and contractible, and diffeomorphic to $\mathbb{R}^6$ [2021-22, p. 65]. To separate such candidates from affine space, one meets all locally nilpotent derivations in their kernels [2021-22, p. 65].

Definition 17.13 ([2021-22, Definition 3.1.1, p. 65])**.** Let R be a ring of characteristic zero. The *Makar-Limanov invariant* of R is the intersection of the kernels of all locally nilpotent derivations of R:

$$\mathrm{ML}(R) = \bigcap_{D \in \mathrm{LND}(R)} \ker D.$$

If W is an algebraic variety with coordinate ring $A(W)$, one sets $\mathrm{ML}(W) = \mathrm{ML}(A(W))$.

For a field k of characteristic zero the invariant of the polynomial ring is k [2021-22, Example 3.1.2, p. 66], and that holds in every number of variables.

Makar-Limanov showed that the element x of the Koras-Russell hypersurface lies in every kernel [2021-22, p. 66]. It turned out that $\mathrm{ML}(V) = \mathbb{C}[x]$ [2021-22, p. 66]. The invariant of affine three-space is $\mathbb{C}$ [2021-22, p. 66]. So the hypersurface V is not isomorphic to $\mathbb{C}^3$ [2021-22, p. 66].

Chapter 12 returns to this invariant for polynomial rings [2021-22, p. 65]. The cancellation problem behind these invariants is treated in the cited book [2021-22, p. 65].

The invariant is the tool of the cancellation problem, and a recent survey follows that use through three settings [2025-19, pp. 1–3]. The survey states the problem of Zariski in the commutative case, then in noncommutative algebras, then in skew extensions [2025-19, pp. 1–3]. The key technical input there is the stability of the invariant under adjoining a variable. For a two-dimensional unique factorization domain it gives a dichotomy [2025-19, Theorem 2.2, p. 7].

Theorem 17.14 ([2025-19, Theorem 2.2, p. 7])**.** *Let k be an algebraically closed field of characteristic zero and let R be a two-dimensional affine k-domain that is a unique factorization domain. Then exactly one of the following holds:* $\mathrm{ML}(R) = R$*, or* $\mathrm{ML}(R) = k$ *and* $R \cong k[x, y]$.

So in dimension two the invariant separates the rigid algebras from the polynomial ring itself. The survey reads the cancellativity of the affine plane off this dichotomy [2025-19, Theorem 2.3, p. 7]. It also reports where the method fails: in the skew case the stability of the invariant can break down [2025-19, pp. 2–3].

Beyond cancellation, the locally nilpotent derivations of the plane have constrained symmetry. Cid and Veloso compute the group of automorphisms that commute with a locally finite derivation [2026-13, p. 1]. Every nonzero

locally finite derivation of $K[X, Y]$ has a nontrivial such group [2026-13, Proposition 3.1, p. 3].

Theorem 17.15 ([2026-13, Proposition 3.1, p. 3])**.** *Let K be an algebraically closed field of characteristic zero and let B be an affine K-algebra. If $D \in \mathrm{LFD}(B)$ with $D \neq 0$, then the group of automorphisms of B that commute with D is nontrivial.*

The proof takes the exponential $\exp(tD)$, a well defined automorphism because D is locally finite [2026-13, p. 4]. For each t this exponential commutes with D, and for some t it differs from the identity [2026-13, p. 4]. So a simple derivation of $K[X, Y]$ cannot be locally finite, because a simple derivation of the plane has a trivial commuting group [2026-13, p. 4]. Theorem 12.10 of Chapter 12 states that triviality. The same authors compute the full commuting group for each of the four normal forms of a locally finite derivation of $K[X, Y]$ [2026-13, Theorems 3.3–3.9, pp. 4–8].

The same authors also compute the tame part of the group, generated by the elementary automorphisms that commute with D [2026-14, p. 1]. For a locally nilpotent derivation the tame group and the tame group of its exponential agree [2026-14, Proposition 4.1, p. 9].

Theorem 17.16 ([2026-14, Proposition 4.1, p. 9])**.** *Let D be a locally nilpotent derivation of $K[X, Y]$. Then the tame isotropy group of D equals the tame isotropy group of* $\exp(D)$.

For a derivation with a nontrivial semisimple part the two full groups can differ, while the tame groups still agree [2026-14, pp. 1–2].

Bianchi, Freitas and Veloso introduce the same tame group for triangular derivations in up to three variables and for simple Shamsuddin derivations in any number of variables [2025-3, Abstract, p. 1].

Chunaev and Evdokimova study the subgroup of locally nilpotent derivations that commute with a homogeneous one. They find its maximal homogeneous subgroups on affine trinomial hypersurfaces, and give when such a subgroup is maximal [2026-12, Abstract, p. 1].

A derivation is locally finite when the vector space spanned by its powers on each element is finite dimensional [2026-11, Definition 2.1, p. 2]. A set of derivations is locally finite when one finite-dimensional space serves every member of the set, and weakly locally finite when every finite subset is [2026-11, Definition 2.1, p. 2]. Chitayat, Daigle and Regata ask when a Lie algebra of locally finite derivations is weakly locally finite [2026-11, pp. 1–2]. They answer for the solvable case on a quasi-affine variety [2026-11, Corollary 4, p. 2].

Theorem 17.17 ([2026-11, Corollary 4, p. 2]). *Let X be a quasi-affine variety over a field k. If L is a solvable Lie subalgebra of $\operatorname{Der}_k \mathcal{O}_X(X)$ whose elements are locally finite derivations, then L is weakly locally finite. Moreover, L admits a countable increasing filtration by locally finite Lie subalgebras.*

The proof reduces the question to a derivation-finite algebra, one fixed by a finite set of values [2026-11, p. 2]. The algebra of regular functions on a quasi-affine variety is of this kind [2026-11, p. 4]. So the result applies to the coordinate rings of the varieties met in this chapter.

The same local nilpotency carries over to modules, and there the finite generation can fail. Tanaka works with a finitely generated module M over an affine algebra B, with a module derivation δ_M that is locally nilpotent [2010-8, p. 1]. Let A be the kernel of δ on B and M_0 the kernel of δ_M on M [2010-8, p. 1]. He asks whether M_0 is a finitely generated A-module [2010-8, p. 2].

Theorem 17.18 ([2010-8, Theorem 4.6, p. 10]). *Let δ be a locally nilpotent derivation of an affine k-algebra B and let M be a finitely generated torsion-free (B, δ)-module. Suppose that $A = \ker \delta$ is a noetherian domain. Then $M_0 = \ker \delta_M$ is a finitely generated A-module.*

The theorem settles the question in dimension at most three, since the kernel of a locally nilpotent derivation in that range is finitely generated [2010-8, Corollary 4.7, p. 10]. In higher dimension the answer can be negative, and the counterexamples come from the counterexamples to the fourteenth problem of Hilbert [2010-8, pp. 2–3]. So the module question is as hard as the invariant question behind it.

Locally nilpotent derivations also form Lie algebras of their own, and Petravchuk, Shevchyk and Sysak study the locally nilpotent subalgebras of the derivations of an integral domain [2017-11, p. 1]. They measure such a subalgebra by its rank over the fraction field [2017-11, p. 1].

Theorem 17.19 ([2017-11, Theorem 1, p. 5]). *Let A be an integral domain over a field K of characteristic zero, let R be its fraction field, and let L be a locally nilpotent subalgebra of $R \operatorname{Der}_K A$ of rank n over R. Then L contains a series of ideals $0 = L_0 \subset L_1 \subset \cdots \subset L_n = L$ with $\operatorname{rk}_R L_i = i$ for all i, and every quotient L_i / L_{i-1} is abelian.*

So a locally nilpotent Lie algebra of derivations is built from abelian pieces, one for each unit of rank. The result holds for the triangular derivations of a polynomial ring, which are the model case [2017-11, p. 2].

On the affine plane the local finiteness can be read off the algebra itself. Zaidenberg takes a solvable Lie subalgebra of the Lie algebra of the automorphism group of the plane, generated by locally finite derivations [2026-30, p. 1]. He proves that it is triangulable and weakly locally finite [2026-30, Theorem 1.2, p. 2].

Theorem 17.20 ([2026-30, Theorem 1.2, p. 2])**.** *Let* $\mathfrak{h} = \langle \partial_1, \dots, \partial_k, \dots \rangle_{\mathrm{Lie}}$ *be a solvable Lie subalgebra of the Lie algebra of the automorphism group of the affine plane* $\mathbb{A}^2$*, generated by locally finite derivations* ∂_i*. Then* $\mathfrak{h}$ *is triangulable and weakly locally finite. If* $\mathfrak{h}$ *is generated by a finite set of locally finite derivations, then it is locally finite.*

The proof uses Newton polygons on the plane [2026-30, pp. 10–16]. So the plane answers the question that Corollary 17.17 answers in general, and it adds triangulability, which the general statement does not give.

Locally nilpotent derivations of the polynomial ring also leave two questions open, and this section states both.

Problem 17.21 ([2017-7, Section 3.9, p. 108])**.** Let D be a standard homogeneous locally nilpotent derivation of $k[x_1, \dots, x_n]$ with $D^2 x_i = 0$ for every i. Is the rank of D always strictly less than n?

Freudenburg reports that this problem is open only in dimension 5 [2017-7, p. 108]. A companion question asks whether a homogeneous locally nilpotent derivation of degree two can have full rank, and that question is open in every dimension [2017-7, Remark 3.51, p. 112].

Problem 17.22 ([2017-7, Question 11.4, p. 288])**.** Let a unipotent group G act on the affine space $\mathbb{A}^n_k$. Can the action be triangularized in the rational sense, that is, can one write $k\langle x_1, \dots, x_n \rangle = k\langle y_1, \dots, y_n \rangle$ so that each subfield $k\langle y_1, \dots, y_i \rangle$ is invariant under G?

Bass asked this question in 1984, and Freudenburg records that it is still open in general [2017-7, p. 288]. The same epilogue asks whether the kernel of a sum of two commuting locally nilpotent derivations is finitely generated when both kernels are [2017-7, Question 11.11, p. 291].

Section 17.4 follows another open problem of polynomial rings, the Jacobian conjecture [2021-22, p. 91], and asks what that conjecture says about derivations.

The LFED conjecture asks when the Lie algebra generated by a set of locally finite derivations is itself locally finite. Lv and Yan settle it for a class of E-derivations [2020-18, Abstract, p. 1].

Over a unique factorization domain Daigle gives the structure of a locally nilpotent derivation and applies it to the rank-two case on $k[X_1, \ldots, X_n]$ [1998-2, pp. 353–371].

17.4 The Jacobian conjecture

Section 17.3 ended with open problems of polynomial rings. The most famous of them is the Jacobian conjecture [2021-22, p. 91]. The story of derivations meets it directly [2021-22, p. 91].

Call a polynomial map $F\colon k^n \to k^n$ a *Keller map* when $\det JF = 1$ [2021-22, p. 106]. The conjecture asks whether every Keller map is invertible [2021-22, p. 106].

The conjecture has many equivalent forms, several of which speak the language of this book [2021-22, p. 91]. Van den Essen, Kuroda and Koenig open their chapter on the new results with the words "there are various equivalent formulations of the Jacobian Conjecture" [2021-22, p. 91].

This section recalls three of them that involve derivations [2021-22, p. 91]. Each one turns the statement about polynomial maps into a statement about one derivation, or one Lie algebra of derivations [2021-22, p. 91].

The first lives on the plane [2021-22, Example 5.2.14, pp. 130–131]. The divergence of a derivation is the object of study here [2021-22, p. 130]. For a k-derivation $D = a\,\partial_x + b\,\partial_y$ of $k[x, y]$, the *divergence* of D is $\operatorname{div} D = \partial_x a + \partial_y b$ [2021-22, p. 130].

A *Mathieu-Zhao space*, or MZ-space, is a subspace whose radical and strong radical coincide [2021-22, Definition 5.2.1, p. 123]. Here the reader only needs one property from the proof of Example 17.23, stated next. An MZ-space that contains 1 is the whole ring [2021-22, Example 5.2.11, pp. 129–130].

Proposition 17.23 ([2021-22, Example 5.2.14, pp. 130–131])**.** *The following two conditions are equivalent.*

1. *For every k-derivation D of $k[x, y]$ such that 1 lies in the image of D and $\operatorname{div} D = 0$, the image of D is an MZ-space of $k[x, y]$.*

2. *The two-dimensional Jacobian conjecture is true.*

The proof recalled in the source builds a derivation from the map itself [2021-22, pp. 130–131]. For $F = (f, g)$ with $\det JF = 1$ it sets

$$D = g_y\,\partial_x - g_x\,\partial_y.$$

It notes that $\operatorname{div} D = 0$ and $1 = \det JF = Df$ [2021-22, pp. 130–131].

It then shows that D is locally nilpotent with kernel $k[g]$ [2021-22, pp. 130–131]. So a Keller map in two variables is invertible exactly when the image of this derivation is a Mathieu space of the ring [2021-22, pp. 130–131].

The second formulation uses a Lie algebra of the kind developed in Section 17.5 [1998-4, p. 211]. Let K be an algebraically closed field of characteristic zero. A polynomial vector field has the form

$$\sum_i a_i \, \partial_{x_i}, \qquad a_i \in K[x_1, \ldots, x_n].$$

Let $\mathcal{P}$ be the Lie algebra of these fields [1998-4, p. 211]. Let $\mathcal{D}$ be the Lie algebra of formal vector fields, whose coefficients are formal power series [1998-4, p. 211].

Proposition 17.24 ([1998-4, Proposition 1, p. 211])**.** *The following statements are equivalent.*

1. *Every endomorphism φ of $K[x_1, \ldots, x_n]$ with invertible Jacobian matrix is an automorphism.*

2. *Every injective endomorphism of $\mathcal{P}$ is an automorphism.*

3. *Every automorphism Φ of $\mathcal{D}$ with $\Phi(\mathcal{P}) \subset \mathcal{P}$ satisfies $\Phi(\mathcal{P}) = \mathcal{P}$.*

Hauser and Müller credit the equivalence of the first two statements to Kulikov [1998-4, p. 211]. Thus the Jacobian conjecture becomes a property of one infinite-dimensional Lie algebra of derivations [1998-4, Proposition 1, p. 211].

The polynomial vector fields must be preserved [1998-4, Proposition 1, p. 211]. They are preserved under the automorphisms of the formal ones that map them into themselves [1998-4, Proposition 1, p. 211].

The third formulation links the conjecture to two other great open problems [2021-22, Theorem 4.4.1, p. 101]. The Dixmier conjecture asks whether every endomorphism of a Weyl algebra is an automorphism [2021-22, Theorem 4.4.1, p. 101]. The Poisson conjecture makes the same demand of the Poisson algebra [2021-22, Theorem 4.4.1, p. 101].

Theorem 17.25 ([2021-22, Theorem 4.4.1, p. 101])**.** *For each $n \geq 1$ one has the implications* $\mathrm{DC}(n, \mathbb{C}) \Rightarrow \mathrm{JC}(n, \mathbb{C})$, $\mathrm{JC}(2n, \mathbb{C}) \Rightarrow \mathrm{P}(n, \mathbb{C})$ *and* $\mathrm{P}(n, \mathbb{C}) \Rightarrow \mathrm{DC}(n, \mathbb{C})$*; that is, the Dixmier, Jacobian, and Poisson conjectures are equivalent.*

The implication from the Dixmier conjecture to the Jacobian conjecture was classical [2021-22, p. 91]. Tsuchimoto proved the converse in 2005, and Belov-Kanel and Kontsevich gave an independent proof [2021-22, p. 91].

The conjecture of this section is open, and this section states it in the form the book uses.

Problem 17.26 ([2017-7, p. 81])**.** Let $F\colon k^n \to k^n$ be a polynomial map over a field k of characteristic 0 whose Jacobian determinant is a nonzero constant. Is F then an automorphism?

Freudenburg records the problem in the language of derivations [2017-7, p. 81]. A jacobian derivation of $k[x_1, \ldots, x_{n-1}]$ has a slice exactly when it is locally nilpotent and its kernel is a polynomial ring in $n-1$ variables [2017-7, p. 81]. The problem is open for every $n \geq 2$ [2007-2, pp. 1–2].

Theorem 17.25 also ties the Jacobian story to the Weyl algebra, and Chapter 12 of this book develops that side. Section 12.2 states the Dixmier problem. Section 17.5 follows the derivations of $k[x_1, \ldots, x_n]$ to their smooth limit, the vector fields on a manifold.

17.5 Derivations as vector fields

Section 17.4 pushed the derivations of the polynomial ring toward their formal limit. This section makes a parallel move in geometry.

Let M be a smooth manifold and $C^\infty(M)$ its algebra of smooth real functions. A vector field on M takes directional derivatives of functions [2020-20, p. 122]. So it acts as an operator on $C^\infty(M)$, and Nestruev takes this action as the definition [2020-20, p. 122].

Definition 17.27 ([2020-20, p. 122])**.** Let M be a smooth manifold. A *smooth vector field* on M is an $\mathbb{R}$-linear operator $X\colon C^\infty(M) \to C^\infty(M)$ that satisfies the Leibniz rule

$$X(fg) = X(f)\,g + f\,X(g), \qquad f, g \in C^\infty(M).$$

In the language of this book, this is a definition [2020-20, p. 122]: a smooth vector field is exactly a derivation of the $\mathbb{R}$-algebra $C^\infty(M)$.

The classical direction is also true [2020-20, p. 122]. A family of tangent vectors X_z gives the operator $X(f)(z) = X_z(f)$. Nestruev checks that this operator satisfies the Leibniz rule [2020-20, p. 122].

Vector fields are local operators [2020-20, Section 9.40, p. 123]. If f and g coincide on an open set, then so do $X(f)$ and $X(g)$ [2020-20, Section 9.40,

p. 123]. They can also be read as sections of the tangent bundle [2020-20, p. 123]. So the dictionary between vector fields and derivations is complete.

The strongest evidence that the algebraic side controls the geometric side is a theorem of Shanks and Pursell from 1954 [1954-1, p. 468]. They state the aim of their paper on the first page [1954-1, p. 468].

They prove a statement about a smooth manifold [1954-1, p. 468]. The Lie algebra of all tangent vector fields on it, with compact supports, determines the manifold [1954-1, p. 468]. Two manifolds with isomorphic Lie algebras are diffeomorphic [1954-1, p. 468].

Their proof recovers the points of M from the Lie algebra [1954-1, p. 470]. Shanks and Pursell describe each maximal ideal of this Lie algebra: it consists of the fields that vanish at one fixed point, together with all their derivatives [1954-1, Theorem 1, p. 470].

This gives a one-to-one correspondence between the points of M and the maximal ideals [1954-1, Theorem 2, p. 471]. With the Stone topology on the set of ideals, the correspondence is a homeomorphism [1954-1, Theorem 2, p. 471].

Their main theorem closes the section, and it is stated for the Lie algebra of tangent vector fields with compact support [1954-1, Theorem 3, p. 471].

Theorem 17.28 ([1954-1, Theorem 3, p. 471])**.** *Let X and Y be smooth manifolds whose Lie algebras of tangent vector fields with compact support are isomorphic. Then the natural homeomorphism of X onto Y given by the correspondence of maximal ideals is infinitely differentiable.*

In other words, the derivations of $C^\infty(M)$ determine the manifold up to diffeomorphism [1954-1, Theorem 3, p. 471]. That holds when the derivations are viewed only as a Lie algebra [1954-1, Theorem 3, p. 471].

So the opening question of this chapter is answered here in the strongest form [1954-1, Theorem 3, p. 471]. For the algebra of smooth functions, the derivations carry all the geometry.

The same hope, stated as a conjecture for local rings of algebraic varieties, was the Zariski–Lipman problem of Section 17.2 [2014-9, p. 270]. That problem is still open [2014-9, p. 270].

17.6 Geometry: tangent vectors and connections

The same rule appears in differential geometry, acting on the commutative algebra of smooth functions [2023-14, p. 70]. Majumdar and Bhattacharyya

give the modern definition of a tangent vector at a point of a smooth manifold as a derivation at that point [2023-14].

Definition 17.29 ([2023-14, p. 70])**.** Let M be a smooth manifold and let $p \in M$. Denote by $F(p)$ the space of control test functions at p. A tangent vector X_p at p is a real linear functional $X_p \colon F(p) \to \mathbb{R}$ satisfying, for all $a, b \in \mathbb{R}$ and all $f, g \in F(p)$,

$$X_p(af + bg) = aX_p(f) + bX_p(g), \qquad X_p(fg) = X_p(f)\, g(p) + f(p)\, X_p(g).$$

The second equation is the Leibniz product rule at p. So a tangent vector is a point derivation of the smooth function algebra at p, in the sense of this chapter. The notation of the book matches ours: [2023-14, equation (2.17), p. 70] writes the product rule as $X_p(fg) = (X_p f)g(p) + f(p)X_p g$.

The set of all tangent vectors at a point forms a vector space $T_p(M)$, and the coordinate functions give its natural basis [2023-14, equation (2.19), p. 70]. Majumdar and Bhattacharyya construct the partial-derivation basis [2023-14, equation (2.19), p. 70].

Definition 17.30 ([2023-14, equation (2.19), p. 70])**.** If $(x_1, \dots, x_n)$ is a local coordinate system in a neighbourhood of $p \in M$, the maps

$$\left(\frac{\partial}{\partial x_i}\right)_p : f \longmapsto \frac{\partial f}{\partial x_i}(p), \qquad i = 1, \dots, n,$$

are tangent vectors at p and they form a basis of $T_p(M)$.

A vector field on M is a smooth choice of one such derivation at every point [2023-14, p. 70]. In the algebraic setting of this chapter, the derivations of the coordinate ring are the algebraic vector fields.

Chapter 18

Differential algebra and differential Galois theory

Chapter 1 ended with the school of Ritt. Ritt opened the algebraic study of differential equations in the 1920s. He asked the question Galois had asked for polynomial equations: given a differential equation, what abstract object controls its symmetries and its solvability in closed form. This chapter reports the two answers his school and the next generations gave.

Section 18.1 tells the first part, the Ritt–Kolchin theory of differential ideals. Ritt proved that every differential algebraic variety is a finite union of irreducible ones. Raudenbush rebuilt the theory as an ideal theory of perfect differential ideals, and proved that any such ideal is an intersection of primes [1934-1, p. 365]. Later authors made the method of Ritt constructive with characteristic sets. Section 4 of [2011-9] develops the modern textbook picture. That section also states the axioms of Kolchin for a differential field (Section 18.2). It then reports the reading of an input–output system as a differential field extension (Section 18.3).

Section 18.5 tells the second part, the Picard–Vessiot theory of 1948. Kolchin attached an algebraic matric group to a homogeneous linear differential equation over a differential field. He proved that the intermediate differential fields of the Picard–Vessiot extension correspond bijectively to the Zariski-closed subgroups of this group [1948-1, p. 32]. The example of the figure below already displays the mechanism. The same theory runs with an iterative derivation in place of a derivation. Michel and Srinivasan extend it to a central simple algebra over a field of positive characteristic [2026-26, Theorem 2.2, p. 3].

Section 18.6 reports the model-theory side. Robinson and Blum had

axiomatized the differentially closed fields of characteristic zero. Wood proved in 1973 that the theory of differential fields of positive characteristic also has a model completion, and that this theory eliminates quantifiers [1973-3, p. 582]. A later section of this chapter reports a different kind of model theory: the automorphism of a difference field replaces the derivation, and the same algebraic argument returns [2003-3, p. 579]. Alongside it, Section 18.7 states the Ax–Schanuel inequality for the exponential differential equation on an algebraic variety [2018-6, Theorem 2.2, p. 6].

Section 18.8 develops that difference side. Replacing the derivation by an automorphism gives the difference algebra of Ritt and Cohn and the $\sigma\partial$-theory of Hardouin and Singer. It runs in parallel to the differential theory [1989-2, p. 912]. The two theories meet in a single Galois correspondence [2008-6, Theorem 2.6, p. 6].

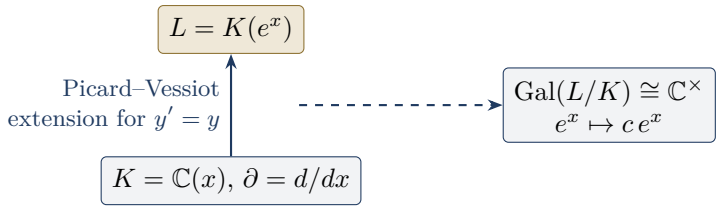


Figure 18.1: The simplest Picard–Vessiot extension: adjoining a solution of $y' = y$ to $\mathbb{C}(x)$. Its differential Galois group is the multiplicative group.

18.1 The Ritt–Kolchin theory

Chapter 1 defined the differential field and the differential ideal with Raudenbush (Definition 1.3). Ritt asked how far the ideal theory of these rings goes [1934-1, p. 361]. Raudenbush rebuilt the theory in 1934 to give the theorem of Ritt a clean proof [1934-1, p. 361]. He replaced the differential ideal by its radical inside the differential category: the perfect differential ideal [1934-1, p. 361].

A *perfect* differential ideal is one that contains an element whenever it contains a positive power of it [1934-1, p. 362]. In the algebraic theory the perfect differential ideals are the radicals of differential ideals [1934-1, p. 362]. Raudenbush first shows that the perfect ideal determined by a set is the radical of the differential ideal it determines.

Theorem 18.1 ([1934-1, Theorem 1, p. 363])**.** *Let R be a differential ring of characteristic zero and let α be a set of its elements. If t is any element*

of the perfect differential ideal $\{\alpha\}$ *determined by* α*, then some positive integral power of* t *is in the differential ideal* $[\alpha]$ *determined by* α*.*

The proof uses the product rule [1934-1, p. 363]. From ab in the ideal, Raudenbush first shows every product $a_p b_q$ of a derivative of a by a derivative of b is in the ideal [1934-1, p. 363]. Multiplying the linear expressions of the two powers of t against each other then places each term in the smaller ideal [1934-1, p. 363]. The same device gives the identity $\{\alpha, a\} \cap \{\alpha, b\} = \{\alpha, ab\}$ for any two elements of R [1934-1, Theorem 2, p. 363], and this identity drives the decomposition theory [1934-1, p. 365].

A perfect differential ideal π is *reducible* if it is the intersection of two perfect differential ideals that both contain it properly [1934-1, p. 365]. Otherwise it is *irreducible* [1934-1, p. 365]. The relations between primality and irreducibility are the first structural facts [1934-1, p. 365].

Theorem 18.2 ([1934-1, Theorems 4 and 5, p. 365])**.** *Let* R *be a differential ring with a basis theorem: every perfect differential ideal has a finite basis.*

1. Irreducible implies prime*. A perfect differential ideal which is irreducible is prime.*

2. Decomposition*. Any perfect differential ideal is the intersection of a finite set of irreducible, hence prime, perfect differential ideals.*

The decomposition theorem follows by noetherian induction [1934-1, p. 365]. Suppose the claim fails; the failing ideal is reducible, one component still fails, and so on. An ascending chain of perfect differential ideals appears, and the basis theorem forbids it [1934-1, p. 365]. Uniqueness comes next: delete in turn each member of the finite set that contains a different member, and the remainder is determined by the ideal [1934-1, pp. 365–366].

Theorem 18.3 ([1934-1, Theorem 6, p. 365])**.** *If a perfect differential ideal is the intersection of each of two essential sets of prime perfect differential ideals, then the two sets have the same number of members and coincide after a suitable rearrangement.*

All this applies to the differential ring that motivates the whole subject: the ring of forms (differential polynomials) in a finite set of indeterminates over a differential field [1934-1, p. 361]. Ritt had proved his basis theorem for this ring [1934-1, p. 361]. Raudenbush gives a second, shorter proof and remarks that its simplicity reflects the strength of the theory [1934-1, p. 362].

Theorem 18.4 ([1934-1, Theorem 7, pp. 366–367])**.** *Let F be a differential field of characteristic zero. Then the differential ring of forms in a finite set of indeterminates over F is a differential ring with a basis theorem.*

With the basis theorem in hand, the preceding theorems give the intersection theorem on primes [1934-1, Theorems 5, 6 and 7, pp. 365–367]. Every perfect differential ideal of the ring of differential polynomials over a characteristic-zero differential field is a finite intersection of prime perfect differential ideals, unique up to rearrangement [1934-1, Theorems 5, 6 and 7, pp. 365–367]. Combined with Theorem 18.1 this recovers the finite basis theorem of Ritt for differential polynomials, and the finite decomposition of differential algebraic varieties into irreducible components [1934-1, p. 361].

The method of Ritt himself is more explicit than the ideal theory, because it comes from elimination [1989-5, p. 176]. For a finite system of differential polynomials he produces an *ascending set*, a finite ordered list of differential polynomials of increasing class. He also produces a *characteristic set* of the differential ideal of the system, whose remainder theory decides membership [1989-5, p. 176]. Wu Wen-tsun rebuilt the whole theory constructively around this device [1989-5, pp. 175–184]. His decomposition theorem makes the irreducible components computable [1989-5, p. 179]. It is the computational form of the prime decomposition of a perfect differential ideal stated above this paragraph [1989-5, p. 179].

Theorem 18.5 ([1989-5, p. 179])**.** *Let* (DP) *be a finite system of differential polynomials and G a differential polynomial. There is a mechanical procedure which decides in a finite number of steps whether the zero set* $\mathrm{Zero}(\mathrm{DP}/G)$ *is empty, and in the non-empty case furnishes a decomposition*

$$\mathrm{Zero}(\mathrm{DP}/G) \;=\; \bigcup_i \mathrm{Zero}(\mathrm{ASC}_i/I_i, G_i)$$

where each (ASC_i) *is a d-irreducible ascending set, I_i is the product of the initials of* (ASC_i)*, and G_i is reduced with respect to* (ASC_i)*.*

The characteristic-set method is the computational side of the Ritt–Kolchin theory [1989-5, pp. 175–184]. Modern textbook treatments [2011-9, pp. 245–261] develop exactly this interface between the abstract ideal theory and the explicit elimination [2011-9, pp. 245–250]. Theorem 18.5 is also what runs the mechanical theorem proving in differential geometry that Wu proposed [1989-5, pp. 175–184]. The analogue on the Galois side of the chapter is the algebraic group computation of Section 18.5.

18.2 Differential equations: fields with a derivation

The algebra of smooth functions on a manifold from Section 17.6 is a differential ring. Differential algebra takes a differential equation as data and studies its solutions by algebra, as Kolchin axiomatized it [1973-2]; Kolchin began the theory in [1973-2, §I.1].

Definition 18.6 ([1973-2, §I.1, p. 58])**.** Let $\mathfrak{R}$ be a ring. An operator δ on $\mathfrak{R}$ is a *derivation operator* if $\delta(a+b) = \delta a + \delta b$ and $\delta(ab) = (\delta a)b + a\,\delta b$ for all $a, b \in \mathfrak{R}$. A *differential ring* is a ring $\mathfrak{R}$ equipped with a finite set Δ of derivation operators on $\mathfrak{R}$ such that $\delta\delta' a = \delta'\delta a$ for every $a \in \mathfrak{R}$ and every $\delta, \delta' \in \Delta$. If $\mathfrak{R}$ is a field, it is a *differential field*.

Kolchin attaches to a differential ring the free commutative semigroup Θ generated by Δ [1973-2, §I.1, p. 59]. An element $\theta \in \Theta$ is a *derivative operator*, and its *order* $\operatorname{ord}\theta$ is the sum of the exponents [1973-2, §I.1, p. 59]. The objects of differential algebra are then the differential subrings, differential ideals and differential extensions, parallel to rings, ideals and field extensions [1973-2, §I.1, p. 59].

The differential equation itself becomes a statement about field extensions [1971-1]. Ax showed that the transcendence of a solution of the exponential differential equation is bounded by the underlying derivation [1971-1, Theorem 3]. We recall his theorem from the account of Aslanyan [2018-6].

Theorem 18.7 ([2018-6, Theorem 1.1, p. 1])**.** *Let $K = (K; +, \cdot, D)$ be a differential field with field of constants C. If $(x_1, y_1), \ldots, (x_n, y_n)$ are non-constant solutions to the exponential differential equation $Dy = y\,Dx$, then*

$$\operatorname{td}_C C(\bar{x}, \bar{y}) - \operatorname{l.dim}_{\mathbb{Q}}(\bar{x}/C) \geq 1,$$

where $\operatorname{l.dim}_{\mathbb{Q}}(\bar{x}/C)$ is the dimension of the $\mathbb{Q}$-span of $x_1, \ldots, x_n$ in the quotient vector space K/C.

This is the Ax–Schanuel theorem, the differential analogue of the open conjecture of Schanuel in number theory; Ax [1971-1] proved it as an application of the differential algebra of Kolchin [1973-2]. The field of definition of a solution of a differential equation is read from the bound [2018-6, §2]. Section 18.7 comes back to it in the model-theoretic form the theorem of Ax started.

18.3 Control theory: differential fields and dynamic feedback

The engineer who designs an automatic controller writes a dynamical system as $\dot{x} = f(x, u)$ with u an input and x a state [1989-1]. Fliess observed in 1989 that the input–output behaviour of the system lives in a finitely generated extension of a differential field, in the sense of Section 18.2 [1989-1].

Definition 18.8 ([1989-1, §1.2, p. 228])**.** A *systemic field* $\mathcal{K}$ is a finitely generated differential extension of the base differential field k. There exists a finite tuple $w = (w_1, \dots, w_s)$ of elements of $\mathcal{K}$ with $\mathcal{K} = k\langle w\rangle$, the differential field generated by k and w. An *input* (or *control*, *command*) is any set containing a differential transcendence basis of $\mathcal{K}/k$; if the set is itself such a basis, the input is *independent*.

There is thus a partition $w = (u, y)$, not necessarily unique, into an independent input u and a remaining tuple y [1989-1, §1.2, p. 228]. The input u is a differential transcendence basis of $\mathcal{K}/k$, and the components of y are differentially algebraic over $k\langle u\rangle$ [1989-1, §1.2, p. 228]. The state-space realisation question then asks for a writing of the extension with a finite non-differential transcendence basis [1989-1, §1.2, p. 228]. Fliess shows that in general one cannot realise a nonlinear system globally in explicit first-order form [1989-1, Introduction, p. 227].

Definition 18.9 ([1989-1, §1.3, p. 228])**.** An *input-output system* consists of an input $u = (w_1, \dots, w_m)$ and an output $y = (y_1, \dots, y_p)$ with components in a universal differential field, such that $y_1, \dots, y_p$ are differentially algebraic over $k\langle u\rangle$. If $w_1, \dots, w_m$ are differential indeterminates, the input is *independent*.

The framework applies to feedback synthesis, input-output inversion and realization, and Fliess uses it to define a feedback as a specialisation of differential fields [1989-1, Introduction, p. 227].

18.4 Automatic differentiation: derivations in computer code

The topological refinement of Section 20.6 computed derivations as homology of a chain complex. The engineering reading of the same rule makes a computer program evaluate the derivative of the function it implements

[2020-23, Introduction, p. 2]. The method is automatic differentiation, and the program runs the chain rule and the Leibniz rule step by step [2020-23]. Yuan, Thomson and Anderson apply the method to the flight dynamics of a tilt-rotor aircraft [2020-23], where automatic differentiation gives the stability derivatives.

The authors define automatic differentiation (AD) as a set of techniques for evaluating numerically the derivative of a function implemented in a computer program [2020-23, Introduction, p. 2].

Definition 18.10 ([2020-23, Introduction, p. 2])**.** *Automatic differentiation* (AD) is a set of techniques for numerically evaluating the derivative of a function implemented within a computer program. The method is based on the chain rule as applied to the differentiation process.

Every program decomposes into elementary operations [2020-23, §II.B]. An arithmetic operator or an intrinsic function applied to intermediate values is one elementary step [2020-23, §II.B]. A step $g = h(a, b)$ of the program differentiates by the chain rule as

$$dg \;=\; \frac{\partial h}{\partial a}\, da \;+\; \frac{\partial h}{\partial b}\, db,$$

[2020-23, equation (2)]. Repeated use of the formula simplifies the differentiation to a series of algebraic operations, with derivative results coming from the elementary operations [2020-23, §II.B]. In the forward mode of [2020-23, §II.B], the propagation of the derivative of the state $\dot{x}_i$ with respect to an initial condition x_j is evaluated. The operation of the forward mode is

$$\frac{\partial \dot{x}_i}{\partial x_j} \;=\; \sum_{k=1}^{n} \frac{\partial \dot{x}_i}{\partial x_k}\, \frac{\partial x_k}{\partial x_j} \;+\; \sum_{l=1}^{m} \frac{\partial \dot{x}_i}{\partial u_l}\, \frac{\partial u_l}{\partial x_j},$$

with $n = \dim(x)$ and $m = \dim(u)$ [2020-23, equation (3)].

The derivative of the composed function is thus read from the Jacobian products, never from a difference quotient [2020-23, Introduction, p. 2]. Numerical differentiation by finite differences would carry a truncation error and is avoided [2020-23, Introduction, p. 2]. The Leibniz rule of a derivation on an algebra is one of the elementary operator identities that the overloaded algebra of AD satisfies [2020-23, §II.B]. The semantics of the overloaded arithmetic are those of the pair $(x, \dot{x})$, the state and its derivative, as in [2020-23, §II.B]. The composition is a chain rule on the first-order jets of the elementary steps [2020-23, equations (2)–(3)].

18.5 Picard–Vessiot and differential Galois theory

Section 18.1 attached algebraic varieties to systems of differential equations. The Galois question is different: pick one homogeneous linear differential equation over a differential field, and ask which group describes the algebraic relations among its solutions [1948-1, pp. 1–6]. Picard and Vessiot saw what the answer should look like [1948-1, pp. 1–2]. Kolchin built it in 1948 [1948-1, pp. 1–6].

Kolchin works over an ordinary differential field F of characteristic zero whose field of constants C is algebraically closed [1948-1, p. 29]. To a homogeneous linear differential polynomial $L(y) = y^{(n)} + p_1 y^{(n-1)} + \cdots + p_n y$ with $p_i \in F$ he attaches an extension of F generated by a full system of solutions, with no new constant [1948-1, p. 29].

Definition 18.11 ([1948-1, §17, p. 29])**.** A differential extension field G of F is a *Picard–Vessiot extension* of F for the homogeneous linear differential polynomial $L(y)$ if

1. $L(y)$ has a fundamental system of solutions $t_1, \ldots, t_n$ with $G = F\langle t_1, \ldots, t_n \rangle$;
2. the field of constants of G is C.

The group of interest is the group of differential automorphisms of G over F [1948-1, pp. 29–31]. Acting on a fundamental system of solutions, such an automorphism sends each t_i to a C-linear combination of the t_j [1948-1, p. 29]. So the group embeds in $\mathrm{GL}_n(C)$ [1948-1, p. 29]. Kolchin shows that its image is closed in the Zariski topology.

Theorem 18.12 ([1948-1, §17, Theorem, p. 31])**.** *The group of differential automorphisms of a Picard–Vessiot extension G over F is isomorphic to an algebraic matric group with a manifold in C^{n^2}.*

The group therefore inherits the structure theory of algebraic matric groups of the first chapter of the same paper: normal chains, solvability, Lie reducibility to triangular form [1948-1, pp. 16–24]. Hubbard and Lundell state the same fact for the numerical setting [2011-9, p. 250].

Theorem 18.13 ([2011-9, Theorem 4.3, p. 250])**.** *The differential Galois group of a linear differential operator is an algebraic subgroup of the general linear group of its solution space: a subset defined by finitely many polynomial equations.*

The very first example already displays the algebraicity [2011-9, pp. 249–250].

Example 18.14 ([2011-9, Example 4.2, pp. 249–250])**.** Let $L(u) = u' - u$ over $\mathbb{C}(t)$. A differential automorphism of its splitting field sends e^t to $c\,e^t$ for some $c \in \mathbb{C}^\times$, and c determines it. The differential Galois group is $\mathrm{GL}_1(\mathbb{C})$, the multiplicative group of the complex numbers.

This is the case drawn in Figure 18.1.

The heart of the theory is the Galois correspondence [1948-1, pp. 29–33]. Kolchin first shows that a Picard–Vessiot extension is normal in his sense [1948-1, Theorem 1, p. 32]. The intermediate differential fields then determine the Zariski-closed subgroups [1948-1, Theorem, p. 32].

Theorem 18.15 ([1948-1, §19, Theorem, p. 32])**.** *Let G be a Picard–Vessiot extension of F. The map sending an intermediate differential field F_1 to the group of differential automorphisms of G over F_1 is a bijection between the intermediate differential fields of G/F and the algebraic subgroups of the Galois group.*

Cassidy and Singer package the full result, including the behaviour of normal subgroups, in one statement [2007-4, Theorem 2.1, pp. 5–6].

Theorem 18.16 ([2007-4, Theorem 2.1, pp. 5–6])**.** *Let k be a Δ-field with algebraically closed field of constants C, and let an integrable system of linear differential equations over k be given.*

1. *There exists a Picard–Vessiot extension K of k associated with the system, and this extension is unique up to differential k-isomorphism.*

2. *The Δ-Galois group $\mathrm{Gal}_\Delta(K/k)$ equals $G(C)$, where G is a linear algebraic group defined over C.*

3. *The map that sends a differential subfield F with $k \subset F \subset K$ to $\mathrm{Gal}_\Delta(K/F)$ is a bijection between the differential subfields of K containing k and the algebraic subgroups of $\mathrm{Gal}_\Delta(K/k)$. Its inverse sends a Zariski-closed group H to the fixed field $K^H = \{z \in K \mid \sigma(z) = z \text{ for all } \sigma \in H\}$.*

4. *A Zariski-closed subgroup H of $\mathrm{Gal}_\Delta(K/k)$ is normal if and only if the field K^H is invariant under $\mathrm{Gal}_\Delta(K/k)$ as a set. Then K^H is a Picard–Vessiot extension of k with group $\mathrm{Gal}_\Delta(K/k)/H$.*

The gain for solvability questions is immediate [1948-1, pp. 5–6]. Kolchin calls the differential analogue of the radical extension a *Liouvillian* extension: one built by adjunction of integrals, exponentials of integrals, and algebraic functions, with no new constant [1948-1, pp. 5–6].

Theorem 18.17 ([1948-1, Chapter V, p. 5])**.** *A Picard–Vessiot extension is contained in a Liouvillian extension of the base differential field if and only if the component of the identity of its Galois group is solvable.*

This is the analogue of the criterion of Galois: the Galois group of the equation decides whether its solutions can be expressed in closed form [1948-1, pp. 5–6]. In the multiplicative example of Figure 18.1 the group is abelian, hence solvable, and the solution e^t is literally obtained by exponentiating an integral [2011-9, pp. 249–250]. In general the criterion of Theorem 18.17 separates the elementary from the non-elementary.

The same theory runs with an iterative derivation in place of a derivation. An iterative derivation on a ring R is a sequence of additive maps $\delta^{(n)}$ [2026-26, p. 1]. The first term $\delta^{(0)}$ is the identity, the Leibniz rule holds in every degree, and the terms satisfy the binomial rule $\delta^{(n)} \circ \delta^{(m)} = \binom{m+n}{n} \delta^{(m+n)}$ [2026-26, p. 1]. In characteristic zero such a sequence is determined by its first term [2026-26, p. 2]. Michel and Srinivasan ask when a derivation of the base field extends to a central simple algebra [2026-26, p. 1].

Theorem 18.18 ([2026-26, Theorem 2.2, p. 3])**.** *Let F be a field of characteristic $p > 0$ and let A be a central simple F-algebra whose exponent in the Brauer group is not divisible by p. Then every iterative derivation on F extends to an iterative derivation on A.*

The extension fails in general without the hypothesis on the exponent [2026-26, p. 2]. With the extension in hand the authors build the Picard–Vessiot theory for the algebra: a splitting field exists and is unique up to isomorphism [2026-26, p. 1]. The Galois group then describes the algebra through its right δ-ideals [2026-26, p. 1]. So the Galois correspondence of this section survives into the noncommutative coefficient ring.

Section 18.6 gives a different reading of the same algebra, with model-theoretic tools.

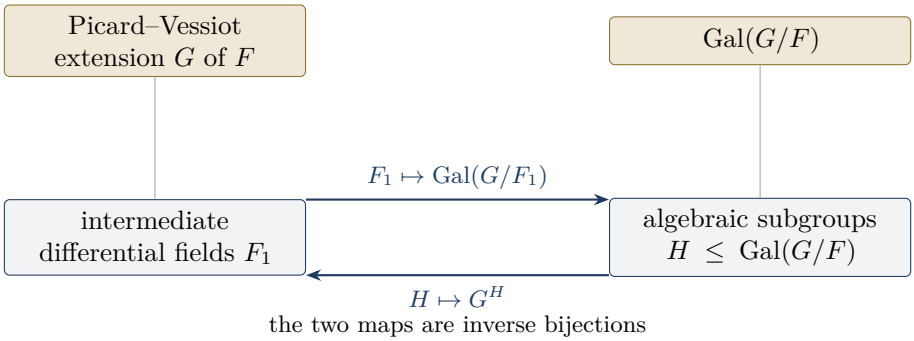


Figure 18.2: The Galois correspondence of Kolchin [1948-1, §19, Theorem, p. 32], in the form of Cassidy and Singer [2007-4, Theorem 2.1, pp. 5–6]. An intermediate differential field F_1 of the Picard–Vessiot extension G of F corresponds to the algebraic subgroup $\mathrm{Gal}(G/F_1)$, and a subgroup H to its fixed field G^H. The two maps are inverse bijections and run in opposite directions.

18.6 Differential fields in model theory

The abstract theories of Sections 18.1 and 18.5 say what a differential field is and what its automorphisms can look like. They do not say that large enough differential fields exist. Existence is a model-theoretic fact [2000-4, pp. 129–132]. Robinson and Blum proved it in characteristic zero [2000-4, pp. 129–132]: the theory of differential fields has a model completion, the theory DCF of differentially closed fields. The axioms of Robinson say that any system of differential equations and inequations that has a solution in some extension already has one in the field [2000-4, p. 131]. Wood states the equivalent reformulations that are used today [2000-4, pp. 129–134].

Definition 18.19 ([2000-4, §1, pp. 129–131])**.** A *differentially closed field* is a differential field K of characteristic zero such that every finite system of differential equations and inequations over K which has a solution in some differential field extension of K already has a solution in K.

The theory DCF has the two model-theoretic properties that make the algebraic machine of Section 18.5 run [2000-4, pp. 129–134].

Theorem 18.20 ([2000-4, Theorem 1.8, p. 132])**.** *The theory* DCF *of differentially closed fields is complete and admits elimination of quantifiers, hence is model complete.*

Quantifier elimination gives the analogue of the Nullstellensatz of Hilbert [2000-4, pp. 129–132].

Corollary 18.21 ([2000-4, Corollary 1.9, p. 132])**.** *Let S be a finite system of differential equations and inequations in n differential indeterminates over a differentially closed field K. If S has a solution in some differential field containing K, then S has a solution in K.*

Inside a differentially closed field, definability is just differential-algebraic definability [2000-4, pp. 132–134]. The theory is stable in the strongest sense [2000-4, p. 134].

Theorem 18.22 ([2000-4, Theorem 2.1, p. 134])**.** *The theory* DCF *of differentially closed fields is ω-stable.*

A further property concerns quotients [2000-4, pp. 133–134].

Theorem 18.23 ([2000-4, Theorem 1.13, p. 133])**.** *The theory* DCF *of differentially closed fields has elimination of imaginaries.*

These four results make DCF a clean subject [2000-4, pp. 129–134]. Every differential field embeds in a differentially closed one (Corollary 18.21). Every formula is a Boolean combination of differential equations (Theorem 18.20). Definable sets are controlled by Morley rank (Theorem 18.22), and quotients by definable equivalence relations stay inside the theory (Theorem 18.23).

Characteristic p changes the picture less than one might fear [1973-3, pp. 577–584]. Wood builds the theory T_p of differential fields of characteristic p [1973-3, pp. 577–584]. He shows that T_p has no model completion itself [1973-3, p. 579], but it still has a model companion T^* of differentially closed fields [1973-3, p. 582]. The reason for the failure of a model completion is the failure of amalgamation [1973-3, Theorem 2, p. 579]. A system of equations in one variable can be solvable over one extension and not over another [1973-3, pp. 579–580]. The theory only sees differential properties [1973-3, pp. 579–580], so it does not see extensions that are not differentially perfect [1973-3, pp. 579–580].

Theorem 18.24 ([1973-3, Theorems 8 and 9, p. 582])**.** *The theory T^* of differentially closed fields of characteristic p is the model completion of the theory of differentially perfect differential fields of characteristic p, and is the model companion of the theory of all differential fields of characteristic p. The theory T^* is complete.*

The quantifier elimination survives as well [1973-3, Theorem 11, p. 583]: every formula over a differentially perfect field is equivalent to one existential differential condition. But stability breaks: unlike the characteristic zero case of Theorem 18.22, T^* is not ω-stable [1973-3, p. 577]. Separably algebraic closure replaces algebraic closure for the constants and for the base field [1973-3, Theorem 10, p. 582].

The difference-field theory of Chatzidakis, Hrushovski and Peterzil answers the same question for an automorphism in place of a derivation. In a difference field the derivation is replaced by an endomorphism σ, and the theory of difference fields was developed in their earlier paper. That paper proved its deepest result, the trichotomy of Zil'ber, only in characteristic 0 [2002-4, p. 257]. The 2002 paper gives a new proof of the trichotomy that holds in all characteristics [2002-4, p. 257]. It works with a sequence of reducts of the theory, each replacing σ by σ^n, and passes to a limit structure that is a Zariski geometry [2002-4, p. 257]. The trichotomy is obtained for that limit structure and then inherited by the original theory [2002-4, p. 257].

18.7 Transcendence and model theory: Ax–Schanuel on a variety

The Ax–Schanuel theorem of Section 18.2 bounds the transcendence of solutions of the exponential differential equation. The derivation D enters the statement only through the exponential differential equation $Dy = y\,Dx$; the conclusion is a transcendence inequality, the kind of statement the transcendence theorist wants [2018-6, §2]. Ax [1971-1] gave the inequality a geometric reading, and Aslanyan explains how [2018-6, §2, p. 5]. Write G_a for the additive group of a field and G_m for its multiplicative group, and set $G_n = G_a^n \times G_m^n$. An algebraic subgroup H of G_m^n has a tangent space T_eH at the identity, which is an algebraic subgroup of G_a^n [2018-6, §2, p. 6].

Theorem 18.25 ([2018-6, Theorem 2.2, p. 6])**.** *Let $K = (K; +, \cdot, D)$ be a differential field with algebraically closed field of constants C, and let $V \subseteq G_n(K)$ be an algebraic variety defined over C with $\dim(V) \leq n$. If $(\bar{x}, \bar{y}) \in V(K)$ and $Dy_i = y_i\,Dx_i$ for each i, then there is a proper algebraic subgroup H of G_m^n such that $(\bar{x}, \bar{y})$ lies in a C-coset of TH, that is, $\bar{y} \in \gamma H$ and $\bar{x} \in \gamma' + T_eH$ for some constant points $\gamma \in G_m^n(C)$ and $\gamma' \in G_a^n(C)$.*

A variety of this kind with a point $(\bar{x}, \bar{y}) \in V(K)$ satisfying the exponential differential equation is said to have an *exponential point.* The

Ax–Schanuel theorem is thus a necessary condition for a variety to have one [2018-6, §2, p. 6]. A compactness argument gives a uniform version. For a parametric family of varieties $V(\bar{c})$ over C there is a bound N; whenever $\dim V(\bar{c}) \leq n$, the integers m_i of Theorem 18.7 may be taken with $|m_i| \leq N$ [2018-6, §2, p. 6]. The theory T_{Exp} of the exponential differential equation has an axiomatisation built from this uniform inequality and a list of elementary axioms [2018-6, §2, pp. 6–7].

The inequality hence divides the possibility space for solutions. A variety meeting the graph of the exponential map in too small a dimension does so inside an algebraic subgroup [2018-6, Theorem 2.2].

Section 18.8 returns to the algebraic side and replaces the derivation by an automorphism.

18.8 Difference and differential–difference algebra

So far this chapter has used one operator, the derivation [1989-2, p. 912]. A small change replaces the whole theory: use an automorphism σ of the ring in place of the derivation [1989-2, p. 912]. Ritt named the resulting theory *difference algebra*. The monograph of Cohn is its standard reference [1989-2, p. 912]. Pillay and Ziegler describe the change in geometric language, following Section 18.6. Difference algebraic geometry is the analogue of differential algebraic geometry "when the derivation is replaced by an automorphism" [2003-3, p. 579].

The two theories run in parallel [1989-2, p. 912]. Each main result of differential algebra has a difference analogue, which appears a little later [1989-2, p. 912]. The table gives the dictionary of the two sides, as Mikhalev and Pankrat'ev set it out [1989-2, p. 912].

differential side	**difference side**
derivation ∂	automorphism σ.
differential field	difference field.
differential ideal	difference ideal.
Ritt–Raudenbush basis	difference basis theorem.
Picard–Vessiot extension	Hardouin–Singer ring.
integration in finite terms	summation in finite terms.

Mikhalev and Pankrat'ev warn that the dictionary is not a translation [1989-2, p. 912]. Their words are that "all these results in difference

algebra are essentially different from the corresponding results in differential algebra" [1989-2, p. 912]. A σ-ideal need not be radical when its differential analogue is, and σ^{-1} plays a role the derivation has no analogue of [1989-2, p. 912]. This section reports the definitions and the Galois theorem on the difference side, as Hardouin and Singer state them [2008-6, pp. 5–6].

The clearest formulation uses both operators and lets them commute [2008-6, Definition 2.1, p. 5].

Definition 18.26 ([2008-6, Definition 2.1, p. 5])**.** A *$\sigma\partial$-ring* is a commutative ring R with unit together with an automorphism σ and a derivation ∂ satisfying $\sigma(\partial(r)) = \partial(\sigma(r))$ for all $r \in R$. A *$\sigma\partial$-field* is a $\sigma\partial$-ring which is a field.

The three basic examples already appear in the source [2008-6, Example 2.2, p. 5].

Example 18.27 ([2008-6, Example 2.2, p. 5])**.** The following are $\sigma\partial$-fields.

1. $\mathbb{C}(x)$ with $\sigma(x) = x + 1$ and $\partial = d/dx$;
2. $\mathbb{C}(x)$ with $\sigma(x) = qx$ $(q \in \mathbb{C} \setminus \{0\})$ and $\partial = x\, d/dx$;
3. $\mathbb{C}(x, t)$ with $\sigma(x) = x + 1$, $\sigma(t) = t$ and $\partial = \partial/\partial t$.

The equation of interest is the difference equation $\sigma(Y) = AY$, taken over a $\sigma\partial$-field k with A an invertible matrix over k [2008-6, pp. 5–6]. To it one attaches a *$\sigma\partial$-Picard–Vessiot ring*. This is a ring containing k with no proper $\sigma\partial$-ideal [2008-6, Definition 2.3, p. 5]. The ring holds a fundamental solution matrix [2008-6, Definition 2.3, p. 5]. It is generated over k, as a ∂-ring, by the entries of that matrix and by the inverse of its determinant [2008-6, Definition 2.3, p. 5]. The $\sigma\partial$-Galois group is the group of its automorphisms [2008-6, Definition 2.5, p. 6].

Definition 18.28 ([2008-6, Definition 2.5, p. 6])**.** The *$\sigma\partial$-Galois group* $\mathrm{Aut}_{\sigma\partial}(R/k)$ of a $\sigma\partial$-Picard–Vessiot ring R over a $\sigma\partial$-field k is the group of all $\sigma\partial$-k-automorphisms of R.

When ∂ vanishes identically, this is the Galois group of the difference equation alone, which is a linear algebraic group [2008-6, pp. 5–6]. When ∂ is present, the group carries an additional structure and lives among the linear differential algebraic groups [2008-6, Theorem 2.6, p. 6]. That is the content of the main theorem of Hardouin and Singer [2008-6, Theorem 2.6, p. 6].

Theorem 18.29 ([2008-6, Theorem 2.6, p. 6])**.** *Let k be a $\sigma\partial$-field whose fixed field k^σ is a differentially closed field, and let $R = k\{Z, \det Z^{-1}\}_\partial$, $\sigma(Z) = AZ$, be a $\sigma\partial$-Picard–Vessiot extension of k.*

1. *The Galois group $\mathrm{Aut}_{\sigma\partial}(R/k)$ identifies with the set of k^σ-points of a linear ∂-differential algebraic group G defined over k^σ.*

2. *The ring R is reduced and is the coordinate ring of a G-torsor V defined over k; the action of G on V induces the action of $\mathrm{Aut}_{\sigma\partial}(R/k)$ on R.*

So σ and ∂ together give back exactly the two theories that they extend [2008-6, pp. 5–6]. Take σ trivial, and Theorem 18.29 reduces to the Picard–Vessiot Theorem 18.16. Take ∂ trivial, and it returns the algebraic-group Galois theory of a pure difference equation [2008-6, pp. 5–6].

The pure difference side has a recent theorem of its own [2025-20, Theorem 1.2, p. 2]. Wibmer and Bachmayr ask when every torsor for a linear difference algebraic group is trivial after a finite extension of the difference field [2025-20, Theorem 1.2, p. 2]. They answer with three equivalent conditions, and the third names the obstruction [2025-20, Theorem 1.2, p. 2].

Theorem 18.30 ([2025-20, Theorem 1.2, p. 2])**.** *For an ordinary difference field k of characteristic zero the following statements are equivalent.*

1. *For every linear difference algebraic group G over k and every G-torsor X, there is an integer $n \geq 1$ with $X(k^{[n]}) \neq \emptyset$.*

2. *The direct limit of the first difference cohomology sets $H^1_{\sigma^n}(k, G_n)$ vanishes for every linear difference algebraic group G over k.*

3. *The field k is algebraically closed, and the matrix equation $\sigma(Y) = AY$ and the one-relator equation $y^{\alpha_0}\sigma(y)^{\alpha_1}\cdots\sigma^m(y)^{\alpha_m} = a$ are each solvable in some $k^{[n]}$.*

Wibmer and Bachmayr note what the theorem does not assume [2025-20, p. 3]. No condition is placed on the constants k^σ of k [2025-20, p. 3]. The one-relator subgroups of the multiplicative group are, in their words, the only obstruction to the immediate difference analogue of the differential statement [2025-20, p. 3].

This is how the chapter ends: differential, model-theoretic, and difference algebra are three expressions of one algebraic method, applied to three kinds

of operator [1989-2, p. 912]. The difference side also carries its own recent results, of which Theorem 18.30 is one.

Part VII

Analysis and cohomology

Chapter 19

Derivations of Banach and operator algebras

Šilov proved in 1947 that the algebra of all infinitely differentiable functions on an interval cannot be normed so as to be a Banach algebra [1955-1, p. 260]. Kaplansky proposed that the reason was a theorem about derivations: a commutative semisimple Banach algebra should carry no nonzero derivation [1955-1, p. 260]. Singer and Wermer proved the conjecture for bounded derivations in 1955 [1955-1, Theorem 1, p. 260]. Their paper opened the study of derivations on Banach and operator algebras. Kaplansky then asked whether a derivation of a C^*-algebra is automatically continuous. Sakai settled that question in 1960 [1960-1, p. 31]. The same circle of ideas led Kadison and Sakai in 1966 to the theorem that every derivation of a von Neumann algebra is inner [1966-2, Theorem 1, p. 274].

The chapter tells this story in the order of the questions. Section 19.1 reports the commutative theory of Singer and Wermer and its completion by Thomas. Section 19.2 gives the Kadison–Sakai theorem on von Neumann algebras and the spatial theorem of Kadison for C^*-algebras on a Hilbert space. It adds the extended derivations of the Murray–von Neumann algebra of a finite von Neumann algebra [2014-13, Theorem 5, p. 2090] and the nonlinear Lie derivations of a factor [2012-16, Theorem 2.1, p. 1980]. It closes with the 2-local derivations of Ayupov, Arzikulov and Kudaybergenov, which turn out to be derivations on every von Neumann algebra [2011-3, Theorem 5, p. 7]. It also gives the local derivations of Hadwin, Li, Li and Ma on measurable operators [2013-8, Theorem 1, p. 2]. On the measurable operators themselves the derivations that are linear over the centre are spatial [2007-1, Theorem 3.9, p. 11]. Section 19.3 explains why derivations

of semisimple Banach algebras are continuous, the theorem of Johnson and Sinclair, and the module form of Ringrose. It adds the continuity ideals of Bade and Curtis [1974-1, Corollary 2.8, p. 379]. It adds too the continuity of strongly generalized derivations [2023-13, Theorem 2.7, p. 5], and the failure of inner derivations to extend from a dense ideal [2026-29, Theorem 4.1, p. 3]. Section 19.4 passes to unbounded derivations, the generators of one-parameter automorphism groups in quantum dynamics, as Sakai began the theory. It records the kernel stabilization of weakly defined derivations [2018-12, Theorem 3.5, p. 7]. It records too that a derivation with a range that is not dense can be closed [1983-5, Theorem 1, p. 181]. Section 19.5 moves to nest algebras and additive derivations, the work of Christensen and Han. It adds the Lie-type derivations of Zhang and Wei [2017-16, Theorem 3.1, p. 6], the additive derivations of a full operator algebra [1991-8, Theorem 2.3, p. 236] and the Lie derivations at a fixed product [2025-14, Theorem 2.2, p. 81]. Section 19.6 studies the range of a commutator map, from Kleinecke and Shirokov to the Wintner–Wielandt–Halmos statement. It adds the ranges of normal Jordan $*$-derivations [1994-4, Theorem 1, p. 692] and the P-symmetric pairs of Bouhafsi, Ech-chad and Zouaki [2022-3, Theorem 3.4, p. 149]. Section 19.7 closes with approximate derivations and the Hyers–Ulam question of whether an almost derivation is near an exact one. It carries the question to Jordan triple higher derivations [2009-10, Theorem 2.6, p. 1984] and to the derivation equation for C^k-functions [2014-16, Theorem 1, p. 407].

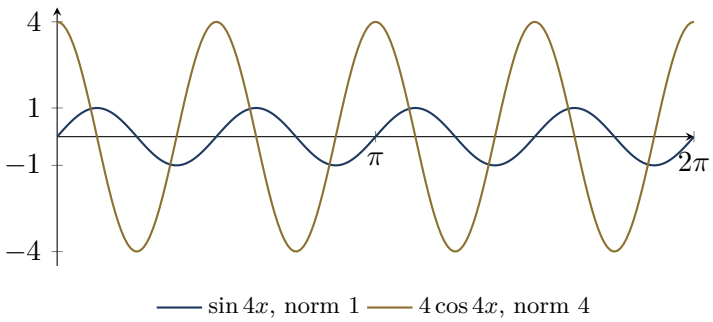


Figure 19.1: On $C[0, 2\pi]$ with the sup norm, $\sin nx$ has norm 1 and its derivative has norm n. Differentiation is an unbounded derivation, here $n = 4$.

19.1 Singer–Wermer and commutative Banach algebras

The chapter opened with the observation of Šilov that the algebra of infinitely differentiable functions on an interval carries no Banach algebra norm [1955-1, p. 260]. Singer and Wermer took up the explanation of Kaplansky of this phenomenon and worked with the following objects [1955-1, p. 260].

Definition 19.1 ([1955-1, p. 260])**.** A *derivation* D on an algebra is a transformation on the algebra such that $D(a+b) = D(a) + D(b)$, $D(\lambda a) = \lambda D(a)$ for any scalar λ, and $D(ab) = D(a)b + aD(b)$. A derivation on a Banach algebra is *bounded* if $\sup_{\|a\|=1} \|D(a)\| < \infty$. The *radical* of a commutative Banach algebra A is the intersection of all maximal ideals M in A such that A/M has a unit; if the radical reduces to the zero element, A is called *semisimple*.

Their main theorem answers the conjecture of Kaplansky in the bounded case [1955-1, p. 260].

Theorem 19.2 ([1955-1, Theorem 1, p. 260])**.** *Let A be a commutative Banach algebra and D a bounded derivation on A. Then D maps A into its radical. In particular, if A is semisimple, $D = 0$.*

The proof of Theorem 19.2 uses the exponential. For a multiplicative linear functional f and a complex number λ, the map $a \mapsto f(e^{\lambda D}a)$ is again multiplicative. Gelfand bound on multiplicative functionals then forces $f(D(a))$ to vanish [1955-1, p. 260]. The result of Šilov is its first consequence [1955-1, p. 260].

Corollary 19.3 ([1955-1, Corollary 1.2, p. 262])**.** *Let C^∞ denote the algebra of all infinitely differentiable complex-valued functions on an interval. Then there exists no norm under which C^∞ is a Banach algebra.*

For the proof, differentiation $f \mapsto f'$ would be a bounded derivation by the closed graph theorem, and C^∞ is semisimple, so Theorem 19.2 would give $D = 0$ [1955-1, p. 262]. Singer and Wermer added two observations that fed the later commutator theory of Section 19.6. A bounded commutator $ab - ba$ inside a commutative Banach algebra is a generalized nilpotent [1955-1, p. 261]. The theorem of Putnam, that a commutator $C = AB - BA$ of bounded Hilbert space operators commuting with both A and B has spectrum $\{0\}$, follows from Theorem 19.2 [1955-1, p. 264].

Derivations into larger algebras behave differently, and the obstruction lives at one point of the maximal ideal space [1955-1, p. 262].

Definition 19.4 ([1955-1, p. 262])**.** A *bounded point derivation* of a commutative Banach algebra A is a bounded linear functional d_φ associated with a multiplicative linear functional φ such that $d_\varphi(a_1 a_2) = d_\varphi(a_1)\varphi(a_2) + \varphi(a_1)d_\varphi(a_2)$.

The next theorem relates point derivations to derivations into a larger algebra.

Theorem 19.5 ([1955-1, Theorem 2, p. 262])**.** *If there exists a nonzero bounded point derivation d_φ of A, then there exists a commutative extension B of A and a nonzero bounded derivation D of A into B. If A is semisimple, B can be taken to be semisimple. If $A \subseteq B$ and if D is a nonzero bounded derivation of A into B but not into the radical of B, then there exists a nonzero bounded point derivation of A.*

On a compact Hausdorff space the point derivation option disappears [1955-1, p. 264].

Corollary 19.6 ([1955-1, Corollary 2.2, p. 264])**.** *The algebra $C(X)$ of all continuous functions on a compact Hausdorff space X has no nonzero derivations into any semisimple commutative extension B.*

Singer and Wermer wrote that the boundedness hypothesis of Theorem 19.2 seemed superfluous [1955-1, p. 260]. The statement became the Singer–Wermer conjecture [2006-6, p. 523]. Thomas proved it in 1988: a linear derivation on a commutative Banach algebra maps into the radical even if it is discontinuous [1988-4, pp. 435–460]. As a direct consequence, there are no nonzero linear derivations on a semisimple commutative Banach algebra, a fact that Johnson had proved earlier [2006-6, p. 523]. That consequence is the commutative end of the continuity story told in Section 19.3. In the noncommutative algebras of Section 19.2, Singer showed that a commutative C^*-algebra has only the zero derivation [1955-1, pp. 260–261]. Kaplansky proved every derivation of a type I von Neumann algebra inner [1966-1, p. 280].

The noncommutative form of the conjecture is open, and this section states it as a problem.

Problem 19.7 ([2002-3, p. 710])**.** Let A be a Banach algebra and let D be a derivation of A, with no continuity assumed. Does D leave every primitive ideal of A invariant?

Brešar and Villena call this the noncommutative Singer–Wermer conjecture and describe it as one of the hardest open problems of Banach algebra theory [2002-3, p. 710]. They prove that it is equivalent to a statement about invertible values of a derivation [2002-3, Theorem 5.2, p. 719]. They also report that the second assertion of their Theorem 5.2 is known for derivations but not for φ-derivations [2002-3, Remark 5.3, p. 720].

Mathieu surveyed the state of the conjecture in 1994 and put it in two forms [1994-3, p. 238]. A derivation is *commuting* when $[a, \delta(a)] = 0$ for every element a [1994-3, p. 238]. The conjecture then reads as follows: every derivation that is commuting modulo the radical is inner modulo the radical [1994-3, Conjecture 1, p. 238]. The second form is a statement about the image of one element [1994-3, p. 242].

Conjecture 19.8 ([1994-3, Conjecture 2, p. 242])**.** *Let δ be a derivation on a Banach algebra A. For each a in A, $[a, \delta(a)] = 0$ implies that $\delta(a)$ is quasinilpotent.*

Mathieu calls Conjecture 19.8 the unbounded Kleinecke–Shirokov theorem and proves that it follows from the conjecture on primitive ideals [1994-3, p. 242]. Three results of that survey bound the image from the other side [1994-3, pp. 243–245]. A derivation that maps into the centre maps into the radical [1994-3, Theorem 7, p. 243]. No derivation of a unital Banach algebra reaches an invertible central value [1994-3, Theorem 8, p. 245]. In particular, no derivation of a unital Banach algebra takes the value 1 [1994-3, Corollary 9, p. 245]. Mathieu notes that the last statement was the key step in the proof of Thomas of the commutative conjecture [1994-3, p. 245].

The conjecture also has a form for twisted product rules [2024-11, p. 1]. Eisaei and Moghimi work with a weighted space of functions on the half-line, and with its dual [2024-11, p. 2]. That dual is a Banach algebra under the first Arens product, and the paper writes it $L_0^\infty(w)^*$ [2024-11, p. 2]. They twist the rule by a homomorphism θ of that algebra. A linear map D is a *θ-derivation* when $D(mn) = D(m)\theta(n) + \theta(m)D(n)$ for all m, n [2024-11, p. 1]. The radical of that algebra equals its right annihilator [2024-11, p. 3]. For an isomorphism θ the range of D lies in the radical [2024-11, Theorem 2.1, p. 4]. If θ is also continuous, then D is continuous exactly when its restriction to the right annihilator is continuous [2024-11, Proposition 2.3, p. 5].

Eisaei, Mehdipour and Moghimi treat the one-sided rule in the same algebra [2024-12, p. 2]. A linear map δ is a *left θ-derivation* when $\delta(st) =$

$\theta(s)\delta(t) + \theta(t)\delta(s)$ for all s, t [2024-12, p. 2]. Such a map is already a θ-derivation, so the one-sided rule adds nothing in this algebra [2024-12, Theorem 2.2, p. 4]. Continuity behaves as before: if a right identity ν satisfies $\|\theta(\nu)\| < 1$, then the zero map is the only continuous left θ-derivation [2024-12, Corollary 2.5, p. 6]. When θ is injective, the range of δ lies in the radical exactly when $\delta = 0$ [2024-12, Theorem 2.7, p. 7].

Russell treats the automorphisms and derivations of continuous trace C^*-algebras together, which is where the inner derivations of such an algebra are classified [1980-1, Abstract, p. 1].

19.2 Kadison–Sakai: derivations of von Neumann algebras

Section 19.1 ended with the commutative C^*-algebras: a derivation of a commutative C^*-algebra is zero [1966-1, p. 280]. On that basis Kaplansky proved that every derivation of a type I von Neumann algebra is inner [1966-1, p. 280]. He proposed that every derivation of a C^*-algebra is continuous [1966-1, p. 280]. Sakai proved the continuity in 1960 [1960-1, p. 31]. Miles then noted that each derivation of a C^*-algebra is induced by an operator in the weak closure of some faithful representation [1966-2, pp. 273–274]. The remaining question was whether a von Neumann algebra can have an outer derivation at all [1966-2, p. 273]. The vocabulary is that of Sakai [1966-2, pp. 273–274].

Definition 19.9 ([1966-2, pp. 273–274])**.** Let A be a C^*-algebra and D a derivation of A. The derivation D is an *inner derivation* if there is an element a in A such that $D(x) = [a, x] = ax - xa$ for x in A. Otherwise it is an *outer derivation*.

A derivation of A is *spatial* when a bounded operator B on H implements it, so that $\delta(A) = BA - AB$ for each A in A [1966-1, p. 281]. Kadison began with the centre [1966-1, Theorem 2, p. 281]. The first step controls how a derivation behaves there [1966-1, Theorem 2, p. 281].

Theorem 19.10 ([1966-1, Theorem 2, p. 281])**.** *Each derivation of a C^*-algebra annihilates its center.*

Two facts drive the proof [1966-1, p. 281]. A pure state is multiplicative on the centre, and a derivation kills every element on which a definite state vanishes to second order [1966-1, p. 281]. With the centre out of the way, extension to the weak closure gives the spatial theorem [1966-1, p. 281].

Theorem 19.11 ([1966-1, Theorem 4, p. 283]). *Each derivation δ of a C^*-algebra A acting on the Hilbert space H is spatial. We may choose B commuting with an assigned maximal abelian subalgebra of A' or with an assigned mixing von Neumann subalgebra of A' so that* $\operatorname{ad} B \mid A = \delta$.

For hyperfinite von Neumann algebras the implementing operator already lies in the algebra.

Theorem 19.12 ([1966-1, Theorem 7, p. 285]). *Each derivation of a mixing von Neumann algebra is inner. In particular, each derivation of a hyperfinite von Neumann algebra is inner.*

Sakai then closed the question [1966-2, p. 274]. The argument takes a least element of a convex set in the double commutant [1966-1, p. 290]. The paper of Kadison presents that step as the completion of its own program [1966-1, p. 290].

Theorem 19.13 ([1966-2, Theorem 1, p. 274]). *Every derivation of a W^*-algebra is inner.*

Theorem 19.13 answers the conjecture of Kadison in the affirmative [1966-2, p. 274]. Kadison states the same result in the language of von Neumann algebras, and attributes it to Sakai, who completed the arguments of his paper [1966-1, Theorem 15, p. 290].

Theorem 19.14 ([1966-1, Theorem 15, p. 290]). *Each derivation of a von Neumann algebra is inner.*

A von Neumann algebra is a W^*-algebra [1991-7, p. 2], so the statement of Theorem 19.14 and the statement of Theorem 19.13 agree. Kadison also records the consequences for quantum physics [1966-1, p. 291]. Take a strongly continuous one-parameter unitary group whose spectrum is bounded below. If it induces automorphisms of a von Neumann algebra, the automorphisms are inner [1966-1, p. 291]. In physical language, energy and momentum are observable without the assumption of a vacuum state [1966-1, p. 291].

The C^*-algebra form places the implementing element in the weak closure [1966-2, Theorem 2, p. 274].

Theorem 19.15 ([1966-2, Theorem 2, p. 274]). *Let A be a C^*-algebra on a Hilbert space H, D a derivation on A, A^- the weak closure of A on H. Then there is a bounded operator b belonging to A^- such that $D(x) = [b, x]$ for all x in A.*

The theorems of this section lean on the automatic continuity of Sakai from 1960 [1966-1, p. 283]. The proof of Kadison for Theorem 19.11 opens by citing it for boundedness [1966-1, p. 283]. Continuity has a story of its own, told in Section 19.3.

A recent line of work drops the assumption that the map is linear at all [2025-15, p. 909]. The definition is due to Šemrl and is recalled by Liu in the form that follows [2025-15, Definition 1.2, p. 909].

Definition 19.16 ([2025-15, Definition 1.2, p. 909])**.** Let A be an algebra. A map $\delta\colon A \to A$, not assumed linear, is a *2-local derivation* when for all $A, B \in A$ there is a linear derivation $\delta_{A,B}\colon A \to A$, depending on the two elements, with

$$\delta(A)\,B + A\,\delta(B) = \delta_{A,B}(AB).$$

This definition generalizes the earlier one, in which only $\delta(A) = \delta_{A,B}(A)$ and $\delta(B) = \delta_{A,B}(B)$ are asked for [2025-15, p. 909]. Molnár proved that every 2-local derivation of a standard operator algebra into the bounded operators is a linear derivation [2025-15, p. 910]. He then asked whether the same holds on an arbitrary von Neumann algebra [2025-15, p. 910]. Liu answers the question on the semi-finite factors [2025-15, Theorem 2.1, p. 910].

Theorem 19.17 ([2025-15, Theorem 2.1, p. 910])**.** *Let A be a semi-finite factor von Neumann algebra, and let $\delta\colon A \to A$ be a map such that for all $A, B \in A$ there is a linear derivation $\delta_{A,B}\colon A \to A$ with $\delta(A)B + A\delta(B) = \delta_{A,B}(AB)$. Then δ is a linear derivation.*

The proof uses the trace [2025-15, p. 910]. Every linear derivation of such an algebra is inner [2025-15, p. 910], which is the theorem of Sakai stated below as Theorem 19.15. Take a trace-class element M. The inner form makes $\delta(A)M + A\delta(M)$ have trace zero [2025-15, p. 910]. Linearity then follows because the trace is faithful [2025-15, p. 910].

On von Neumann algebras the answer is now complete. Ayupov and Arzikulov work with the original notion of Šemrl, where the two values alone are matched [2011-3, p. 1]. For a von Neumann algebra every derivation is inner, so the condition reads as follows [2011-3, p. 2]. For each pair x, y of elements there is an element a of the algebra with $\Delta(x) = ax - xa$ and $\Delta(y) = ay - ya$ [2011-3, p. 2].

Theorem 19.18 ([2011-3, Theorem 5, p. 7])**.** *Let M be a von Neumann algebra of type I and let $\Delta\colon M \to M$ be a 2-local derivation. Then Δ is a derivation.*

The proof splits the algebra into its homogeneous summands, each of the form $C(X) \otimes M_n(\mathbb{C})$ [2011-3, p. 7]. On one summand the additivity follows from the matrix units [2011-3, Theorem 1, p. 3]. A Jordan derivation on a semiprime algebra is a derivation, so additivity is enough [2011-3, p. 3].

The same authors carry the result to matrix algebras over a semiprime Banach algebra [2014-4, p. 1]. An algebra has the inner derivation property when every derivation of it is inner [2014-4, p. 2].

Theorem 19.19 ([2014-4, Theorem 2.1, p. 2])**.** *Let A be a unital semiprime Banach algebra with the inner derivation property and let $M_{2^n}(A)$ be the algebra of $2^n \times 2^n$ matrices over A, with $n \geq 2$. Then every 2-local derivation of $M_{2^n}(A)$ is a derivation.*

The matrix case passes to AW*-algebras [2014-4, p. 11]. Every AW*-algebra has a central decomposition into summands that are matrix algebras over abelian ones, and the theorem applies to each summand [2014-4, p. 11]. So every 2-local derivation of an AW*-algebra is a derivation [2014-4, Theorem 3.1, p. 11]. Together with Theorem 19.18 this settles the question of Molnár for von Neumann algebras [2014-4, p. 2].

The local version, where each single value is matched by a derivation, was settled for von Neumann algebras earlier. Hadwin, Li, Li and Ma answer a question of Kadison on the algebra of measurable operators [2013-8, p. 1].

Theorem 19.20 ([2013-8, Theorem 1, p. 2])**.** *Let M be a von Neumann algebra on a Hilbert space H. If the abelian summand M_1 of M is discrete, then every local derivation on the algebra of closed densely defined operators affiliated with M is a derivation.*

The proof uses a separating family of central idempotents [2013-8, Lemma 2, p. 2]. A local derivation that is a derivation on each piece is a derivation on the whole [2013-8, p. 2]. On a minimal projection Q of a discrete abelian summand the algebra QM is one-dimensional, so the condition holds there [2013-8, p. 3].

The local derivations of an algebra need not be derivations one by one, but as a set they again form a Lie algebra. Hao and Chen prove this for every algebra over a field [2026-17, p. 1].

Theorem 19.21 ([2026-17, Theorem 2.8, p. 4])**.** *Let A be an algebra over a field. Then the set* $\mathrm{LocDer}(A)$ *of local derivations of A is a Lie algebra under the commutator bracket.*

The proof passes through a reflexive hull, a single object that covers the local derivations and the almost inner ones [2026-17, p. 1]. The same argument works for a family of multilinear operations, so the statement is not tied to the associative product [2026-17, p. 5]. The authors also prove the structural conjecture of Ayupov, Elduque and Kudaybergenov on local derivations [2026-17, p. 1].

A parallel line drops boundedness instead of linearity. For a type I von Neumann algebra M with center Z, let $LS(M)$ be the algebra of locally measurable operators affiliated with M [2007-1, p. 1]. Albeverio, Ayupov and Kudaybergenov ask for the derivations of $LS(M)$ that are linear over Z [2007-1, p. 1].

Theorem 19.22 ([2007-1, Theorem 3.9, p. 11])**.** *Let M be a type I von Neumann algebra with center Z, and let A be a subalgebra of $LS(M)$ that contains M. Then every Z-linear derivation on A is spatial and implemented by an element of $LS(M)$.*

The measurable operators and the totally measurable ones are such subalgebras, so their Z-linear derivations are spatial too [2007-1, Corollary 3.10, p. 11]. When the lattice of projections of M is atomic, every derivation of the τ-measurable operators is spatial, hence continuous in the measure topology [2007-1, Corollary 3.11, p. 11].

For the τ-compact operators the same authors give the full description [2008-2, p. 1]. Let $S_0(M,\tau)$ be that algebra.

Theorem 19.23 ([2008-2, Theorem 3.7, p. 10])**.** *Let M be a type I_∞ von Neumann algebra with a faithful normal semi-finite trace τ. Then every derivation of $S_0(M,\tau)$ is spatial and implemented by an element of $S(M,\tau)$.*

Outside type I_∞ the centre of $S_0(M,\tau)$ can be trivial, and the methods of the earlier papers do not apply [2008-2, p. 10].

The same circle of ideas returns for finite von Neumann algebras, where the algebras of affiliated operators carry the quantum-mechanical picture that opened this chapter [2014-13, p. 2087]. Kadison and Liu work with a finite von Neumann algebra R and its Murray–von Neumann algebra $A_f(R)$ of affiliated operators [2014-13, p. 2089]. They study the derivations of $A_f(R)$ that map R into R [2014-13, p. 2089].

Theorem 19.24 ([2014-13, Theorem 5, p. 2090])**.** *Let δ be an extended derivation of $A_f(R)$. Then there is an operator B in R such that $\delta(A) = A \cdot B - B \cdot A$ for every A in $A_f(R)$.*

The proof restricts δ to R, applies Theorem 19.13 to obtain B, and then shows that $\delta - \operatorname{ad} B$ kills all of $A_f(R)$ [2014-13, p. 2090]. For a factor of type II_1 the paper goes further: the only derivation of $A_f(M)$ with values in M is zero [2014-13, Theorem 12, p. 2091]. Kadison and Liu record that the full statement, that every derivation of $A_f(R)$ is inner, is still open [2014-13, p. 2090].

Nonlinear maps on a factor von Neumann algebra give another extension of the same theorem [2012-16, p. 1979]. Yu and Zhang drop linearity and the involution is twisted in the rule [2012-16, p. 1979].

Theorem 19.25 ([2012-16, Theorem 2.1, p. 1980])**.** *Let M be a factor von Neumann algebra on a complex Hilbert space H with $\dim H \geq 2$. Suppose that $\varphi\colon M \to M$ is a nonlinear $*$-Lie derivation, that is, $\varphi([A,B]^*) = [\varphi(A), B]^* + [A, \varphi(B)]^*$ for all A, B in M, where $[A,B]^* = AB - BA^*$. Then φ is an additive $*$-derivation.*

The proof splits each element over the two Peirce spaces of a projection and checks the derivation rule on the four blocks, then uses Theorem 19.13 for the inner form [2012-16, pp. 1990–1991].

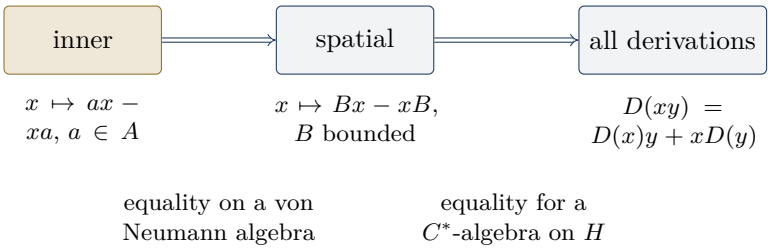


Figure 19.2: The derivation spaces of a C^*-algebra A on a Hilbert space H. An inner derivation $x \mapsto ax - xa$ has a in A, so it is spatial, and a spatial derivation is a derivation [1966-2, pp. 273–274]. Kadison proves that every derivation of a C^*-algebra on H is spatial [1966-1, Theorem 4, p. 283], and Sakai that every derivation of a von Neumann algebra is inner [1966-2, Theorem 1, p. 274], so on a von Neumann algebra all three spaces agree [1966-1, Theorem 15, p. 290].

19.3 Automatic continuity

Section 19.2 used one input at every step: a derivation of a C^*-algebra is continuous. Kaplansky had stated this as a conjecture, and Sakai proved it

in 1960 [1960-1, p. 31].

Theorem 19.26 ([1960-1, p. 31]). *Any derivation of a C^*-algebra is automatically continuous.*

The proof applies the closed graph theorem to the self-adjoint portion [1960-1, p. 31]. Johnson and Sinclair then widened the algebra but widened the map as well: they worked with derivations which are additive but not necessarily homogeneous [1968-3, p. 1067]. For such a map there is an obstruction, and it is finite dimensional.

Theorem 19.27 ([1968-3, Theorem 4.1, p. 1071]). *Let A be a semi-simple Banach algebra over the reals or the complexes. Let D be an additive derivation on A. Then A contains a central idempotent e such that eA and $(1-e)A$ are closed under D, the restriction of D to $(1-e)A$ is continuous, and eA is finite dimensional.*

The idempotent can vanish: examples built with Steinitz' theorem on algebraically closed field extensions give additive derivations of the complex numbers which are not real-linear [1968-3, p. 1072]. For a linear derivation the finite dimensional part is continuous anyway, so linearity removes the exception.

Corollary 19.28 ([1968-3, Remark 4.3, p. 1072]). *A linear derivation on a semi-simple Banach algebra over the reals or the complexes is continuous. If the algebra has no finite dimensional two-sided ideals, the same holds for every additive derivation.*

Corollary 19.28 answers the wider problem of Kaplansky for linear maps [1968-3, p. 1067]. Ringrose carried the question one step further, from the algebra to its modules [1972-5, p. 432]. A *Banach A-module* is a Banach space M with bounded bilinear maps sending (a, m) to am and to ma, associative in each triple product [1972-5, p. 432].

Theorem 19.29 ([1972-5, Theorem 2, p. 434]). *If A is a C^*-algebra and M is a Banach A-module, then every derivation of A into M is automatically norm continuous.*

For a dual module there is also continuity in the weak topologies.

Theorem 19.30 ([1972-5, Theorem 4, p. 435]). *For a faithful representation of a C^*-algebra A and a suitable dual normal A-module M, a derivation of A into M is continuous also relative to the ultraweak topology on A and the weak* topology on M.*

The theorems of this section say that boundedness comes for free on semisimple algebras [1968-3, Remark 4.3, p. 1072].

Bade and Curtis asked a wider question in 1974, with the module left free: which semisimple algebras and which modules make every derivation bounded [1974-1, p. 373]. Their answer separates the points at which continuity can fail. For a derivation D of a Banach algebra into a Banach module they define a left and a right continuity ideal [1974-1, Definition 2.1, p. 374].

Definition 19.31 ([1974-1, Definition 2.1, p. 374])**.** Let A be a Banach algebra, M a Banach A-module and $D\colon A \to M$ a derivation. Put $N(x) = \max(\|x\|, \|D(x)\|)$. Then I_L is the set of y in A for which the map $x \mapsto N(yx)$ is continuous on A, and I_R is the set of y for which $x \mapsto N(xy)$ is continuous.

Both sets are closed two-sided ideals [1974-1, Theorem 2.3, p. 375]. Their size controls the derivation: the socle of a semisimple Banach algebra lies in $I_L \cap I_R$ [1974-1, Theorem 2.7, pp. 378–379].

Corollary 19.32 ([1974-1, Corollary 2.8, p. 379])**.** *Let A be a Banach algebra with dense socle. Then $I_L = I_R = A$. If in addition A has a bounded left or right approximate identity, then every derivation of A is bounded.*

Corollary 19.32 covers the compact operators, which have a bounded approximate identity, and also the trace-class and Hilbert–Schmidt operators, which have a dense socle but no bounded approximate identity [1974-1, p. 379]. The general criterion counts sequences with $x_n x_m = 0$ and reads as follows [1974-1, Corollary 2.6, p. 378].

Corollary 19.33 ([1974-1, Corollary 2.6, p. 378])**.** *Let A be a semisimple Banach algebra. Suppose that every closed two-sided ideal of infinite codimension contains an infinite sequence x_n with $x_n x_m = 0$ for $n \neq m$ and $x_n^2 \notin K$, and that every closed two-sided ideal of finite codimension has a bounded left or right approximate identity. Then every derivation of A is bounded.*

The same paper reports discontinuous derivations on algebras that fail these conditions [1974-1, p. 375]. Take a Banach algebra A whose square A^2 is not closed [1974-1, p. 375]. Take a discontinuous linear functional f on A that vanishes on A^2. Take an A-module M with an element $m_0 \neq 0$ killed by A on both sides [1974-1, p. 375]. Then $D(x) = f(x)m_0$ is a discontinuous derivation. On the Hilbert–Schmidt operators the square is the trace-class ideal, so those operators carry a discontinuous derivation [1974-1, p. 375].

The continuity question for one-sided maps was settled the same way by Ali, for generalized left derivations [2011-2, p. 209].

Theorem 19.34 ([2011-2, Theorem 6.1, pp. 223–224])**.** *Let A be a semisimple Banach algebra and let $G\colon A \to A$ be a generalized Jordan left derivation with associated Jordan left derivation δ. Then G is continuous.*

The proof splits G into the two pieces of its definition [2011-2, p. 223]. If δ is zero, then G is a Jordan right multiplier and Lemma 2.9 of the paper makes it continuous [2011-2, p. 223]. If δ is nonzero, then δ is a derivation and Lemma 2.3 of the paper makes it continuous [2011-2, p. 223]. The difference $G - \delta$ is then a right centralizer, which is continuous by the same lemma [2011-2, p. 223].

A further class is that of the strongly generalized derivations. Hosseini and Park call a linear map $D\colon A \to B$ a strongly generalized derivation of order n when there are families of linear maps E_k, F_k, G_k, H_k, $k = 1, \ldots, n$, with $D(ab) = \sum_{k=1}^{n} [E_k(a)F_k(b) + G_k(a)H_k(b)]$ for all a, b [2023-13, Definition 2.1, p. 3]. For $n = 1$ the map is an (E, F, G, H)-derivation [2023-13, p. 3]. Every higher derivation is of this kind [2023-13, Example 2.3, p. 4].

Theorem 19.35 ([2023-13, Theorem 2.7, p. 5])**.** *Let A be a unital C^*-subalgebra of $B(H)$ and let $D\colon A \to B(H)$ be a $*$-(E, F, G, H)-derivation, so that D, E, F, G, H are $*$-maps. Suppose that $E(I) = H(I) = I$, where I is the identity map on H, that $H(a)D(I)E(b) = -E(a)D(I)H(b)$ for all $a, b \in A$, and that $E + H$ is D-continuous. Then D is automatically continuous. The same holds with F, G in place of E, H.*

Hosseini proves the same continuity for Banach algebras and simple Banach algebras [2023-13, Theorems 2.11–2.12, p. 7]. He later restricts the order to one and works into a Banach bimodule [2025-9, Theorem 2.1, p. 5]. That theorem extends the earlier results on generalized derivations [2025-9, p. 1].

Three uses of the term generalized derivation circulate in the literature, and Hosseini, Peralta and Su show that on a C^*-algebra they coincide [2025-8, p. 1]. The first type is a map F with a derivation d into the second dual M^{**} satisfying $F(ab) = F(a)b + ad(b)$ [2025-8, p. 1]. The second type is a map with an element ξ of M^{**} satisfying $F(ab) = F(a)b + aF(b) - a\xi b$ [2025-8, p. 1]. The third type is a map with two arbitrary maps G, H satisfying $F(ab) = G(a)b + aH(b)$ [2025-8, p. 1]. The third type is the weakest and needs no linearity in the two auxiliary maps [2025-8, p. 2].

Theorem 19.36 ([2025-8, Theorem 2.8, p. 10])**.** *Let A be a C^*-algebra and M a Banach A-bimodule. Let $F\colon A \to M$ be a generalized derivation of the third type. Then F is continuous. Consequently F is also a generalized derivation of the second type.*

The same paper treats a generalized Jordan derivation of the third type [2025-8, p. 5]. It is continuous on a C^*-algebra, and it is of the second type when the module is essential [2025-8, Theorem 3.1, p. 5].

Continuity also governs whether an inner derivation on a dense ideal extends. Let A be a Banach algebra and I a dense ideal of it [2026-29, p. 1]. Shafieasl and Tavakkoli ask whether every derivation $A \to I$ being inner forces every derivation $A \to A$ to be inner [2026-29, p. 1]. They answer no [2026-29, p. 1]. On a separable infinite-dimensional Hilbert space take $A = K(H)$, the compact operators, and $I = F(H)$, the finite-rank operators [2026-29, p. 3].

Theorem 19.37 ([2026-29, Theorem 4.1, p. 3])**.** *Let H be a separable infinite-dimensional Hilbert space, $A = K(H)$ and $I = F(H)$. Then every derivation $D\colon K(H) \to F(H)$ is inner and implemented by an element of $F(H)$, while $K(H)$ admits outer derivations.*

So the inner derivations into the small ideal do not reach the outer ones on the algebra. The same construction works for the Schatten p-classes [2026-29, Theorem 5.1, p. 4]. The authors check their main statements in the Lean proof assistant [2026-29, p. 1].

Chapter vocabulary of the next section is different: the derivations of quantum dynamics are densely defined and unbounded, as Section 19.4 reports.

19.4 Unbounded derivations and dynamics

Section 19.3 closed with continuity for free: an everywhere defined derivation of a C^*-algebra is bounded and weakly inner. Mathematical physics forces a wider notion: the derivations of quantum dynamics are defined as infinitesimal generators of one-parameter subgroups of $*$-automorphisms, and they are unbounded [1976-2, p. 427]. The basic example already sets the pattern [1976-2, pp. 427–428].

Example 19.38 ([1976-2, pp. 427–428])**.** Let $A = C_0(-\infty, \infty)$ be the C^*-algebra of all continuous functions on $(-\infty, \infty)$ vanishing at infinity. The only bounded derivation of A is the identically zero mapping. On the other

hand, the differential operator d/dt corresponding to the one-parameter subgroup of translation operators is an unbounded derivation of A. It cannot be approximated by the zero mapping in any reasonable sense.

The generator behind Example 19.38 has a general definition [1976-2, p. 428].

Definition 19.39 ([1976-2, p. 428])**.** Let $\rho(t)$, $-\infty < t < \infty$, be a strongly continuous one-parameter subgroup of $*$-automorphisms on a C^*-algebra A. The strong derivative of $\rho(t)$ at $t = 0$ is a closed, densely defined linear operator δ called the *infinitesimal generator* of the one-parameter subgroup: $\delta(a) = \lim_{t\to 0}(\rho(t)(a) - a)/t$, and the domain $D(\delta)$ consists of all elements a for which this limit exists.

An element a is *analytic* if the map $t \mapsto \rho(t)(a)$ is analytic, and the analytic elements form a dense linear subset of A [1976-2, p. 428]. Innerness survives in a restricted form on the uniformly hyperfinite algebras [1976-2, Theorem 2, p. 435].

Theorem 19.40 ([1976-2, Theorem 2, p. 435])**.** *Let A be a uniformly hyperfinite C^*-algebra and let δ be a derivation in A whose domain $D(\delta)$ is a dense $*$-subalgebra of A. Suppose that $D(\delta)$ contains a finite type I-subfactor F; then there is an element d in A such that $\delta(a) = [d, a]$ for all a in F.*

The whole theory became the book of Sakai on operator algebras in dynamical systems, devoted to unbounded derivations in C^*-algebras [1991-7]. The book also fixes the vocabulary used across this chapter [1991-7, p. 2]. A C^*-algebra is a *W^*-algebra* if it is a dual space as a Banach space. The W^*-algebras are exactly the von Neumann algebras [1991-7, p. 2].

One question stayed open [1981-4, p. 202]. As a quantization of spaces, Sakai asked whether one can find a simple C^*-algebra A and a non approximately bounded pregenerator δ_0 of A. Every $*$-derivation with the same domain should then be a scalar multiple of δ_0 plus an approximately bounded $*$-derivation [1981-4, p. 202]. Takai answered a two dimensional form of it on the irrational rotation algebra A_θ. There are two non approximately bounded pregenerators δ_1, δ_2 with the same domain. Each $*$-derivation of A_θ with that domain is a combination $k\delta_1 + l\delta_2$ plus an approximately bounded $*$-derivation [1981-4, p. 202].

Ismert studies the kernel of an unbounded derivation [2018-12, p. 1]. A derivation δ has kernel stabilization when $\ker \delta^n = \ker \delta$ for every n

[2018-12, p. 1]. She works with the weakly defined derivation δ_w of $B(H)$ that commutes with a self-adjoint operator D [2018-12, p. 1]. Its domain consists of the operators that map the domain of D into itself [2018-12, p. 3].

Theorem 19.41 ([2018-12, Theorem 3.5, p. 7])**.** *Let D be a self-adjoint operator on a Hilbert space H. Then for every $n \geq 1$ one has* $\ker(\delta_w^D)^n = \ker \delta_w^D$.

The proof compares the kernel with the commutant of D [2018-12, p. 7]. Two consequences follow [2018-12, p. 7]. The $*$-derivations of Bratteli and Robinson have kernel stabilization [2018-12, p. 8]. Also, a pair of operators that satisfies the Heisenberg commutation relation cannot both be bounded [2018-12, p. 8].

For a simple C^*-algebra Niknam connects two of the notions above, the separating space and the range, through the question of when a derivation can be closed [1983-5, p. 181]. He takes a derivation δ defined on a dense $*$-subalgebra of the algebra, and calls the set of limits of $\delta(a_n)$ along sequences $a_n \to 0$ its separating space. The derivation is closable exactly when that space is the zero subspace [1983-5, p. 181].

Theorem 19.42 ([1983-5, Theorem 1, p. 181])**.** *Let A be a simple C^*-algebra and let δ be a derivation of A. Then δ is closable if the range of δ is not dense in A.*

The proof shows that the separating space is a closed two-sided ideal, hence either $\{0\}$ or all of A, and the density assumption rules out the second case [1983-5, p. 182]. A $*$-derivation is closable as soon as one of the two sets $\{a \pm \delta(a)\}$ is not dense [1983-5, Corollary 2, p. 182]. On a fixed domain, a second $*$-derivation is closed exactly when it is bounded relative to the first [1983-5, Theorem 3, p. 182].

The next section returns to bounded maps on subalgebras of $B(H)$.

19.5 Derivations on operator spaces and nest algebras

Section 19.4 closed with an answer on one simple C^*-algebra. This section looks at subalgebras of the algebra of all bounded operators on a Hilbert space, and the key family is built from a chain of projections. The vocabulary is taken from the paper of Han Deguang [1993-8, p. 1165].

Definition 19.43 ([1993-8, p. 1165])**.** A *nest* N is a complete totally ordered family of selfadjoint projections on a Hilbert space H that contains 0 and I. The algebra $\operatorname{alg} N = \{T \in L(H) : P^{\perp}TP = 0 \text{ for all } P \in N\}$ is called the *nest algebra* with respect to the nest N.

The next definition states the derivations on a subalgebra of the bounded operators.

Definition 19.44 ([1993-8, p. 1165])**.** For a subalgebra W of $L(H)$, an additive (linear) map $D : W \to L(H)$ is said to be an *additive* (*linear*) *derivation* if $D(AB) = AD(B) + D(A)B$ holds for all A, B in W. The map D is said to be *inner* if there exists an operator T in $L(H)$ such that $D(A) = AT - TA$ holds for all A in W.

The linear case was settled by Christensen in 1977: if W is a nest algebra on H, then every linear derivation from W into $L(H)$ is inner [1993-8, p. 1165]. On the full algebra $L(H)$ the additive case follows from Section 19.3. The Johnson and Sinclair idempotent theorem, Theorem 19.27, implies that every additive derivation from $L(H)$ into $L(H)$ is inner, provided that H is infinite dimensional [1993-8, p. 1165].

Finite dimensions spoil the additive story. If $\dim H < \infty$, then there exist additive derivations of the nest algebra which are not inner [1993-8, p. 1165]. The theorem of Han closes the infinite dimensional case [1993-8, Theorem 2.5, p. 1168].

Theorem 19.45 ([1993-8, Theorem 2.5, p. 1168])**.** *Let N be a nest acting on an infinite-dimensional Hilbert space H. Then every additive derivation from* $\operatorname{alg} N$ *into $L(H)$ is inner.*

Theorem 19.45 extends a relative result for linear derivations of nest algebras [1993-8, p. 1165]. It also carries over to certain bimodules [1993-8, Corollary 2.6, p. 1169].

Corollary 19.46 ([1993-8, Corollary 2.6, p. 1169])**.** *Let N be a nest acting on an infinite-dimensional Hilbert space and X' an ultraweakly closed bimodule of W which contains the nest algebra. Then every additive derivation from X' into X is inner.*

The same nest algebras carry a wider class of maps that act like derivations on commutators. Take n elements $x_1, \ldots, x_n$ of an algebra and form

the $(n-1)$-th commutator $p_n(x_1, \ldots, x_n)$ [2017-16, p. 1]. A *Lie n-derivation* is a linear map L that satisfies

$$L(p_n(A_1, \ldots, A_n)) = \sum_{k=1}^{n} p_n(A_1, \ldots, L(A_k), \ldots, A_n)$$

for all $A_1, \ldots, A_n$ [2017-16, p. 1]. Zhang and Wei describe every such map on a nest algebra [2017-16, Theorem 3.1, p. 6].

Theorem 19.47 ([2017-16, Theorem 3.1, p. 6])**.** *Let X be a Banach space, let N be a nontrivial nest on X, and let $\operatorname{Alg} N$ be its nest algebra. Then every Lie-type derivation $L\colon \operatorname{Alg} N \to B(X)$ has the standard form: there are a derivation $D\colon \operatorname{Alg} N \to B(X)$ and a linear map $H\colon \operatorname{Alg} N \to \mathbb{C}I$ that vanishes on every $(n-1)$-th commutator, with $L = D + H$. If L is continuous, then D and H are continuous as well.*

So the Lie-type maps add only a scalar functional to the derivations of the previous theorems. On the Banach space, the scalar part is what makes the difference from the Hilbert space case [2017-16, p. 6].

Two later papers widen the question of Han in the same direction [1991-8, 2025-14, pp. 234, 81]. Šemrl drops linearity and keeps additivity, on a standard operator algebra [1991-8, p. 234]. He reduced every such map to one additive derivation of the base field [1991-8, p. 235].

Theorem 19.48 ([1991-8, Theorem 2.3, p. 236])**.** *Let X be an infinite-dimensional Hilbert space. Then every additive derivation $D\colon B(X) \to B(X)$ has the form $D(A) = AT - TA$ for some operator T in $B(X)$.*

The finite-dimensional case has a second term [1991-8, Theorem 2.2, p. 236]. On the $n \times n$ matrices over F, a map is an additive derivation if and only if it has two parts [1991-8, Theorem 2.2, p. 236]. The first is the inner derivation by a matrix (b_{ij}) [1991-8, Theorem 2.2, p. 236]. The second is the action of an additive derivation f of F on each matrix entry [1991-8, Theorem 2.2, p. 236]. Such an f exists for a transcendental extension, so finite dimensions carry additive derivations that are not inner [1991-8, p. 235].

Liu and Hou relax the rule instead of the linearity [2025-14, p. 81]. They ask for $\varphi([A, B]) = [\varphi(A), B] + [A, \varphi(B)]$ only on those pairs whose product equals a fixed operator F [2025-14, p. 81].

Theorem 19.49 ([2025-14, Theorem 2.2, p. 81])**.** *Let X be a Banach space of dimension greater than 2. Let F be an operator for which there*

is a nontrivial idempotent P with $F = PF$. Suppose that the linear map $\varphi\colon B(X) \to B(X)$ satisfies $\varphi([A,B]) = [\varphi(A), B] + [A, \varphi(B)]$ whenever $AB = F$. Then there are an operator T and a linear map $\gamma\colon B(X) \to \mathbb{C}I$ that vanishes on each commutator $[A,B]$ with $AB = F$, such that $\varphi(A) = AT - TA + \gamma(A)$ for all A.

With $F = 0$ the theorem returns the Lie derivable maps at zero, and with F a nontrivial idempotent it returns the maps derivable at an idempotent [2025-14, Remark 2.4, p. 81]. When the range of F is the whole space and complemented, F is a Lie all-derivable point of $B(X)$ [2025-14, Corollary 2.3, p. 81].

Inner derivations $A \mapsto AT - TA$ are commutator maps, and Section 19.6 studies what such a map can and cannot produce.

Park builds a Lie bracket on the space of derivation-derivations of a Banach algebra, so that the derivations themselves form a Lie algebra in a second way [2019-14, Abstract, p. 1].

19.6 Ranges of derivations and commutators

Section 19.5 ended with inner derivations, the maps $X \mapsto AX - XA$. This section asks what such a commutator map produces. The first answer is the Kleinecke–Shirokov theorem as Lin states it [1971-3, p. 529]. Take bounded linear operators P and Q on a Hilbert space, $T = PQ - QP$; if T commutes with P, then the spectrum of T is $\{0\}$ [1971-3, p. 529]. Lin relaxed the commutation of T with P to a compact error [1971-3, Theorem 1, p. 529].

Theorem 19.50 ([1971-3, Theorem 1, p. 529])**.** *If P and Q are bounded linear operators on a Banach space X, $T = PQ - QP$, and T almost commutes with P, that is, $PT - TP$ is a compact operator, then T is a Riesz operator. In particular, if X is a Hilbert space, then there exists a compact normal operator S such that $T - S$ is a quasinilpotent operator.*

The proof of Theorem 19.50 passes to the Calkin quotient $B(X)/K(X)$ and applies the Leibniz formula to the induced bounded derivation [1971-3, p. 529]. The same paper gives the Wintner–Wielandt–Halmos statement on scalars moved by compact operators.

Theorem 19.51 ([1971-3, Theorem 2, p. 530])**.** *If b is a nonzero scalar and S is a compact operator on a Banach space X, then $b + S$ is not a commutator.*

With S the zero operator, Theorem 19.51 is precisely the Wintner–Wielandt theorem: the only scalar commutator is 0 [1971-3, p. 530]. On the algebra $B(H)$ of bounded operators on a complex separable infinite-dimensional Hilbert space, every derivation is inner, $\delta = \delta_A$ with $\delta_A(X) = AX - XA$ [1971-4, p. 273]. Williams studied the range $R(\delta_A)$ of this map, starting from products of two derivations [1971-4, Theorem 1, pp. 273–274].

Theorem 19.52 ([1971-4, Theorem 1, pp. 273–274])**.** *Let $A, B \in B(H)$. Then $\delta_A\delta_B$ is a derivation if and only if A or B is a scalar multiple of the identity operator.*

Two corollaries of Theorem 19.52 pin down how small the commutant of the range is.

Corollary 19.53 ([1971-4, Corollaries 1 and 2, p. 275])**.** *If $A \in B(H)$ is not a scalar multiple of the identity, then the commutant of $R(\delta_A)$ is the scalars, $R(\delta_A)' = \mathbb{C}$. If A and B are not scalar multiples of the identity, then $R(\delta_A) \cap R(\delta_B)$ is nonzero.*

The second half of Corollary 19.53 was proposed by Joel Anderson [1971-4, p. 275]. Density is more delicate: Stampfli showed that $R(\delta_A)$ is never norm dense [1971-4, p. 275], and weak density has an exact answer.

Theorem 19.54 ([1971-4, Theorem 2, p. 275])**.** *Let $A \in B(H)$. Then the range of δ_A is dense in the weak operator topology if and only if the commutant $\{A\}'$ of A does not contain a nonzero operator of finite rank.*

Many familiar operators do lie in the range [1975-2, p. 495]. The proof is the Brown–Halmos–Pearcy lemma as corrected by Dash [1975-2, p. 495].

Lemma 19.55 ([1975-2, p. 495])**.** *If the matrix (a_{ij}), $i, j = 0, 1, \dots$ of an operator A on H with respect to an orthonormal basis $\{x_0, x_1, \dots\}$ is such that $a_{ij} = 0$ whenever $i + j$ is an even integer, then A is a commutator.*

For Lemma 19.55, with D the diagonal operator with diagonal $\{(-1)^n\}$, one has $DAD = -A$, hence $A = (-\frac{1}{2}DA)D - D(-\frac{1}{2}DA)$ [1975-2, p. 495]. It follows that every weighted unilateral shift and weighted bilateral shift is a commutator, and analogous results hold with operator weights [1975-2, p. 495].

Molnár studied the same range question for Jordan $*$-derivations instead of derivations [1994-4, p. 691]. On $B(H)$ each of those maps has the form $J_A(T) = TA - AT^*$ for a fixed operator A, and R_A denotes its range [1994-4,

p. 692]. The spectrum of the generating operator decides both questions about the range [1994-4, p. 692].

Theorem 19.56 ([1994-4, Theorem 1, p. 692])**.** *Let A in $B(H)$ be a normal operator. Then the following statements are equivalent: the closure of R_A is self-adjoint; R_A itself is self-adjoint; the spectrum of A lies in $\mathbb{R} \cup i\mathbb{R}$.*

The next theorem treats the case of a self-adjoint operator.

Theorem 19.57 ([1994-4, Theorem 2, p. 693])**.** *Let A in $B(H)$ be self-adjoint. Then R_A is closed in the operator norm topology if and only if 0 is not a limit point of the spectrum of A.*

Molnár adds that Theorem 19.57 fails for a normal operator that is not self-adjoint, with the bilateral shift as the example [1994-4, Remark 2, p. 694]. He leaves the normal case open [1994-4, p. 694].

The commutator can also be read with two different operators at its ends [2022-3, p. 145]. For the generalized commutator $X \mapsto AX - XB$ the range carries a symmetry of its own.

Definition 19.58 ([2022-3, Definition 3.2, p. 149])**.** Let A, B in $L(H)$ and let J be a two-sided ideal of $L(H)$. The pair (A, B) has the *Fuglede–Putnam property* $(FP)_J$ when $AT = TB$ with T in J implies $A^*T = TB^*$. The pair (A, B) is *P-symmetric* when $\overline{R(\delta_{A,B})}^{W^*} = \overline{R(\delta_{A^*,B^*})}^{W^*}$, where $\delta_{A,B}(X) = AX - XB$.

Bouhafsi, Ech-chad and Zouaki describe the P-symmetric pairs through that property [2022-3, p. 149].

Theorem 19.59 ([2022-3, Theorem 3.4, p. 149])**.** *Let A, B in $L(H)$. Then the pair (A, B) is P-symmetric if and only if (A, B) has $(FP)_{C_1(H)}$ and (B, A) has $(FP)_{C_1(H)}$.*

Two P-symmetric operators whose spectra do not meet form a P-symmetric pair [2022-3, Theorem 3.7, p. 151]. The paper also shows that a rationally cyclic operator that restricts a normal operator is D-symmetric. That means its range equals the range of its adjoint [2022-3, Theorem 3.12, p. 152].

Section 19.7 changes the question once more: it asks whether a map that almost satisfies the derivation identity is near an exact derivation.

19.7 Approximate derivations and the stability of functional equations

Section 19.6 studied exact derivations. This section asks whether a map that satisfies the equations only approximately is near a map that satisfies them exactly [1941-1, p. 222]. Ulam posed the question at the University of Wisconsin [1941-1, p. 222]. Let E and E' be Banach spaces and δ a positive number; a transformation f of E into E' with $\|f(x+y)-f(x)-f(y)\| < \delta$ for all x, y in E is called *δ-linear*. Must it stay within a fixed distance of a linear one [1941-1, p. 222]? Hyers proved that one always exists in 1941 [1941-1, Theorem 1, p. 223].

Theorem 19.60 ([1941-1, Theorem 1, p. 223])**.** *Let E and E' be Banach spaces and let $f(x)$ be a δ-linear transformation of E into E'. Then the limit $l(x) = \lim_{n\to\infty} f(2^n x)/2^n$ exists for each x in E, $l(x)$ is a linear transformation, and the inequality $\|f(x) - l(x)\| \le \delta$ is true for all x in E. Moreover $l(x)$ is the only linear transformation satisfying this inequality.*

Hyers added a continuity observation [1941-1, Theorem 2, p. 224].

Theorem 19.61 ([1941-1, Theorem 2, p. 224])**.** *If under the hypotheses of Theorem 19.60 we suppose that $f(x)$ is continuous at a single point y of E, then $l(x)$ is continuous everywhere in E.*

Theorem 19.60 is called the Hyers–Ulam stability of the additive Cauchy equation [2006-6, p. 524]. Later writers widened the control function, and Miura, Hirasawa and Takahasi brought the idea to ring derivations [2006-6, p. 524].

Definition 19.62 ([2006-6, p. 524])**.** Let A be a Banach algebra, $p \ge 0$ and $\varepsilon \ge 0$. A mapping $f : A \to A$ is an *approximate ring derivation* if $\|f(a+b) - f(a) - f(b)\| \le \varepsilon(\|a\|^p + \|b\|^p)$ for all a, b in A, and $\|f(ab) - af(b) - f(a)b\| \le \varepsilon\|a\|^p\|b\|^p$ for all a, b in A.

The history of such control functions runs through Rassias, with error $\varepsilon(\|x\|^p + \|y\|^p)$ for $0 \le p < 1$, and Gajda, who settled $1 < p$ and gave an example that stability fails at $p = 1$; Bourgin proved the superstability of ring homomorphisms, where the map is exactly a homomorphism [2006-6, pp. 524–526]. For derivations on general algebras, an approximate derivation need not be an exact one. On $C(X)$ over a compact Hausdorff space X without isolated points, the map $f(a)(x) = a(x)\log|a(x)|$ satisfies the derivation identity [2006-6, pp. 523–524]. It is a slight modification of an example

of Semrl whose idea is due to Johnson, and it differs from every additive map by an unbounded term [2006-6, pp. 523–524]. Superstability holds once the algebra has no zero divisor from either side [2006-6, Theorem 1, p. 526]. A Banach algebra is *without order* if no nonzero x_0 and y_0 satisfy $x_0 A = A y_0 = \{0\}$; semisimple commutative Banach algebras and Banach algebras with approximate identities are without order [2006-6, p. 526].

Theorem 19.63 ([2006-6, Theorem 1, p. 526])**.** *Suppose A is a Banach algebra without order. If $f : A \to A$ is a mapping satisfying the two inequalities of Definition 19.62 for some $\varepsilon \geq 0$ and $p \geq 0$, $p \neq 1$, then f is a ring derivation.*

Theorem 19.63 answers a question raised by Watanabe [2006-6, p. 526]. On functions it has a zero consequence [2006-6, Corollary 3, p. 526].

Corollary 19.64 ([2006-6, Corollary 3, p. 526])**.** *Suppose A is a semisimple commutative Banach algebra with the maximal ideal space M_A without isolated points. If $f : A \to A$ satisfies the two inequalities of Definition 19.62 for some $\varepsilon \geq 0$, $p \geq 0$, $p \neq 1$, then f is identically zero.*

Corollary 19.64 is the approximate form of the Singer–Wermer theorem of Section 19.1: where the exact theory gives only the zero derivation, the approximate theory gives only the zero map. The chapter closed as it opened, on commutative Banach algebras [2006-6, Corollary 3, p. 526].

The same question was carried to higher derivations and to the Jordan triple product [2009-10, p. 1975]. Roh, Jung and Chang control a sequence of maps by an inequality for the Jordan triple product xyx [2009-10, p. 1976]. The rule reads $f_n(xyx) = \sum_{i+j+k=n} f_i(x) f_j(y) f_k(x)$.

Theorem 19.65 ([2009-10, Theorem 2.6, p. 1984])**.** *Let A be a commutative semiprime Banach algebra. Suppose that a sequence $F = \{f_0, f_1, \ldots, f_m\}$ of continuous operators on A, with f_0 the identity, satisfies the inequalities of the paper for the Jordan triple rule. Then F maps A into its radical.*

On a commutative semisimple algebra the same hypotheses force the map to be zero [2009-10, Corollary 2.8, p. 1986]. This is the higher-order form of Corollary 19.64 [2009-10, p. 1986].

Continuity also enters the stability question directly [2014-16, p. 405]. König and Milman ask which operators on $C^k(\mathbb{R})$ satisfy the Leibniz rule up to a small error term, and the answer is a rigidity theorem [2014-16, p. 405]. The error term is a function B of the derivatives of order below k only [2014-16, p. 407].

Theorem 19.66 ([2014-16, Theorem 1, p. 407])**.** *Let k be a positive integer, $T\colon C^k(\mathbb{R}) \to C(\mathbb{R})$ an operator, and B a function such that $T(fg) = Tf \cdot g + f \cdot Tg + B(x, f, \ldots, f^{(k-1)}, g, \ldots, g^{(k-1)})$ for all f, g and all x. Suppose that T annihilates every polynomial of degree at most $k-1$. Then T is a multiple of the k-th derivative, $Tf = d\, f^{(k)}$ for a continuous function d, and B is the corresponding sum from the k-th order Leibniz rule.*

The theorem is a stability result for the k-th derivative [2014-16, p. 407]. For $k = 2$ on $\mathbb{R}^n$ the same authors describe the Laplacian by a similar equation [2014-16, p. 405]. Two further requirements enter there: the operator is unchanged under maps that preserve the inner product, and it kills the polynomials of lower order [2014-16, p. 405].

Two later works measure how far a derivation can reach. Kumar and Wirth use derivations as the tool for coarse rigidity of von Neumann algebras [2026-19, Abstract, p. 1]. Lupini compares the length a derivation needs with the length an automorphism needs on a unital C^*-algebra [2026-22, Abstract, p. 1]. Park, in the same setting, treats permuting triderivations together with the permuting trihomomorphisms they extend [2020-21, Abstract, p. 1].

Chapter 20

Hochschild cohomology and deformations

Chapter 1 placed derivations inside a cohomology theory. In 1945 Hochschild set out to connect the structure of an algebra to the vanishing of its cohomology groups [1945-1, p. 58]. His 1-cocycles are derivations, his 1-coboundaries are inner derivations, and the separable algebras are exactly those on which every derivation is inner. This chapter follows the theory from that point.

$$M \xrightarrow[u \mapsto (a \mapsto au - ua)]{\delta} \operatorname{Hom}(A, M) \xrightarrow{\delta} \operatorname{Hom}(A \otimes A, M)$$

$$\ker \delta = \text{derivations}, \quad \operatorname{im} \delta = \text{inner derivations}$$
$$H^1(A, M) = \operatorname{Der}(A, M) / \operatorname{Inn}(A, M)$$

Figure 20.1: The Hochschild complex in low degrees: derivations are the one-cocycles, inner derivations the one-coboundaries.

Section 20.1 gives the first group H^1 its modern form, derivations modulo inner derivations, and recalls why it controls the rest. Section 20.2 computes the theory for tensor algebras, commutative algebras and matrix rings. It also reports the theory built for an algebra together with a derivation. This covers the generalized derivation extensions of n-Lie algebras [2021-12, Proposition 2.13, pp. 25–26], the 3-LieDer pairs of Xu and Liu [2021-23, p. 1], and the non-abelian cohomology of Nijenhuis Lie algebras [2025-5,

Theorem 3.4, p. 6]. Section 20.6 reads the same rules as Hochschild and cyclic homology of a simplicial object, and compares them with de Rham cohomology. Section 20.3 turns to deformations, where H^2 and H^3 govern how an algebra can be changed. Section 20.7 then keeps a differential calculus when the functions no longer commute. Section 20.4 passes to operator algebras, where only bounded cochains are admitted and the theory becomes trivial. Section 20.5 closes the chapter with the parallel theory of groups, where crossed homomorphisms play the role of derivations.

20.1 The first Hochschild cohomology group

Chapter 1 defined the Hochschild complex for an associative algebra A over a field with coefficients in an A-bimodule M [1945-1, p. 60]. The definition follows Hochschild [1945-1, p. 60].

The complex has a one-cocycle and a one-coboundary [1945-1, p. 60]. A one-cocycle is a derivation of A into M, and a one-coboundary is an inner derivation [1945-1, p. 60]. So the first group measures how far the derivations are from being inner.

Weibel gives the modern statement of that measurement [1994-6, p. 301]. It gives the first group as the derivations modulo the inner ones, and this book uses that form.

Lemma 20.1 ([1994-6, Lemma 9.2.1, p. 301])**.** $H^1(R, M) = \mathrm{Der}_k(R, M)/\,\mathrm{PDer}_k(R, M)$, *the derivations of R into M modulo the principal derivations, where* PDer *denotes the principal, that is inner, derivations.*

Weibel notes a similarity here [1994-6, p. 301]. The low-dimensional Hochschild groups have the same shape as the cohomology of groups, and that is the situation of Theorem 20.16 below.

So Lemma 20.1 gives one half of a story. The other half is for groups, and Section 20.5 completes it.

The first group also controls the higher ones [1945-1, Theorem 3.1, p. 60]. Theorem 1.8 of Chapter 1 gives the reduction of Hochschild. For every bimodule M it produces a bimodule $(A : M)$ that raises the degree by one:

$$H^m(A, M) \cong H^{m-1}(A, (A : M)) \quad \text{for every } m \geq 2$$

[1945-1, Theorem 3.1, p. 60]. Two consequences follow from that reduction [1945-1, Theorem 3.1, p. 60].

The first consequence is that one group decides all the others, so the groups of dimension one determine the rest of the theory [1945-1, Theorem 3.1, p. 60].

The second consequence concerns the algebras where the theory vanishes [1945-1, p. 58 and § 4]. An algebra is separable exactly when every derivation of it into every bimodule is inner [1945-1, p. 58 and § 4]. That is the same as the vanishing of all the cohomology groups [1945-1, p. 58 and § 4], and it is the statement of Theorem 1.8 in Chapter 1.

The identification of Lemma 20.1 has a recent generalization [2023-10, Definition 2.1, p. 1]. Bennett, Heil and Laubacher work over a commutative triple, in which the second algebra is commutative and its map into the first one lands in the centre [2023-10, Definition 2.1, p. 1]. They then define a *secondary derivation* on that triple [2023-10, Definition 3.1, p. 5].

Definition 20.2 ([2023-10, Definition 3.1, p. 5])**.** A *secondary derivation* of the commutative triple $\mathcal{T} = (A, B, \varepsilon)$ with values in M is a k-linear map $D\colon B \otimes A \to M$ with the symmetric bimodule structure $(\alpha \otimes a) \cdot D(\beta \otimes b) = a\varepsilon(\alpha)D(\beta \otimes b)$ such that

1. $D(\lambda(\alpha \otimes a) + \mu(\beta \otimes b)) = \lambda D(\alpha \otimes a) + \mu D(\beta \otimes b)$,
2. $D((\alpha \otimes a)(\beta \otimes b)) = a\varepsilon(\alpha)D(\beta \otimes b) + D(\alpha \otimes a)b\varepsilon(\beta)$,
3. $D(\alpha \otimes 1) + D(\alpha \otimes 1) = D(1 \otimes \varepsilon(\alpha))$

for all $a, b \in A$, $\alpha, \beta \in B$ and $\lambda, \mu \in k$. The set of such secondary derivations is denoted $\mathrm{Der}_k(\mathcal{T}, M)$.

When $B = k$ this reduces to the ordinary derivations of A [2023-10, Remark 3.3, p. 5]. The recent theorem then reads as the analogue of Lemma 20.1.

Theorem 20.3 ([2023-10, Theorem 3.6, pp. 5–6])**.** *For a commutative triple* $\mathcal{T} = (A, B, \varepsilon)$ *one has* $HH^1(A, B, \varepsilon) \cong \mathrm{Der}_k(\mathcal{T}, A^*)$.

Theorem 20.3 keeps the shape of Lemma 20.1: the first group is again a set of derivations, and they now live over $B \otimes A$ rather than over A alone [2023-10, Theorem 3.6, pp. 5–6].

Further work in the cohomology line, held in the collection: *The Lie module structure on the Hochschild cohomology groups of monomial algebras with radical square zero* [2008-1]. *The BV-algebra structure of the Hochschild cohomology of the group ring of cyclic groups of prime order* [2016-8]. *Bounded Hochschild cohomology of Banach algebras with a matrix-like*

structure [2006-1]. *Gerstenhaber brackets on Hochschild cohomology of quantum symmetric algebras and their group extensions* [2016-4]. *Central extensions of the algebra of formal pseudo-differential symbols via Hochschild cohomology and quadratic symplectic Lie algebras* [2018-1]. *The first module σ τ -cohomology group of triangular Banach algebras of order three* [2018-4]. *The first and second Hochschild cohomology groups of Banach algebras with coefficients in special symmetric bimodules* [2020-4]. *On the cohomology of the spaces of differential operators acting on skewsymmetric tensor fields or on forms as modules of the Lie algebra of vector fields* [2004-1].

20.2 Computations for classes of algebras

The theory is computable on the standard classes, and Weibel collects the answers [1994-6, §9].

The tensor algebra is the first class, and its Hochschild homology vanishes above dimension one [1994-6, Proposition 9.1.6, pp. 303–304].

Proposition 20.4 ([1994-6, Proposition 9.1.6, pp. 303–304])**.** *Let $T = T(V)$ be the tensor algebra of a k-module V, and let M be a T-T bimodule. Then $H_i(T, M) = 0$ for $i \neq 0, 1$, and there is an exact sequence*

$$0 \longrightarrow H_1(T, M) \longrightarrow M \otimes V \xrightarrow{b} M \longrightarrow H_0(T, M) \longrightarrow 0,$$

where b is the map $b(m \otimes v) = mv - vm$.

Polynomial rings sit inside this class [1994-6, Exercise 9.1.3, p. 304]. For $R = k[x_1, \ldots, x_n]$ one has $H_i(R, M) = 0$ for $i > 0$ over a field [1994-6, Exercise 9.1.3, p. 304]. Weibel marks this as a special case of Theorem 20.6 below.

For a commutative algebra the first group is built from the differentials of Section 17.1.

Proposition 20.5 ([1994-6, Proposition 9.2.2, pp. 306–307])**.** *Let R be a commutative k-algebra, and let M be a right R-module made into an R-R bimodule by $rm = mr$. Then $H_0(R, M) \cong M$ and $H_1(R, M) \cong M \otimes_R \Omega_{R/k}$.*

The Hochschild–Kostant–Rosenberg theorem computes the whole graded theory for a smooth commutative algebra [1994-6, Theorem 9.4.7, pp. 322–323]. Its Hochschild homology is then the exterior algebra on the Kähler differentials.

Theorem 20.6 (Hochschild–Kostant–Rosenberg, [1994-6, Theorem 9.4.7, pp. 322–323])**.** *Let R be a commutative algebra, essentially of finite type over a field k. If R is smooth over k, then ψ is an isomorphism of graded R-algebras $H_*(R,R) \cong \Lambda^* \Omega_{R/k}$.*

The degree one case of Theorem 20.6 gives $H_1(R,R) \cong \Omega_{R/k}$ [1994-6, Theorem 9.4.7, p. 322]. That is Proposition 20.5 read on $M = R$, and it feeds the de Rham comparison of Section 20.6 [1994-6, Theorem 9.8.13, p. 351]. The derivation side of the same degree is the universal derivation of Section 17.1.

Morita equivalence leaves the theory unchanged [1994-6, Theorem 9.5.6, p. 325]. A theorem of R. K. Dennis says that this unchanged value holds across the equivalence [1994-6, Theorem 9.5.6, p. 325], and it comes next.

Theorem 20.7 (R. K. Dennis, [1994-6, Theorem 9.5.6, p. 325])**.** *Hochschild homology is Morita invariant: if R and S are Morita equivalent rings and M is an R-R bimodule, then $H_*(R,M) \cong H_*(S, Q \otimes_R M \otimes_R P)$.*

So the Hochschild theory of a matrix ring agrees with that of the ring itself [1994-6, Theorem 9.5.6, p. 325].

The theory also runs for an algebra together with a derivation, and the derivation then enters the complex. Ateşli, Esen and Sütlü build such a cohomology for an n-Lie algebra together with a representation [2021-12, p. 2]. Their complex carries the generalized derivation extensions, and for $n = 3$ it agrees with the known cohomology of 3-Lie algebras [2021-12, p. 1].

Theorem 20.8 ([2021-12, Proposition 2.13, pp. 25–26])**.** *Let L be an n-Lie algebra and let $D\colon L^{n-1} \to L$ satisfy the generalized derivation identity (2.10) of the cited paper, with $D^\sharp\colon L^{n-2} \to \mathfrak{gl}(L)$ the map assigned to it. Then $D^\sharp$ is a 1-cocycle of L with coefficients in $\mathfrak{gl}(L)$.*

So a generalized derivation is a cocycle in degree one, which is the statement the paper aims at [2021-12, p. 2]. The abelian extensions and the infinitesimal deformations are read from the usual cohomology of n-Lie algebras [2021-12, p. 1]. The paper also carries the Hochschild–Serre spectral sequence to n-Lie algebras [2021-12, p. 1].

For 3-Lie algebras Xu and Liu build the same theory from a 3-LieDer pair, a 3-Lie algebra with a distinguished derivation [2021-23, p. 1]. Their complex controls the deformations and the abelian extensions of the pair [2021-23, p. 1]. The degree -1 bracket on the complex makes it a graded Lie algebra, and its Maurer–Cartan elements are exactly the 3-LieDer pairs [2021-23, p. 2].

A non-abelian cohomology also exists for such pairs. Das works with a Nijenhuis Lie algebra, a Lie algebra with an operator N that satisfies $[N(x), N(y)] = N([N(x), y] + [x, N(y)] - N[x, y])$ [2025-5, Definition 2.1, p. 3]. The non-abelian second group of one such algebra with values in another classifies the non-abelian extensions [2025-5, p. 2].

Theorem 20.9 ([2025-5, Theorem 3.4, p. 6])**.** *Let* $(\mathfrak{g}, N)$ *and* $(\mathfrak{h}, S)$ *be two Nijenhuis Lie algebras. Then there is a bijection between the equivalence classes of non-abelian extensions of* $(\mathfrak{g}, N)$ *by* $(\mathfrak{h}, S)$ *and the non-abelian cohomology group* $H^2_{\mathrm{nab}}((\mathfrak{g}, N); (\mathfrak{h}, S))$*.*

With abelian values the statement specializes to the usual second group [2025-5, Theorem 3.7, p. 7]. Das then turns to derivations. Given an abelian extension of a Nijenhuis Lie algebra, a pair of Nijenhuis Lie algebra derivations can be lifted exactly when it is compatible and its Wells map vanishes [2025-5, Theorem 5.4, p. 19]. The Wells map here fits a short exact sequence [2025-5, Theorem 5.5, p. 20].

20.3 Gerstenhaber structure and deformations

The first group $H^1(A, A)$ is the group of infinitesimal automorphisms of A, that is the derivations modulo the inner ones [1964-1, pp. 59–60]. Gerstenhaber asked what the higher groups do, and his 1964 paper answers for the second one [1964-1, pp. 59–60]. The group $H^2(A, A)$ is the group of infinitesimal deformations [1964-1, pp. 59–60]. It behaves toward deformations of A as H^1 behaves toward automorphisms [1964-1, pp. 59–60]. From this he reads a rigidity criterion.

Gerstenhaber states the conclusion in his introduction [1964-1, p. 60]. The vanishing of $H^2(A, A)$ is a sufficient condition for the rigidity of A [1964-1, p. 60]. Whether it is also necessary is open [1964-1, p. 60]. It follows at once that separable semi-simple algebras are rigid [1964-1, pp. 59–60].

The deformation theory comes with obstructions [1964-1, §6]. An infinitesimal deformation need not integrate to a true deformation [1964-1, §6]. Gerstenhaber defines a cohomology operation, the square, mapping $H^1(A, A)$ into $H^2(A, A)$ [1964-1, §6]. The obstruction to integrating an infinitesimal deformation is a 3-cocycle [1964-1, pp. 60–61, §6]. He develops this in his section on the additivity and integrability of the square [1964-1, pp. 60–61, §6].

For fields the commutative theory gives a clean answer.

Theorem 20.10 ([1964-1, Theorem 3, p. 78])**.** *Let K be an extension of a field k. Then K is separable over k if and only if the second commutative cohomology group of K, considered as an algebra over k, with coefficients in itself vanishes.*

The higher Hochschild groups carry more structure than the groups alone suggest [1983-2, p. 144]. Gerstenhaber and Schack later identified a graded Lie product on $H^*(A, A)$, possibly non-abelian [1983-2, p. 144]. They found that it had gone unnoticed on the simplicial side, for a simple reason. Their main theorem identifies the simplicial cohomology of a locally finite complex C with the Hochschild cohomology of the algebra kC. It also preserves the cup product [1983-2, pp. 143–144].

Theorem 20.11 (SC = HC, [1983-2, pp. 143–144])**.** *Let k be a commutative unital ring. There is a functor $C \mapsto kC$ from locally finite simplicial complexes to associative unital k-algebras inducing an isomorphism $H^*(C, k) \cong H^*(kC, kC)$ between the simplicial cohomology of C and the Hochschild cohomology of kC, preserving the cup product.*

When the algebra is kC this graded Lie product is abelian, which explains why no such product had been recorded on $H^*(C, k)$ [1983-2, p. 144].

20.4 Kadison–Ringrose and bounded cohomology

For operator algebras the continuous structure matters [1971-2, pp. 227–228]. Kadison and Ringrose admit only bounded maps as cochains, and the bound is what changes the theory [1971-2, pp. 227–228]. These maps give cohomology groups, written

$$H^n_w(\mathcal{R}, \mathcal{M}),$$

and the letter w stands for weak [1971-2, pp. 227–228].

Their first theorem of this kind is a vanishing statement, and it comes next [1971-2, Theorem 4.4, p. 239]. On the type I von Neumann algebras the whole bounded theory vanishes [1971-2, Theorem 4.4, p. 239].

Theorem 20.12 ([1971-2, introduction and Theorem 4.4, pp. 227–228, 239])**.** *$H^n_w(\mathcal{R}, \mathcal{R}) = 0$ for every $n \geq 1$ when $\mathcal{R}$ is a type I von Neumann algebra.*

In degree one this is an older statement [1971-2, pp. 227–228]. A von Neumann algebra has no outer derivations [1971-2, pp. 227–228]. So

every derivation of it is inner [1971-2, pp. 227–228]. Read through the Hochschild theory, that says the first group vanishes [1971-2, pp. 227–228]. So Theorem 20.12 extends the vanishing of derivations to every degree.

The theory of Johnson for amenable algebras is the bounded analogue of the separability of Hochschild [1971-2, pp. 227–228]. The memoir that reports it is recorded in the bibliography [1972-4]. No copy of that memoir could be secured, so no statement of it is quoted here. A reader who has the memoir will find the bounded cohomology of an amenable algebra treated there.

20.5 Derivations of groups and the first cohomology of groups

Section 1.9 traced the identity $D(gh) = gD(h) + D(g)$ from Hilbert through Speiser to Eilenberg and Mac Lane. This section states the modern theory, in which a crossed homomorphism is a derivation of a group. Section 20.1 gave the ring half; this is the group half of the same story.

Definition 20.13 ([1994-6, Definition 6.4.1, p. 174])**.** A *derivation* (or *crossed homomorphism*) of a group G in a left G-module A is a set map $D\colon G \to A$ satisfying $D(gh) = gD(h) + D(g)$.

The principal derivations are the image of the elements of A [1994-6, Example 6.4.2, p. 174].

Example 20.14 ([1994-6, Example 6.4.2, p. 174])**.** For $a \in A$, the map $D_a(g) = ga - a$ is a derivation; these D_a are the principal derivations of G in A.

The derivations are computed by the augmentation ideal IG of the group ring [1994-6, Lemma 6.4.4, p. 175].

Lemma 20.15 ([1994-6, Lemma 6.4.4, p. 175])**.** *The map $\varphi \mapsto D_\varphi$ is a natural isomorphism of abelian groups* $\operatorname{Hom}_G(IG, A) \cong \operatorname{Der}(G, A)$.

Modulo the principal derivations, they give the first cohomology of the group [1994-6, Theorem 6.4.5, p. 175].

Theorem 20.16 ([1994-6, Theorem 6.4.5, p. 175])**.** $H^1(G; A) \cong \operatorname{Der}(G, A)/\operatorname{PDer}(G, A)$, *where* PDer *denotes the principal derivations.*

Theorem 20.16 is the group analogue of Lemma 20.1. One object carries three names drawn from three traditions: the principal derivation of a group, the principal crossed homomorphism of Section 1.9, and the inner derivation of a ring.

20.6 Algebraic topology: Hochschild and cyclic homology

The derivations of an algebra are one-cycles of a chain complex, the Hochschild complex of this chapter. The complex is itself a topological object: Hochschild homology is built from a simplicial object [1994-6, §9.6], and the extra structure of its cyclic permutation defines a second homology, the cyclic homology of Connes. Weibel recalls the computation for a smooth commutative algebra, where the answer is algebraic geometry.

The starting point is the shuffle product on the chains [1994-6, §9.4.1]. For a commutative k-algebra R, the Hochschild chain complex $C(R \otimes R^{\otimes *})$ is a graded-commutative differential graded k-algebra, and the Hochschild homology $H_*(R, R)$ is a graded-commutative k-algebra [1994-6, Proposition 9.4.2, p. 321]. Since $H_1(R, R)$ is isomorphic to the module $\Omega_{R/k}$ of Kähler differentials, the universal property of the exterior algebra extends the isomorphism to a graded R-algebra map $\psi\colon \Omega^*_{R/k} \to H_*(R, R)$ [1994-6, Corollary 9.4.4, p. 322]. For a smooth algebra the map is an isomorphism, which is Theorem 20.6 above.

The theorem is the Hochschild–Kostant–Rosenberg theorem [1994-6, p. 9.4.7]. Hochschild homology in low degrees already has the geometric answer: for $R = k[x_1, \ldots, x_n]$ one computes $H_p(R, R) \cong \Lambda^p \Omega_{R/k}$ for every p [1994-6, Exercise 9.1.3, p. 304].

The cyclic refinement adds one operator [1994-6, §9.8]. Weibel recalls the definition of Kassel.

Definition 20.17 ([1994-6, Definition 9.8.1, p. 344])**.** A *mixed complex* (M, b, B) in an abelian category $\mathcal{A}$ is a graded object $(M_m\colon m \geq 0)$ endowed with two families of morphisms $b\colon M_m \to M_{m-1}$ and $B\colon M_m \to M_{m+1}$ such that $b^2 = B^2 = bB + Bb = 0$. Thus a mixed complex is both a chain and a cochain complex.

A cyclic object A, in particular the simplicial object $A_n = R \otimes R^{\otimes n}$ of a k-algebra R, gives rise to a mixed complex (A, b, B) [1994-6, §9.8, p. 345]. Here b is the Hochschild differential and B is Connes' operator. The *cyclic homology* $HC_*(A)$ is the homology of the total complex of

the bicomplex built from the pair, the double complex of Tsygan [1994-6, Definition 9.6.7, p. 334]. When R is commutative and $\mathbb{Q} \subset R$, the projection $\varepsilon(r_0 \otimes \cdots \otimes r_n) = r_0\, dr_1 \wedge \cdots \wedge dr_n/n!$ is a morphism of mixed complexes from $R^{\otimes *+1}$ to the de Rham complex $(\Omega^*_{R/k}, 0, d)$ [1994-6, 9.8.12, p. 351], and one has the following theorem.

Theorem 20.18 ([1994-6, Theorem 9.8.13, p. 351])**.** *If R is a smooth commutative algebra, essentially of finite type over a field k of characteristic 0, then ε induces natural isomorphisms on Hochschild homology HH_*, cyclic homology HC_* and periodic cyclic homology HP_*.*

The target of the comparison is de Rham cohomology, the homology of the complex $(\Omega^*_{R/k}, 0, d)$ [1994-6, 9.8.9, p. 350]. In characteristic zero, Lemma 9.6.10 of the same chapter computes cyclic homology as the homology of the quotient of the Hochschild complex by the cyclic action [1994-6, Lemma 9.6.10, p. 335]. The cyclic refinement is the topological reading of the same rule.

20.7 Noncommutative geometry and differential calculi

The differential calculi of this chapter start from functions that commute. Noncommutative geometry drops that assumption. The derivations themselves pay for it. The derivations of a noncommutative ring A fail to form an A-module [2007-11, Abstract, p. 1].

The Q-valued derivation of a noncommutative K-ring A is a K-module morphism $u\colon A \to Q$ with the Leibniz rule $u(ab) = u(a)b + au(b)$ [2007-11, p. 1]. Several definitions of a differential operator compete in this setting, and none of them agrees with another [2007-11, Abstract, p. 1]. The commutative side of the comparison is Theorem 12.23.

Remark 20.19 ([2007-11, Abstract, p. 1])*.* In contrast with differential operators on modules over commutative and graded commutative rings, there is no satisfactory notion of a differential operator in noncommutative geometry [2007-11, Abstract, p. 1].

The Chevalley–Eilenberg differential calculus still provides an important example, and on smooth functions it is the de Rham complex [2007-11, p. 1]. This is the smooth thread of Section 20.2 kept alive.

Ginzburg and Schedler give one answer on the operator side [2010-4, Abstract, p. 673]. They replace the derivations of a commutative algebra

by the bimodule $\mathbb{D}\mathrm{er}(A)$ of *double derivations* of an associative algebra A. Their operators act on a Fock space $\mathcal{F}(A)$ built from A [2010-4, Abstract, p. 673].

Remark 20.20 ([2010-4, Abstract, p. 673]). The algebra $\mathcal{D}(\mathcal{F}(A))$ of Ginzburg and Schedler is filtered, and its associated graded algebra is commutative in a wheeled sense. For a smooth algebra A it satisfies $\operatorname{gr}\mathcal{D}(\mathcal{F}(A)) \cong \mathcal{F}(T^A(\mathbb{D}\mathrm{er}(A)))$. A Ricci-flat, torsion-free bimodule connection on $\mathbb{D}\mathrm{er}(A)$ gives a second-order wheeled differential operator, a noncommutative analogue of the Koszul–Batalin–Vilkovisky delta operator [2010-4, Abstract, p. 673].

The derivation therefore survives in noncommutative geometry three times over. As an object of study it loses its module structure [2007-11, Abstract, p. 1]. As a double derivation it becomes the building block of a new calculus [2010-4, Abstract, p. 673]. On smooth functions it also keeps the de Rham calculus of the manifold [2007-11, p. 1].

Part VIII

Appendices

Appendix A

Atlas of the literature

The bibliography printed with this edition holds works from 1684 to this year. This appendix describes what that collection looks like, so that a reader can see the shape of the field at a glance. The counts cover the works the text cites. They do not measure the field itself, which is larger. They measure the part of it this edition has read.

A.1 Works by decade

The collection is thin before 1900 and grows through the twentieth century. Table A.1 gives the exact count, and Figure A.1 draws it.

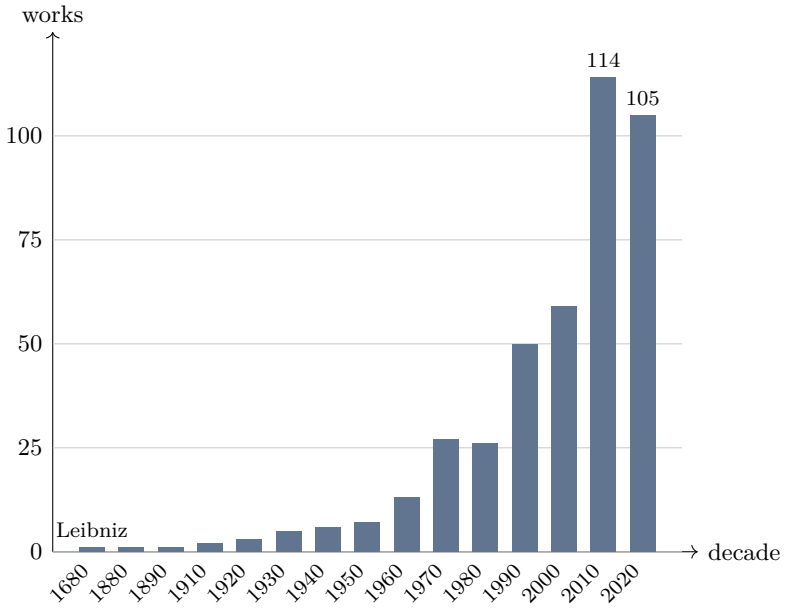


Figure A.1: Works per decade. The single 1680s entry is Leibniz, where the product rule is stated; the 2010s is the peak at 114.

Three things in the figure are worth naming. The first bar is alone for two centuries: Leibniz is the only work this edition cites from the 1600s, and nothing follows until the 1880s. The figure then rises unevenly, doubling about every two decades after 1950. And the last two bars are nearly equal, which is what a decade still in progress looks like.

Table A.1: Works in the bibliography, by decade of publication.

Decade	Works	Decade	Works
1680s	1	1960s	13
1880s	1	1970s	27
1890s	1	1980s	26
1910s	2	1990s	50
1920s	3	2000s	59
1930s	5	2011s	114
1940s	6	2020s	105
1950s	7		

Three features stand out. The first is the single entry in the 1680s, the work of Leibniz where the product rule is stated. The second is the slow start: fewer than ten works a decade before 1950. The third is the recent weight. Of the 517 works, 360 appeared after 2000, and the 2010s alone hold more works than every decade before 1950 together.

The 2020s count is close behind the 2010s, and the decade is not over.

A.2 Works by language

Table A.2 gives the language of the collection, and Figure A.2 shows when the collection changed language.

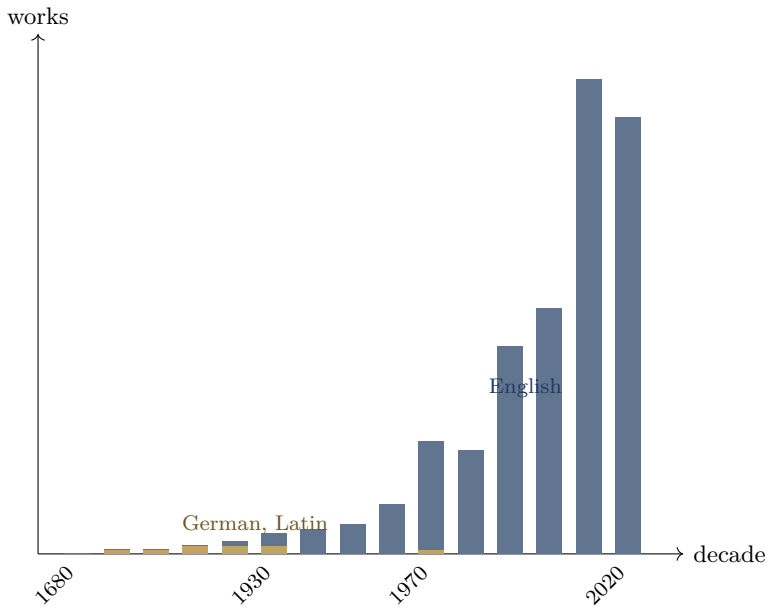


Figure A.2: Works per decade by language. The gold stratum is German and Latin, which dominates before 1940; the blue is English, which is nearly all of the collection after it.

The two strata barely overlap. German and Latin carry the subject to 1940, with one late German work in the 1970s; English takes over in the 1930s and is the whole collection from 1990. The two French works and the single Latin one sit at the ends of that movement rather than outside it.

Table A.2: Works in the bibliography, by language.

Language	Works
English	408
German	9
French	2
Latin	1

Most of the collection is in English. The German and Latin works are the older ones, from the period when the subject was written in those languages, and the two French works are cited for the same reason. That distribution follows the history in Chapter 1, where the thread passes from Latin to German and then to English as the centres of the subject move.

A.3 Kinds of derivation-like map

The kind census holds 200 rows. Twenty-seven of them are tokenizer fragments rather than names, and they are listed with their reason in the kind ledger of the book, so the book names 173 kinds of derivation-like map. A number in that table or in the index of kinds is the number of works in this atlas whose title names the kind. A title that names three kinds counts once under each of the three, so the numbers do not add up to a number of works. Table A.3 gives the eight that the literature treats most.

Table A.3: The eight most studied kinds, with the number of works in the census that name each.

Kind	Works
generalized derivation	592
skew derivation	179
higher derivation	145
Jordan derivation	135
Lie derivation	107
triple derivation	98
nilpotent derivation	85
$*$-derivation	82

The ordering is not a ranking of importance. It reflects how the literature itself distributes its attention, and it explains the length of the chapters in

this book. Generalized derivations, which include the skew and the twisted cases, occupy Chapter 6 for that reason. Local and 2-local derivations, which ask when a map that is a derivation at each point is a derivation everywhere, occupy Chapter 8.

A.4 Kinds by chapter

Table A.4 groups the same census by the chapter that treats each kind.

Table A.4: Works in the kind census, grouped by the chapter that treats the kind.

Chapter	Works	Chapter	Works
Generalized and twisted	993	Basic theory	30
Prime and semiprime	306	Polynomial and Weyl	24
Banach and operator	301	Matrix and triangular	13
Lie-type	230	Functional identities	13
Other systems	184	Unclassified	12
Higher and multi	181	Cohomology	5
Local maps	176	Origins	3
Commutative	163	Group and Hopf	2
Jordan-type	144		
Nonassociative	136		
Lie algebras	66		
Applications	37		

The two largest groups are the generalized and the prime-ring kinds. Together they hold 1,299 of the 3,019 named occurrences in the census, which is 43 per cent of it. A work can name more than one kind, so the census counts occurrences rather than works, and its total is larger than the number of works in the bibliography. The smallest are the group and Hopf kinds, with 2 works, which are the most recent arrivals in the literature and which Chapter 13 treats as an open frontier.

A.5 How the collection has grown

Figure A.3 draws the same dates cumulatively, which is the clearer way to see the shape.

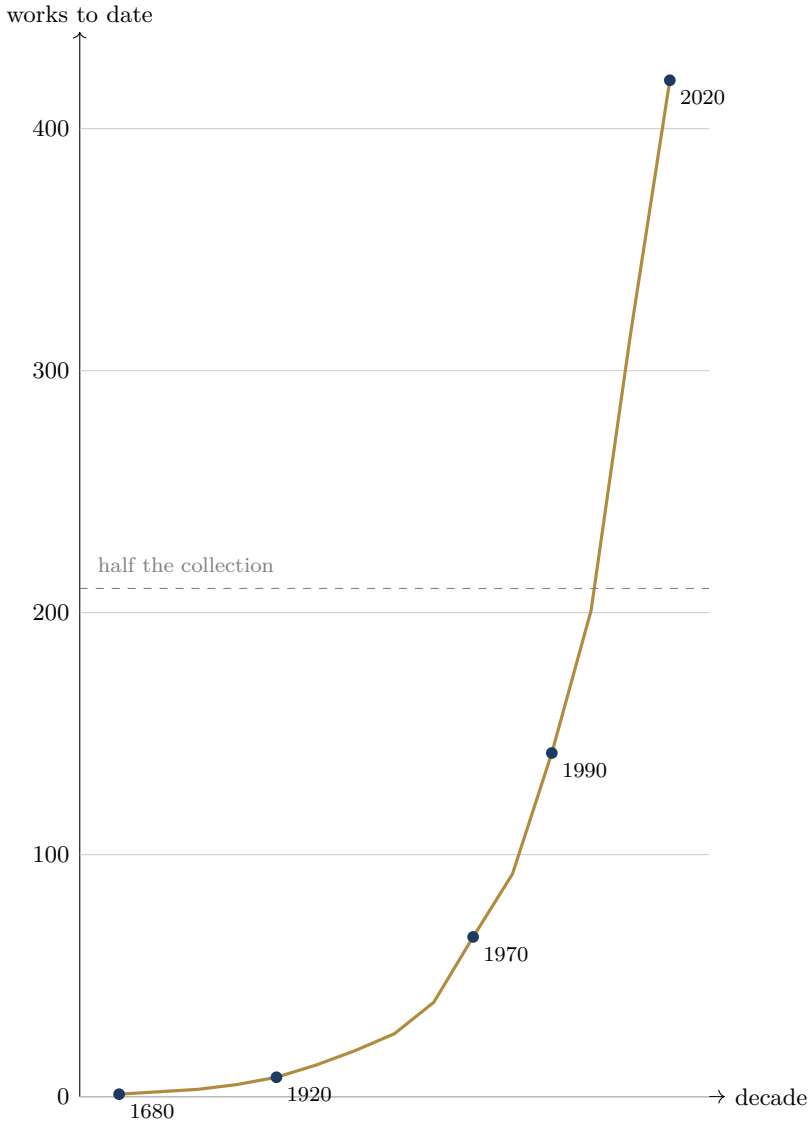


Figure A.3: Works in the collection up to each decade. The curve is flat for two centuries, then steep; the halfway mark falls at the end of the 2000s.

The curve has an elbow, and the elbow is the point of the section. Half of the bibliography was published after the median year, so this edition has read mostly recent work. That is a choice the subject forces: the questions collected in Section 2.10 are being worked on now, and a encyclopaedia

that described only settled work would miss them. The cost is visible in the same curve, which barely leaves the axis before 1950.

The dates in the bibliography describe a subject that grew slowly and then quickly. Table A.5 gives four measures of that growth.

Table A.5: Four measures of the collection, computed from the dates in the bibliography.

Measure	Value
Works with a date	517 of 517
Earliest and latest	1684 and 2026
Median year	2012
Mean year	1998
Works published after 2000	360
Works published before 1950	19
Works published since 2010	219 (half the collection)

The mean lies eight years below the median, so the distribution has a long early tail: a few very old works pull the mean back without moving the middle. The median year of 2010 means that half the collection was published in the last twenty years. The peak decade is the 2010s with 114 works, and the three most productive decades hold 278 of the 420 works, which is 66 per cent of them.

Every entry carries a year, so the table describes the whole collection and no correction is needed for undated works.

A.6 How the atlas grows

The collection is not fixed. Each edition adds the works read since the last one, and the counts above are counted again when it does. A work enters the bibliography only when its text has been read from the page, so the atlas records the reading, not the field.

The counts are produced by one program from two records, the bibliography and the kind census, so a new download that adds a work changes the tables at the next rebuild without any hand editing. That is what makes the appendix a living part of the edition rather than a snapshot taken once.

The gap between the collection and the field is the work of later editions, and Appendix B lists the questions that gap leaves open.

Appendix B

Question tables

Section 2.9 fixed eight questions that the book asks of every class of algebra. The chapters answer them one class at a time. This appendix collects the answers in one place, family by family. An answer is one of three: known, with the theorem that settles it; false, with a counterexample; or open, with the source that states it as a question. Each open row cites the problem number, and the problem is stated in full in the section that asks its question.

This chapter collects the questions that no chapter of the book could answer. Section 2.9 fixed eight questions for the book to ask of every class of algebra. It asked for one of three answers: known, false with a counterexample, or open. The open answers are collected here, so that a reader can see the frontier of the subject in one place.

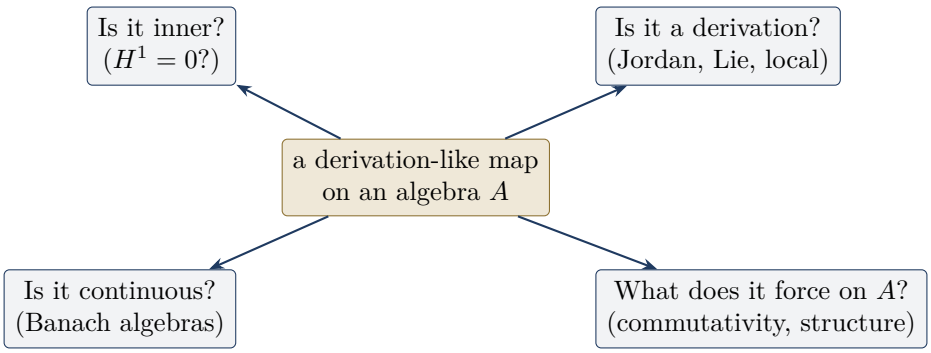


Figure B.1: The question families that organise the open problems.

Most of these problems have a named author, and the chapters met

them one at a time. The Zariski–Lipman conjecture carries the names of Zariski and of Lipman, and Lipman proved that a free module of derivations forces the local ring to be normal [2014-9, p. 270]. The Jacobian conjecture stands behind the theory of locally nilpotent derivations, and it can be stated in the language of derivations [2017-7, p. 81]. The conjecture of Dixmier has been open since 1968 for every dimension [2017-3, pp. 1–2]. Larson asks which algebras have a reflexive space of derivations [2004-5, p. 703]. Šemrl introduced 2-local derivations in 1997 and asked when such a map is a derivation [2020-7, p. 1]. Singer and Wermer proved their radical theorem in 1955 and at the same time stated the conjecture that continuity is not needed [2002-3, p. 710]. Martindale studied the Lie derivations of a primitive ring in 1964 and said that his result falls far short of the conjecture of Herstein [1964-2, p. 183].

Each of those answers holds for the moment the source states it. A problem can move from open to known over decades.

Example B.1 ([2002-3, p. 710])**.** Singer and Wermer proved in 1955 that a continuous derivation on a commutative Banach algebra takes its values in the radical. At the same time they stated the conjecture that the continuity hypothesis is superfluous. Johnson proved that conjecture for semisimple algebras in 1969. Thomas settled it for every commutative Banach algebra more than thirty years later [2002-3, p. 710]. The noncommutative form of the same question is still open, and Problem 19.7 states it.

Each open problem is stated in the chapter that asks it, with the page where the literature makes it. This appendix gathers them instead of restating them: every row names the problem and the chapter that carries the discussion. A problem that a chapter settles for one class of algebras is recorded in Section 2.10 under that class. The list is a snapshot of the book as it now stands.

The problems the chapters state, in the order that Section 2.9 fixed: the Zariski–Lipman problem, the Jacobian and Dixmier problems, and the questions of polynomial rings about finite generation; then the question of Larson on reflexivity, the 2-local problem, the noncommutative Singer–Wermer conjecture, the Lie problem of Herstein, and the four problems the chapters of the book raise. Section 2.10 sorts the same list by where the algebra lives: polynomial algebras, finite-dimensional algebras, C^*-algebras, Lie algebras, prime rings and triangular algebras. For each class it records which problem is settled there and which is not.

B.1 Family 1: which kind implies which

The book records the implications between kinds of derivation-like map. Table B.1 gives the ones the chapters settle.

Table B.1: Answers for the family "which kind of derivation implies which".

Class of algebra	Answer	Source
Triangular algebra	f-derivations are not all determined; the question of which f admit only inner ones is open	Problem 6.66
Prime ring, characteristic not 2	symmetric biderivations: the expected description fails without a further hypothesis	open, Problem 7.24
Semiprime ring	additive maps with $[f(x), x]_n = 0$: the commuting case is settled, the general one is not	open, Problem 9.3

B.2 Family 2: when two definitions are equivalent

Section 2.8 named the kinds that carry more than one definition in the literature. Where the definitions agree, a chapter says so and cites both sources; where they do not, the row below records it.

B.3 Family 3: is every derivation inner

This is the question of H^1, and the book answers it class by class.

B.4 Family 4: is every derivation continuous

Continuity is the question of this family, and it is settled for some classes and open for others.

Table B.2: Answers for the family "when are two definitions equivalent".

Class of algebra	Answer	Source
Rings with an f	three notions carry the name f-derivation; the one used here is fixed by a defining identity	the census of Section 2.8

Table B.3: Answers for the family "is every derivation inner".

Class of algebra	Answer	Source
Central simple algebra	known: every derivation into a bimodule is inner	Theorem 2.11
Simple ring	Lie derivations: whether each is a derivation plus a map into the centre is open	Problem 5.5
Local ring, characteristic 0	known for a free module of derivations; the general case is the Zariski–Lipman conjecture	open, Problem 17.7
Projective surface	the logarithmic case is open	Problem 17.8

Table B.4: Answers for the family "is every derivation continuous".

Class of algebra	Answer	Source
Banach algebra	whether a derivation with no continuity assumed leaves every primitive ideal invariant is open	Problem 19.7
Algebra of bounded operators	whether $\mathrm{der}(A)$ is reflexive is open for general A	Problem 8.6
Lie algebra, characteristic 0	whether every 2-local derivation is a derivation is open in infinite dimension	Problem 8.14

B.5 Family 5: does the algebra have to be commutative

the theorem of Posner and the functional identities answer this for prime rings. The noncommutative cases that remain open are collected here.

Table B.5: Answers for the family "does the algebra have to be commutative".

Class of algebra	Answer	Source
Prime ring	known: a nonzero derivation d with $ad(a) - d(a)a$ central for every a forces the ring to be commutative	Theorem 3.2
C^*-algebra	Jordan derivations into a bimodule: the general case is open	Problem 4.35

B.6 Family 6: can all derivations be described

A class whose derivations can be listed has a structure theorem. The classes where the list is still unknown carry the open rows.

Table B.6: Answers for the family “can all derivations of the class be described”.

Class of algebra	Answer	Source
Polynomial ring	locally nilpotent derivations: the homogeneous case with $D^2 x_i = 0$ is open beyond the known dimensions	Problem 17.21
Weyl algebra	whether every endomorphism is an automorphism, the Dixmier conjecture, is open since 1968	Problem 12.22
Polynomial map, characteristic 0	the Jacobian conjecture is open	Problem 17.26

B.7 Family 7: does a derivation extend or lift

Section 2.7 fixed this question. The rows below record where the extension question is still open.

Table B.7: Answers for the family “does a derivation extend or lift to a larger algebra”.

Class of algebra	Answer	Source
Unipotent group acting on affine space	whether the action can be triangularized, the question of Bass, is open	Problem 17.22

B.8 Family 8: where does an expected implication fail

A counterexample answers this family. The counterexamples the book records are in the chapters named below, and the one open case is listed last.

Table B.8: Answers for the family "where does an expected implication fail".

Class of algebra	Answer	Source
Prime ring, prime characteristic	false: the Zariski–Lipman statement fails	Chapter 17
Simple Lie algebra, characteristic 0	false: outer derivations exist where the characteristic-zero answer predicts none	Chapter 14

B.9 Reading the tables

Every row above names a chapter or a problem number, and each of those carries the located citation in its own place. No row is a summary of a summary: a reader who wants the statement turns to the chapter or the problem and finds it there, with its page. The tables are an index to those statements, and they are refreshed with each edition as the chapters change.

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